SHARP LOCAL CYCLICITY OF THE I_2^1 GRAPHIC IN A FIVE-PARAMETER QUADRATIC FAMILY

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Abstract. The DRR graphic I_2^1 is a loop through a nilpotent saddle–node at infinity. We prove that its local cyclicity is exactly three in a five-parameter quadratic family, for every fixed organizing value $C_0 > 1$. The upper bound holds throughout a full parameter neighborhood and counts periodic orbits close to the whole graphic in Hausdorff distance. We parameterize both halves of the singular passage by one central height and compare their endpoint derivatives after the exterior transition. Drift-weighted errors preserve this comparison when boundary roots merge or passage times diverge. With $C = C_0$ fixed, direct action matching and a transverse variation of the connection splitting yield three limit cycles converging to the whole graphic. They are hyperbolic and persist on open parameter sets approaching the organizing value. Thus the local upper bound is attained within the same quadratic family.

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1. Local cyclicity of I_2^1

The Dumortier–Roussarie–Rousseau program reduces the finiteness problem for quadratic limit cycles to local cyclicity questions for specified limit-periodic sets [4]. We study the loop I_2^1 through a nilpotent saddle–node at infinity. Our aim is to count the cycles approaching the entire loop, including parameter directions in which boundary roots merge or coefficient scales vanish independently, and to attain the resulting bound in the original family.

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Fix $C_0 > 1$. We consider the five-parameter quadratic family

$$\begin{aligned} \dot{x} &= Ax - y + x^2 + (\mu_2 + \mu_3)xy + \mu_1 y^2, \\ \dot{y} &= Cx + x^2 + xy + \mu_3 y^2 \end{aligned} \tag{1.1}$$

near the parameter

$$(A, C, \mu_1, \mu_2, \mu_3) = (1, C_0, 0, 0, 0). \tag{1.2}$$

At this parameter the invariant parabola

$$y = (x^2 - C_0)/2 \tag{1.3}$$

and its resolved nilpotent saddle–node passage at infinity form the directed graphic $\Gamma(C_0)$, denoted I_2^1 in the DRR classification [4, Proposition 4.2.2(6), Figure 8(6)] (see Section 2 and Figure 1).

We measure proximity to this graphic on the Poincaré sphere $\mathbb{S}^2$, where the Poincaré compactification of (1.1) is an analytic vector field. Let d_{H} denote Hausdorff distance for a fixed metric on $\mathbb{S}^2$. When resolved or scaled coordinates are used below, the sets are projected back to $\mathbb{S}^2$ before their Hausdorff distance is taken.

For $\varepsilon > 0$ and a neighborhood $V \subset \mathbb{R}^5$ of the organizing parameter (1.2), let $N(V, \varepsilon)$ be the supremum over $p \in V$ of the number of limit cycles γ of (1.1) with parameter p such that $d_{\mathrm{H}}(\overline{\gamma}, \Gamma(C_0)) < \varepsilon$. Following the Hausdorff formulation of cyclicity in [4, Definition 2.2(2)] (see also [10, §2.2.1]), the cyclicity of $\Gamma(C_0)$ in the family (1.1) is

$$\mathrm{Cycl}(\Gamma(C_0)) = \inf_{V,\,\varepsilon} N(V, \varepsilon). \tag{1.4}$$

The Hausdorff condition requires proximity to the whole graphic.

Theorem 1.1 (Local upper bound)**.** *For every fixed $C_0 > 1$, there exist a neighborhood $V \subset \mathbb{R}^5$ of $(1, C_0, 0, 0, 0)$ and a number $\varepsilon > 0$ such that, for every $p \in V$, the vector field (1.1) with parameter p has at most three periodic orbits γ with*

$$d_{\mathrm{H}}(\overline{\gamma}, \Gamma(C_0)) < \varepsilon.$$

Each of these periodic orbits is a limit cycle. In particular, $\mathrm{Cycl}(\Gamma(C_0)) \le 3$.

Proposition 1.2 (Three periodic orbits)**.** *For every fixed $C_0 > 1$, there is a sequence of parameters $p_n \to (1, C_0, 0, 0, 0)$, with $C = C_0$, for which (1.1) has three distinct isolated periodic orbits $\gamma_{j,n}$, $j = 1, 2, 3$, satisfying*

$$d_{\mathrm{H}}(\overline{\gamma_{j,n}}, \Gamma(C_0)) \longrightarrow 0.$$

These orbits eventually lie in every fixed sufficiently small neighborhood of the whole graphic, and in particular satisfy the Hausdorff condition of Theorem 1.1.

Corollary 1.3 (Cyclicity three)**.** *For every fixed $C_0 > 1$,* $\mathrm{Cycl}(\Gamma(C_0)) = 3$.

The three orbits of Proposition 1.2 are also hyperbolic and persist on open sets in the original five-parameter space arbitrarily close to the organizing parameter (Corollary 12.3).

The neighborhood V and the radius ε depend on C_0. Their construction uses, in particular, the positive speed (2.2) of the organizing orbit and the limiting contraction multiplier $\Lambda_0 < 1$ of Lemma 2.5. The speed bound $(C_0 - 1)/2$ tends to zero as $C_0 \downarrow 1$, and the margin $1 - \Lambda_0$ tends to zero as $C_0 \to \infty$.

Relation to previous work. The DRR program develops normal forms near saddle–nodes and applies them to finite-cyclicity questions; see Dumortier, Ilyashenko, and Rousseau [3]. Related parts of the program treat the degenerate graphics DF_{1a} and DF_{2a} [5], with the latter study completed in the remaining chart by Huzak [7]. For graphics through triple nilpotent points, Roussarie and Rousseau study the center graphic I_{14}^1 and distinguish it from the blown-up boundary limit-periodic sets appearing in the same analysis [11]; Rousseau, Shan, and Zhu treat the saddle-type graphics I_{12}^1 and I_{13}^1 [12].

Huzak and Kristiansen develop entry–exit formulas for degenerate turning points and apply them to the present five-parameter family [8, Theorem 2.4 and §6]. Their theorem constructs a local Dulac map, with finite-order remainder estimates in a logarithmic parameterization, in a principal root-free region where the scaled boundary drift has a uniform sign margin. Their Section 6 derives the entry–exit relation for this family and leaves the cyclicity analysis to further work. The passage problem here also includes merging boundary roots, roots approaching the central section, and independently vanishing coefficient scales. We compare the two endpoint maps on the full interval of admissible central heights, compose them with the exterior transition, and use geometric capture to count the periodic orbits satisfying the whole-graphic Hausdorff condition.

The root-free calculation enters our buffered compact-exit analysis through the leading relation between normalized exit radii,

$$f_\Lambda(\rho) = \frac{\Lambda\rho}{1+(1+\Lambda)\rho}.$$

Here ρ is the normalized positive exit radius and Λ is the boundary-action multiplier. This relation is the inverse, after a sign change and positive rescaling, of the base-point relation in Huzak–Kristiansen [8, (2.13) and (6.8)]. The change of variables is given after (9.41).

Uniform C^3 matching transfers this relation to the endpoint maps used for counting. Drift-weighted estimates treat the other passage regions, and a scattering comparison treats contact with the central section. The parameter cover combines these comparisons into the upper bound. A direct matching construction and variation of the connection splitting then attain it with $C = C_0$.

Finite-order passage analysis for canard cycles with singular slow dynamics, including divergence comparisons and exponential-flatness estimates, was developed by De Maesschalck and Dumortier [1]. For limit-periodic sets associated with singular slow dynamics, their later work obtains cycle bounds by studying suitable derivatives of the slow-divergence integral, even when that integral is unbounded [2]. Marín and Villadelprat give parameter-dependent hyperbolic-saddle Dulac expansions, including the near-resonant case [9]. These works provide the background for the passage estimates and derivative comparisons developed here.

Proof strategy. The proof has two counting steps. For fixed parameters, we control one return displacement Δ_p on its full interval $\mathcal{H}_p$ of central heights. A finite cover of parameter space then selects an applicable estimate throughout a neighborhood of the organizing value. Geometric capture connects these zeros to the periodic orbits in the theorem.

A common height for the two halves. In normal-rate coordinates (η, z), the field is

$$\dot\eta = q(\eta,p) - zB(\eta,z,p), \qquad \dot z = -\eta z.$$

Here $q(\eta,p) = q_0+q_1\eta+q_2\eta^2+\eta^3R_q(\eta,p)$ is the boundary drift. At fixed nearby A, C, its coefficients (q_0,q_1,q_2) are local coordinates in (μ_1,μ_2,μ_3) (Appendix A). The central height h is the coordinate z on $\eta = 0$. From $(0,h)$, follow the positive half-orbit backward and the negative half-orbit forward. Their endpoint functions $I_\pm(h)$ are defined on the same interval $\mathcal{H}_p$. The regular exterior map Π_p maps the negative endpoint window to the positive one. The return displacement is

$$\Delta_p(h) = \Pi_p(I_-(h)) - I_+(h).$$

A zero expresses the matching of the two endpoints through the exterior transition.

From endpoint derivatives to a zero count. Both endpoint derivatives with respect to h are positive. The logarithmic slope ratio $\mathcal{L} = \log[\Pi_p'(I_-)I_{-,h}/I_{+,h}]$ satisfies

$$\Delta_{p,h} = I_{+,h}(\mathrm{e}^{\mathcal{L}} - 1).$$

The zeros of $\mathcal{L}$ are therefore the critical points of Δ_p. To bound them on the full height interval, we combine crossing directions on the outer height ranges with negative curvature at critical points on the middle range. The order of these ranges allows at most two zeros of $\mathcal{L}$; Rolle's theorem then gives at most three zeros of Δ_p.

Uniformity near degenerate passages. The comparison must remain uniform when roots degenerate and coefficient scales vanish independently. Each endpoint derivative retains the tail multiplier $(z_*/t)^{-2q_2}$, even though the common quadratic term cancels from their ratio; this allows unbounded $q_2 \log t$. Near a boundary root, the leading curvature vanishes with the drift. The boundary-flow coordinate of Lemma 4.3 absorbs the height-dependent radial term, and Lemma 4.8 makes the phase correction smaller than any prescribed finite drift power after a fixed positive boundary-time advance. The resulting drift-weighted curvature formula (4.27) preserves every fixed sign margin.

The exterior transition supplies the remaining contraction when the singular slopes nearly agree. Its organizing derivative tends to

$$\Lambda_0 = \exp\left(-\frac{4\pi}{\sqrt{C_0 - 1}}\right) < 1$$

as the sections approach infinity. Fixed sections retain a contraction margin, while the nonconstant exterior term carries the same drift weights as the passage terms. At central contact, a decreasing scattering derivative ratio with a small C^2 error gives at most one middle critical point. The outer derivative signs then give at most two contact zeros on the full height interval.

Realizing three cycles. For the lower bound, keep $C = C_0$ and choose a coefficient path in the root-free region with negative quadratic drift. Comparing the two boundary actions gives a compact exit profile with a strict negative minimum. The parameter A leaves the boundary drift unchanged and splits the organizing connection transversely, so it can adjust the additive constant of this profile. Three test exits then give two displacement zeros.

This compact exit comparison alone does not locate the third cycle. Returning to the common central height, the organizing contraction gives the opposite displacement sign at a larger height. This supplies a third zero beyond the heights tested by the compact profile. Graph order and regular continuation then show that all three orbits converge to the whole graphic. The coefficient path, action matching, and splitting calculation are developed in Section 12; hyperbolicity and the separation of the third height follow in Corollaries 12.3 and 12.4.

The proof is organized around these comparisons. The geometry and the common-height return problem are established in Section 2, and Sections 3 and 4 give the counting principles and uniform passage estimates. Contact and positive offsets are treated in Section 5. For negative offsets, Section 6 develops the height-range comparisons used on both coefficient scales; the primary-scale analysis is in Section 7, and the secondary-scale cases begin in Section 8. These are estimates on overlapping height intervals for fixed parameters. Their coverage of a full parameter neighborhood is proved separately in Section 11, where Table 3 identifies the applicable comparison and bound in each region.

2. The graphic and its central displacement

To count periodic orbits near the whole graphic, we first show that each of them makes one passage through a fixed system of sections. Its central height then parameterizes both halves of the singular passage. We use this common parameter to construct the displacement and its interval of admissible heights.

2.1. The organizing orbit and the projective field.

At the parameter (1.2), set

$$H_0(x, y) = y - \frac{x^2}{2} + \frac{C_0}{2}.$$

A direct calculation gives

$$\dot{H}_0 = 2xH_0. \tag{2.1}$$

Thus (1.3) is invariant. On this parabola,

$$\dot{x} = \frac{(x+1)^2 + C_0 - 1}{2} > 0. \tag{2.2}$$

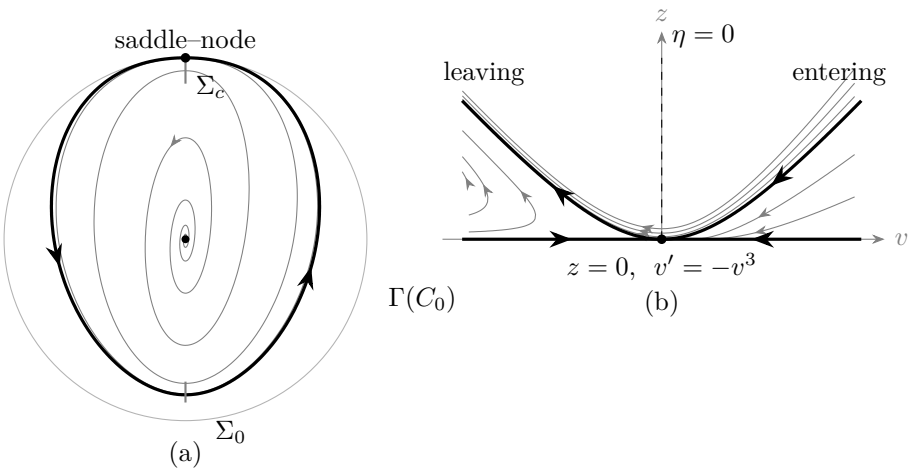


FIGURE 1. The organizing field of (1.1) for $C_0 = 10$, computed. (a) The Poincaré disk, through $(x, y) \mapsto (x, y)/\sqrt{9 + x^2 + y^2}$: the graphic $\Gamma(C_0)$, the unstable focus at the origin with an orbit spiralling to $\Gamma(C_0)$, the section Σ_0, and the central section Σ_c below the nilpotent saddle–node at infinity. (b) The chart (2.3)–(2.4) near the saddle–node, with the vertical scale enlarged: the line at infinity $z = 0$, the branches of $\Gamma(C_0)$ entering the saddle–node from $v > 0$ and leaving it into $v < 0$, the central section $\eta = 0$, and orbits in the three sectors of $z \geq 0$.

It follows that the parabola contains one regular orbit, directed from $x = -\infty$ to $x = +\infty$.

Near the positive y-direction at infinity, put

$$x = \frac{v}{z}, \qquad y = \frac{1}{z}, \qquad z > 0, \tag{2.3}$$

and multiply the transformed vector field by the positive factor z. We obtain

$$\begin{aligned} v' &= P(v) - (1 - Av + Cv^2)z, \\ z' &= -z\eta(v, z), \end{aligned} \tag{2.4}$$

where

$$\begin{aligned} P(v) &= \mu_1 + \mu_2 v - v^3, \\ \eta(v, z) &= \mu_3 + v + v^2 + Cvz. \end{aligned} \tag{2.5}$$

The boundary drift q below therefore depends on the three μ-parameters, whereas the positive factor B retains the dependence on (A, C). This separation is used later when A is varied to change the connection without changing the boundary action. After shrinking the (A, C)-neighborhood of $(1, C_0)$, $A^2 < 4C$ and hence

$$1 - Av + Cv^2 > 0 \qquad (v \in \mathbb{R}). \tag{2.6}$$

At the organizing parameter, $(v, z) = (0, 0)$ is the nilpotent saddle–node where the two ends of the affine orbit meet. Together with the affine orbit, it forms the compact directed graphic $\Gamma(C_0)$.

2.2. **A fixed neighborhood of the whole graphic.** The Hausdorff condition must force passage through the sections as well as proximity to the graphic. Removing a neighborhood of the finite repelling equilibrium gives an isolating block and sections valid for every nearby parameter.

Proposition 2.1 (Isolation and whole-graphic capture)**.** *There exist a compact set $B_{\mathrm{iso}} \subset \mathbb{S}^2$, a compact annular neighborhood $B_{\mathrm{cap}} \Subset \operatorname{int} B_{\mathrm{iso}}$ of $\Gamma(C_0)$, a parameter neighborhood V_{cap}, a transverse affine section Σ_0, and $\rho_{\mathrm{cap}} > 0$ with the following properties.*

(i) *The maximal invariant subset of* B_{iso} *for each field with parameter in* V_{cap} *is contained in* $\operatorname{int} B_{\mathrm{iso}}$.
(ii) *If a periodic orbit* γ *satisfies* $d_{\mathrm{H}}(\overline{\gamma}, \Gamma(C_0)) < \rho_{\mathrm{cap}}$, *then* $\overline{\gamma} \subset B_{\mathrm{cap}}$ *and* γ *crosses* Σ_0 *and the entrance and exit faces of a fixed projective box exactly once per minimal period, in the order of* $\Gamma(C_0)$.

A periodic orbit as in (ii) is called *captured*. The reverse Hausdorff inclusion forces section crossings; in a thin tubular annulus, their common orientation and the intersection number of an embedded circle make each crossing unique. Appendix B.1 gives the neighborhood construction and proof.

Remark 2.2. The bound in Theorem 1.1 also applies after restricting the orbits to a compact set: for each $p \in V$ and each compact $B \subset \mathbb{S}^2$, there are at most three limit cycles γ with $\overline{\gamma} \subset B$ and $d_{\mathrm{H}}(\overline{\gamma}, \Gamma(C_0)) < \varepsilon$. One such set is B_{iso} of Proposition 2.1, whose maximal invariant subset lies in its interior.

2.3. **The normal-rate coordinate.** The negative logarithmic rate of z identifies the central section and the two directions of passage. Following [8, Lemma 2.1], we take this rate, $\eta = -\dot{z}/z$, as a coordinate. In the projective time of (2.4), it is

$$\eta = \mu_3 + v + v^2 + Cvz. \tag{2.7}$$

Since $\partial_v \eta = 1 + 2v + Cz$, this is an analytic coordinate together with z on a fixed projective box and a sufficiently small parameter neighborhood.

Lemma 2.3 (Normal form)**.** *In the coordinates* (η, z), *the projective field has the exact form*

$$\dot{\eta} = q(\eta, p) - zB(\eta, z, p), \qquad \dot{z} = -\eta z, \tag{2.8}$$

where $p = (A, C, \mu_1, \mu_2, \mu_3)$, B *and* q *are analytic, and*

$$B > 0, \qquad \partial_z\{zB(0, z, p)\} > 0 \tag{2.9}$$

on a fixed smaller box. On $z = 0$,

$$q(\mu_3 + v + v^2, p) = (1 + 2v)P(v). \tag{2.10}$$

At the organizing parameter,

$$B(0, 0) = 1, \qquad B_\eta(0, 0) = 1, \qquad q(\eta) = -\eta^3 + O(\eta^4). \tag{2.11}$$

Consequently, after one more reduction of the box and parameter neighborhood, there are positive constants b_0, b_1, b_2 *such that*

$$B \geq b_0, \qquad B_\eta \geq b_1, \qquad 3R_q + \eta\partial_\eta R_q \leq -b_2, \tag{2.12}$$

where

$$q(\eta, p) = q_0 + q_1\eta + q_2\eta^2 + \eta^3 R_q(\eta, p). \tag{2.13}$$

Proof. The equation for z in (2.8) follows directly from (2.4) and (2.7). Differentiating η along the projective vector field (2.4) gives an analytic first component. Its restriction to $z = 0$ is $(1+2v)P(v)$, which proves (2.10). The remainder vanishes on $z = 0$ and therefore factors as $-zB$. Direct evaluation at the organizing parameter gives (2.11). The inequalities follow by continuity on a compact smaller box. □

For fixed (A, C), the coefficients in (2.13) are local coordinates on (μ_1, μ_2, μ_3), as shown in Appendix A.

2.4. **Regular transitions on fixed flow boxes.**

Lemma 2.4 (Regular transitions)**.** *Fix a finite order k. Let $\mathcal{X}_p$ be a C^{k+1} family with parameters in a neighborhood of a compact subset of a finite-dimensional parameter space. Consider transitions between transverse sections, defined on neighborhoods of the closed section windows under consideration. Suppose that their orbit segments remain in a common compact regular region and have flight times bounded by a fixed $t_{\max}$. Suppose also that the section parameterizations, their inverse charts, and their defining functions have uniform C^{k+1} bounds, and that the input and output transverse fluxes have absolute value at least a fixed $c > 0$. Then the transition maps and their local inverses have uniform C^k bounds on the section windows.*

Proof. Smooth flow dependence [13, Theorem 2.11], on finitely many flow neighborhoods, gives uniform derivatives up to time $t_{\max}$. The implicit function theorem for the hitting equation $g_p(\varphi_p^t(x)) = 0$, whose t-derivative is the nonzero transverse flux, gives the transition bounds; reversed time gives the inverse bounds. □

2.5. **The exterior transition.** Fix a small level $z = z_*$. At the organizing parameter the two branches of the graphic meet this level in two points with opposite signs of η. Let W_+ and W_- be compact transverse windows around the positive and negative endpoints. The finite part of the graphic defines, for nearby parameters, a regular first-hit map

$$\Pi_p : W_- \longrightarrow W_+. \tag{2.14}$$

On both windows we use the auxiliary energy coordinate

$$I = z_* - \frac{\eta^2}{2}. \tag{2.15}$$

Choose a bounded normal-rate graph box which is downward closed in the energy variable $U = \eta^2/2$, for example $0 \le U \le U_{\max}$, $0 < z \le z_*$, with the organizing branches away from its upper radius face. It contains the two strips of B_{cap} along the organizing branches with $0 < z \le z_*$. Choose the organizing endpoints in the interiors of $W_\pm$. After reducing the capture data of Proposition 2.1 and the parameter neighborhood of (2.14), every captured periodic orbit stays inside the graph box, away from its upper radius face, during its central passage and meets the interiors of both windows. This follows by narrowing the tubular neighborhood at the two regular section crossings.

Lemma 2.5 (Exterior multiplier)**.** *Put*

$$r_* = \sqrt{2z_* + C_0 z_*^2}, \qquad a_* = 1 + C_0 z_*, \qquad R_* = \frac{r_*}{z_*},$$

and take z_* small enough that $a_* > 2r_*$. Define the finite-section transverse multiplier

$$m_{\mathrm{aff}}(R_*) = \exp\left(\int_{-R_*}^{R_*} \frac{4x}{x^2 + 2x + C_0}\, \mathrm{d}x \right).$$

Let I_-^0 and I_+^0 be the two endpoint states of the organizing orbit in the coordinate (2.15). The derivative of the organizing exterior transition at that orbit is

$$(\Pi_0)'(I_-^0) = m_{\mathrm{aff}}(R_*) \frac{(a_* + r_*)(a_* + 2r_*)}{(a_* - r_*)(a_* - 2r_*)}, \qquad \Lambda_0 = \exp\left(-\frac{4\pi}{\sqrt{C_0 - 1}} \right). \tag{2.16}$$

In particular, $(\Pi_0)'(I_-^0) = \Lambda_0\{1 + O(\sqrt{z_*})\}$ as $z_* \downarrow 0$. Let $\Lambda_0 < \Lambda_+ < 1$ and fix z_* sufficiently small. There are enlarged compact transverse windows $\widetilde{W}_\pm$ at $z = z_*$ around the organizing endpoints, in whose interiors the windows $W_\pm$ are chosen, and $\Lambda_- > 0$ such that, for nearby parameters, Π_p extends to a neighborhood of $\widetilde{W}_-$ with values in the positive section, has uniformly bounded derivatives of every fixed finite order there, and satisfies

$$0 < \Lambda_- \le \Pi_p'(I) \le \Lambda_+ < 1 \qquad (I \in \widetilde{W}_-). \tag{2.17}$$

In particular, endpoints uniformly $o(1)$-close to $W_\pm$ lie in $\widetilde{W}_\pm$ for sufficiently small parameters.

Proof. Along the invariant parabola, use x as time and let $\mathfrak{k}$ be a transverse variation measured by H_0. Equations (2.1)–(2.2) give

$$\frac{d}{dx}\log \mathfrak{k} = \frac{4x}{x^2+2x+C_0}.$$

The finite-section multiplier in this affine transverse coordinate is $m_{\mathrm{aff}}(R_*)$. Since

$$\mathrm{PV}\int_{-\infty}^{\infty} \frac{4x}{x^2+2x+C_0}\,\mathrm{d}x = -\frac{4\pi}{\sqrt{C_0-1}},$$

and direct integration on the symmetric finite interval gives

$$m_{\mathrm{aff}}(R_*) = \Lambda_0\{1+O(R_*^{-1})\},$$

we have $m_{\mathrm{aff}}(R_*) = \Lambda_0\{1+O(\sqrt{z_*})\}$. We now convert the affine multiplier to the endpoint coordinate I. At fixed $z = z_*$ the organizing parabola meets the section at $v = \pm r_*$. Direct differentiation of the energy coordinate with respect to the affine transverse coordinate H_0 gives the two section factors

$$z_*^2(a_*+r_*)(a_*+2r_*), \qquad z_*^2(a_*-r_*)(a_*-2r_*),$$

respectively. Their ratio, multiplied by $m_{\mathrm{aff}}(R_*)$, yields (2.16); the ratio tends to one as $z_* \downarrow 0$. Since $\Lambda_0 < \Lambda_+$, fix z_* with this derivative below Λ_+. Smooth dependence then gives the asserted extension and (2.17) on sufficiently small windows $\widetilde{W}_\pm$ around the organizing endpoints, for nearby parameters. The nearby exterior arcs have uniformly bounded flight times through regular flow boxes and uniformly transverse endpoints, so Lemma 2.4 gives the finite-order derivative bounds on $\widetilde{W}_-$. The last assertion holds because $W_\pm$ lie in the interiors of $\widetilde{W}_\pm$. □

2.6. **One central height.** The central section separates the incoming and outgoing halves of the projective passage. Put

$$\Sigma_c(p) = \{\eta = 0 : 0 < z < z_*\}. \tag{2.18}$$

The height $h = z|_{\Sigma_c(p)}$ gives a common analytic coordinate on this parameter-dependent section in the original (v, z) chart. We call the point $(\eta, z) = (0, h)$ of $\Sigma_c(p)$ a *central point* and h its *central height*.

Lemma 2.6 (Unique central crossing)**.** *Every periodic orbit captured by Proposition 2.1 meets $\Sigma_c(p)$ exactly once during its projective passage, and this intersection is transverse.*

Proof. At $\eta = 0$, set

$$f_p(z) = q(0,p) - zB(0,z,p).$$

By (2.9), f_p is strictly decreasing. If $f_p(z) = 0$, both components of (2.8) vanish, so the point is an equilibrium and cannot lie on a periodic orbit. Hence every central crossing of a captured periodic orbit is transverse.

Suppose a crossing at z_1 goes from $\eta < 0$ to $\eta > 0$. Until a hypothetical second crossing, z decreases. The second crossing would therefore occur at $z_2 < z_1$ and would require $f_p(z_2) < 0$, whereas strict decrease gives $f_p(z_2) > f_p(z_1) > 0$. The opposite orientation is treated by reversing all inequalities. There is at most one crossing. Existence follows because the entrance and exit faces have opposite signs of η. □

For the directed passage considered here the crossing is from $\eta > 0$ to $\eta < 0$; hence a passing height satisfies

$$f_p(h) < 0. \tag{2.19}$$

2.7. **The interval of central returns.** For a passing height h, parameterize both half-orbits by increasing z: follow the positive half-orbit backwards and the negative half-orbit forwards. Put

$$U_\sigma(z;h) = \frac{\eta_\sigma(z;h)^2}{2}, \qquad \eta_\sigma = \sigma\sqrt{2U_\sigma}, \qquad \sigma \in \{+,-\}. \tag{2.20}$$

With z increasing, both functions satisfy

$$\frac{dU_\sigma}{dz} = \mathcal{G}_{\sigma,p}(U_\sigma, z) := B(\sigma\sqrt{2U_\sigma}, z, p) - \frac{q(\sigma\sqrt{2U_\sigma}, p)}{z}, \qquad U_\sigma(h;h) = 0. \tag{2.21}$$

The smooth field selects a unique branch through the initial square-root singularity of this coordinate.

Proposition 2.7 (Central displacement on one interval)**.** *After reducing the capture data, there is, for every parameter p, an interval $\mathcal{H}_p \subset (0, z_*)$, possibly empty, such that both graphs (2.21) reach W_+ and W_-. The endpoint functions*

$$I_\sigma(h) = z_* - U_\sigma(z_*;h) \tag{2.22}$$

are continuous and strictly increasing on $\mathcal{H}_p$. The central first-return domain is an interval $\mathcal{D}_p \subset \mathcal{H}_p$, and its fixed points are exactly the zeros on $\mathcal{H}_p$ of

$$\Delta_p(h) = \Pi_p(I_-(h)) - I_+(h). \tag{2.23}$$

For each fixed p, the maps $I_\pm$ and Δ_p are real analytic on $\operatorname{int}\mathcal{H}_p$. Every captured periodic orbit gives one zero of (2.23), and every zero gives one periodic orbit in the selected return neighborhood. This correspondence preserves isolation.

We call a central height $h \in \mathcal{H}_p$ *admissible*, and use the same adjective for its half-orbits and their endpoints in $W_\pm$.

Proof. We first compare the graphs to determine which heights reach the endpoint windows. We then determine the domain and fixed points of the first return. Let

$$\mathcal{Z}_p = \{0 < h < z_* : f_p(h) < 0\}.$$

This is an interval because f_p is strictly decreasing. For $h \in \mathcal{Z}_p$ and $z > h$, condition (2.19) and the strict decrease of f_p give

$$\mathcal{G}_{\sigma,p}(0,z) = \frac{zB(0,z,p) - q(0,p)}{z} > 0. \tag{2.24}$$

Thus a graph which has become positive cannot return to $U = 0$ at a larger height.

Let $h_1 < h_2$ be passing heights and suppose first that the graph starting at h_1 exists up to h_2. Then $U_\sigma(h_2;h_1) > 0 = U_\sigma(h_2;h_2)$. Scalar uniqueness away from $U = 0$ gives

$$U_\sigma(z;h_1) > U_\sigma(z;h_2) \tag{2.25}$$

for every common later height. If the first graph stays in the fixed downward-closed graph box until z_*, the second stays between it and zero and also reaches z_* by standard continuation. Consequently, for each sign, the set $\mathcal{Z}_{\sigma,p} \subset \mathcal{Z}_p$ of heights whose graph reaches z_* inside that box is an interval which contains, with each of its heights, every larger height of $\mathcal{Z}_p$.

To show that $zB - q > 0$ along these graphs, note that initially $\mathcal{G}_{\sigma,p} > 0$. At the first zero of $\mathcal{G}_{\sigma,p}(U(z), z)$, if one exists, one has $U > 0$ and $q = zB$, so

$$\frac{d}{dz}\mathcal{G}_{\sigma,p}(U(z),z) = B_z + q/z^2 = \frac{B + zB_z}{z} > 0. \tag{2.26}$$

This contradicts a first crossing from positive values. The strict inequality $B + zB_z > 0$ holds on the fixed normal-rate box after it is reduced.

To determine how the endpoints vary with the central height, differentiate the moving initial condition at an interior height. This gives

$$\partial_h U_\sigma(z_*;h) = \frac{f_p(h)}{h}\exp\!\left(\int_h^{z_*} \partial_U \mathcal{G}_{\sigma,p}(U_\sigma(z;h),z)\,\mathrm{d}z\right) < 0, \tag{2.27}$$

where $f_p(h)/h = q(0,p)/h - B(0,h,p) < 0$. The possible initial $O((z-h)^{-1/2})$ singularity is integrable; equivalently one may differentiate the smooth flow before passing to U. Hence $U_\sigma(z_*;h)$ is continuous and strictly decreasing, and $I_\sigma(h)$ is continuous and strictly increasing, on $\mathcal{Z}_{\sigma,p}$.

We can now impose the endpoint windows. Each W_σ is a compact interval in the energy coordinate, so

$$\mathcal{H}_{\sigma,p} = \{h \in \mathcal{Z}_{\sigma,p} : I_\sigma(h) \in W_\sigma\}$$

is an interval, because it is the inverse image of an interval under a strictly monotone map. Define

$$\mathcal{H}_p = \mathcal{H}_{+,p} \cap \mathcal{H}_{-,p}.$$

Every captured periodic orbit has its unique central height in $\mathcal{H}_p$, by the choice of these strips and of the windows above. In fact its height belongs to $\operatorname{int}\mathcal{H}_p$. Its central crossing is regular, its endpoint crossings are transverse and lie in $\operatorname{int} W_\pm$, and its two compact half-orbits avoid the radius face. Smooth flow dependence therefore continues both half-orbits for all central heights in an open neighborhood of that height, with endpoints still in the two windows.

For fixed p and $h_0 \in \operatorname{int}\mathcal{H}_p$, the analytic field has a regular central point, since $f_p(h_0) < 0$. The selected half-orbits reach $z = z_*$ in finite time: $\ddot{z}(0) = -h_0 f_p(h_0) > 0$ near the central point, and $U_\sigma > 0$ on the remaining compact graph. At the endpoint, $\dot{z} = -\eta z_* \neq 0$. Analytic flow dependence and the analytic implicit function theorem for these hitting equations therefore give analytic I_σ. The same argument applies to the exterior transition, whose section coordinate has derivative $-\eta \neq 0$. Thus Δ_p is analytic on $\operatorname{int}\mathcal{H}_p$.

It remains to identify the return domain within the admissible interval. We restrict the central return to heights whose return height lies in $\mathcal{H}_p$. The images $I_\sigma(\mathcal{H}_p)$ are intervals, and this return is defined precisely when $\Pi_p(I_-(h))$ belongs to $I_+(\mathcal{H}_p)$. Since $\Pi_p \circ I_-$ is strictly increasing, its inverse image of this interval is an interval $\mathcal{D}_p$. On $\mathcal{D}_p$ the return is

$$\mathcal{P}_p(h) = I_+^{-1}\big(\Pi_p(I_-(h))\big).$$

The equality $\mathcal{P}_p(h) = h$ is equivalent to (2.23). Every zero on $\mathcal{H}_p$ belongs to $\mathcal{D}_p$, since its common endpoint lies in $I_+(\mathcal{H}_p)$. Joining the negative half-orbit, the exterior arc, and the positive half-orbit proves the correspondence with periodic orbits. The unique central crossing proves injectivity, and a transverse flow box preserves isolation in both directions. □

Returns based at other transverse sections are conjugate to $\mathcal{P}_p$; they have the same number of fixed points, and the conjugacy preserves isolation.

2.8. **Paired variations of the central passage.** The two graphs start at the same central height. Pairing their variations at equal z-levels cancels the quadratic boundary term from the endpoint derivative ratio. We use the remaining terms to prove contraction when $q_1 \le 0$, and then at sufficiently large central heights when $q_1 > 0$.

Reduce the capture data of Proposition 2.1 so that

$$0 < h < h_0 < \frac{z_*}{2}. \tag{2.28}$$

This is possible because the central crossing of the organizing graphic is $(\eta, z) = (0,0)$.

Let L_p denote the local map from the positive endpoint window to the negative endpoint window obtained by following the central passage. Expressing both endpoints in terms of the same central

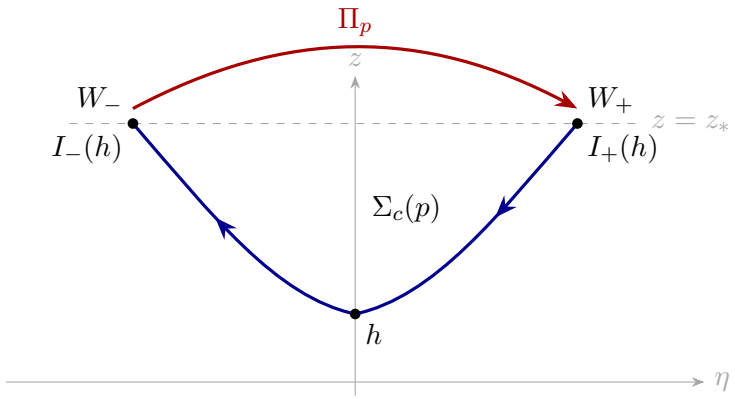


FIGURE 2. The central return in normal-rate coordinates. The blue curves are the two halves of the orbit through $(0, h)$; the red arrow is the regular exterior transition. Positions and curvature are schematic. The closing condition is $\Pi_p(I_-(h)) = I_+(h)$.

height gives, in the coordinate $I = z_* - \eta^2/2$,

$$L_p'(I_+(h)) = \frac{I_-'(h)}{I_+'(h)} > 0. \tag{2.29}$$

Lemma 2.8 (Paired-variation identity)**.** *Let $U_\pm$ be the graph solutions in (2.21). Define*

$$J_\sigma(z) = B_\eta(\eta_\sigma, z, p) - \frac{q_1}{z} - \frac{\eta_\sigma^2}{z}(3R_q(\eta_\sigma, p) + \eta_\sigma \partial_\eta R_q(\eta_\sigma, p)). \tag{2.30}$$

Then

$$\log L_p'(I_+(h)) = -\int_h^{z_*} \left(\frac{J_-(z)}{\sqrt{2U_-(z;h)}} + \frac{J_+(z)}{\sqrt{2U_+(z;h)}} \right) \mathrm{d}z. \tag{2.31}$$

The integral is improper only at $z = h$ and is convergent.

Proof. Differentiate the right-hand side of (2.21) with respect to U. Since $\eta_\sigma = \sigma\sqrt{2U}$, direct differentiation of (2.13) gives

$$\partial_U \mathcal{G}_{\sigma,p} = -\frac{2q_2}{z} + \frac{\sigma}{\sqrt{2U}} \left\{ B_\eta - \frac{q_1}{z} - \frac{\eta^2}{z}(3R_q + \eta R_{q,\eta}) \right\}. \tag{2.32}$$

We compare the endpoint variations using (2.27). Their initial factor $f_p(h)/h$ is common to both signs and cancels in the quotient. The term $-2q_2/z$ in (2.32) cancels at each common z-level. Subtracting the two remaining integrands gives (2.31). As in (2.27), the singularity at $z = h$ is integrable. □

Proposition 2.9 (Signed contraction)**.** *If $q_1 \le 0$, then*

$$0 < L_p' < 1 \tag{2.33}$$

on the whole central return interval. The return is strictly contracting in the endpoint coordinate I, and Δ_p is strictly decreasing in h. In particular, Δ_p has at most one zero.

Proof. The sign conditions (2.12) and $q_1 \le 0$ give $J_\sigma \ge b_1$ along both halves. The right-hand side of (2.31) is therefore strictly negative, which proves (2.33). Composing with the exterior transition in the endpoint coordinate and using (2.17) gives $0 < (\Pi_p \circ L_p)' < 1$. In terms of the central height, this is

$$\Delta_p'(h) = I_+'(h) \left\{ \Pi_p'(I_-(h)) \frac{I_-'(h)}{I_+'(h)} - 1 \right\} < 0.$$

Thus the displacement has at most one zero. □

For $q_1 > 0$, the term $-q_1/z$ in (2.30) may reverse the sign of J_σ near the central crossing. The next lemma controls this contribution at central heights that are large relative to the boundary coefficients.

Lemma 2.10 (Large central heights)**.** *After fixing z_* sufficiently small, there are constants $\theta_0, c_* > 0$ such that the following holds. Suppose $q_1 > 0$, $0 < h < h_0$, and*

$$\frac{|q_0|}{h} + \frac{q_1}{\sqrt{h}} + |q_2| \le \theta_0. \tag{2.34}$$

Then both halves through $(0, h)$ reach the endpoint windows and

$$0 < L_p' \le \exp(-c_*\sqrt{z_*}) < 1. \tag{2.35}$$

The return is strictly contracting in I, and $\Delta_p'(h) < 0$ if $h \in \mathcal{H}_p$.

Proof. We first continue both graphs in a cone, then place their endpoints in the windows, and finally estimate the derivative ratio. A first-contact comparison in (2.21), first in a provisional fixed box and then in the resulting cone, gives constants $c, C > 0$ such that

$$c(z-h) \le U_\sigma(z;h) \le C(z-h), \qquad \mathcal{G}_{\sigma,p}(U_\sigma, z) \ge c, \qquad h \le z \le z_*. \tag{2.36}$$

Indeed, in this cone the constant, linear, quadratic, and cubic parts of q/z are bounded, respectively, by

$$\frac{|q_0|}{h}, \qquad C\frac{q_1}{\sqrt{h}}, \qquad C|q_2|, \qquad C\sqrt{z_*}.$$

Choose z_* and then θ_0 so that these quantities fit the strict cone margins. This proves continuation to $z = z_*$ and (2.36).

To place the endpoints in the prescribed windows, choose a fixed intermediate height $\zeta \ll z_*$, then reduce h_0 so that $h_0 < \zeta/2$. The cone gives $U_\sigma(\zeta) = O(\zeta)$, and the organizing boundary graph has the same bound. On $[\zeta, z_*]$, continuous dependence and the first-variation estimate show that the endpoint difference between these graphs is $O(\zeta) + o_p(1)$. The organizing variational coefficient on its positive cone is $O(z^{-1/2})$, whose integral is uniformly bounded; for fixed ζ, the perturbed coefficients converge uniformly on this interval. Choose ζ small relative to the fixed window widths, and then reduce the parameter neighborhood.

With continuation and the endpoint windows established, we return to the derivative ratio. The sign margins (2.12) give $J_\sigma \ge b_1 - q_1/z$. In (2.31), use the upper cone bound for the negative contribution and the lower cone bound for the positive contribution. Since

$$\int_h^{z_*} \frac{\mathrm{d}z}{\sqrt{z-h}} = 2\sqrt{z_*-h}, \qquad \int_h^{z_*} \frac{\mathrm{d}z}{z\sqrt{z-h}} \le \frac{\pi}{\sqrt{h}},$$

we obtain

$$\log L_p' \le -c\sqrt{z_*-h} + C\frac{q_1}{\sqrt{h}} \le -c'\sqrt{z_*} + C\theta_0, \tag{2.37}$$

where $h < h_0 < z_*/2$. Decreasing θ_0 absorbs the positive term and proves (2.35). Composition with (2.17) proves the last assertion. □

3. Logarithmic slopes and zero-counting tools

To pass from local derivative estimates to a bound on the whole height interval, we must retain the order and direction of zero crossings. We first give the scalar counting tools for this step. The coefficient scales then identify which passage estimates supply their hypotheses.

3.1. **Counting zeros of the return displacement.** We count distinct zeros of Δ_p on an interval of central heights h. Multiplication by a nonvanishing factor or a regular monotone change of height coordinate preserves their number. To relate the critical points of the displacement to the endpoint slopes, put $m_\sigma = hI_{\sigma,h} > 0$ and define

$$\mathcal{L} = \log\left[\Pi_p'(I_-)\frac{m_-}{m_+}\right], \qquad \Delta_p'(h) = I_+'(h)(\mathrm{e}^{\mathcal{L}} - 1). \tag{3.1}$$

We express the crossing directions in a logarithmic coordinate that increases as the central height decreases:

$$T = -c_T \log(h/h_{\mathrm{ref}}), \qquad \partial_T = -\frac{h}{c_T}\partial_h, \qquad m_\sigma = -c_T I_{\sigma,T}, \tag{3.2}$$

where $c_T > 0$ and $h_{\mathrm{ref}} > 0$ depend only on the parameters and are held fixed when differentiating in h. The height-range covers are ordered by increasing T, hence decreasing h. Changing these scales replaces T by $c_1T + c_2$, with $c_1 > 0$ constant for fixed parameters. This preserves zero-crossing directions and the sign of curvature at a critical point. We write $\mathcal{L}(T)$ for the same slope ratio in this coordinate.

By (3.1), a bound of n for the zeros of $\mathcal{L}$ on one interval of central heights gives at most $n+1$ zeros of Δ_p, by Rolle's theorem. This bound is uniform in the additive constant of Δ_p. It remains to combine estimates from adjacent height ranges without losing their crossing directions. The next lemma also allows a zero at a strict maximum.

Lemma 3.1 (Ordered zero crossings)**.** *Let $f \in C^2(J)$, where J is an interval. If every critical point of f satisfies $f'' < 0$, then f has at most one critical point and at most two distinct zeros. More generally, suppose J has an ordered cover by an initial, middle, and final interval, with overlaps allowed and any of these intervals possibly empty. Assume that*

(i) every zero in the initial interval has $f' > 0$;
(ii) every critical point in the middle interval has $f'' < 0$;
(iii) every zero in the final interval has $f' < 0$.

Then f has at most two distinct zeros on J. If every zero on J has the same strict derivative sign, there is at most one. An interval on which f never vanishes may be included in either end interval.

Proof. At a zero of f' with $f'' < 0$, the derivative crosses strictly from positive to negative. Two such crossings would require an intervening zero of f' that is not a downward crossing. Thus f is monotone or increases and then decreases. It has at most two zeros; when there are two, an upward crossing is followed by a downward crossing. A zero at the critical point is a strict maximum and is the only zero. The same argument gives at most one zero when f' has the same strict sign at every zero of f.

For the ordered cover, choose cuts in the overlaps and assign each shared endpoint to only one adjacent interval. In the initial interval zeros cross upward; in the middle interval they occur in the order just proved; in the final interval they cross downward. Two upward crossings require an intervening downward crossing, and two downward crossings require an intervening upward crossing. Either configuration contradicts the permitted crossing directions and their order. A zero at a middle maximum likewise excludes zeros in both end intervals. Hence the only possible pair consists of an upward crossing followed by a downward crossing. □

Corollary 3.2 (Ordered cover count)**.** *Let J be an interval of central heights on which $\mathcal{L}$ is C^2, and let T be a logarithmic height coordinate as in (3.2). If J has an ordered cover, in increasing T, satisfying the hypotheses of Lemma 3.1 for $f = \mathcal{L}$, then Δ_p has at most three distinct zeros on J. If every zero of $\mathcal{L}$ on J has the same strict crossing direction, then Δ_p has at most two.*

Proof. By Lemma 3.1, $\mathcal{L}$ has at most two zeros on J, and at most one in the second case. Since $\Delta_T = -(m_+/c_T)(\mathrm{e}^{\mathcal{L}} - 1)$ with $m_+ > 0$, the critical points of Δ_p are the zeros of $\mathcal{L}$. Rolle's theorem gives the assertion. □

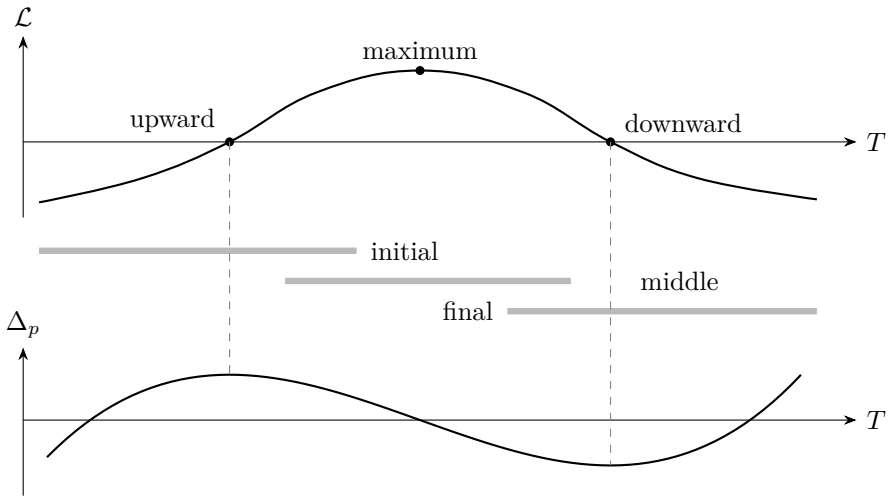


FIGURE 3. Schematic of Corollary 3.2, in increasing T, that is, decreasing central height: $\mathcal{L}$ with an upward zero, a strict maximum and a downward zero; the initial, middle and final intervals of an ordered cover; and Δ_p, whose critical points (dashed) are the zeros of $\mathcal{L}$.

Figure 3 shows the configuration allowed by Corollary 3.2. In the applications, thresholds on the half-orbits define the pieces of the cover. The following lemma uses the graph order (2.25) to place these pieces in their order in T.

Lemma 3.3 (Order of central heights)**.** *Let $h_1 < h_2$ be central heights whose half-orbits of the same sign σ are defined up to a common height z. Then $|\eta_\sigma(z;h_1)| > |\eta_\sigma(z;h_2)|$, and at every common radius the half-orbit through h_2 has the larger height. Consequently, for fixed $z_e, R > 0$, the central heights with $h \geq z_e$ or $|\eta_+(z_e;h)| \leq R$ form an interval that contains all larger heights, that is, an initial interval in T. For a fixed height z_o and an increasing function g, the central heights with $g(|\eta_+(z_o;h)|) \geq c_{\mathrm{th}}$ form a final interval in T. The intersection of such intervals with $\mathcal{H}_p$ is an interval.*

Proof. The first assertion is (2.25), since $|\eta_\sigma| = \sqrt{2U_\sigma}$. Each graph is strictly increasing in z: the first-zero argument (2.26) gives $zB - q > 0$, using $B + zB_z > 0$ on the reduced normal-rate box. Hence, at a common radius, the graph through the larger central height has the larger height. The remaining assertions follow because $T = -c_T \log(h/h_{\mathrm{ref}})$ decreases with h and $\mathcal{H}_p$ is an interval. □

Table 1 lists the ordered covers used in Sections 5 to 10. The remaining regional statements reduce to Proposition 6.9.

Near a boundary root, taking the logarithm of a power-law derivative reduces the slope comparison to a constant, one logarithm, and a remainder. The next lemma uses the strict negative constant to control the zeros even as the coefficient of the logarithm tends to zero.

Lemma 3.4 (One logarithm)**.** *For $c_0, C_e, \vartheta > 0$ there is $x_c \in (0,1)$ with the following property. Let $G \in C^1((0,x_c))$ have the form*

$$G(x) = c + \alpha \log x + e(x), \qquad c \leq -c_0, \quad \alpha \in \mathbb{R}, \quad |e(x)| + |xe'(x)| \leq C_e x^\vartheta.$$

If $\alpha \geq 0$, then $G < 0$. In general $xG'(x) < 0$ at every zero of G, so G has at most one zero. If $f \in C^1((0,x_c))$ and f' has the sign of G, then f has at most two zeros.

Proof. Choose x_c so that $C_e x^\vartheta < c_0/2$ and $C_e x^\vartheta |\log x| < c_0/4$ for $0 < x < x_c$. If $\alpha \geq 0$, then $G \leq c + e < -c_0/2$. If $\alpha < 0$, every zero satisfies $|\alpha||\log x| = |c + e(x)| \geq c_0/2$, and hence

TABLE 1. Ordered covers of $\mathcal{H}_p$, in increasing T. "Up" and "down" mean that every zero of $\mathcal{L}$ in the range crosses upward, respectively downward; "max" means that every critical point of $\mathcal{L}$ in the range is a strict maximum. The bounds follow from Corollary 3.2 unless another lemma is named.

Proposition	Ranges and the property of $\mathcal{L}$	Zero bound for Δ_p
5.1	large heights, $\mathcal{L}<0$; compact heights, up, by Lemmas 3.6 and 5.2; small heights, $\mathcal{L}>0$	2
5.3	heights above the saddle collar, $\mathcal{L}<0$; the collar, up, by Lemma 3.4	2
6.9	early heights and the root interval, up	2
7.1	early heights, $\mathcal{L}<0$; bounded radii, max, by Lemma 4.10; large exits, $\mathcal{L}<0$	3
8.1, compact drift	early heights, up; the root interval, max, by Lemma 4.10	3
8.1, small b	early heights and the root collar, up, by Lemma 3.4 on the collar	2
9.5	early heights, up; annular exits, max, or $\mathcal{L}<0$ for small b; later exits, down	3
10.1, 10.2	early heights, up; bounded outputs and escaped heights, max	3
10.3	early heights, up; the root interval, max	3

$xG'(x)=\alpha+xe'(x)<-|\alpha|+|\alpha|/2<0$. All zeros are crossings in the same direction, so there is at most one. Rolle's theorem gives the bound for f. □

Remark 3.5. The auxiliary sections used in a local representation are held fixed during differentiation. If each point of an interval J has a neighborhood on which $f\in C^1(J)$ is represented with auxiliary sections fixed for that neighborhood, then a sign of f' proved in these representations holds on all of J. Endpoint preparations formed from the same $I_\pm$ and Π_p on $\widetilde{W}_\pm$ of Lemma 2.5 represent the same $\mathcal{L}$ in (3.1), including shared cuts.

3.2. **Derivative-ratio comparisons on compact intervals.** For analytic functions, the classical Wronskian criterion [6, Lemma 2.3(b)] identifies the extended complete Chebyshev property by nonvanishing successive Wronskians. For the three functions $1,\phi,\psi$, the relevant identities are

$$W(1,\phi)=\phi',\qquad W(1,\phi,\psi)=(\phi')^2(\psi'/\phi')'.$$

For the compact-interval comparisons below, we need this argument to remain valid after adding a small C^2 remainder. The next lemma records the strict margins on the denominator and the derivative of the ratio that make this possible.

Lemma 3.6 (Stable derivative-ratio count)**.** *On a fixed compact interval J, let $\phi,\psi\in C^2(J)$, with $\phi'\ge c_*>0$ and a uniform C^2 bound on ϕ. Let $0<\lambda_0\le\lambda\le\lambda_1$. If*

$$|(\psi'/\phi')'|\ge c_*\quad \text{on } J,$$

then, for every constant D and every sufficiently small C^2 remainder R, the function $g=D+\phi-\lambda\psi+R$ has at most two zeros on J, and $(g'/\phi')'$ has the sign of $-(\psi'/\phi')'$ on J. If instead $1-\lambda\psi'/\phi'\ge c_*$, then $g'/\phi'>0$, and g has at most one zero. The remainder threshold depends only on the displayed uniform bounds.

Proof. Dividing the derivative by $\phi'>0$ gives $g'/\phi'=1-\lambda\psi'/\phi'+R'/\phi'$. The derivative of the remainder term is bounded by $C\|R\|_{C^2}$. In the first case, the continuous function $(\psi'/\phi')'$ has one sign and $|\lambda(\psi'/\phi')'|\ge\lambda_0c_*$. Consequently, $(g'/\phi')'=-\lambda(\psi'/\phi')'+O(\|R\|_{C^2})$ has the sign of $-(\psi'/\phi')'$ when the remainder is sufficiently small. The quotient is strictly monotone, so Rolle's theorem gives at most two zeros of g. In the second case, the quotient remains positive, which gives at most one zero. □

3.3. **Coefficient scales.** There are two covers in the argument. The ordered height cover above combines estimates for one fixed parameter on subintervals of $\mathcal{H}_p$. The parameter cover selects a regional result that already bounds the zeros on that whole interval, so its bound is the maximum of the regional bounds. Signed contraction treats $q_1 \le 0$; the scales below distinguish the regions for $q_1 > 0$. We use the coefficient scales

$$H = \max\{|q_0|^{1/3}, \sqrt{q_1}, |q_2|\}, \qquad \rho_1 = \sqrt{q_1}.$$

For $q_0 < 0$ and $q_2 \neq 0$, the secondary parameters are

$$s = |q_2|, \qquad r = \max\{\sqrt{|q_0|/s^3}, q_1/s^2\}.$$

The central-height unit reflects the balance between the boundary drift q and zB in (2.8); the output-height unit reflects the comparability of $U = \eta^2/2$ and z in the common cone. On the primary scale, $q(HX) = O(H^3)$ for bounded X, whereas $|\eta| = O(H)$ gives $U = O(H^2)$, yielding the height units H^3 and H^2, respectively. Table 2 lists the coordinate normalizations; the rescaled central height ranges over the corresponding image of $\mathcal{H}_p$. Near a weak root, whose radius tends to zero relative to the current coefficient scale, we rescale again.

For $q_1 > 0$, the contact strip $|q_0| \le Lq_1^2$ is treated by the scattering comparison. Positive offsets outside this strip are treated by the interior saddle.

For negative offsets, we use the primary scale when $|q_0|/H^3$ has a fixed positive lower bound. At smaller ratios, the secondary scale treats small r, separately for the two signs of q_2. The remaining parameters are treated either on the linear scale ρ_1, by the weak-root comparison, or in the contact strip. After the regional estimates, Section 11 fixes L and the overlaps needed to make this classification exhaustive.

TABLE 2. Coordinate scales for $|\eta|$, h and z, respectively. Output heights are fixed positive multiples of the listed scale.

Region	Radius scale	Height scale	Output scale
Contact	q_1	q_1^2	q_1^2
Positive offset	$\sqrt{q_0}$	q_0	q_0
Primary negative drift	H	H^3	H^2
Secondary negative offset	sr	s^3r^2	s^2r^2

Table 3 in Section 11 lists the parameter regions and the bound each provides on the whole height interval $\mathcal{H}_p$. Appendix C collects the recurring notation and passage terminology.

4. Uniform passage estimates

Two degeneracies govern the endpoint comparison. At exponentially small incoming heights, the quadratic boundary coefficient can produce an unbounded power-law multiplier in each endpoint derivative. Near a boundary root, the drift and the leading curvature both tend to zero, so an unweighted error estimate cannot preserve the curvature sign. We retain the common tail multiplier explicitly, remove the height-dependent term by a boundary-flow coordinate, and fix the phase of both projected orbits using the same central height. After a fixed positive boundary-time advance, the phase error carries arbitrarily high powers of the drift. These estimates give the curvature criterion of Lemma 4.10, including the exterior contribution, and the one-sided slope comparison in the weak-root range of Corollary 4.11.

In all passage estimates below, derivatives are taken in the stated state variables, with parameters and all parameter-only scales and actions fixed. Bounds are uniform on the indicated parameter regions, and each finite order k is fixed before choosing the small parameter threshold.

4.1. **Weighted tail estimates.** We first estimate the map from an incoming height t to the fixed section $z = z_*$, keeping its power-law multiplier explicit. Write $\mathcal{G}_\sigma(U,z) = B(\sigma\sqrt{2U},z) - q(\sigma\sqrt{2U})/z$, as in (2.21). Fix a compact interval of positive incoming energies $\mathcal{U} = U(t)/t$. For a sufficiently small fixed z_*, the conditions

$$\frac{|q_0|}{t} + \frac{|q_1|}{\sqrt{t}} + |q_2| \quad \text{sufficiently small}, \qquad 0 < t < z_*, \tag{4.1}$$

put all those solutions in a slightly enlarged invariant cone $c_- z \le U(z) \le c_+ z$. Indeed, on this cone,

$$\frac{|q(\sigma\sqrt{2U})|}{z} \le C\left(\frac{|q_0|}{z} + \frac{|q_1|}{\sqrt{z}} + |q_2| + \sqrt{z}\right).$$

Choose $c_- < b_0/4$ and $c_+ > 4\sup B$, containing the incoming energies in their interior. Reduction of z_* and (4.1) makes $b_0/2 < \mathcal{G}_\sigma < 2\sup B$. The vector field points into the cone along both boundary lines. This proves continuation to z_*, also for nearby incoming energies needed for differentiation.

To make the coefficient of $\sqrt{z_*}$ independent of later scaled height thresholds, fix a common cone using coefficient bounds on a preliminary box in (η, z). Enlarge it once to contain the moving-entrance energy $1/(2m_{\mathrm{land}})$ used in the adaptive and annular landings, with m_{land} fixed before the coefficient and height thresholds. A scaled entrance with a larger energy-to-height ratio is first continued to this cone. Constants for this finite continuation may depend on the thresholds; the coefficient C_{cone} remains independent of them.

For a fixed positive scaled entrance outside the common cone, put $z = \ell^2 Z$, $U = \ell^2 \mathcal{U}$, with fixed $Z_0 > 0$ and $\mathcal{U}(Z_0) = \mathcal{U}_0$ in a compact positive interval, and set

$$\epsilon_{\mathrm{in}} = \ell + |q_0|/\ell^2 + |q_1|/\ell + |q_2|.$$

Write $B_0 = B(0,0,p)$. On any fixed positive-height interval the graph equation is $\mathcal{U}_Z = B_0 + O_{C^k}(\epsilon_{\mathrm{in}})$ on a fixed enlarged positive energy interval. Integration and the finite variational equations give

$$\mathcal{U}(Z_1;\mathcal{U}_0) = \mathcal{U}_0 + B_0(Z_1 - Z_0) + O_{C^k}(\epsilon_{\mathrm{in}}). \tag{4.2}$$

Choose Z_1 large after the entrance interval, and then ϵ_{in} small; the ratio $\mathcal{U}(Z_1)/Z_1$ lies in the common cone. The constants in this finite continuation may depend on Z_0, Z_1 and the entrance interval. Its derivative with respect to $\mathcal{U}_0$ is $1 + O(\epsilon_{\mathrm{in}})$, and replacing the tail multiplier based at $\ell^2 Z_0$ by that based at $\ell^2 Z_1$ introduces only $(Z_1/Z_0)^{2q_2} = 1 + O(|q_2|)$, with Z_0 and Z_1 fixed.

Lemma 4.1 (Weighted tail)**.** *For every fixed k, set*

$$\varpi_t(z) = (z/t)^{-2q_2}, \qquad \varpi_t = \varpi_t(z_*), \qquad e(t) = \sqrt{t} + |q_1|/\sqrt{t}.$$

If $|q_2| < 1/8$ and (4.1) holds, then the centered endpoint map satisfies

$$\frac{U(z_*; t\mathcal{U}) - U(z_*; t\mathcal{U}_\circ)}{t\varpi_t} = a_{t,p}(\mathcal{U} - \mathcal{U}_\circ) + R_{t,p}(\mathcal{U}), \qquad c < a_{t,p} < C, \quad \|R_{t,p}\|_{C^k} \le C_k e(t). \tag{4.3}$$

Its inverse in these centered coordinates has uniformly bounded derivatives through order k. If the incoming energies lie in the common cone, then

$$|\log a_{t,p}| \le C_{\mathrm{cone}}\left(\sqrt{z_*} + \frac{|q_1|}{\sqrt{t}}\right). \tag{4.4}$$

Here C_{cone} depends only on the common cone and the uniform bounds for B, R_q and their first derivatives on the preliminary normal-rate box. The same constant applies for all admissible t, z_* in that box, independently of the later coefficient and height thresholds. For another fixed cone the

same bound holds with a constant depending on that cone. For a moving entrance $U(t) = ct$, with fixed $c > 4 \sup B$, let $U_{\mathrm{end}}(t)$ be its energy at z_*. Then

$$\begin{aligned} U'_{\mathrm{end}}(t) &= \varpi_t a(t), \qquad c_1 < a(t) < C_1, \\ |(t\partial_t)^j a(t)| + |(t\partial_t)^j \log a(t)| &\le C_j \left(e(t) + \frac{|q_0|}{t} \right), \quad 1 \le j \le k. \end{aligned} \tag{4.5}$$

Here the cone is enlarged to contain c before the smallness constants are chosen.

Proof. We separate the quadratic term in the first energy derivative from the coefficients that control the higher variations. On the cone,

$$\begin{aligned} \mathcal{G}_U &= -2q_2/z + R_1, \qquad |R_1| \le C(z^{-1/2} + |q_1| z^{-3/2}), \\ |\mathcal{G}_{U^j}| &\le C_j(z^{1/2-j} + |q_1| z^{-j-1/2}), \qquad j \ge 2. \end{aligned} \tag{4.6}$$

The constant drift disappears on differentiation, and the constant in the first line depends only on the stated cone and coefficient bounds. Hence the first variation is

$$J(z;t) = \varpi_t(z) \exp \int_t^z R_1(U(\zeta), \zeta)\, d\zeta. \tag{4.7}$$

Its quotient by $\varpi_t(z)$ is positive and bounded; integrating the bound for R_1 in (4.6) bounds its logarithm by the right side of (4.4).

To estimate the higher variations, factor out the same multiplier. Put $V_j = t^{-1} \partial_{\mathcal{U}}^j U = \varpi_t(z) W_j$. Here $W_1(t) = 1$ and $W_j(t) = 0$ for $j \ge 2$. After cancellation of the quadratic linear term, their equations are

$$(W_j)_z = R_1 W_j + \sum_{m=2}^{j} t^{m-1} \varpi_t(z)^{m-1} \mathcal{G}_{U^m} \sum_{\substack{i_1 + \cdots + i_m = j \\ i_\ell \ge 1}} c_{i_1, \ldots, i_m} W_{i_1} \cdots W_{i_m}.$$

Every variation in the inner sum has order less than j, so it remains to bound the coefficients of these products. The two required integrals obey

$$\begin{aligned} t^{m-1} \int_t^{z_*} z^{1/2-m} (z/t)^{2|q_2|(m-1)}\, dz &\le C_m \sqrt{t}, \\ |q_1| t^{m-1} \int_t^{z_*} z^{-m-1/2} (z/t)^{2|q_2|(m-1)}\, dz &\le C_m |q_1| / \sqrt{t}. \end{aligned} \tag{4.8}$$

After $z = t\zeta$, the first integral has decay margin $m - 3/2 - 2|q_2|(m-1) \ge 1/4$; the second has a larger margin. Variation of constants and induction give bounded W_1 and $W_j = O(e(t))$ for $j \ge 2$. Choose $a_{t,p}$ as the first normalized derivative at $\mathcal{U}_\circ$ and integrate the second derivative bound. Constructing one extra derivative if necessary proves (4.3). Positivity of the first derivative gives the inverse bounds.

For the moving entrance, differentiation includes the change of initial height as well as that of initial energy. It gives

$$U'_{\mathrm{end}}(t) = J(z_*; t)\, f_c(t), \qquad f_c(t) = c - \mathcal{G}_\sigma(ct, t) > c/2.$$

Thus $a(t) = f_c(t) J(z_*; t)/\varpi_t$. To justify its derivatives, write $\mathcal{E}_t = t\partial_t$, holding z fixed inside the flow. The differentiated equations and (4.8) give

$$|\mathcal{E}_t^j U(z;t)| \le C_j t \varpi_t(z), \qquad j \ge 1. \tag{4.9}$$

For $j = 1$ this is $\mathcal{E}_t U = tJ f_c$. To compute higher moving boundary data, set $\mathcal{B}_j(t) = (\mathcal{E}_t^j U)(t;t)$. Then

$$\mathcal{B}_0 = ct, \qquad \mathcal{B}_j = t \frac{d\mathcal{B}_{j-1}}{dt} - t(\partial_z \mathcal{E}_t^{j-1} U)(t;t).$$

In the local variables $z = t\zeta,\ U = tW$, near $(\zeta, W) = (1, c)$, the equation has bounded finite jets in $\sqrt{t}, q_1/\sqrt{t}, q_0/t, q_2$. The positive-order Euler derivatives of these quantities are bounded under (4.1), so $\mathcal{B}_j = O_j(t)$. The same triangular induction then proves (4.9).

These moving-flow bounds allow us to differentiate the logarithm in (4.7). The lower-boundary terms are $O(e(t))$. At order j, the chain rule gives a finite sum of terms with m flow variations, $1 \le m \le j$. By (4.8) with $m+1$ in place of m, each is bounded by $C_j e(t)$. Repeated boundary differentiation has the same estimate. Finally $\mathcal{E}_t^j f_c = O(e(t) + |q_0|/t)$ for $j \ge 1$. In particular its quadratic contribution $2q_2c$ is constant in t. These estimates bound the Euler derivatives of a; since a has a positive lower bound, they also bound those of $\log a$. This proves (4.5). □

For $|q_2| < 1/8$ and $0 < t \le z_*$, the tail amplitude satisfies

$$t\varpi_t = z_*^{1/4} t^{3/4} (t/z_*)^{1/4+2q_2} \le z_*^{1/4} t^{3/4}. \tag{4.10}$$

When the scaled entrance height also tends to zero, we first continue the graph to a fixed scaled height where the weighted-tail estimate applies. The following lemma supplies this continuation and the resulting endpoint expansion. Its additional symbols and those of its proof are local.

Lemma 4.2 (Continuation from a small scaled height)**.** *Suppose $0 < \ell \le \delta$ and $|q_0|/(\delta\ell^2) + |q_1|/(\delta\ell) + |q_2|/\delta \le C$. Put $z = \ell^2 Z$, $U = \ell^2 \mathcal{U}$ and $K_\sigma(x) = -q(\sigma\ell x)/(\delta\ell^2)$. Let a graph enter at $Z = \delta\zeta_0$ with $\mathcal{U} = \mathcal{U}_0 \in [0, \mathcal{U}_{\max}]$, where $0 < c \le \zeta_0 \le C$ and $K_\sigma(\sqrt{2\mathcal{U}_0}) \ge k_0 > 0$. For every fixed $Z_1 > 0$ and sufficiently small δ, it continues with positive flux to*

$$\mathcal{U}(Z_1) = \mathcal{U}_0 + B_0 Z_1 + O(\delta(1 + |\log\delta|)).$$

A large fixed Z_1, chosen after $\mathcal{U}_{\max}$ and before δ, puts $\mathcal{U}(Z_1)/Z_1$ in the common cone. If $\mathcal{U}_0 = x^2/2$, with x in a fixed compact positive interval and $\|\zeta_0(x)\|_{C^3} \le C$, set $t = \ell^2 Z_1$, $E = \ell^2 \varpi_t$ and

$$b_\sigma = z_* - U(z_*; t, B_0 t), \qquad A_\sigma = \varpi_t^{-1}\, \partial_u U(z_*; t, u)|_{u = B_0 t}.$$

Then the endpoint satisfies

$$I_\sigma(x) = b_\sigma - \tfrac{1}{2} E A_\sigma x^2 + E e_\sigma(x), \qquad c < A_\sigma < C, \quad \|e_\sigma\|_{C^3} \le C\delta(1 + |\log\delta|). \tag{4.11}$$

For two branches with the same ℓ, Z_1, $|\log(A_-/A_+)| \le 2C_{\text{cone}}\sqrt{z_*} + C\delta$. The constants depend only on the displayed bounds, the compact radius interval and Z_1.

Proof. We first prove continuation, then retain the radius derivatives needed for the endpoint expansion. The coefficient bounds and $\ell \le \delta$ bound all radius derivatives of K_σ on compact intervals. Choose $\mathcal{U}_1 > \mathcal{U}_{\max} + 2Z_1 \sup B$, then $\zeta_1 > \sup\zeta_0$ so large that $|K_\sigma| \le b_0\zeta_1/2$ for $0 \le \mathcal{U} \le \mathcal{U}_1$. In $Z = \delta\zeta$, the graph equation is

$$\mathcal{U}_\zeta = \delta\{B(\sigma\ell\sqrt{2\mathcal{U}}, \delta\ell^2\zeta) + K_\sigma(\sqrt{2\mathcal{U}})/\zeta\}.$$

Up to ζ_1, the energy changes by $O(\delta)$ and the radius by $O(\sqrt{\delta})$, preserving $K_\sigma \ge k_0/2$ and positive flux. For $Z \ge \delta\zeta_1$, $|\delta K_\sigma/Z| \le b_0/2$ while $\mathcal{U} \le \mathcal{U}_1$, so $b_0/2 \le \mathcal{U}_Z \le 2\sup B$, and this energy interval persists to Z_1. Integrating $B - B_0 = O(\ell)$ and $\delta K_\sigma/Z$ gives the displayed continuation estimate. Since B_0 is interior to the common cone, large Z_1 gives entry.

For the radius-dependent entrance, the energies stay bounded away from zero, so the preceding continuation can be differentiated in the entrance radius. The flow in ζ has uniformly bounded finite jets; composition with $(\zeta_0(x), x^2/2)$ gives an $O_{C^3}(\delta)$ change up to ζ_1. Thereafter the coefficient error and its energy derivatives are bounded by $C_j(\delta + \delta/Z)$. Integration of the finite variational equations gives the same continuation estimate in C^3. At t, the smallness expression in (4.1) and $e(t)$ are $O(\delta)$. Apply (4.3) with center $\mathcal{U}_\circ = B_0$ to the incoming energy $\mathcal{U}(Z_1; x)/Z_1$. The coefficient of this centered map is A_σ, the normalized center derivative, and $t\varpi_t = Z_1 E$; this proves (4.11). Finally (4.4), with $|q_1|/\sqrt{t} = O(\delta)$, gives the slope-ratio bound. □

4.2. **A coordinate following the boundary flow.** The passage to a fixed scaled height combines boundary motion with a height-dependent term. Dividing by the boundary drift would be singular at its roots. We instead construct a coordinate that absorbs the height-dependent term and follows the boundary flow, uniformly through those roots. On a fixed positive radius interval, write the scaled height equation as

$$\frac{dX}{d\log Y} = \delta F(X) + YG(X,Y), \qquad \delta > 0, \tag{4.12}$$

where F has bounded derivatives through the required order and $G(X,Y) = B_0/X + O_{C^k}(\delta)$, with B_0 in a compact positive interval.

Lemma 4.3 (Boundary-flow coordinate)**.** *On a smaller positive radius interval and $0 \le Y \le Y_0$, for every fixed finite k and all sufficiently small δ, there is a uniformly C^k coordinate*

$$\mathcal{W}(X,Y) = X + Y\mathcal{W}_1(X,Y), \qquad \mathcal{W}_X > 0, \quad \mathcal{W}_Y < 0,$$

which satisfies $\mathcal{W}_{\log Y} = \delta F(\mathcal{W})$ along (4.12). Its inverse $X = \chi(w,Y)$ has

$$\frac{\chi(w,Y)^2}{2} = \frac{w^2}{2} + B_0 Y + \delta Y E_\chi(w,Y), \qquad \|E_\chi\|_{C^k} \le C_k, \quad \chi_Y > 0. \tag{4.13}$$

The construction is uniform through simple or multiple zeros of F. The height bound Y_0 depends only on the bounds for B_0 and on the lower ends of the two radius intervals; the threshold for δ and the constants C_k may depend on the whole interval.

Proof. We construct the coordinate by following the full passage backwards and then undoing the boundary motion. Extend the coefficients smoothly to a larger interval and then to bounded functions, preserving positive G. Let ϕ^u be the flow of δF. Follow (4.12) backward for logarithmic time u, writing its radius as $X_b(u)$ and its height as $Y\mathrm{e}^{-u}$. Define

$$\mathcal{W}(X,Y) = \lim_{u\to\infty} \phi^u(X_b(u)).$$

The derivative of the expression inside this limit is

$$-Y\mathrm{e}^{-u}(\phi^u)'(X_b(u))G(X_b(u), Y\mathrm{e}^{-u}). \tag{4.14}$$

To differentiate the limit, we compare growth of the flow derivatives with the height decay in this integrand. For each fixed order, boundary-flow derivatives grow at most as a polynomial in u times $\mathrm{e}^{C_k\delta u}$. The backward equation is

$$\partial_u X_b = -\delta F(X_b) - Y\mathrm{e}^{-u}G(X_b, Y\mathrm{e}^{-u}).$$

Its state variations have linear coefficient $O(\delta) + O(Y\mathrm{e}^{-u})$. In each higher variational equation, the remaining terms contain lower variations; coefficients from F carry δ, and those from G carry e^{-u}. Variation of constants therefore gives the same polynomial-times-exponential bounds at every fixed order. Consequently

$$\mathcal{W} = X - Y\int_0^\infty \mathrm{e}^{-u}(\phi^u)'(X_b(u))G(X_b(u), Y\mathrm{e}^{-u})\,du.$$

Every prescribed derivative of this integral is bounded by an integrable function $C_k(1+u)^{N_k}\mathrm{e}^{-(1-C_k\delta)u}$, after reducing δ. This proves convergence and $\mathcal{W} = X + Y\mathcal{W}_1$ with uniform derivative bounds for $\mathcal{W}_1$. The flow identity gives $\mathcal{W}_{\log Y} = \delta F(\mathcal{W})$.

The signs and the inverse energy expansion follow by comparison with $\delta = 0$. At that parameter, the energy identity is $\mathcal{W}_0^2 = X^2 - 2B_0 Y$. Choose Y_0 so that this backward graph remains inside the positive interval; since $\mathcal{W}_0 \le X$, this constrains only the lower ends of the intervals. Write $G = B_0/X + \delta g$, and interpolate by $\lambda\delta F,\ B_0/X + \lambda\delta g$, $0 \le \lambda \le 1$, using compatible extensions. A

λ-variation has forcing of size δ and introduces only additional polynomials in u. Differentiating the preceding integral therefore gives

$$\mathcal{W} - \mathcal{W}_0 = \delta Y R(X, Y), \qquad \|R\|_{C^k} \le C_k.$$

The construction gives two additional derivatives whenever needed. Since $\mathcal{W}_{0,X} > c > 0$ and $\mathcal{W}_{0,Y} < -c < 0$, the same signs and uniform inverse bounds hold for $\mathcal{W}$. If $\chi_0 = (w^2 + 2B_0 Y)^{1/2}$, the identity

$$\mathcal{W}_0(\chi, Y) - \mathcal{W}_0(\chi_0, Y) = -\delta Y R(\chi, Y)$$

and the positive lower bound on $\mathcal{W}_{0,X}$ give $\chi - \chi_0 = \delta Y O_{C^k}(1)$. Squaring proves (4.13), while $\chi_Y = -\mathcal{W}_Y/\mathcal{W}_X > 0$. □

Remark 4.4 (Continuation of the projected state)**.** If the projected orbit starts in a component where $F > 0$, it cannot cross the first zero of F in finite boundary-flow time. Its inverse is an outward graph of (4.12), since

$$X_{\log Y} = \delta F(w)\chi_w + Y\chi_Y > 0. \tag{4.15}$$

This remains valid as the projected state approaches a double root.

The projected state moves with bounded speed in the boundary-flow time $\delta \log Y$. This bounds the height at which a fixed radial advance can start.

Lemma 4.5 (Height before a radial advance)**.** *In Lemma 4.3, let $|F| \le C_F$, $\delta \sup |E_\chi| \le B_0$ and $X_1 < X_2$. Suppose a solution of (4.12) lies in the domain of $\mathcal{W}$ at the heights $Y_1 \le Y \le Y_2$, with radius at most X_1 at Y_1 and at least X_2 at Y_2. If $8B_0 Y_2 \le X_2^2 - X_1^2$, then*

$$Y_1 \le Y_2 \exp\Big(-\frac{X_2 - X_1}{2C_F\delta}\Big).$$

Proof. At a point of the solution with radius X, height Y and projected state w, (4.13) gives $0 \le X^2 - w^2 = 2Y\{B_0 + \delta E_\chi(w, Y)\} \le 4B_0 Y$. Hence $w \le X_1$ at Y_1. At Y_2 the hypothesis gives $w^2 \ge (X_1^2 + X_2^2)/2$, so $w \ge (X_1 + X_2)/2$. Since $w_{\log Y} = \delta F(w)$ along the solution, w changes by at most $C_F\delta \log(Y_2/Y_1)$ between the two heights, and this change is at least $(X_2 - X_1)/2$. □

4.3. **Contracting height coordinates and the exact phase.** The projected boundary-flow times must now be expressed in the common central height h. We first linearize the height equation over a possibly multidimensional base flow. We then normalize the passage from the central section and use that normalization to fix the phase of each projected orbit. The following height normalization uses contraction in backward τ-time.

Lemma 4.6 (Contracting height coordinate)**.** *Consider a skew system*

$$x_\tau = g(x), \qquad y_\tau = \mathfrak{a}(x)y + y^2\mathcal{N}(x, y)$$

on a base domain $\mathcal{O}$ and $|y| \le 2y_0$, with $\mathfrak{a}, \mathcal{N}$ and their C^{k+1} derivatives bounded for a prescribed finite k. Suppose $\partial_y(\mathfrak{a}y + y^2\mathcal{N}) \ge 2c_y > 0$ there. Fix a set of initial base points $\mathcal{O}_0 \subset \mathcal{O}$, and suppose that for x in a neighborhood of $\mathcal{O}_0$ the backward base flow $x_b(t; x)$ stays in $\mathcal{O}$ with uniform bounds

$$|D_x^j x_b(t; x)| \le C_j(1 + t)^{M_j} \mathrm{e}^{jc_x t}, \qquad 1 \le j \le k + 1,$$

where $(k + 1)c_x < c_y$. After reducing y_0, the function

$$\Theta(x, y) = -\int_0^\infty y_b(t; x, y)\mathcal{N}(x_b(t; x), y_b(t; x, y))\, dt$$

is uniformly C^{k+1} on $\mathcal{O}_0 \times [-y_0, y_0]$, is divisible by y with a bounded C^k quotient, and $\Psi(x, y) = y\mathrm{e}^{\Theta(x,y)}$ has positive height derivative and uniformly bounded inverse height derivatives through order k. Along the system, $\Psi_\tau = \mathfrak{a}(x)\Psi$.

Proof. The backward height satisfies $|y_b(t)| \le |y|\mathrm{e}^{-2c_y t}$. In its variational equations the highest height jet has a negative coefficient bounded away from zero. Each pure base derivative of $\mathfrak{a}y + y^2\mathcal{N}$ contains the decaying height; the other forcings are products of lower height jets and base-flow jets. Variation of constants and induction therefore bound every derivative of the integrand through order $k+1$ by $C_j(1+t)^{m_j}\mathrm{e}^{-(2c_y - jc_x)t}$, after a reduction of the decay margin. The stated inequality makes these bounds integrable. Differentiation under the integral gives $\Theta = O(y)$ and its uniform derivatives; one additional derivative gives the bounded quotient Θ/y. Thus $\Psi_y > c > 0$ on a smaller height interval, with the asserted inverse bounds. Finally the flow property gives $\Theta_\tau = -y\mathcal{N}$, and hence $\Psi_\tau = \mathfrak{a}\Psi$. □

For the passage from the central section, we next normalize the height at a fixed regular radius. On a regular central prefix, the scaled height equation is

$$y_x = \frac{xy}{\delta(K(x) + B(x,y)y)}, \qquad y(0) = h/Q, \qquad K(x) \ge k_0 > 0 \tag{4.16}$$

on a fixed interval. Here $Q > 0$ depends only on parameters. The required derivatives of K, B are uniformly bounded. Use $t = \delta^{-1}\mathcal{A}(x)$, where $\mathcal{A}(x) = \int_0^x u/K(u)\,du$. The height equation becomes

$$y_t = y + y^2\mathcal{N}_t(y), \qquad \mathcal{N}_t(y) = -B/(K + By).$$

The inverse of $x \mapsto t$ is continuous at zero; the normalization below differentiates only the height.

Lemma 4.7 (Exact central-height normalization)**.** *At each fixed terminal radius $x_i > 0$, on a sufficiently small fixed height interval there is a coordinate*

$$\Psi_i(y) = y\exp\Theta_i(y), \quad \Theta_i(0) = 0, \quad \Psi_i'(y) > c, \quad \|\Psi_i\|_{C^k} + \|\Psi_i^{-1}\|_{C^k} \le C_k$$

such that

$$\Psi_i(y(x_i)) = \mathrm{e}^{\mathcal{A}(x_i)/\delta}h/Q. \tag{4.17}$$

These bounds hold on arbitrarily long intervals of boundary time. The same conclusion holds if $K(x) \asymp K_* + x$, $K'(x) \ge c$, $K_*/\delta \ge 1$, with $0 < K_* \le K_{\max}$ and fixed $K_{\max}$, and the other coefficient derivative bounds are uniform.

Proof. Let $T_i = \mathcal{A}(x_i)/\delta$, and let $x_b(u)$ and $y_b(u;y)$ be the radius and height at backward time u. Set

$$Z(u) = \mathrm{e}^u y_b(u;y), \qquad a_\delta(u) = \frac{\mathrm{e}^{-u}}{K(x_b(u))}.$$

The backward height equation becomes

$$Z' = \frac{a_\delta(u)Z^2B(x_b(u), \mathrm{e}^{-u}Z)}{1 + a_\delta(u)ZB(x_b(u), \mathrm{e}^{-u}Z)} =: \mathcal{N}_u(Z).$$

The long-time estimate reduces to a uniform bound on this coefficient and its integral. In both cases of the lemma,

$$\sup_{0\le u\le T_i} a_\delta(u) + \int_0^{T_i} a_\delta(u)\,du \le C.$$

For $K \ge k_0$, this follows from $a_\delta \le \mathrm{e}^{-u}/k_0$. For $K \asymp K_* + x$, let u_δ be the backward time to $x_i/2$. The fixed terminal segment gives $u_\delta \ge c/\delta$ and $K(x_b) \ge c_i > 0$ before that time. Afterwards $K \ge cK_* \ge c\delta$, so

$$\sup_{u\ge u_\delta} a_\delta(u) + \int_{u_\delta}^{T_i} a_\delta(u)\,du \le C\delta^{-1}\mathrm{e}^{-c/\delta}.$$

These bounds are independent of the positive lower endpoint of the allowed K_*-range.

On a sufficiently small fixed Z-interval the denominator in $\mathcal{N}_u$ is at least $1/2$, and

$$|\mathcal{N}_u(Z)| \le Ca_\delta|Z|^2, \qquad |\partial_Z\mathcal{N}_u(Z)| \le Ca_\delta|Z|, \qquad |\partial_Z^j\mathcal{N}_u(Z)| \le C_j a_\delta \quad (j \ge 2).$$

Grönwall's inequality and the finite variational equations, with the uniform integral bound for a_δ, give $|Z(u;y)| \le 2|y|$ and uniform height derivatives of every prescribed finite order on a smaller fixed interval. The first variation is positive and bounded above and below. Set $\Psi_i(y) = Z(T_i;y)$; then $y_b(T_i;y) = h/Q$ gives (4.17), and

$$\Theta_i(y) = \log \frac{\Psi_i(y)}{y} = \int_0^{T_i} \frac{a_\delta Z B(x_b, \mathrm{e}^{-u} Z)}{1 + a_\delta Z B(x_b, \mathrm{e}^{-u} Z)}\, du.$$

The same bounds, with one extra derivative, give $\Theta_i = O(y)$ with uniformly bounded derivatives. Thus $\Psi_i'(0) = 1$, and the inverse function theorem gives the asserted inverse bounds. □

The preceding normalization fixes the terminal radius. We next control the dependence of the central height on that radius. Suppose x ranges in a fixed compact positive interval with $K \ge k_0$, and fix a small terminal height $y_f > 0$. Then the corresponding central height satisfies

$$\delta \log(h(x)/Q) = -\mathcal{A}(x) + \delta M_p(x), \qquad \|M_p\|_{C^k} \le C_k. \tag{4.18}$$

Choose a smaller fixed $x_0 > 0$ with a positive action gap to all terminal radii. On $x \ge x_0$, use time $\tau = x/\delta$. The system is

$$x_\tau = \delta, \qquad y_\tau = \mathfrak{a}(x) y + y^2 \mathcal{N}(x,y), \qquad \mathfrak{a}(x) = x/K(x) > c.$$

Extend the coefficients by fixed smooth cutoffs preserving the positive rate, and reduce the height box to preserve its derivative margin. The backward base $x_b(u) = x - \delta u$ has bounded state derivatives, so Lemma 4.6 applies with $c_x = 0$. It gives a uniformly regular coordinate $\Phi(x,y) = y\mathrm{e}^{\Theta(x,y)}$, with the required state derivative and inverse bounds, satisfying $\Phi_\tau = \mathfrak{a}(x)\Phi$. Compose it at x_0 with Lemma 4.7, whose coordinate there is Ψ_0. The unit

$$W_p(\xi) = \frac{\Phi(x_0, \Psi_0^{-1}(\xi))}{\xi}, \qquad W_p(0) = 1,$$

has uniform derivative bounds, using one extra order at zero. The composition gives

$$\Phi(x, y_f) = \mathrm{e}^{\mathcal{A}(x)/\delta} \frac{h}{Q} W_p(\xi_0), \qquad \xi_0 = \mathrm{e}^{\mathcal{A}(x_0)/\delta} h/Q, \qquad W_p(0) = 1.$$

Rearrange this as

$$\xi_0 W_p(\xi_0) = \mathrm{e}^{-[\mathcal{A}(x) - \mathcal{A}(x_0)]/\delta} \Phi(x, y_f).$$

The positive action gap bounds the right side and its j-th terminal-radius derivative by $C_j \delta^{-j} \mathrm{e}^{-c/\delta}$. The uniformly regular inverse of $\xi \mapsto \xi W_p(\xi)$ preserves these finite-order bounds. Taking logarithms gives (4.18) with $M_p = \log \Phi(x, y_f) - \log W_p(\xi_0)$.

The central-height normalization and the boundary-flow coordinate now give the phase of the projected orbit. At a fixed regular entry x_L, combine Lemmas 4.3 and 4.7. In the notation $Y_i = \delta y_i$, let $\xi = \mathrm{e}^{\mathcal{A}(x_L)/\delta} h/Q$, so that $Y_i = \delta \xi \Upsilon(\xi)$, where $\Upsilon(0) = 1$ and Υ has uniform derivative bounds. Because $F(x_L) > c > 0$, the coordinate shift at entry corresponds to the boundary-flow time

$$\alpha_i(\xi) = \int_{x_L}^{\mathcal{W}(x_L, \delta\xi\Upsilon(\xi))} \frac{dx}{F(x)} = \delta O(\xi).$$

The integral lies in a regular entry collar. At a fixed terminal height Y_o, the resulting formula is

$$w = \Phi_F^{\tau(T)}(x_L), \quad \tau = T + C_p + \delta \Xi(\xi), \quad T = -\delta \log(h/Q), \quad C_p = \delta \log(Y_o/\delta) - \mathcal{A}(x_L). \tag{4.19}$$

Here $\Xi = \alpha_i/\delta - \log \Upsilon$ vanishes at zero and has uniform derivative bounds. Both phase origins are now fixed by the same central height h. Near a boundary root, $F(w)$ may tend to zero; the next lemma bounds the derivatives of the phase correction relative to this drift.

Lemma 4.8 (Phase error at a boundary root)**.** *Suppose the phase in (4.19) satisfies $\tau \ge \tau_0 > 0$, its orbit has $F > 0$, and $F(x_L) \ge f_0 > 0$, $|F'| \le L_F$. For every fixed $N, j \ge 0$, writing $\theta = \tau_T$, one has*

$$|\theta - 1| + \sum_{\ell=1}^{j} |\partial_T^\ell \theta| \le \varepsilon_{N,j}(\delta) F(w)^N, \qquad \varepsilon_{N,j}(\delta) \longrightarrow 0. \tag{4.20}$$

Proof. In terms of ξ, (4.19) gives

$$\xi = (Y_o/\delta) \exp[-\tau/\delta + \Xi(\xi)], \qquad \xi_T = -\xi/\delta, \qquad \theta = 1 - \xi \Xi'(\xi).$$

The left side of (4.20) is at most $C_j \delta^{-j} \xi$. To compare this bound with the drift, use $d \log F / d\tau = F'$ along the boundary flow; hence $F(w) \ge f_0 \mathrm{e}^{-L_F \tau}$. After division by $F(w)^N$, the error is bounded by

$$C_{N,j} \delta^{-j-1} \exp[-(1/\delta - N L_F)\tau] \le C_{N,j} \delta^{-j-1} \mathrm{e}^{-\tau_0/(2\delta)}.$$

Reduction of δ proves the assertion, uniformly up to simple or multiple boundary roots. □

For a no-root bottleneck, we also need height derivatives that remain uniform as its action grows. On a fixed interval $[x_L, x_R]$ suppose $F > 0$, with positive margins only in fixed collars of its endpoints. Define

$$\alpha_j(Y) = \int_{x_j}^{\mathcal{W}(x_j, Y)} \frac{dx}{F(x)}, \qquad \Psi_j(y) = y \exp[-\alpha_j(\delta y)/\delta], \quad j = L, R.$$

These are uniformly regular height coordinates, since $\alpha_j(Y) = Y \mathcal{A}_j(Y)$. The exact identity is

$$\Psi_R(y_R) = \exp\left(\frac{1}{\delta} \int_{x_L}^{x_R} \frac{dx}{F(x)} \right) \Psi_L(y_L). \tag{4.21}$$

For a small outgoing height, the backward projected flow and (4.15) reconstruct the outward passage. Let $\mathcal{A}_L$ be the action of the preceding regular prefix and write its inverse height coordinate as $y_L = P_p(\zeta)$, where $\zeta = \mathrm{e}^{\mathcal{A}_L/\delta} h/Q$. By Lemma 4.7, $P_p(0) = 0$, $P_p'(0) = 1$, and its derivatives through the prescribed order are uniformly bounded on a fixed small interval. Put

$$\mathcal{A}_f = \int_{x_L}^{x_R} \frac{dx}{F(x)}, \qquad \beta = \mathrm{e}^{-\mathcal{A}_f/\delta} \in (0, 1],$$
$$\xi = \mathrm{e}^{(\mathcal{A}_L + \mathcal{A}_f)/\delta} h/Q, \qquad \mathcal{C}_p = \Psi_L \circ P_p.$$

Thus $\zeta = \beta\xi$ and $\mathcal{C}_p(x) = x \Upsilon_p(x)$, with $\Upsilon_p(0) = 1$. Composition with the prefix gives

$$\begin{aligned} \Psi_R(y_R) &= \beta^{-1} \mathcal{C}_p(\beta\xi) = \xi \Upsilon_p(\beta\xi), \\ y_R &= \Psi_R^{-1}(\xi \Upsilon_p(\beta\xi)). \end{aligned} \tag{4.22}$$

The factor β^{-1} can be large, but differentiation of the argument $\beta\xi$ contributes powers of β. With β fixed, the derivatives in the expanded height ξ are

$$\partial_\xi^j [\beta^{-1} \mathcal{C}_p(\beta\xi)] = \beta^{j-1} \mathcal{C}_p^{(j)}(\beta\xi), \qquad j \ge 1.$$

Its first derivative has a uniform positive lower bound, and all prescribed higher derivatives are bounded because $\beta \le 1$. Choose a fixed $\xi_* > 0$ so that $\beta\xi$ lies in the prefix domain and $\xi \Upsilon_p(\beta\xi)$ lies in the common domain of Ψ_R^{-1} for $0 \le \xi \le \xi_*$. The resulting map $\xi \mapsto y_R$ has a uniformly positive first derivative; its image contains a fixed small terminal-height interval, and its inverse has uniformly bounded derivatives there. Constructing one extra derivative before dividing by ξ gives $y_R = \xi V_p(\xi)$, where $V_p(0) = 1$, and both V_p and its reciprocal have uniform derivatives through the prescribed order. These bounds for derivatives in the expanded height ξ are uniform in $\mathcal{A}_f$.

4.4. **Endpoint derivatives and the logarithmic slope ratio.** We now transfer the drift estimates to the endpoints. Composing the inverse boundary projection with the weighted tail gives the derivative in the projected radius; the phase formula then converts it to a derivative in T. Keeping the drift factor through this conversion allows comparison with the exterior contribution to the logarithmic slope ratio.

Lemma 4.9 (Endpoint after a boundary-flow passage)**.** *Let $\ell > 0$, and write $|\eta| = \ell X$ and $z = \ell^2 Y$ on one branch. Suppose that on a fixed positive radius interval the height equation has the form (4.12) with small parameter δ, and let w be the projected state of Lemma 4.3. Fix a small height $Y_o > 0$ and a compact interval $J_w \subset (0,\infty)$ on which (4.13) holds at $Y = Y_o$, and put*

$$t = \ell^2 Y_o, \qquad \varepsilon_o = \ell + \frac{|q_0|}{t} + \frac{|q_1|}{\sqrt{t}} + |q_2|, \qquad E = \ell^2 \Big(\frac{z_*}{t}\Big)^{-2q_2}.$$

For δ and ε_o sufficiently small, the endpoint I of the graph whose projected state at height t is $w \in J_w$ satisfies

$$-I_w = E A_p w\, U_p(w), \qquad c < A_p < C, \qquad U_p = 1 + O_{C^3}(\delta + \varepsilon_o), \tag{4.23}$$

where A_p depends only on the parameters and U_p is a dimensionless positive unit factor. The constants depend on J_w, Y_o and the fixed boundary-flow data. If the two branches use the same ℓ and Y_o, then

$$\left|\log \frac{A_-}{A_+}\right| \le 2C_{\mathrm{cone}}\sqrt{z_*} + O(\delta + \varepsilon_o), \tag{4.24}$$

where $A_\pm$ are the values of A_p on the negative and positive branches, respectively. The branch labels are fixed throughout the passage; later contraction estimates use only the upper one-sided consequence of this absolute comparison.

Proof. By (4.13), the energy at height t is $U(t) = \ell^2\{w^2/2 + B_0 Y_o + \delta Y_o E_\chi(w, Y_o)\}$, and its w-derivative is $\ell^2 w\{1 + O_{C^3}(\delta)\}$ on J_w. The normalized energies $U(t)/t$ lie in a fixed compact positive interval. If this interval is not inside the common cone, continue the graphs first by (4.2) from Y_o to a fixed larger Y_1, whose energies lie in that cone. There the error is $O_{C^k}(\varepsilon_o)$ and the first derivative is $1 + O(\varepsilon_o)$, since $\varepsilon_{\mathrm{in}} \le C\varepsilon_o$ for fixed Y_o, Y_1. Lemma 4.1 at $t_1 = \ell^2 Y_1$ gives a centered endpoint map with slope $t_1(z_*/t_1)^{-2q_2} a$, where $c < a < C$ and, by (4.4), $|\log a| \le C_{\mathrm{cone}}(\sqrt{z_*} + |q_1|/\sqrt{t_1})$, and with a relative C^3 error $O(e(t_1)) = O(\varepsilon_o)$. Since $(t_1/t)^{2q_2} = (Y_1/Y_o)^{2q_2} = 1 + O(\varepsilon_o)$, the chain rule gives (4.23). The factors ℓ^2, $(z_*/t)^{-2q_2}$ and $(Y_1/Y_o)^{2q_2}$ are common to the two branches, which gives the bound for A_-/A_+. □

To compute the logarithmic rate in the common height coordinate, consider an endpoint map on a projected-radius interval contained in a fixed compact subset of $(0,\infty)$, of the form

$$-I_w = E A_p w\, U_p(w), \qquad c < A_p < C, \quad U_p = 1 + O_{C^3}(\delta), \quad 0 < E \longrightarrow 0, \tag{4.25}$$

as given by Lemma 4.9 when $\varepsilon_o = O(\delta)$; its amplitude satisfies $E \le C\ell^{3/2}$ by (4.10) with $t = \ell^2 Y_o$. Here $E = E(p)$ and A_p are independent of the central height. Use the phase (4.19), so that $w_T = \theta F(w)$, with $\theta = \tau_T$ and $F = K/w > 0$. Put $m = hI_h = -\delta I_T$, $\mathfrak{H} = \partial_T \log m$. In the identities below, primes on $\theta, \mathfrak{H}$ denote T-derivatives, and primes on $K, F, \log U_p, \mathfrak{H}_0$ denote w-derivatives:

$$\begin{aligned} m &= \delta E A_p K(w)\theta U_p(w), \\ \mathfrak{H} &= \theta\mathfrak{H}_0(w) + \theta'/\theta, \qquad \mathfrak{H}_0 = K'/w + F(\log U_p)', \\ \mathfrak{H}' &= \theta^2 F\mathfrak{H}_0' + \theta'\mathfrak{H}_0 + \theta''/\theta - (\theta'/\theta)^2, \\ \mathfrak{H}_0' &= (wK'' - K')/w^2 + F'(\log U_p)' + F(\log U_p)''. \end{aligned} \tag{4.26}$$

By Lemma 4.8, these identities imply

$$\mathfrak{H}' = F(w)\left\{\frac{wK'' - K'}{w^2} + o(1)\right\}, \qquad |I_T| + |I_{TT}| \le CEF(w). \tag{4.27}$$

The errors are uniform up to a limiting boundary root, with the drift factor retained in both estimates.

For the two endpoints use the same logarithmic height coordinate $T = -\delta \log(h/Q)$ in the logarithmic slope ratio (3.1). The exterior contribution is

$$\partial_T^2[(\log \Pi_p')(I_-)] = (\log \Pi_p')''(I_-)I_{-,T}^2 + (\log \Pi_p')'(I_-)I_{-,TT}. \tag{4.28}$$

By (4.27), this term is $O(EF_- + E^2F_-^2)$.

The exterior contribution has the same drift weight as the negative endpoint estimate and carries the vanishing amplitude E. This leaves the coefficients $Q_\sigma = (wK_\sigma'' - K_\sigma')/w^2$ of the two drifts to determine the principal curvature of $\mathcal{L}$.

Lemma 4.10 (Curvature criterion)**.** *On an interval of central heights, suppose that both endpoints have the form (4.25) in the common logarithmic height coordinate $T = -\delta \log(h/Q)$, that their phases satisfy the hypotheses of Lemma 4.8, and that the negative projected state stays in a fixed compact subset of $(0, \infty)$. If $Q_- \le -c$ and $Q_+ \ge c$ there, then $\mathcal{L}'' < 0$ for all sufficiently small parameters, uniformly up to simple or multiple boundary roots.*

Proof. By (4.27), $\mathfrak{H}_\sigma' = F_\sigma(w_\sigma)\{Q_\sigma + o(1)\}$. The exterior term (4.28) is $O(EF_- + E^2F_-^2) = o(1)F_-$, since F_- is bounded and $E \to 0$. Hence

$$\mathcal{L}'' = \mathfrak{H}_-' - \mathfrak{H}_+' + \partial_T^2[(\log \Pi_p')(I_-)] \le -\tfrac{c}{2}(F_- + F_+) < 0. \qquad \square$$

For a drift $K(w) = K(0) + K'(0)w + \frac{1}{2}K''(0)w^2 + w^3R_K(w)$, direct differentiation gives

$$\frac{wK'' - K'}{w^2} = -\frac{K'(0)}{w^2} + 3R_K(w) + 5wR_K'(w) + w^2R_K''(w). \tag{4.29}$$

In the weak-root range, the first rate identity gives a one-sided slope comparison. The next result expresses it in the common height coordinate, retaining the change of scale from the positive branch's coordinate.

Corollary 4.11 (Weak-root slope comparison)**.** *Let $I_\pm(h)$ be the C^2 endpoint maps on one interval of central heights, with $I_{\pm,h} > 0$, and let $a, \kappa > 0$. Suppose the positive endpoint map satisfies $-I_{+,w} = EA_pwU_p(w)$ with $E > 0$, $c < A_p < C$ and $U_p = 1 + O_{C^2}(\kappa)$, where E, A_p are independent of the central height. The radius interval satisfies $0 < w_* \le w_a \le w < \alpha \le w^* < \infty$, with fixed w_*, w^*, and its exact phase uses $\widehat{T} = -\kappa \log h$. Assume the weighted-phase lemma applies there, $F = K/w > 0$, the C^2 bounds for K are uniform, and $K'(w)/w \le -c_0 < 0$. In the common logarithmic height coordinate $T = a\widehat{T}$, put $\mathfrak{H}_\sigma = \partial_T \log(hI_{\sigma,h})$. Then, for sufficiently small κ,*

$$\mathfrak{H}_+ \le -\frac{c_0}{2a}. \tag{4.30}$$

If also $\mathfrak{H}_- \ge -C_n$, $|I_{-,T}| \le \varepsilon_p \to 0$, and the C^2 exterior map on these endpoint windows has positive slope with $|(\log \Pi_p')'| \le M$, then, for sufficiently small a, $\mathcal{L}_T > 0$ on this interval, and the return displacement has at most two zeros there. All bounds, the phase-lemma constants, and the convergence $\varepsilon_p \to 0$ are uniform on a fixed preliminary parameter region with small a.

Proof. With $\theta = \partial_{\widehat{T}}\tau$, (4.26) gives

$$a\mathfrak{H}_+ = \theta\{K'/w + F(\log U_p)'\} + \theta_{\widehat{T}}/\theta.$$

The bounded radius interval and the stated phase estimates make the error $o(1)$, proving (4.30). In the coordinate T,

$$\mathcal{L}_T = (\log \Pi_p')'(I_-)I_{-,T} + \mathfrak{H}_- - \mathfrak{H}_+ \geq \frac{c_0}{2a} - C_n - M\varepsilon_p > 0.$$

Fix the constants on the preliminary region, reduce a, and then choose the phase, tail, and parameter thresholds. Since $\mathcal{L}_T > 0$, (3.1) gives the two-zero count. □

5. Contact with the central section and positive offsets

We begin with two regions in which Δ_p has at most two zeros: the contact scale $q_0 = \widehat{q}_0 q_1^2$ with bounded $\widehat{q}_0$, and positive offsets with small $q_1/\sqrt{q_0}$. In the first, a scattering comparison controls the derivative ratio on a compact range of central heights, while the paired identity controls the two ends of the passing interval. In the second, the Dulac map of an interior saddle determines the crossing direction of the logarithmic slope ratio near the separatrices; exterior contraction controls the remaining heights.

In this section and in Sections 6 to 10 and 12, the height z_* of the fixed section, the exterior windows, the common cone, and the landing level m_{land} are those chosen in the first row of Table 4. All scaled thresholds and parameter bounds below are chosen after these fixed data.

The constant C_{cone} in (4.4) is independent of the contact width and the later coefficient margins and regional cuts. The continuation constants and higher exterior derivatives may depend on these cuts and on z_*; they enter only the final parameter thresholds, with the sections fixed. The section requirements are justified in the proof of Proposition 11.1 and the derivation of (12.17).

5.1. The contact strip.

We treat the contact strip, with fixed $L < \infty$,

$$q_1 > 0, \qquad q_0 = \widehat{q}_0 q_1^2, \qquad |\widehat{q}_0| \leq L. \tag{5.1}$$

The limits of q_1 and q_2 are independent, including $q_2 = 0$; constants may depend on L and C_0. The difficulty is to compare the two passages uniformly through contact, without relating the sizes of q_1 and q_2. The paired identity gives opposite derivative signs at the two ends of the height interval. On the compact interval between them, Lemma 5.2 gives a strictly decreasing ratio of the limiting scattering derivatives. We transfer its derivative margin to the endpoint maps by a C^2 estimate. Every middle zero of $\mathcal{L}$ then crosses upwards in T, and the outer signs complete the count on $\mathcal{H}_p$.

Proposition 5.1 (Bounded contact offsets)**.** *For every fixed L, the return displacement (2.23) has at most two distinct zeros on the height interval $\mathcal{H}_p$ in (5.1), for all sufficiently small parameters.*

5.1.1. *Barrier and tail estimates.*

With the contact scales of Table 2, put

$$d_{\mathrm{c}} = \max\{|q_2|, \sqrt{q_1}\}, \qquad z = q_1^2 Z, \qquad h = q_1^2 \widehat{h}, \qquad |\eta_\sigma| = q_1 u_\sigma, \qquad \mathcal{U}_\sigma = u_\sigma^2/2.$$

The rescaled equations are

$$(\mathcal{U}_\sigma)_Z = B(\sigma q_1 u_\sigma, q_1^2 Z) + \frac{-\widehat{q}_0 - \sigma u_\sigma - q_2 u_\sigma^2 - \sigma q_1 u_\sigma^3 R_q(\sigma q_1 u_\sigma)}{Z}. \tag{5.2}$$

They are the graphs of the analytic outward time equations

$$(u_\sigma)_\tau = -\widehat{q}_0 - \sigma u_\sigma - q_2 u_\sigma^2 - \sigma q_1 u_\sigma^3 R_q + ZB, \qquad Z_\tau = u_\sigma Z. \tag{5.3}$$

The central point is $u_\sigma = 0,\ Z = \widehat{h}$. Its radial flux is

$$f(\widehat{h}) = \widehat{h} B(0, q_1^2 \widehat{h}) - \widehat{q}_0, \qquad f_{\widehat{h}} = B(0, q_1^2 \widehat{h}) + q_1^2 \widehat{h} B_z(0, q_1^2 \widehat{h}) \geq b_0.$$

Thus all $\widehat{h} > 0$ pass when $\widehat{q}_0 \leq 0$; when $\widehat{q}_0 > 0$ the passing interval is $\widehat{h} > \widehat{h}_0$, where $f(\widehat{h}_0) = 0$.

The negative graph has the uniform upper barrier

$$U_-(z) \leq C(d_{\mathrm{c}}^2 + z), \qquad |\eta_-(z)| \leq C(d_{\mathrm{c}} + \sqrt{z}). \tag{5.4}$$

Indeed, for $x \geq Kd_{\mathrm{c}}$, the coefficient bounds give

$$q(-x) \geq cx^3 - d_{\mathrm{c}}x^2 - q_1 x - Lq_1^2 \geq 0.$$

After division by x^3, the three adverse terms are at most $K^{-1}, K^{-2}, Ld_{\mathrm{c}}/K^3$. Choose K large and then d_{c} small. On the line $U = C(d_{\mathrm{c}}^2 + z)$, with $\sqrt{2C} > K$, the graph derivative $B - q(-\sqrt{2U})/z$ is at most $\sup B < C$. The line starts above the central point, so first contact proves (5.4).

To continue from bounded rescaled central heights to the fixed section, we first obtain the following cone bounds. For each fixed bounded set of rescaled central heights, they hold for a sufficiently large fixed Z_0:

$$cz \leq U_{\pm}(z), \qquad U_{+}(z) \leq Cz, \qquad z_0 := q_1^2 Z_0 \leq z \leq z_*. \tag{5.5}$$

For continuation on a fixed interval $\widehat{h} \leq Z \leq Z_1$, first bound the positive radius by M_{L,Z_1}. At a fixed $M > 2(L + (\sup B)Z_1 + 1)$, the first right side of (5.3) is negative, after reducing $|q_2|$ and q_1. Hence the graph cannot first reach M. Choose Z_1 larger than twice these rescaled central heights and so large that $L/Z_1 < b_0/4$. On the lower comparison line $\mathcal{U} = m(Z - Z_1)$, the adverse terms in (5.2) are bounded by

$$L/Z_1 + C\sqrt{m/Z_1} + 2d_{\mathrm{c}}m + Cm^{3/2}\sqrt{z_*}.$$

Choose $m > 0$ small after Z_1, and use the common-cone choice of z_* already fixed before the regional thresholds. Reducing d_{c} and q_1 after these choices makes the graph derivative greater than m. This proves a lower bound for both signs. For the positive upper bound use $W = C_2 Z + C_3$, choosing C_3 to dominate the energy at Z_1 and to have $\sqrt{2W} \geq 2L$. The term $-\widehat{q}_0 - \sqrt{2W}$ is then nonpositive, and the remaining derivative is at most

$$\sup B + 2d_{\mathrm{c}}W/Z + Cq_1 W^{3/2}/Z < C_2$$

for C_2 fixed large and the same successive reductions. Enlarge Z_0 relative to Z_1, C_3. These comparisons give (5.5), with fixed cone constants, up to z_*. Together with (5.4) they keep both graphs inside the box in the coordinates (η, z).

The negative graph may remain above the upper cone for part of the passage. To estimate the resulting loss in the paired identity, set $v_0 = |\eta_-(z_0)|/q_1$ and $t_h = \log(Z_0/\widehat{h})$. For $z \geq z_0$, discard the favorable negative cubic term and use $q_1\sqrt{2U} \leq U/8 + 4q_1^2$. Taking $Z_0 \geq L + 4$ and then d_{c} small yields

$$U_-' \leq C + \tfrac{1}{2}U_-/z, \qquad U_-(z) \leq Cz + U_-(z_0)(z/z_0)^{1/2}. \tag{5.6}$$

The second term is at most z once $z \geq z_0 \max\{1, (U_-(z_0)/z_0)^2\}$. Below that height use (5.4); above it use the resulting upper cone. Splitting the integral, also when this height exceeds z_*, gives

$$\int_{z_0}^{z_*} \frac{|\eta_-(z)|}{z}\, dz \leq C\sqrt{z_*} + Cd_{\mathrm{c}} \log(1 + v_0). \tag{5.7}$$

The tail loss must be smaller than the gain in the paired identity at small central heights. We therefore express its last logarithm in terms of the central height. In the negative equation use $t = \log(Z/\widehat{h})$ and $\mathcal{V} = \sqrt{1 + u_-^2}$. Dropping the favorable cubic term, and using $|\widehat{q}_0| \leq L$, gives

$$\mathcal{V}_t \leq C_L + d_{\mathrm{c}}\mathcal{V} + (\sup B)\widehat{h}\mathrm{e}^t, \qquad \mathcal{V}(t) \leq \mathrm{e}^{d_{\mathrm{c}}t}(1 + C_L t) + C\widehat{h}\mathrm{e}^t.$$

In particular,

$$\log(1 + v_0) \leq C_{L,Z_0} + d_{\mathrm{c}}t_h + \log(1 + t_h). \tag{5.8}$$

This bounds the tail loss in terms of t_h, uniformly in $d_{\mathrm{c}}|\log q_1|$.

5.1.2. *The two ends of the height interval.* For the fixed contact width L, choose Z_0 after the bounded-height comparison, then fix the cone and barrier constants. Reduce q_1 so that

$$q_1 Z_0 \le \frac{b_1}{4C_B}, \qquad z_0 = q_1^2 Z_0 \le \frac{q_1}{4C_B}, \tag{5.9}$$

where C_B bounds $|B_\eta|$ on the comparison box and b_1 is the positive coefficient margin in (2.12). With z_* already fixed by the common-cone construction, reduce $|q_2|$ (and hence d_c) relative to this z_*, after q_1. The absorptions below are then uniform in the independent limits of q_1 and q_2.

For $\widehat{q}_0 \le 0$, choose initially $\widehat{h} \le 1/2$. On $h \le z \le z_0$, the terms $B_\eta/|\eta_\sigma|$ in the paired identity (2.31) are absorbed by half of the positive terms $q_1/(|\eta_\sigma|z)$: indeed, (5.9) gives $z \le z_0 \le q_1/(4C_B)$, so $|B_\eta|/|\eta_\sigma| \le q_1/(4|\eta_\sigma|z)$. The fixed positive-radius bound gives a gain at least $c_L t_h$. By (5.4), the cubic loss in this prefix is at most $C_L d_c t_h$. On $z \ge z_0$, the lower cones bound the $B_\eta/|\eta_\sigma|$ integrals; the positive upper cone bounds the cubic integral on the positive branch, and (5.7) bounds it on the negative branch. Therefore

$$\log \frac{I_{-,h}}{I_{+,h}} \ge (c_L - C_L d_c)t_h - C\sqrt{z_*} - C d_c \log(1+v_0) \ge \tfrac{1}{2} c_L t_h - C_L.$$

The last step uses (5.8) and $\log(1+t_h) \le t_h$, reducing d_c after L, Z_0 and (5.9) so that the coefficient of t_h remains at least $c_L/2$. Choose a fixed $\widehat{h}_{\mathrm{lo}} > 0$ so small that this bound exceeds $\log(1/\Lambda_-)$ for $\widehat{h} \le \widehat{h}_{\mathrm{lo}}$. Then $\Delta_h > 0$ there.

For $\widehat{q}_0 \ge 0$, the lower end of the passing interval is controlled by the radial flux. Consider central heights with small positive flux $0 < f \le 1$. Then

$$\widehat{h} \le (L+1)/b_0, \qquad \widehat{h} \ge f/\sup B.$$

For $0 \le t \le 1$, subtracting the value of ZB at the central point in (5.2) gives

$$(u_-^2/2)_t \le C_L u_- + d_c u_-^2 + f + C_L t.$$

The explicit upper comparison

$$\bar{v} = K(t + \sqrt{ft}), \qquad (\bar{v}^2/2)' = K^2(t + \tfrac{3}{2}\sqrt{ft} + f/2)$$

proves $u_- \le \bar{v}$. Choose K large first to dominate the linear, f, and t terms, then d_c small to absorb $d_c \bar{v}^2$. A fixed provisional radius box larger than $4K$ completes the first-contact continuation argument. Thus the favorable negative-branch term in the paired identity satisfies

$$\int_0^1 \frac{dt}{u_-(t)} \ge \frac{2}{K} \log(1 + f^{-1/2}) \ge c_L \log(1/f) - C_L.$$

Choose $Z_0 > \mathrm{e}(L+1)/b_0$, so this interval lies in the prefix. The other prefix losses and the whole tail are bounded as above. Since $t_h \le \log(C_L/f)$, the result is

$$\log \frac{I_{-,h}}{I_{+,h}} \ge \tfrac{1}{2} c_L \log(1/f) - C_L. \tag{5.10}$$

Choose $f_{\mathrm{lo}} > 0$ so small that the right side exceeds $\log(1/\Lambda_-)$ for $f \le f_{\mathrm{lo}}$. Thus $\Delta_h > 0$ on this interval, whose connectedness follows from $f_{\widehat{h}} > 0$. This estimate is uniform as $f/\widehat{h} \to 0$.

At the other end let θ_0 be the constant of Lemma 2.10, and choose $\widehat{h}_{\mathrm{hi}} > (L+1)/b_0 + \widehat{h}_{\mathrm{lo}}$ so large that $L/\widehat{h}_{\mathrm{hi}} + \widehat{h}_{\mathrm{hi}}^{-1/2} \le \theta_0/2$. For $h \ge q_1^2 \widehat{h}_{\mathrm{hi}}$,

$$|q_0|/h \le L/\widehat{h}_{\mathrm{hi}}, \qquad q_1/\sqrt{h} \le \widehat{h}_{\mathrm{hi}}^{-1/2},$$

and $|q_2| \le \theta_0/2$ for small parameters. Hence Lemma 2.10 applies and gives $\Delta_h < 0$ at these central heights.

Write $\widehat{h}_{\mathrm{a}} = \widehat{h}_{\mathrm{lo}}$ when $\widehat{q}_0 \le 0$, and otherwise let $\widehat{h}_{\mathrm{a}}$ be the unique solution of $f(\widehat{h}_{\mathrm{a}}) = f_{\mathrm{lo}}$. Put $\widehat{h}_{\mathrm{b}} = \widehat{h}_{\mathrm{hi}}$. Taking $f_{\mathrm{lo}} \le 1$, the choice of $\widehat{h}_{\mathrm{b}}$ gives $f(\widehat{h}_{\mathrm{b}}) > 1$ for $\widehat{q}_0 > 0$, so $\widehat{h}_{\mathrm{a}} < \widehat{h}_{\mathrm{b}}$ in both cases. At

central heights in $\mathcal{H}_p$, the displacement has $\Delta_h > 0$ for $\widehat{h} \le \widehat{h}_{\rm a}$ and $\Delta_h < 0$ for $\widehat{h} \ge \widehat{h}_{\rm b}$, including the height thresholds whenever they are admissible. The middle interval is $[\widehat{h}_{\rm a}, \widehat{h}_{\rm b}]$. On bounded heights,

$$f(\widehat{h}) - (B_0\widehat{h} - \widehat{q}_0) = O(q_1^2\widehat{h}^2), \qquad B_0 = B(0,0,p).$$

Consequently the central heights of the middle interval belong, after parameter reduction, to

$$\mathcal{K}_{\rm c} = \left\{ (\widehat{h}, \widehat{q}_0, B_0) : \begin{array}{c} b_0 \le B_0 \le B_{\max}, \quad |\widehat{q}_0| \le L, \\ h_* \le \widehat{h} \le \widehat{h}_{\rm hi}, \quad B_0\widehat{h} - \widehat{q}_0 \ge f_* \end{array} \right\}, \tag{5.11}$$

where

$$h_* := \min\{\widehat{h}_{\rm lo}, f_{\rm lo}/(2B_{\max})\} > 0, \qquad f_* := \min\{b_0\widehat{h}_{\rm lo}, f_{\rm lo}/2\} > 0.$$

Here $B_{\max}$ is a fixed upper bound for B_0. Every fixed $(\widehat{q}_0, B_0)$ fibre of $\mathcal{K}_{\rm c}$ is an interval, possibly empty. This compact set lies strictly inside the model passing domain

$$|\widehat{q}_0| \le L, \qquad b_0 \le B_0 \le B_{\max}, \qquad \widehat{h} > \max\{0, \widehat{q}_0/B_0\}. \tag{5.12}$$

For the estimates below, choose one fixed $\widehat{h}_{\rm ref} > \widehat{h}_{\rm hi} + 1 + 2L/b_0$, and let $\mathcal{K}_{\rm c}^+$ be the compact set obtained from $\mathcal{K}_{\rm c}$ by replacing its upper height bound by $\widehat{h}_{\rm ref} + 1$ and its lower height and flux bounds by $h_*/2$ and $f_*/2$. Every fibre contains the reference height $\widehat{h}_{\rm ref}$ and is still an interval. By the estimate of f above, the radial flux at the central point is positive on a neighborhood of this enlarged set for small parameters.

5.1.3. *Scattering and a strictly decreasing derivative ratio.* The outer signs leave only the compact passing set $\mathcal{K}_{\rm c}$. On this set we compare the derivatives of the limiting scattering maps, and then transfer their strict ratio margin to the endpoint maps. For fixed contact heights the limiting equations are

$$(\mathcal{U}_\sigma^0)_Z = B_0 - \frac{\widehat{q}_0 + \sigma\sqrt{2\mathcal{U}_\sigma^0}}{Z}, \qquad \mathcal{U}_\sigma^0(\widehat{h}; \widehat{h}) = 0. \tag{5.13}$$

Define

$$S_\sigma(\widehat{h}; \widehat{q}_0, B_0) = \lim_{Z\to\infty} [\mathcal{U}_\sigma^0(Z; \widehat{h}_{\rm ref}) - \mathcal{U}_\sigma^0(Z; \widehat{h})]. \tag{5.14}$$

The limit exists with local C^2 convergence of order $O(Z^{-1/2})$ in $\widehat{h}$, uniformly on compact passing sets. Fix a height beyond all the rescaled central heights of such a set. The time equations (5.3) with $q_1 = q_2 = 0$ have positive initial radial flux, and their solutions meet this height section transversely. The resulting finite-height maps are analytic. The limiting barriers put both graphs in $cZ \le \mathcal{U}_\sigma^0 \le CZ$ above a sufficiently large fixed cut. There

$$\partial_{\mathcal{U}}\mathcal{G}_\sigma^0 = -\frac{\sigma}{Z\sqrt{2\mathcal{U}_\sigma^0}}, \qquad |\partial_{\mathcal{U}}\mathcal{G}_\sigma^0| \le CZ^{-3/2}, \qquad |\partial_{\mathcal{U}}^2\mathcal{G}_\sigma^0| \le CZ^{-5/2}.$$

The mean-value equation for two graphs and the triangular first- and second-variation equations have these integrable coefficients. Hence the graph difference and its first two $\widehat{h}$-derivatives remain bounded, and their Z-derivatives are $O(Z^{-3/2})$. Integration to infinity gives the asserted C^2 convergence and rate. The same majorants give continuous dependence on $(\widehat{q}_0, B_0)$, and $S_\sigma(\widehat{h}_{\rm ref}) = 0$.

The variation formula for a moving initial point gives

$$S_\sigma'(\widehat{h}) = \left(B_0 - \frac{\widehat{q}_0}{\widehat{h}}\right) \exp\left[-\sigma \int_{\widehat{h}}^{\infty} \frac{dZ}{Z\sqrt{2\mathcal{U}_\sigma^0(Z; \widehat{h})}}\right] > 0. \tag{5.15}$$

Indeed, the initial factor of $\partial_{\widehat{h}}\mathcal{U}_\sigma^0$ is $-(B_0 - \widehat{q}_0/\widehat{h})$ for both signs. The integral in (5.15) converges both at the central point and at infinity.

Lemma 5.2 (Scattering derivative ratio)**.** *For fixed $(\widehat{q}_0, B_0)$, the function*

$$\mathscr{R}(\widehat{h}) := S'_-(\widehat{h})/S'_+(\widehat{h})$$

is strictly decreasing throughout the passing interval (5.12). On each fixed compact passing set there are constants $c_1, c_2 > 0$ such that

$$S'_+ \geq c_1, \qquad -\mathscr{R}' \geq c_2. \tag{5.16}$$

Proof. Put $t = \log(Z/\widehat{h})$ and $W_\sigma(t;\widehat{h}) = \mathcal{U}^0_\sigma(\widehat{h}\mathrm{e}^t;\widehat{h})$. The initial time is now fixed, and

$$(W_\sigma)_t = B_0\widehat{h}\mathrm{e}^t - \widehat{q}_0 - \sigma\sqrt{2W_\sigma}, \qquad W_\sigma(0;\widehat{h}) = 0.$$

Take $V_\sigma = \partial_{\widehat{h}} W_\sigma$ at fixed t. With $f^0 = B_0\widehat{h} - \widehat{q}_0 > 0$, the regular time equations at the central point give, uniformly on compact passing sets,

$$W_\sigma = f^0 t + O(t^{3/2}), \qquad V_\sigma = B_0 t + O(t^{3/2}).$$

For example, in the analytic time variable τ, $u_\sigma = f^0\tau + O(\tau^2)$ and $t = f^0\tau^2/2 + O(\tau^3)$; inversion in $\sqrt{t}$ gives these expansions with their derivatives in $\widehat{h}$. For $t > 0$ the variation equation is

$$(V_\sigma)_t = B_0\mathrm{e}^t - \frac{\sigma V_\sigma}{\sqrt{2W_\sigma}}, \qquad V_\sigma(0) = 0.$$

Its coefficient is $O(t^{-1/2})$ at zero. Variation of constants, first from positive time and then by passage to zero, yields

$$V_\sigma(t) = B_0 \int_0^t \mathrm{e}^{\bar{t}} \exp\left[-\sigma \int_{\bar{t}}^t \frac{du}{\sqrt{2W_\sigma(u)}}\right] d\bar{t} > 0. \tag{5.17}$$

For large t, the cone estimates give $W_\sigma \asymp \mathrm{e}^t$. The tail coefficient of this variation equation is integrable, so (5.17) gives $V_\sigma = O(\mathrm{e}^t)$. Thus both $(2W_\sigma)^{-1/2}$ and $V_\sigma(2W_\sigma)^{-3/2}$ are bounded by $Ct^{-1/2}$ near zero and by $C\mathrm{e}^{-t/2}$ at infinity, uniformly on every compact passing set. Dominated differentiation gives

$$\mathcal{T}_\sigma(\widehat{h}) := \int_0^\infty \frac{dt}{\sqrt{2W_\sigma(t;\widehat{h})}}, \qquad \mathcal{T}'_\sigma = -\int_0^\infty \frac{V_\sigma(t;\widehat{h})}{(2W_\sigma(t;\widehat{h}))^{3/2}}\, dt < 0,$$

and $\mathcal{T}'_\sigma$ is continuous in the passing parameters. The common positive factor $B_0 - \widehat{q}_0/\widehat{h}$ in (5.15) cancels, giving

$$\mathscr{R} = \exp(\mathcal{T}_- + \mathcal{T}_+), \qquad \mathscr{R}' = \mathscr{R}(\mathcal{T}'_- + \mathcal{T}'_+) < 0.$$

Continuity and compactness give (5.16). □

5.1.4. *Endpoint preparation and the two-zero bound.* To transfer the scattering comparison to the endpoint maps, we must retain the quadratic tail factor even when its logarithm is unbounded. The common endpoint amplitude is

$$E = q_1^2(z_*/q_1^2)^{-2q_2}. \tag{5.18}$$

By (4.10) with $t = q_1^2$, $E \leq z_*^{1/4} q_1^{3/2} \to 0$ for $|q_2| < 1/8$, while $q_2 \log q_1$ may be unbounded. Let $b_\sigma = I_\sigma(q_1^2\widehat{h}_{\mathrm{ref}})$ and set

$$A_\sigma(p) = \frac{\partial_{\widehat{h}} I_\sigma(q_1^2\widehat{h})|_{\widehat{h}=\widehat{h}_{\mathrm{ref}}}}{E S'_\sigma(\widehat{h}_{\mathrm{ref}})}.$$

On the enlarged compact passing set above,

$$I_\sigma(q_1^2\widehat{h}) = b_\sigma + EA_\sigma(p)[S_\sigma(\widehat{h}) + R_\sigma(\widehat{h}, p)], \qquad \|R_\sigma\|_{C^2} \longrightarrow 0, \tag{5.19}$$

where A_σ and A_σ^{-1} are bounded and positive. The signs $\sigma = -$ and $\sigma = +$ label the negative and positive branches, as in (5.2). The coefficient ratio satisfies

$$A_-/A_+ \le \exp(2C_{\mathrm{cone}}\sqrt{z_*}) + o(1). \tag{5.20}$$

Choose a large fixed matching height $Z_c > 2\widehat{h}_{\mathrm{ref}}$, and put

$$D_{\sigma,c}(\widehat{h}) = \mathcal{U}_\sigma(Z_c;\widehat{h}_{\mathrm{ref}}) - \mathcal{U}_\sigma(Z_c;\widehat{h}).$$

At each fixed Z_c, the finite-height analytic flow (5.3) converges, with all state derivatives through any prescribed finite order, to (5.13). The scattering estimate therefore gives

$$\|D_{\sigma,c} - S_\sigma\|_{C^2} \le CZ_c^{-1/2} + C(Z_c)(q_1 + |q_2|).$$

After fixing Z_c and reducing the parameters, these differences have bounded C^2 norms. Integration of the comparison equation gives $\mathcal{U}_\sigma^0(Z;\widehat{h}) = B_0 Z + O(\sqrt{Z} + \log Z)$, uniformly on each fixed compact set of rescaled central heights. Since $B_0 \in [b_0, \sup B]$ lies in the interior of the common cone, a sufficiently large Z_c puts the normalized energies $\mathcal{U}_\sigma^0(Z_c;\widehat{h})/Z_c$, and after parameter reduction also $\mathcal{U}_\sigma(Z_c;\widehat{h})/Z_c$, in that cone. At $t = q_1^2 Z_c$, moreover,

$$|q_0|/t \le L/Z_c, \qquad q_1/\sqrt{t} = Z_c^{-1/2}.$$

Apply Lemma 4.1 at $t = q_1^2 Z_c$, retaining $(z_*/t)^{-2q_2}$. Its centered map is $a_{\sigma,c}x + N_\sigma(x)$, where $N_\sigma(0) = N_\sigma'(0) = 0$, $\|N_\sigma\|_{C^4} \le Ce(t)$, and $e(t) = q_1\sqrt{Z_c} + Z_c^{-1/2}$. Here $x = -D_{\sigma,c}/Z_c = O_{C^2}(Z_c^{-1})$, so Taylor's formula gives $\|N_\sigma(x)\|_{C^2} \le Ce(t)/Z_c^2$. Since $t(z_*/t)^{-2q_2}/E = Z_c^{1+2q_2}$, the resulting endpoint is

$$\frac{I_\sigma(q_1^2\widehat{h}) - b_\sigma}{E} = Z_c^{2q_2} a_{\sigma,c}[D_{\sigma,c}(\widehat{h}) + n_{\sigma,c}(\widehat{h})], \qquad \|n_{\sigma,c}\|_{C^2} \le Ce(t)/Z_c,$$

with $n_{\sigma,c} = n_{\sigma,c}' = 0$ at $\widehat{h}_{\mathrm{ref}}$. Normalize by the positive reference derivative

$$A_\sigma = Z_c^{2q_2} a_{\sigma,c} \frac{D_{\sigma,c}'(\widehat{h}_{\mathrm{ref}})}{S_\sigma'(\widehat{h}_{\mathrm{ref}})}.$$

The scattering margin bounds the denominator below, and the preparation (5.19) follows with

$$\|R_\sigma\|_{C^2} \le CZ_c^{-1/2} + C(Z_c)(q_1 + |q_2|) + C(q_1/\sqrt{Z_c} + Z_c^{-3/2}).$$

Choose Z_c large and then $q_1, |q_2|$ small. With this fixed Z_c, $q_2 \log Z_c \to 0$, so $A_\sigma^{\pm 1}$ are bounded.

Since the normalized energies at t lie in the common cone, (4.4) gives

$$|\log a_{\sigma,c}| \le C_{\mathrm{cone}}(\sqrt{z_*} + Z_c^{-1/2}).$$

The factor $Z_c^{2q_2}$ cancels from A_-/A_+, and the two reference derivative ratios tend to one in the successive choices above, proving (5.20).

We now combine this preparation with the outer derivative signs on the same admissible height interval. Let $\widehat{\mathcal{H}}_p = q_1^{-2}\mathcal{H}_p$ be its rescaling in the coordinate $\widehat{h}$. If $\widehat{\mathcal{H}}_p \cap [\widehat{h}_{\mathrm{a}}, \widehat{h}_{\mathrm{b}}]$ is empty, connectedness puts $\widehat{\mathcal{H}}_p$ in one outer interval, where the strict derivative sign already gives at most one zero. Otherwise, the endpoint graphs are defined on the whole fibre of $\mathcal{K}_{\mathrm{c}}^+$, and every endpoint on this fibre, including the reference endpoint, is within $O(E)$ of the corresponding admissible endpoint. By Lemma 2.5, all these endpoints lie in $\widetilde{W}_\pm$ for small parameters. By (5.20) and the choice of z_*, a reduction of parameters gives

$$0 < \lambda_0 \le \lambda(p) := \Pi_p'(b_-)A_-/A_+ \le \bar{\lambda} < 1.$$

Put $G(\widehat{h};\widehat{q}_0, B_0, \lambda) = \lambda S_-(\widehat{h}) - S_+(\widehat{h})$. Taylor's formula for the exterior map and (5.19) give

$$\frac{\Delta(q_1^2\widehat{h})}{EA_+} = D(p) + G(\widehat{h};\widehat{q}_0, B_0, \lambda(p)) + R_\Delta(\widehat{h}, p), \qquad \|R_\Delta\|_{C^2} \to 0. \tag{5.21}$$

Here $D = (\Pi_p(b_-) - b_+)/(EA_+)$. The exterior Taylor remainder is $O_{C^2}(E)$, since the normalized endpoint functions have bounded C^2 norms and the exterior map has bounded derivatives through order four.

Write $\widehat{\Delta} = \Delta(q_1^2\widehat{h})/(EA_+)$. Apply Lemma 3.6 on each fibre of $\mathcal{K}_c$ to $-\widehat{\Delta}$, with $\phi = S_+$, $\psi = S_-$ and $\lambda \in [\lambda_0, \bar{\lambda}]$. The uniform margins of Lemma 5.2 give

$$\left(\frac{\widehat{\Delta}'}{S_+'}\right)' < 0. \tag{5.22}$$

In increasing T, the large-height, compact and small-height intervals are ordered by the cuts $\widehat{h}_b$ and $\widehat{h}_a$. The outer signs exclude slope zeros there, including at the cuts. In the middle, (3.1) writes $\widehat{\Delta}'/S_+'$ as $e^{\mathcal{L}} - 1$ times the positive factor $q_1^2 I_{+,h}/(EA_+S_+')$. Thus the displayed inequality gives $\mathcal{L}_{\widehat{h}} < 0$, or $\mathcal{L}_T > 0$, at every slope zero. All preparations represent the same displacement on the enlarged windows. The second case of Corollary 3.2 gives at most two zeros on the whole interval.

This proves Proposition 5.1, with

$$N_{ct}(L) = 2.$$

5.2. **Positive offsets and the interior saddle.** For positive offsets large relative to q_1^2, rescaling reveals a uniformly hyperbolic interior saddle. Its exponent determines the single logarithmic term of $\mathcal{L}$ near the separatrices, even as that exponent crosses one. Continuing to the fixed sections adds a common, possibly unbounded tail multiplier, which cancels from the endpoint derivative ratio. Away from the saddle, exterior contraction makes $\mathcal{L}$ negative. Together these estimates control all its zeros on $\mathcal{H}_p$.

Proposition 5.3 (Positive offset)**.** *There is a fixed $\kappa_p > 0$ such that the return displacement has at most two distinct zeros when*

$$q_0 > 0, \qquad 0 \le q_1/\sqrt{q_0} \le \kappa_p,$$

for all sufficiently small parameters.

5.2.1. *The saddle scale and its local passage.* Put

$$\ell = \sqrt{q_0}, \qquad \kappa = q_1/\ell, \qquad \eta = \ell X, \quad z = \ell^2 S, \quad \tau = \ell t.$$

The scaled field is

$$\begin{aligned} X_\tau &= 1 + \kappa X + q_2 X^2 + \ell X^3 R_q(\ell X) - SB(\ell X, \ell^2 S), \\ S_\tau &= -XS. \end{aligned} \tag{5.23}$$

On every fixed state box it is $O_{C^k}(\kappa + |q_2| + \ell)$ close to

$$X_\tau = 1 - B_0 S, \qquad S_\tau = -XS, \qquad B_0 = B(0,0,p). \tag{5.24}$$

The comparison field depends on the independent parameters A, C through B_0, which lies in a compact positive interval. Its first integral is

$$\mathscr{H} = B_0 S - \log S - X^2/2. \tag{5.25}$$

The equilibrium of the scaled field is $X = 0$, $S = S_s$, where $S_s B(0, \ell^2 S_s) = 1$. It is unique at positive height in the box in the coordinates (η, z), since $B + zB_z > 0$. Its Jacobian is

$$\begin{pmatrix} \kappa - \ell S_s B_\eta(0, \ell^2 S_s) & -(B + \ell^2 S_s B_z)(0, \ell^2 S_s) \\ -S_s & 0 \end{pmatrix}.$$

The positive quantities S_s, λ_+ and $-\lambda_-$, where $\lambda_+ > 0 > \lambda_-$ are its eigenvalues, are bounded above and below uniformly. Moreover,

$$\nu_p = -\lambda_-/\lambda_+ = 1 + O(\kappa + \ell). \tag{5.26}$$

For passing central heights, the normalized central height is $\widehat{h}_+ = h/q_0 > S_s$.

Choose small fixed incoming and outgoing sections in the passing quadrant of this saddle. On each, choose a positive transverse coordinate equal to $\mathscr{H}-\mathscr{H}_s$ for the comparison field, and translate its zero to the perturbed separatrix. The local passage between these coordinates has the form

$$\begin{aligned} D_p(x) &= u_p x^{\nu_p}(1+r_p(x)), \qquad u_p = 1+O(\kappa+|q_2|+\ell), \\ |(x\partial_x)^j r_p(x)| &\le C_j x^{\vartheta}, \qquad 0\le j\le 2, \end{aligned} \tag{5.27}$$

for a fixed $\vartheta>0$. Parameter-uniform expansions of hyperbolic Dulac maps are developed in [9]. We derive the finite-order estimate and its leading coefficient below, uniformly as ν_p crosses one.

We first derive the power law on saddle sections where the nonlinear term is integrable along the passage. We then transfer it to the chosen sections, keeping the leading coefficient as well as the Euler remainder. The parameter-dependent stable manifold theorem [13, Theorem 9.6], applied and normalized in Appendix B.2, gives on a common saddle box

$$x' = x, \qquad y' = -\nu_p y + xy^2 c_2(x,y). \tag{5.28}$$

The required state derivatives of c_2 are uniformly bounded for ν_p near one.

From input (x,y_i) to output $x=x_o$, set $\tilde{y}(\xi;x)=y(\xi)(\xi/x)^{\nu_p}$, and write

$$k(t,x)=(x/t)^{\nu_p}, \qquad \mathscr{G}(t,x,\tilde{y}) = k(t,x)\tilde{y}^2 c_2(t,k(t,x)\tilde{y}).$$

The solution satisfies the Volterra equation

$$\tilde{y}(\xi;x) = y_i + \int_x^{\xi} \mathscr{G}(t,x,\tilde{y}(t;x))\,dt.$$

For $3/4\le\nu_p\le 5/4$ and fixed $0<\vartheta<1/2$,

$$\int_x^{x_o} k(t,x)\,dt \le x^{3/4}\int_x^{x_o} t^{-3/4}\,dt \le Cx^{3/4}\le Cx^{\vartheta}.$$

Fixed small sections therefore keep $y_i/2\le\tilde{y}\le 2y_i$. For $\mathcal{E}_x=x\partial_x$, with $t,\tilde{y}$ fixed in the integrand, one has $\mathcal{E}_x k=\nu_p k$, hence $|\mathcal{E}_x^i\partial_{\tilde{y}}^j\mathscr{G}|\le C_{i,j}k$ at every required finite order. Euler differentiation of the Volterra equation gives scalar linear variational equations with coefficient $\mathscr{G}_{\tilde{y}}$; their forcing terms are bounded by k times lower variations. Moving-lower-limit terms are $O(x)$: for the first derivative the term is $-xy_i^2c_2(x,y_i)$, and another Euler derivative has the same bound. Variation of constants and the integral estimate give

$$|\tilde{y}(x_o;x)-y_i| + \sum_{j=1}^{2}|\mathcal{E}_x^j\tilde{y}(x_o;x)| \le Cx^{\vartheta}.$$

Division by the fixed positive y_i proves the relative Euler estimates and the leading power on the prepared sections.

The prepared sections and the chosen incoming and outgoing sections lie in common regular collars at a fixed positive distance from the saddle. By Lemma 2.4, the transitions between them, followed by the changes of section coordinates, contribute positive units, with remainders $O(x)$ at input and $O(x^{\nu_p})$ at output. They preserve the stated Euler estimates. For (5.24) the passage in these coordinates is exactly the identity, by its first integral. Uniform dependence of the graph constructions, coordinate changes, and fixed section maps gives $u_p=1+O(\kappa+|q_2|+\ell)$. This proves (5.27).

The orbits through central points with $\widehat{h}_+\downarrow S_s$ meet the incoming and outgoing sections in their directed order. The two eigen-coordinates of these central points are positive multiples of $\widehat{h}_+-S_s$, bounded above and below uniformly. Following backward to the incoming section decreases the unstable coordinate, so its incoming value tends uniformly to zero. The endpoint order in Proposition 2.7, together with the positive regular section derivative, makes this rebasing injective. Thus a sufficiently small preliminary interval $0<\widehat{h}_+-S_s<\rho_0$ maps into a fixed collar $0<x<x_c$.

5.2.2. *Continuation to the fixed sections.* We next continue the maps to the fixed endpoint windows. This step must include central heights tending to the saddle, whose local passage time is unbounded. Choose a large fixed M_h, followed by $M_t > 2M_h + 2$, beyond all saddle heights. For $S_s < \widehat{h}_+ \le M_h$, both outward graphs reach $S = M_t$, with scaled energies $\mathcal{U}_\sigma = X_\sigma^2/2 = U_\sigma/q_0$ bounded above and below there. As $\widehat{h}_+ \downarrow S_s$, the local construction above makes the half-orbits converge to the upper separatrices, whose passage from the fixed saddle box to M_t is regular. For the comparison field their energies are exactly

$$\mathcal{U}_\sigma(M_t; \widehat{h}_+) = B_0(M_t - \widehat{h}_+) - \log(M_t/\widehat{h}_+),$$

including the continuous separatrix value at $\widehat{h}_+ = 1/B_0$. For large M_t their ratios to M_t are close to B_0, uniformly for $\widehat{h}_+ \le M_h$. Since $B_0 \in [b_0, \sup B]$ lies in the interior of the common cone, these ratios lie in that cone. Outside the saddle box these segments, including the limiting separatrices, form a compact regular family with uniformly transverse endpoints at $S = M_t$. For a small fixed $\rho_0 > 0$, use the local Dulac construction on $0 < \widehat{h}_+ - S_s < \rho_0$, followed by Lemma 2.4 from the saddle sections. On the compact complement $S_s + \rho_0 \le \widehat{h}_+ \le M_h$, the same lemma applies from the central section, where the flux has a positive margin. These continuations preserve the energy bounds and common-cone entry at M_t. The regular maps have positive bounded first derivatives and the required higher derivative bounds; the local Dulac map accounts for the unbounded time.

At the common cut $t = M_t q_0$,

$$q_0/t = 1/M_t, \qquad q_1/\sqrt{t} = \kappa/\sqrt{M_t}, \qquad \sqrt{t} = \ell\sqrt{M_t}.$$

Taking M_t large and the parameters small permits Lemma 4.1 to continue both graphs to z_*. Keep the tail multiplier in the common endpoint amplitude:

$$\varpi_{M_t q_0} = (z_*/(M_t q_0))^{-2q_2}, \qquad E = q_0 \varpi_{M_t q_0}. \tag{5.29}$$

By (4.10) with $t = M_t q_0$, $E \le z_*^{1/4} q_0^{3/4} \to 0$ as $q_0 \to 0$, uniformly for $|q_2| < 1/8$.

Let e_σ denote the chosen saddle-section coordinates, with $e_+ = x$ and $e_- = D_p(x)$. For the comparison field the first derivatives of the regular maps from these sections to $S = M_t$ are both one in the normalized I coordinate: at fixed S, a change of $\mathscr{H}$ is exactly a change of $S - X^2/2$. Their quotient is consequently $1 + O_{M_t}(\kappa + |q_2| + \ell)$. The weighted tail gives the endpoint maps

$$I_\sigma(e) = b_\sigma + E\mathcal{F}_\sigma(e), \qquad \mathcal{F}_\sigma(0) = 0, \qquad c \le \mathcal{F}'_\sigma \le C, \qquad \|\mathcal{F}_\sigma\|_{C^3} \le C, \tag{5.30}$$

and

$$\frac{\mathcal{F}'_-(0)}{\mathcal{F}'_+(0)} \le \exp\left(2C_{\mathrm{cone}}\left(\sqrt{z_*} + \kappa/\sqrt{M_t}\right)\right)\left[1 + O_{M_t}(\kappa + |q_2| + \ell)\right]. \tag{5.31}$$

The tail normalization error is $O_{C^3}(\ell\sqrt{M_t} + \kappa/\sqrt{M_t})$, and converting from the cut scale $M_t q_0$ to q_0 only changes fixed constants. Both tails start at the common cut, with normalized energies in the common cone after parameter reduction; their common multiplier $\varpi_{M_t q_0}$ is contained in E, and (4.4) bounds the logarithm of each normalized tail slope by $C_{\mathrm{cone}}(\sqrt{z_*} + \kappa/\sqrt{M_t})$.

If the saddle collar contains a central height in $\mathcal{H}_p$, all its endpoints differ from the admissible endpoints by $O(E)$, so they lie in $\widetilde{W}_\pm$ for small parameters, by Lemma 2.5.

5.2.3. *The return displacement and the remaining central heights.* In the incoming coordinate x, the closing equation is

$$\widetilde{\Delta}_p(x) = \Pi_p(I_-(D_p(x))) - I_+(x) = 0.$$

Its derivative is

$$\widetilde{\Delta}'_p(x) = E\mathcal{F}'_+(x)\left[\exp \mathcal{L}_p(x) - 1\right], \tag{5.32}$$

where

$$\mathcal{L}_p(x) = \log \Pi'_p(I_-(D_p(x))) + \log \mathcal{F}'_-(D_p(x)) - \log \mathcal{F}'_+(x) + \log D'_p(x).$$

The common amplitude E has disappeared from this ratio. The local Dulac derivative supplies the logarithmic term, while the endpoint and exterior factors contribute a bounded constant and a small remainder. By (5.27) and (5.31),

$$\mathcal{L}_p(x) = c_p + (\nu_p - 1)\log x + e_p(x), \qquad |e_p| + |xe_p'| \le Cx^{\vartheta}, \tag{5.33}$$

and, after the fixed choices and parameter reductions,

$$c_p = \log\left[\Pi_p'(b_-)\frac{\mathcal{F}_-'(0)}{\mathcal{F}_+'(0)}u_p\nu_p\right] \le -c_0 < 0.$$

For the local factor differentiate (5.27): its relative derivative is $1 + r_p + (x/\nu_p)r_p'$, with the required Euler bound. The endpoint factors contribute $O(x) + O(D_p(x))$; the nonlinear exterior term contributes $O(ED_p(x))$. Their first Euler derivatives have the same bounds, by (5.30). Decrease ϑ if necessary to obtain (5.33).

Using the full constants in (5.33), choose $x_c < \min\{1, e^{-1/\vartheta}\}$ such that $Cx_c^{\vartheta} < c_0/2$ and $Cx_c^{\vartheta}|\log x_c| < c_0/4$. Decrease the preliminary ρ_0 so that its incoming coordinates lie in $(0, x_c)$; the preceding estimates restrict to this collar. Use this final $\rho_0 > 0$ in the compact complement below, fixing its continuation constants before the final parameter reduction. Then Lemma 3.4 and the negative bound for c_p give $x\mathcal{L}_p'(x) < 0$ at every zero of $\mathcal{L}_p$. To return to the central height, observe that the ratio agrees with (3.1): the factor dx/dh cancels from the quotient of endpoint derivatives. Moreover, $dx/dh = I_{+,h}/(E\mathcal{F}_+'(x)) > 0$. Thus the central heights in $\mathcal{H}_p$ with $0 < \widehat{h}_+ - S_s < \rho_0$ form one interval on which every zero of $\mathcal{L}$ has $\mathcal{L}_h < 0$, that is, $\mathcal{L}_T > 0$.

For the remaining central heights we prove contraction. First, for $h \ge M_h q_0$,

$$|q_0|/h \le 1/M_h, \qquad q_1/\sqrt{h} \le \kappa/\sqrt{M_h}.$$

For a sufficiently large fixed M_h and small $|q_2|$, Lemma 2.10 therefore gives $\Delta_h < 0$ at these central heights when $q_1 > 0$; when $q_1 = 0$, Proposition 2.9 gives the same sign. Second, on $S_s + \rho_0 \le \widehat{h}_+ \le M_h$ the radial flux at the central point has a fixed positive margin. On its compact inner portion, the scaled linear integral in (2.31) is bounded by $C_{\rho_0,M_h}\kappa$, and the other nonquadratic integrals are $O_{\rho_0,M_h}(\ell)$. To make the tail constant independent of these height thresholds, continue first to a fixed large $S = S_1$. For the comparison field its energy is

$$U/q_0 = B_0(S - \widehat{h}_+) - \log(S/\widehat{h}_+).$$

Uniformly on $S_s + \rho_0 \le \widehat{h}_+ \le M_h$, its ratio to S tends to B_0. Choose S_1 to enter the interior of the common cone; the positive radial flux at the central point and finite-height continuation preserve this property for small $\kappa, |q_2|, \ell$. The continuation constants depend on the height thresholds, while the tail constant is fixed by the common cone. By (4.4), the two tails from $S = S_1$ contribute at most $2C_{\mathrm{cone}}(\sqrt{z_*} + \kappa/\sqrt{S_1})$, and hence

$$|\log(I_{-,h}/I_{+,h})| \le 2C_{\mathrm{cone}}\sqrt{z_*} + C_{\rho_0,M_h,S_1}(\kappa + \ell).$$

The fixed common-cone choice of z_*, followed by small $\kappa_{\mathrm{p}}, |q_2|, \ell$, makes this bound smaller than the exterior contraction margin. These two ranges join into the single interval of large central heights $\widehat{h}_+ \ge S_s + \rho_0$, on which $\mathcal{L}$ is strictly negative, including the shared height thresholds. In increasing T, this interval and the saddle collar, intersected with $\mathcal{H}_p$, cover $\mathcal{H}_p$. Since the endpoints of the collar lie in $\widetilde{W}_{\pm}$, the collar represents this same displacement. Hence every zero of $\mathcal{L}$ on $\mathcal{H}_p$ lies in the collar and crosses upward, and the second case of Corollary 3.2 gives at most two zeros of Δ_p, proving Proposition 5.3.

6. Early heights, large exits, and weak roots

For negative constant coefficient, the height interval may extend from nearly regular passages to long passages near a boundary root. We first establish contraction at early heights, then compare the branches while both drifts remain positive. The other two estimates control positive half-orbits that leave every fixed multiple of the coefficient scale and boundary roots that approach the central line relative to that scale. The applications in Sections 7 to 10 verify the hypotheses on the primary and secondary scales and join the resulting intervals for the same logarithmic slope ratio.

6.1. **Early central heights.** The first part of the height interval is controlled by exterior contraction. Besides heights above a coefficient-dependent threshold, it includes smaller central heights whose positive half-orbits still have small normalized radius at that threshold. The estimates below keep the tail error independent of the threshold, so that the common fixed sections suffice for later applications. We use the scales

$$q_0 = -a\delta\ell^2, \qquad q_1 = b\delta\ell, \qquad q_2 = -\bar{c}\delta, \qquad \epsilon_3 = \ell/\delta \le 1, \tag{6.1}$$

where $a \ge a_{\min} > 0$, $b \ge 0$, $\bar{c}$ lie in a fixed compact coefficient set and $\delta > 0$ is sufficiently small. Put $Q = \delta\ell^2$. In $x = |\eta|/\ell$ the outward drifts are

$$K_\sigma(x) = -q(\sigma\ell x)/(\delta\ell^2) = a - \sigma b x + \bar{c}x^2 - \sigma\epsilon_3 x^3 R_q(\sigma\ell x).$$

Lemma 6.1 (Early central heights)**.** *Fix $d_f > 0$ and put $z_e = d_f Q$. There is $r_0 > 0$, depending only on d_f and the compact coefficient set, such that for all sufficiently small δ the central heights in $\mathcal{H}_p$ with*

$$h \ge z_e, \qquad \text{or} \qquad 0 < h < z_e, \quad |\eta_+(z_e; h)|/\ell \le r_0, \tag{6.2}$$

form an initial interval in every logarithmic height coordinate $T = -c_T \log h$, and $\Pi_p'(I_-)I_{-,h}/I_{+,h} < 1$ on it. In particular $\mathcal{L} < 0$ there. The coefficient of $\sqrt{z_*}$ in the estimates is $2C_{\mathrm{cone}}$; the constants that depend on d_f and r_0 enter only the final threshold for δ.

Proof. By Lemma 3.3, the set (6.2) is an initial interval in T. There are three ranges: large heights, the intermediate height band, and smaller heights with small radius at the prescribed cut. The large-height estimate handles the first. In the other two, we bound the paired logarithm $\log(I_{-,h}/I_{+,h})$ of Lemma 2.8. The common-cone tail contributes $2C_{\mathrm{cone}}\sqrt{z_*}$, already below a fixed fraction of the exterior contraction margin. The remaining errors are controlled by choosing the radius cut small where needed, and then reducing δ.

Large heights. Let θ_0 be the constant of Lemma 2.10, and fix $K \ge 2a_{\max}/\theta_0$. For $h \ge KQ$,

$$|q_0|/h \le a/K, \qquad q_1/\sqrt{h} \le b\sqrt{\delta/K}, \qquad |q_2| \le C\delta,$$

so Lemma 2.10 gives $0 < L_p' < 1$ for small δ.

The band $d_f Q \le h \le KQ$. Set $z = Q\zeta$, $U = Q\mathcal{U}$. Since $q_0 < 0$, the constant term of the graph equation is favorable, and the centered lower barrier $U_\sigma \ge m(z - h)$ holds for a small fixed m. On a fixed interval ending at $\zeta = K_1 > 2K$, the graph equation converges uniformly, with its finite variations, to

$$\mathcal{U}_\zeta = B_0 + a/\zeta, \qquad \mathcal{U}(\zeta) = B_0(\zeta - h/Q) + a\log(\zeta Q/h).$$

Choose K_1 large after d_f, K. Since $\log(K_1/d_f)/K_1 \to 0$, the energy ratio at $K_1 Q$ lies in the interior of the common cone, uniformly for $d_f \le h/Q \le K$. Before this height, the positive flux and the centered lower barrier bound the three paired integrals by $C_{d_f,K_1}(\sqrt{Q} + q_1/\sqrt{Q}) = C_{d_f,K_1}(\sqrt{Q} + b\sqrt{\delta})$. Their square-root singularities at $\zeta = h/Q$ are integrable. Beyond this height, Lemma 4.1 applies on the common cone. Combining the two estimates gives

$$|\log(I_{-,h}/I_{+,h})| \le 2C_{\mathrm{cone}}\sqrt{z_*} + C_{d_f,K_1}\sqrt{\delta}. \tag{6.3}$$

Small radius at z_e. Write $x_\sigma = |\eta_\sigma(z_e;h)|/\ell$. Take a fixed large L_0, and then r_0 so small that both K_σ lie between $a/2$ and $2a$ on $0 \le x \le L_0 r_0$. Up to a hypothetical first hit of $L_0 r_0$ by the negative branch, the graph equations give $\mathcal{G}_{-,p}(U_-, z) \le C\mathcal{G}_{+,p}(U_+, z)$: compare $\sup B + 2a\delta\ell^2/z$ with $b_0 + a\delta\ell^2/(2z)$, the supremum being taken on the fixed box. Integration from the common zero energy gives $U_- \le CU_+$, and $L_0^2 > C$ excludes that hit. Thus $x_+ \le r_0$ implies $x_- \le L_0 r_0$.

On the prefix up to z_e, use $dz/d|\eta| = |\eta|z/(zB - q)$. The lower bounds $zB - q \ge a_{\min}\delta\ell^2/2$ and $zB - q \ge b_0 z$ give

$$\int_h^{z_e} \frac{dz}{|\eta_\sigma|} \le C\ell r_0, \qquad q_1 \int_h^{z_e} \frac{dz}{|\eta_\sigma| z} \le C b r_0, \qquad \int_h^{z_e} \frac{|\eta_\sigma|}{z}\, dz \le C\epsilon_3 r_0^3.$$

From z_e onward there are barriers

$$m(z - z_e) \le U_\sigma(z) \le U_\sigma(z_e) + M(z - z_e). \tag{6.4}$$

The initial adverse ratios $q_1\sqrt{U_\sigma(z_e)}/z_e$, $|q_2|U_\sigma(z_e)/z_e$ and $U_\sigma(z_e)^{3/2}/z_e$ are $O(r_0/d_f)$, $O(r_0^2/d_f)$ and $O(\epsilon_3 r_0^3/d_f)$. Choose r_0 small after d_f. Use the strip between the barriers only up to $t = \ell^2 Z_1$, with fixed Z_1. Its added linear, quadratic and cubic losses are $O(\sqrt{\delta})$, $O(\delta M)$ and $O(M^{3/2}\ell\sqrt{Z_1})$, so small δ gives positive flux.

These shifted barriers preserve positive flux, but the endpoint estimate also requires entry into the common cone. For this, apply the continuation part of Lemma 4.2 with $\zeta_0 = d_f$ and $0 \le \mathcal{U}_0 = U_\sigma(z_e)/\ell^2 \le Cr_0^2$. The preceding radius bounds give $K_\sigma(\sqrt{2\mathcal{U}_0}) \ge a_{\min}/2$. Choose the height Z_1 for entry into the cone large after r_0, d_f. The shifted barriers bound the paired integrals up to this height by $C_{d_f,r_0,Z_1}\{\sqrt{\delta} + \ell(1 + |\log\delta|)\}$: the linear integral is $O(\sqrt{\delta})$, and splitting the square root in the upper barrier bounds the cubic integral. This tends to zero uniformly for $0 < \epsilon_3 \le 1$, since $\ell = \epsilon_3\delta$. The remaining common-cone tail contributes $2C_{\text{cone}}\sqrt{z_*} + O(\delta)$. Together with the prefix bounds, this proves contraction for small r_0 and then small δ. □

6.2. **A common-height estimate before a boundary root.** Beyond the contraction range, suppose both drifts still have positive margins. Equal leading actions then relate the two radii at the same central height. The derivative of this relation has a useful cancellation: the quadratic coefficients drop out of the derivative ratio. This determines the crossing direction of $\mathcal{L}$ before the positive branch reaches a boundary root, for either sign of the quadratic coefficient.

Suppose the scales (6.1) hold with δ and $\epsilon_3 = \ell/\delta$ sufficiently small. The drifts K_σ of Section 6.1 have the quadratic comparisons

$$K_\sigma^0(x) = a - \sigma b x + \bar{c}x^2.$$

Choose a positive terminal radius R, possibly varying over the compact coefficient set, such that K_+^0 has a fixed positive margin on $0 \le x \le R$. Assume also that the equal-action negative radii for these terminal actions lie in a fixed compact interval on which K_-^0 has a fixed positive margin.

On these comparison intervals set

$$\mathcal{A}_\sigma^0(x) = \int_0^x \frac{u\, du}{K_\sigma^0(u)}, \qquad \mathcal{A}_-^0(y) = \mathcal{A}_+^0(x), \qquad y = f(x).$$

The drift margins make $\mathcal{A}_-^0$ strictly increasing, so its inverse at the prescribed actions defines f on the chosen compact intervals.

Lemma 6.2 (Equal-action comparison)**.** *For $x, y = f(x) > 0$ in these intervals, put $R_1(x) = K_-^0(y)/K_+^0(x)$. Then*

$$\frac{yy'}{x} = R_1, \qquad \frac{R_1'}{R_1} = \frac{b(1+x/y)}{K_+^0(x)}, \qquad \frac{(D + x^2 - \lambda y^2)'}{2x} = 1 - \lambda R_1 \tag{6.5}$$

for any constants D, λ. In particular, R_1 is nondecreasing when $b \geq 0$.

Proof. Differentiate $\mathcal{A}_-^0(y) = \mathcal{A}_+^0(x)$ to obtain the first identity. Since $(K_-^0)'(y) = b + 2\bar{c}y$ and $(K_+^0)'(x) = -b + 2\bar{c}x$, logarithmic differentiation of R_1 and substitution of the first identity cancel the two quadratic terms and give the second. The third follows by differentiating $D + x^2 - \lambda y^2$. □

Lemma 6.3 (Quadratic prefix)**.** *Fix a small $d_f > 0$ and put $z_e = \delta\ell^2 d_f$. On the central heights in $\mathcal{H}_p$ satisfying*

$$h \geq z_e, \qquad \text{or} \qquad 0 < h < z_e, \quad |\eta_+(z_e; h)|/\ell \leq R,$$

the return displacement has at most two distinct zeros. Moreover, for any logarithmic height coordinate $T = -c_T \log h$, where $c_T > 0$ is independent of the central height, every zero of

$$\mathcal{L}(T) = \log\left[\Pi_p'(I_-(h))\frac{I_{-,h}(h)}{I_{+,h}(h)}\right]$$

on this interval satisfies $\mathcal{L}_T > 0$. The constants are uniform on the compact coefficient and action domains just specified.

Proof. By Lemma 6.1, there is $r_0 \in (0, R)$ such that the central heights with $h \geq z_e$, or with $h < z_e$ and $x_+ = |\eta_+(z_e; h)|/\ell \leq r_0$, form an initial interval in T on which the return contracts; in particular $\mathcal{L} < 0$ there.

Matching the branches. It remains to consider $r_0 \leq x = x_+ \leq R$. On the prescribed compact drift intervals, the central-height identity (4.18), with $Q = \delta\ell^2$, gives

$$\delta \log(h_\sigma(x)/Q) = -\mathcal{A}_\sigma(x) + \delta M_\sigma(x), \qquad \mathcal{A}_\sigma(x) = \int_0^x \frac{u\,du}{K_\sigma(u)}, \qquad \|M_\sigma\|_{C^4} \leq C.$$

The fixed terminal height here is $z_e/Q = d_f$.

The leading action relation must first give two passages with exactly the same central height. The assumed margins and compact negative inverse provide this matching: on the relevant terminal interval, $(\mathcal{A}_-^0)'(y)$ is bounded below. At the endpoints of a slightly enlarged negative radius interval, the trial passages have central heights bracketing the positive passage's central height. The order of central heights (Lemma 3.3) gives a matching negative passage. Implicit differentiation then gives

$$x_-(x) = f(x) + O_{C^3}(\epsilon_3 + \delta). \tag{6.6}$$

Preparing the endpoints. Apply Lemma 4.2 with the scales (6.1), $\mathcal{U}_0 = x_\sigma^2/2$ and the constant entrance height factor $\zeta_0 = d_f$. Both radius intervals are compact and positive, and the assumed drift margins persist for small ϵ_3. Choose the common-cone height $Z_1 = Z_0 \geq 1$. The endpoints are

$$I_\sigma(x_\sigma) = b_\sigma - \tfrac{1}{2}EA_\sigma x_\sigma^2 + Ee_\sigma(x_\sigma), \qquad A_\sigma \asymp 1, \quad \|e_\sigma\|_{C^3} \leq C\delta(1 + |\log\delta|),$$

with $A_-/A_+ \leq \exp(2C_{\text{cone}}\sqrt{z_*} + o(1))$. Here $E = \ell^2(z_*/(\ell^2 Z_0))^{-2q_2}$; by (4.10) with $t = \ell^2 Z_0$, $E \leq z_*^{1/4}\ell^{3/2}$, since $Z_0 \geq 1$ and $|q_2| < 1/8$ for small δ.

Whenever the interval contains a central height in $\mathcal{H}_p$, the endpoints are within $O(E)$ of admissible endpoints and therefore lie in $\widetilde{W}_\pm$, by Lemma 2.5. Composing with the exterior transition gives a Taylor remainder of $O(E)$ after division by E. Inserting (6.6) gives

$$\frac{2\Delta(x)}{EA_+} = D + x^2 - \lambda f(x)^2 + R_\Delta(x), \qquad \|R_\Delta\|_{C^3} \leq C\{\epsilon_3 + \delta(1 + |\log\delta|) + E\}, \tag{6.7}$$

where D depends only on the parameters and $0 < \lambda_0 \leq \lambda \leq \bar{\lambda} < 1$.

Controlling the crossing direction. Apply Lemma 3.6 on $r_0 \le x \le R$ to $\mathcal{F}(x) = 2\Delta(x)/(EA_+)$, with $\phi = x^2$, $\psi = f^2$, and the $\lambda \in [\lambda_0, \bar{\lambda}]$ and remainder R_Δ of (6.7); put $\mathscr{Q} = \mathcal{F}'/(2x)$. Here $\phi' = 2x \ge 2r_0$, $\phi'' = 2$, and $\psi'/\phi' = R_1$ by (6.5).

At $b = 0$, equal actions give $y = x$ and $R_1 = 1$. Uniform smooth dependence on the compact drift domains gives $|R_1 - 1| \le Cb$. Fix $b_* > 0$ so small that $1 - \lambda R_1 \ge (1 - \bar{\lambda})/2$ for $0 \le b \le 2b_*$. The second case of the lemma gives $\mathscr{Q} > 0$, and hence $\Delta_x > 0$ in this range. For $b \ge b_*$, Lemma 6.2 gives $R_1' \ge c_* > 0$, so the first case of the derivative-ratio lemma gives

$$\mathscr{Q}' < 0. \tag{6.8}$$

These bounds and cuts are fixed before δ, ϵ_3 are reduced to meet the remainder threshold.

To express these signs in the common logarithmic coordinate, use the positive central-height formula:

$$\frac{h_x}{h} = -\frac{\mathcal{A}_+'(x)}{\delta} + M_+'(x) < 0.$$

Thus $x_T > 0$ for every such coordinate T. The contraction interval of large central heights and small radii has $\mathcal{L} < 0$, including the cut $x = r_0$. For $b \le 2b_*$, $\Delta_x > 0$ and $h_x < 0$ give $\mathcal{L} < 0$ on the remaining interval as well. For $b \ge b_*$, a zero of $\mathcal{L}$ in the compact-radius part has $\mathcal{F}' = 0$. By (6.8),

$$\Delta_{xx} = EA_+ x\mathscr{Q}' < 0, \qquad \Delta_{TT} = \Delta_{xx} x_T^2 < 0.$$

With $m_+ = hI_{+,h} > 0$, the exact identity

$$\Delta_T = -\frac{m_+}{c_T}(\mathrm{e}^{\mathcal{L}} - 1)$$

therefore gives

$$\mathcal{L} = 0 \quad \Longrightarrow \quad \mathcal{L}_T = -\frac{c_T}{m_+}\Delta_{TT} > 0. \tag{6.9}$$

Finally, the central heights considered here form one interval in $\mathcal{H}_p$, by Lemma 3.3. In increasing T, the contraction interval and the compact-radius part cover it. The two estimates apply to the same displacement and include their shared cut. The compact-radius estimate also includes the terminal cut $x = R$, where the endpoints lie in $\widetilde{W}_\pm$ and the exterior map is smooth. Thus every zero of $\mathcal{L}$ on this interval satisfies $\mathcal{L}_T > 0$. The second case of Corollary 3.2 gives the two-zero bound. □

The compact comparison ends before the positive drift vanishes. The next lemma gives a strict overlap with the boundary-flow estimate near that root. It then combines both estimates for the same logarithmic slope ratio.

Lemma 6.4 (Joining the early and root intervals)**.** *Under (6.1), suppose R ranges over a compact positive interval and K_+^0 has a fixed positive margin on $0 \le x \le R$. The negative equal-action radii then lie in a fixed compact interval where K_-^0 has a positive margin, as required in Lemma 6.3.*

Suppose further that K_+ has a first positive root α in a fixed compact positive interval. Choose $0 < w_a < R < \alpha$, with fixed positive lower bounds for w_a and $R - w_a$, and a fixed small $Y_f > 0$ such that the inverse boundary-flow coordinate is defined on $[w_a, \alpha]$ up to Y_f. Fix a small $d_f > 0$, put $z_e = \delta\ell^2 d_f$, and use a logarithmic height coordinate T as in (3.2). For sufficiently small δ, ϵ_3, the early heights in $\mathcal{H}_p$

$$h \ge z_e, \qquad \text{or} \qquad h < z_e, \quad |\eta_+(z_e; h)|/\ell \le R$$

form an initial interval in T; every zero of $\mathcal{L}$ there satisfies $\mathcal{L}_T > 0$. Every remaining height has its positive projected state at $z = \ell^2 Y_f$ in $[w_a, \alpha)$, with a strict overlap at the early cut. If every critical point of $\mathcal{L}$ in this root range satisfies $\mathcal{L}_{TT} < 0$, then Δ_p has at most three distinct zeros on $\mathcal{H}_p$; if $\mathcal{L}_T > 0$ at every zero there, it has at most two.

Proof. The fixed radius and drift margins bound the positive terminal actions. To obtain a compact negative inverse, note that for $\bar{c} \geq 0$, the action of $a + by + \bar{c}y^2$ increases to infinity; for $\bar{c} < 0$, it diverges at the first positive root. Its inverse at each bounded positive action therefore lies strictly inside the positive-drift interval. Smooth dependence at a larger fixed action and compactness of the coefficient set give the claimed compact interval. Thus Lemma 6.3 applies to the early interval.

For every remaining height, the positive graph hits R below $Y = \delta d_f$. By (4.13), its projected state there lies between $R - C\delta d_f > w_a$ and $R < \alpha$. Remark 4.4 keeps it increasing and below α, and continues the inverse projection to Y_f, also at a double root. The same strict bounds hold at the early cut and give an overlap. Lemma 3.3 orders the two intervals, and Corollary 3.2 gives the stated zero bounds. □

6.3. **Positive passages beyond a coefficient scale.** The bounded-radius estimates do not control positive graphs that leave every fixed multiple of the coefficient scale. To treat these graphs in the primary and secondary regions, we follow them within a small unscaled radius box. An adapted height coordinate gives exact action transport and estimates for the moving exit radius. Continuation from that exit to a cone then permits the weighted-tail estimate to give the positive endpoint slope. The corollary below compares it with the negative endpoint slope to prove contraction after a sufficiently large exit.

Write $\varrho = \eta > 0$ for the positive radius; in the negative quadratic-drift application below, $\varrho = s\rho$. Suppose that

$$-q(\varrho) = \varrho^2(s+\varrho)\mathcal{K}(u, v, \varrho, p), \qquad u = \varrho_0/\varrho, \quad v = \varrho/(s+\varrho), \qquad s, \varrho_0 \geq 0, \tag{6.10}$$

where $\mathcal{K}$ has fixed positive upper and lower bounds and uniform derivative bounds on an enlarged compact box of the displayed base variables. Both s and the unscaled radius are small. The coefficient ϱ_0 is held fixed along a graph. A restriction of the box in v is handled by a smooth extension preserving positivity. The factor $\varrho^2(s+\varrho) = s\varrho^2 + \varrho^3$ joins the quadratic range $\varrho \ll s$ to the cubic range $\varrho \gg s$.

Put

$$\varrho^+ = s + \varrho, \qquad y = \frac{z}{\varrho^2\varrho^+}, \qquad \mathfrak{l} = \log\varrho, \qquad \frac{d\tau}{d\mathfrak{l}} = \frac{1}{\varrho^+}, \qquad \mathcal{B} = B(\varrho, \varrho^2\varrho^+ y, p).$$

The graph equation gives the extended system

$$\begin{aligned} &u_\tau = -(s+\varrho)u, \quad \varrho_\tau = (s+\varrho)\varrho, \qquad v_\tau = sv, \\ &y_\tau = \mathfrak{a}\, y + y^2\mathcal{N}, \quad \mathfrak{a} = \mathcal{K}^{-1} - 2s - 3\varrho, \quad \mathcal{N} = -\frac{\mathcal{B}}{\mathcal{K}(\mathcal{K} + \mathcal{B}y)}. \end{aligned} \tag{6.11}$$

The defining relations among the base variables are invariant. For a small fixed $y_0 > 0$, the quantities $\mathfrak{a}$, y_τ/y, $\partial_y y_\tau$ have fixed positive lower bounds on $|y| \leq 2y_0$.

Lemma 6.5 (Adaptive height and cone landing)**.** *For every fixed finite order k, after reducing s and the radius box, there is a uniformly regular coordinate*

$$\Psi(u, v, \varrho, y, p) = y \exp\Theta(u, v, \varrho, y, p), \qquad \Theta = y\widetilde{\Theta}, \quad \|\widetilde{\Theta}\|_{C^k} \leq C_k,$$

with positive height derivative and bounded inverse derivatives, such that $\Psi_\tau = \mathfrak{a}\Psi$. The adapted unscaled height $\widehat{z} = \varrho^2(s+\varrho)\Psi$ therefore satisfies exactly

$$\widehat{z}(\varrho) = \exp\left(\int_{\varrho_i}^{\varrho} \frac{t\,dt}{-q(t)}\right)\widehat{z}(\varrho_i) \tag{6.12}$$

as long as the passage is thin, $y \leq y_0$.

For an exit at $y = y_0$, let ϱ_e be its radius and $\lambda_{\mathrm{in}} = \log\widehat{z}(\varrho_i)$. Then

$$-C(s+\varrho_e) \leq \frac{d\log\varrho_e}{d\lambda_{\mathrm{in}}} \leq -c(s+\varrho_e), \qquad \left|\frac{d^j\log\varrho_e}{d\lambda_{\mathrm{in}}^j}\right| \leq C_j(s+\varrho_e)^j, \quad 1 \leq j \leq k. \tag{6.13}$$

There is a subsequent first landing on $z = m_{\mathrm{land}}\varrho^2$, for a small fixed $m_{\mathrm{land}} > 0$, with radius ϱ_l satisfying

$$\begin{aligned} \varrho_e \le \varrho_l \le 2\varrho_e, \qquad & \partial_{\log \varrho_e} \log \varrho_l = 1 + O(\epsilon_e) > c > 0, \\ \left|\partial^j_{\log \varrho_e} \log \varrho_l\right| \le C_j \epsilon_e \quad (2 \le j \le k), \qquad & \epsilon_e = m_{\mathrm{land}} + (s + \varrho_e)|\log(s + \varrho_e)|, \end{aligned} \tag{6.14}$$

with constants independent of m_{land}. If a graph has $y \ge y_0$ and $z < m_{\mathrm{land}}\varrho^2$ at a radius $\bar\varrho$ of the box, its first landing after $\bar\varrho$ has radius at most $2\bar\varrho$. The construction applies to admissible graphs before their endpoint.

Proof. The adapted height coordinate. At backward time $t \ge 0$, the base flow of (6.11) is explicitly

$$g_s(t) = \frac{1 - \mathrm{e}^{-st}}{s}, \quad \varrho_b(t) = \frac{\varrho \mathrm{e}^{-st}}{1 + \varrho g_s(t)}, \quad u_b(t) = u\mathrm{e}^{st}(1 + \varrho g_s(t)), \quad v_b(t) = v\mathrm{e}^{-st},$$

where $g_0(t) = t$. Since $0 \le g_s(t) \le t$ and the denominator is at least one, each fixed-order state derivative is bounded by a polynomial in t times e^{jst}. The bounds are uniform for initial base points in the original compact box. To cover backward trajectories that leave it, extend the coefficients by smooth bounded retractions of the base variables into the enlarged positive box, preserving the positive height-rate and height-derivative margins. The extension agrees with the original coefficients and makes $\mathfrak{a}, \mathcal{N}$, together with their required derivatives, bounded on the entire nonnegative base domain and a small negative-height strip. Along the backward trajectories, ϱ decreases.

Choose $c_y > 0$ so that $\partial_y y_\tau \ge 2c_y$ on the extended box, and reduce s so that $(k+1)s < c_y$. The base-flow bounds above verify Lemma 4.6 with $c_x = s$. It supplies Θ with the asserted bounds on Θ/y and the inverse, and $\Psi_\tau = \mathfrak{a}\Psi$. The logarithmic derivative of $\varrho^2(s + \varrho)$ in τ is $2s + 3\varrho$, so $(\log \widehat{z})_{\mathfrak{l}} = 1/(\varrho^+\mathcal{K})$. This is (6.12), by (6.10).

The exit radius. At the thin exit, $\mathfrak{l}_e = \log \varrho_e$ solves

$$\mathcal{M}(\mathfrak{l}_e) = \lambda_{\mathrm{in}}, \qquad \mathcal{M}(\mathfrak{l}) = \log[\varrho^2(s + \varrho)\Psi(u, v, \varrho, y_0, p)] - \int_{\varrho_i}^{\varrho} \frac{t\,dt}{-q(t)}.$$

The transport equation gives

$$\mathcal{M}'(\mathfrak{l}) = -\frac{f_0}{\varrho^+}\,\partial_y \log \Psi(u, v, \varrho, y_0, p) = -\frac{g(u, v, \varrho, p)}{\varrho^+}, \qquad c \le g \le C,$$

where $f_0 = \mathfrak{a}y_0 + y_0^2\mathcal{N}(u, v, \varrho, y_0, p)$ is the height velocity at the exit. Fixed logarithmic radial derivatives of g are bounded. The corresponding base vector field is $-u\partial_u + v(1 - v)\partial_v + \varrho\partial_\varrho$, whose fixed iterates applied to ϱ^+ are $O(\varrho^+)$. Set $\phi_0 = -\varrho^+/g$, so that $\partial^j_{\mathfrak{l}}\phi_0 = O_j(\varrho^+)$. Inverse differentiation gives

$$\frac{d^j \mathfrak{l}_e}{d\lambda_{\mathrm{in}}^j} = \left[(\phi_0\partial_{\mathfrak{l}})^{j-1}\phi_0\right]_{\mathfrak{l}=\mathfrak{l}_e}.$$

Each expanded term contains exactly j factors, each a radial derivative of ϕ_0 of some nonnegative order. It is therefore $O_j((s + \varrho_e)^j)$. The sign and two-sided bound at first order follow from $c \le g \le C$, proving (6.13). Since the thin height increases and scalar order holds, the exit is unique and its existence set is one interval of incoming heights.

Passage to the cone. After exit use $\mathcal{Y} = z/\varrho^2 = \varrho^+ y$ as the independent height. The equations become

$$\mathcal{Y}_{\mathfrak{l}} = \frac{\mathcal{Y}}{B\mathcal{Y} + \varrho^+\mathcal{K}} - 2\mathcal{Y}, \qquad \frac{d\mathfrak{l}}{d\mathcal{Y}} = \frac{\varrho^+\mathcal{K} + B\mathcal{Y}}{\mathcal{Y}(1 - 2\varrho^+\mathcal{K} - 2B\mathcal{Y})}.$$

Choose m_{land} small and then the radius box and s small, so that the last denominator factor is at least $1/2$ on $\mathcal{Y} \le m_{\mathrm{land}}$. Starting at $\mathcal{Y}_e = (s + \varrho_e)y_0$, bootstrap $\varrho \le 2\varrho_e$. With $\varrho_e^+ = s + \varrho_e$, integration gives

$$0 \le \log(\varrho_l/\varrho_e) \le C\{m_{\mathrm{land}} + \varrho_e^+ \log(m_{\mathrm{land}}/(y_0\varrho_e^+))\} < \log 2.$$

This closes the bootstrap and proves arrival at $\mathcal{Y} = m_{\mathrm{land}}$. For the derivatives, write $x = \log \varrho_e$ and $\mathfrak{l}(\mathcal{Y}; x) = \log \varrho(\mathcal{Y}; x)$. Denote by $R(\mathfrak{l}, \mathcal{Y})$ the right side of the equation for $\mathfrak{l}$ in the variable $\mathcal{Y}$, with parameters fixed, so that

$$\mathfrak{l}_{\mathcal{Y}} = R(\mathfrak{l}, \mathcal{Y}), \qquad \mathfrak{l}(\mathcal{Y}_e(x); x) = x.$$

On the preceding bootstrap range,

$$|\partial_{\mathfrak{l}}^j R| \le C_j(\varrho_e^+/\mathcal{Y} + 1), \qquad \int_{\mathcal{Y}_e}^{m_{\mathrm{land}}} (\varrho_e^+/\mathcal{Y} + 1)\, d\mathcal{Y} \le C\epsilon_e, \quad \epsilon_e = m_{\mathrm{land}} + \varrho_e^+ |\log \varrho_e^+|.$$

Since the last denominator factor is at least $1/2$, the constants C_j depend only on the bounds for $\mathcal{K}$ and B, and not on m_{land}. The moving initial height is treated as in the moving-entrance part of Lemma 4.1: near the entrance set $\mathcal{Y} = \mathcal{Y}_e(x)\zeta$ and $\mathfrak{l} = x + \tilde{\mathfrak{l}}$. The rescaled equation has right side $\mathcal{Y}_e R$, whose required mixed derivatives are $O_j(\varrho_e^+)$ on a fixed ζ-interval. Conversion back to fixed $\mathcal{Y}$ uses $\partial_x|_{\mathcal{Y}} = \partial_x|_\zeta - v_e\zeta\partial_\zeta$, where $v_e = \varrho_e/\varrho_e^+$ has bounded derivatives. Thus the initial data for $P_j = \partial_x^j \mathfrak{l}|_{\mathcal{Y}}$ are

$$P_1(\mathcal{Y}_e) = 1 - R(x, \mathcal{Y}_e) y_0 \varrho_e = 1 + O(\varrho_e^+), \qquad P_j(\mathcal{Y}_e) = O_j(\varrho_e^+) \quad (j \ge 2).$$

Their triangular variational equations have linear coefficient $R_{\mathfrak{l}}$ and higher forcing coefficients $\partial_{\mathfrak{l}}^m R$. The displayed integral bound and variation of constants therefore give $P_1(m_{\mathrm{land}}) = 1 + O(\epsilon_e)$ and $P_j(m_{\mathrm{land}}) = O_j(\epsilon_e)$ for $j \ge 2$, proving (6.14).

For an input above the thin threshold at a radius $\bar{\varrho}$, the same equation in $\mathcal{Y}$, started at a larger value of $\mathcal{Y}$, gives radial growth to $\mathcal{Y} = m_{\mathrm{land}}$ no larger than for an input at that threshold; hence its landing radius is at most $2\bar{\varrho}$. An input with $\mathcal{Y} \ge m_{\mathrm{land}}$ already has the required cone height.

Reaching the cone before the endpoint. The organizing endpoints satisfy $\eta_\pm^0 = \pm a_* r_* + r_*^2$, in the notation of Lemma 2.5, by (2.7). Hence $z_*/(\eta_\pm^0)^2 \to 1/2$ as $z_* \downarrow 0$. For each sufficiently small fixed z_*, the endpoint windows and parameter neighborhood can therefore be chosen so that $\mathcal{Y}_* = z_*/\varrho_*^2 \ge c_{\mathrm{end}} > 0$, with one c_{end} independent of z_*. Thus $m_{\mathrm{land}} < c_{\mathrm{end}}/2$ can be fixed before z_*. The endpoint's normalized thin height is $y_* = \mathcal{Y}_*/(s + \varrho_*) > y_0$ on the small box in the coordinates (η, z). Thus every admissible graph that starts thin exits and lands before the fixed endpoint section. Monotonicity of the positive radius keeps the construction in the chosen radius box. □

The preceding lemma controls the moving exit and its first cone landing. To obtain a return-map estimate, we now differentiate the endpoint reached from this landing. At $t = m_{\mathrm{land}}\varrho_l^2$, the entrance energy is $U = t/(2m_{\mathrm{land}})$. By the choice $1/(2m_{\mathrm{land}}) > 4 \sup B$, the moving-cone part of Lemma 4.1 applies whenever (4.1) holds there. The endpoint energy then has

$$\frac{dU(z_*; t/(2m_{\mathrm{land}}))}{dt} = (z_*/t)^{-2q_2} a(t), \qquad c < a(t) < C.$$

Under this tail condition, suppose also that the adapted incoming height satisfies $0 < c \le d\lambda_{\mathrm{in}}/d\log h \le C$, where $\lambda_{\mathrm{in}} = \log \widehat{z}(\varrho_i)$. Then

$$hI_{+,h} \asymp \varrho_e^2(s + \varrho_e)(z_*/(m_{\mathrm{land}}\varrho_l^2))^{-2q_2}. \tag{6.15}$$

Indeed (6.13) and (6.14) give $-dt/d\log h \asymp t(s + \varrho_e)$, and $I_+ = z_* - U(z_*)$.

Corollary 6.6 (Contraction after a large exit)**.** *In the setting of Lemma 6.5, let $\mathcal{J} \subset \mathcal{H}_p$ be an interval of central heights whose positive half-orbits are thin at ϱ_i, satisfy (4.1) at their landings, and have $c \le d\lambda_{\mathrm{in}}/d\log h \le C$. Let c_1 be the constant of the lower bound in (6.15). Suppose that $q_2 \le C_2 s$ and $0 < m_- \le C_1 s^3$ on $\mathcal{J}$, with fixed C_1, C_2, and fix $M_1 \ge 1$ with $c_1 M_1^2(1 + M_1) > 2\Lambda_+ C_1$. Then, for sufficiently small s, the central heights in $\mathcal{J}$ with $\varrho_e \ge M_1 s$ form an interval on which $\mathcal{L} < 0$. In particular Δ_p is strictly decreasing there and has at most one zero.*

Proof. By (6.13) and the lower bound for $d\lambda_{\mathrm{in}}/d\log h$, the exit radius decreases with h, so these central heights form an interval. The landing precedes the endpoint, so $t = m_{\mathrm{land}}\varrho_l^2 < z_*$, and the multiplier in (6.15) is at least one if $q_2 \le 0$. If $0 < q_2 \le C_2 s$, then $\varrho_l \ge \varrho_e \ge s$ gives $t \ge m_{\mathrm{land}} s^2$, and

$$(z_*/t)^{-2q_2} \ge \exp[-2C_2 s \log(z_*/(m_{\mathrm{land}} s^2))] \ge \tfrac{1}{2}$$

for small s. In both cases $m_+ \ge \frac{1}{2} c_1 \varrho_e^2 (s + \varrho_e)$, and therefore

$$\frac{m_-}{m_+} \le \frac{2C_1}{c_1} \frac{1}{(\varrho_e/s)^2(1 + \varrho_e/s)}.$$

The endpoints of central heights in $\mathcal{H}_p$ lie in the windows, where $\Pi_p' \le \Lambda_+$ by Lemma 2.5. For $\varrho_e \ge M_1 s$, the choice of M_1 gives $\mathrm{e}^{\mathcal{L}} = \Pi_p'(I_-) m_-/m_+ < 1$, and $\Delta_p' = I_+'(\mathrm{e}^{\mathcal{L}} - 1) < 0$ by (3.1). □

6.4. **A boundary root approaching the central line.** When the constant coefficient is small compared with the linear one, the first boundary root on the positive branch approaches the central line relative to the current coefficient scale. Rescaling by its radius keeps this root simple and near $x = 1$. In the resulting logarithmic height coordinate, the positive endpoint has a large negative logarithmic rate. A uniform lower bound for the negative endpoint's rate then makes $\mathcal{L}$ increase throughout the root interval.

Let $a, \kappa > 0$ be small and $\hat{r} \in (0, 1]$, and suppose

$$q_0 = -\kappa\ell^2, \qquad q_1 = \kappa\ell, \qquad q_2 = -\epsilon_2\kappa, \qquad \ell = \hat{r}\kappa a^2, \qquad |\epsilon_2| \le C_2 a, \tag{6.16}$$

with a fixed C_2. In $|\eta| = \ell x$, $z = \ell^2 Y$ the outward drifts and height equations are

$$K_\sigma(x) = -\frac{q(\sigma\ell x)}{\kappa\ell^2} = 1 - \sigma x + \epsilon_2 x^2 - \sigma\hat{r} a^2 x^3 R_q(\sigma\ell x), \qquad x_{\log Y} = \kappa K_\sigma(x)/x + Y B(\sigma\ell x, \ell^2 Y)/x.$$

For a fixed small preliminary upper bound on a, the first positive root of K_+ satisfies $\alpha = 1 + O(a)$ and $K_+'(\alpha) < -c < 0$. Put $x_L = \alpha/4$, $w_a = \alpha/2$ and $R_p = 3\alpha/4$, and use the logarithmic height coordinates $T = -\kappa a \log h$ and $\widehat{T} = T/a = -\kappa \log h$. We compare the branches in T, after estimating the positive branch in $\widehat{T}$. For a fixed small height $Y_f > 0$, the root interval is

$$w = \mathcal{W}_+(x(\ell^2 Y_f; h), Y_f) \in [w_a, \alpha). \tag{6.17}$$

The negative half-orbit need not be at the weak-root radius when the positive half-orbit approaches its root. To estimate it on the same central heights, introduce a variable radius scale: let $d > 0$ and put $\ell_d = \ell d/a$, $\delta_d = \kappa a/d$ and $a_d = a/d$, so that $a_d/\delta_d = 1/\kappa$. In $|\eta| = \ell_d x$, the outward drift on the negative branch is

$$K_d(x) = a_d + x + (\epsilon_2/a) d x^2 + \hat{r} d^2 x^3 R_q(-\ell_d x),$$

so that $K_d(x) = K_1(dx)/d$. Fix $d_{\max} > 0$ such that, for $0 < d \le d_{\max}$,

$$K_d(x) \asymp a_d + x, \qquad K_d'(x) \ge \tfrac{1}{2} \qquad (0 \le x \le 4), \tag{6.18}$$

with constants depending on $d_{\max}$. This holds for every small $d_{\max}$, and for every fixed $d_{\max}$ when $\epsilon_2 \ge 0$ and $\hat{r}$ is small.

Lemma 6.7 (Two branches at the weak root)**.** *There are $c_1, c_d > 0$ and a fixed small $Y_b > 0$ such that, for sufficiently small a and κ, the following hold in the coordinate T.*

(i) *On the root interval (6.17), $\mathfrak{H}_+ \le -c_1/a$.*

(ii) *Let $c_d a \le d \le d_{\max}$, and let $J \subset [1/2, x_{\max}]$ be a compact interval, with a fixed $x_{\max}$, such that $K_d > 0$ on $(0, \max J)$. At the height $\ell_d^2 Y_b$, and for negative projected states in J,*

$$d\,\mathfrak{H}_- \ge \min_{x \in J} \frac{K_d'(x)}{x} - o(1), \qquad |I_{-,T}| \le C E_d/d = o(1), \qquad E_d = \ell_d^2 (z_*/(\ell_d^2 Y_b))^{-2q_2},$$

with $o(1) \to 0$ as $\kappa \to 0$ uniformly in d and J, up to a root of K_d at $\max J$; the constants and the smallness of a, κ depend on $x_{\max}$. In particular, $\mathfrak{H}_- \ge c/d > 0$ for $J = [0.9, 4]$.

(iii) *On* (6.17)*, the negative graph has* $z < (Y_b/2)|\eta|^2$ *at* $|\eta| = \ell c_d$.

Proof. (i) The boundary-flow coordinate applies with small parameter κ, since the error from B is $O(\ell) = O(\kappa)$. At $t = \ell^2 Y_f$ the coefficient ratios are $|q_0|/t = \kappa/Y_f$, $q_1/\sqrt{t} = \kappa/\sqrt{Y_f}$ and $|q_2| \le C_2\kappa a$. Hence Lemma 4.9 gives $-I_{+,w} = E_a A_+ w V_+(w)$, with $E_a = \ell^2(z_*/(\ell^2 Y_f))^{-2q_2}$ and $V_+ = 1 + O_{C^3}(\kappa)$. The fixed positive phase advance from x_L to w_a allows us to apply Lemma 4.8 in the coordinate $\widehat{T}$. Since $K_+'(w)/w$ has a fixed negative margin on $[w_a, \alpha)$, Corollary 4.11 gives (i).

(ii) By (6.18) and $a_d/\delta_d = 1/\kappa \ge 1$, the case of a small central coefficient in Lemma 4.7, with $K_* = a_d$, gives a uniform height unit at the entry $x_i = 1/4$, also as $a_d \to 0$. The boundary-flow projection at Y_b, on a fixed radius interval extending $[1/4, x_{\max}]$, has inverse energy error $O_{C^k}(\delta_d Y)$, since $\ell_d/\delta_d = \hat{r} d^2$ is bounded. By Lemma 4.3, the projection's height bound Y_0 depends only on the lower end of that interval, so Y_b does not depend on $x_{\max}$.

For output states in J, the advance from x_i to $1/2$ takes a fixed positive boundary time, since K_d/x is bounded there. Thus Lemma 4.8 applies uniformly up to a root of K_d. At $\ell_d^2 Y_b$, the coefficient ratios are $\delta_d a_d/Y_b$, $\delta_d/\sqrt{Y_b}$ and $|q_2| \le C_2\kappa a$, and $\ell_d \to 0$. These quantities are small because $\delta_d \le \kappa/c_d$ and $a_d \le 1/c_d$. Lemma 4.9 with $\ell = \ell_d$, $\delta = \delta_d$ and $J_w = [1/2, x_{\max}]$ therefore gives $-I_{-,w} = E_d A_d w(1 + O_{C^3}(\delta_d))$.

To express these estimates in the common coordinate, use $\widehat{T}_d = -\delta_d \log h = T/d$, so $\partial_T = d^{-1}\partial_{\widehat{T}_d}$. In $\widehat{T}_d$, the first two lines of (4.26), with $\theta - 1, \theta' = o(1)$, give $|I_{-,\widehat{T}_d}| \le C E_d$ and $\partial_{\widehat{T}_d} \log(h I_{-,h}) \ge \min_J K_d'(x)/x - o(1)$. Division by d gives the two estimates. On $[0.9, 4]$, $K_d'(x)/x \ge 1/8$ by (6.18).

Finally, $|q_2| \le C_2\kappa a < 1/8$, so (4.10) with $t = \ell_d^2 Y_b$ gives $E_d \le C\ell_d^{3/2}$ and $E_d/d \le C(\ell/a)^{3/2} d^{1/2} \to 0$, uniformly for $0 < d \le d_{\max}$, since $\ell/a = \hat{r}\kappa a \to 0$.

(iii) Choose c_d so small that the negative prefix action to $x = c_d$ in the weak-root scale is less than one quarter of the fixed positive phase advance τ_* from x_L to w_a. On (6.17) this advance gives $h/(\kappa\ell^2) \le (Y_f/\kappa)\mathrm{e}^{-\tau_*/\kappa}$. The negative central-height coordinate then gives

$$\frac{z(-\ell c_d; h)}{(\ell c_d)^2} \le \frac{C Y_f}{c_d^2}\, \mathrm{e}^{-3\tau_*/(4\kappa)} < \frac{Y_b}{2}$$

for small κ: the exponential decay in $1/\kappa$ absorbs the bounded height units and the polynomial prefactors. □

Corollary 6.8 (Negative branch up to its first root)**.** *Assume* (6.16) *and* (6.18)*, with* Y_b *as in Lemma 6.7, and suppose that the negative drift* K_1 *in* $|\eta| = \ell X/a$ *has a first positive root* $\beta \le \beta^*$*, with a fixed* β^**. For sufficiently small* a *and* κ*, every central height whose negative graph has* $z < Y_b|\eta|^2$ *at* $|\eta| = \ell_{d_{\max}}$ *satisfies, in the coordinate* T*,*

$$\mathfrak{H}_- \ge \min_{d_{\max}/2 \le X \le \beta} \frac{K_1'(X)}{X} - o(1), \qquad |I_{-,T}| = o(1),$$

where $o(1) \to 0$ *as* $\kappa \to 0$*, uniformly as the negative projected state at the height* $\ell_{d_{\max}}^2 Y_b$ *approaches the root.*

Proof. By (6.18) at $d = d_{\max}$ and $K_d(x) = K_1(dx)/d$, $K_1 > 0$ on $(0, 4d_{\max}]$. The first positive root of $K_{d_{\max}}$ is therefore $\beta/d_{\max} > 4$. In $|\eta| = \ell_{d_{\max}} x$, $z = \ell_{d_{\max}}^2 Y$, the graph reaches $x = 1$ below $Y = Y_b$. Equation (4.13) puts its projected state there above $1/2$, since Y_b and κ are small. The projected state then increases; by Remark 4.4, it does not cross $\beta/d_{\max}$, and the graph continues to Y_b.

At that height the projected state lies in $J = [1/2, \beta/d_{\max}]$. Apply part (ii) of Lemma 6.7 with $d = d_{\max}$, this J, and $x_{\max} = \beta^*/d_{\max}$. The assertion follows from $K_d'(x)/x = d\, K_1'(dx)/(dx)$, with $d_{\max}$ fixed. □

On the root interval, a negative graph is controlled by Lemma 6.7 if $z/|\eta|^2$ reaches Y_b in the prescribed radius range. The following proposition isolates the additional estimate needed for graphs that remain below this level. Each application will verify that estimate on its own coefficient scale.

Proposition 6.9 (Weak root)**.** *Assume* (6.16) *and* (6.18)*, and let Y_b be as in Lemma 6.7. Suppose there are C_n and $\varepsilon_p \to 0$ such that every central height in the root interval* (6.17) *whose negative graph has $z < Y_b|\eta|^2$ at $|\eta| = \ell_{d_{\max}}$ satisfies*

$$\mathfrak{H}_- \geq -C_n, \qquad |I_{-,T}| \leq \varepsilon_p. \tag{6.19}$$

Then there are $a_\circ, \kappa_\circ > 0$ such that, for $0 < a \leq a_\circ$, $0 < \kappa \leq \kappa_\circ$ and all sufficiently small remaining parameters, every zero of $\mathcal{L}$ on $\mathcal{H}_p$ satisfies $\mathcal{L}_T > 0$. Consequently Δ_p has at most two distinct zeros there.

Proof. The negative branch on the root interval. By Lemma 6.7(iii), the ratio $z/|\eta|^2$ along the negative graph is below $Y_b/2$ at $|\eta| = \ell c_d$. If the ratio reaches Y_b at a radius ℓ_d with $c_d a < d \leq d_{\max}$, choose the first such d for the central height under consideration. At $\ell_d^2 Y_b$, the normalized radius is then one, and small Y_b puts the projected state in $[0.9, 4]$. Hold this section fixed when varying the central height locally. The resulting representation of I_- agrees with $I_-(h)$, so Lemma 6.7(ii) applies as in Remark 3.5. Otherwise, the ratio is still below Y_b at $|\eta| = \ell_{d_{\max}}$, and (6.19) applies. In both cases, $\mathfrak{H}_- \geq -C_n$ and $|I_{-,T}| = o(1)$ throughout (6.17).

The root interval. Combine this bound for the negative branch with Lemma 6.7(i) for the positive branch. Corollary 4.11 gives $\mathcal{L}_T > 0$ throughout (6.17), after $a_\circ$ is reduced. The constants are fixed on the preliminary small-a region.

The height cover. Apply Lemma 6.4 with $\delta = \kappa$, $\epsilon_3 = \ell/\kappa = \hat{r}a^2$, comparison drifts $1 - \sigma x + \epsilon_2 x^2$, and the radii w_a, R_p, α above. The comparison drifts have the required uniform positive margin, the gap $R_p - w_a$ is uniform, and the boundary-flow inverse is defined up to α at Y_f. Together with $\mathcal{L}_T > 0$ on the root interval, the lemma gives at most two distinct zeros of Δ_p on $\mathcal{H}_p$. □

7. Negative offsets on the primary scale

On the primary scale the cubic term remains in the limiting boundary drift. Its contribution gives opposite signs to the curvature coefficient $wK'' - K'$ on the two branches. When the normalized constant coefficient is bounded away from zero, we combine this curvature estimate on bounded radii with contraction at early heights and large exits. In particular, Lemma 4.10 gives concavity of $\mathcal{L}$ through a root-free passage and up to simple or double boundary roots, and Lemma 6.5 treats the positive half-orbits that leave this scale. When the constant coefficient is small compared with $q_1^{3/2}$, a boundary root approaches the central line, and Proposition 6.9 applies on the linear scale.

7.1. Negative drift on the primary scale.

Assume the magnitude of the constant coefficient has a positive normalized lower bound on the primary scale. Positive boundary roots may still merge. At bounded radii the drift-weighted curvature estimate continues through this fold; at early heights and large exits we use contraction instead. We prove that these ranges overlap and cover $\mathcal{H}_p$, so that Corollary 3.2 applies to a single logarithmic slope ratio on the full interval.

Proposition 7.1 (Primary negative drift)**.** *For every fixed $a_0^{\mathrm{pr}} > 0$, the return displacement has at most three distinct zeros when*

$$q_0 = -aH^3, \quad q_1 = bH^2, \quad q_2 = \bar{c}H, \qquad a_0^{\mathrm{pr}} \leq a \leq 1, \quad 0 \leq b \leq 1, \quad |\bar{c}| \leq 1, \tag{7.1}$$

for all sufficiently small $H > 0$ and original parameters.

Write $|\eta_\sigma| = HX$, $z = H^2 Y$. The outward boundary drifts are

$$K_\sigma(X) = -H^{-3} q(\sigma H X) = a - \sigma b X - \bar{c} X^2 - \sigma X^3 R_q(\sigma H X). \tag{7.2}$$

With the central-height scale $Q = H^3$ of Table 2, the common logarithmic height coordinate below is $T = -H\log(h/Q)$. The coefficient calculation in the appendix shows that the two drifts converge, on fixed boxes and in every required state order, to

$$K_+^0 = a - bX - \bar{c}X^2 + X^3, \qquad K_-^0 = a + bX - \bar{c}X^2 - X^3.$$

For $X > 0$, their curvature coefficients are $b/X^2 + 3$ and $-b/X^2 - 3$, respectively, so the endpoint estimates below must preserve these opposite signs uniformly as positive boundary roots merge. The drift K_-^0 has a unique, uniformly simple positive root β separated from zero. Its derivative has at most one positive zero, and

$$-\frac{(K_-^0)'(\beta)}{\beta} = 2a/\beta^2 + b/\beta + \beta > 0.$$

Compactness gives the uniform bounds, which persist under the $O(H)$ perturbation. All positive roots of K_+^0, if present, also lie in a fixed compact interval away from zero.

On every fixed positive radius interval, the height equation is

$$X_{\log Y} = HF_\sigma(X) + YB(\sigma HX, H^2Y)/X, \qquad F_\sigma = K_\sigma/X.$$

Thus Lemma 4.3 applies with $\delta = H$, and (4.13) holds for its inverse projection. Below the first root of F_σ, where $F_\sigma > 0$, Remark 4.4 applies.

7.1.1. *Early central heights.* Choose a small fixed $d_f > 0$ and put $z_e = H^3 d_f$. The present scales are (6.1) with $\delta = \ell = H$ and $\bar{c}$ replaced by $-\bar{c}$. By Lemma 6.1, there is a small fixed $R > 0$ such that the set (6.2), with $r_0 = R$, is an initial interval in T on which the return contracts.

7.1.2. *Bounded radii, including the fold.* For every remaining central height, the positive half-orbit reaches the radius R before z_e. Fix a small $Y_f > 0$ for the central entry and the negative-root box below, and let M be large; M is fixed after M_1 in the covering step. The same Y_f serves every such M: at $H = 0$ the positive projection on $[R/2, M]$ is $w^2 = X^2 - 2B_0Y$, and only its lower-radius margin matters, while a larger M requires only a smaller final H for the derivative bounds. Consider those central heights with $X_+(H^2Y_f; h) \le M$. Their projected state w_+ at this height lies in $[R/2, M]$: its value at z_e, when used, differs from $X_+(z_e)$ by $O(Hd_f)$, and w_+ increases. The phase comparison below also places their negative-branch state w_- in $[d_0, \beta)$ for a fixed $d_0 > 0$.

Choose a small fixed $x_L > 0$ inside the central positive-drift margins of both K_σ. With

$$\mathcal{A}_\sigma(x) = \int_0^x \frac{u\,du}{K_\sigma(u)},$$

(4.19) gives

$$w_\sigma = \Phi_{F_\sigma}^{T + C_\sigma + H\Xi_\sigma(\xi_\sigma)}(x_L), \quad C_\sigma = H\log(Y_f/H) - \mathcal{A}_\sigma(x_L), \quad \xi_\sigma = \mathrm{e}^{\mathcal{A}_\sigma(x_L)/H} h/H^3. \tag{7.3}$$

The phase corrections Ξ_σ vanish at zero and have uniformly bounded derivatives of the required orders. Positive advance to $R/2$ takes a fixed positive boundary time. Take both entry actions smaller than one quarter of this minimum time. The two phases in (7.3) then have a common positive lower bound. The negative scalar flow advances into a fixed interval away from zero and remains below its first root. Thus $d_0 \le w_- < \beta$, and the inverse projection continues the negative graph to Y_f.

At $t = H^2Y_f$, all three coefficient ratios in (4.1) are $O(H)$, and Lemma 4.9 with $\ell = \delta = H$ gives

$$-I_{\sigma,w} = EA_\sigma^* w V_\sigma(w), \qquad E = H^2(z_*/(H^2Y_f))^{-2q_2} \asymp H^2, \qquad A_\sigma^* \asymp 1, \quad V_\sigma = 1 + O_{C^3}(H). \tag{7.4}$$

The negative endpoint estimate must also hold in the large-exit range, before the positive radius cut is chosen. The positive states range over $[R/2, M]$, so the constants of the positive endpoint depend on M. The negative states lie in $[d_0, \beta)$, and the constants of the negative endpoint are independent of M.

The drifts (7.2) have $K_\sigma'(0) = -\sigma b$ and, in the notation of (4.29), $R_{K_\sigma}(w) = -\sigma R_q(\sigma H w)$, which is $\sigma + O_{C^2}(H)$ on the state intervals by the coefficient calculation in the appendix. Hence (4.29) gives

$$\frac{wK_+'' - K_+'}{w^2} = \frac{b}{w^2} + 3 + O_M(H) > 2, \qquad \frac{wK_-'' - K_-'}{w^2} = -\frac{b}{w^2} - 3 + O_M(H) < -2. \tag{7.5}$$

The margins hold also when $b = 0$. Since the negative state stays in $[d_0, \beta)$, Lemma 4.10 gives $\mathcal{L}'' < 0$ in T, uniformly up to either root. The bounded-radius central heights form an interval, by Lemma 3.3.

7.1.3. *Escaping radii and their endpoint slope.* Fix M_0 beyond all positive roots and critical points of the limiting primary polynomial. For the positive radius $\varrho = \eta \geq M_0 H$, write

$$\varrho^+ = H + \varrho, \qquad u = H/\varrho, \qquad v = \varrho/(H + \varrho).$$

Then

$$-q(\varrho) = \varrho^2 \varrho^+ \mathcal{K}, \qquad \mathcal{K} = v\{au^3 - bu^2 - \bar{c}u - R_q(\varrho)\}.$$

For large M_0 this $\mathcal{K}$ is positive and has uniform derivative bounds on an enlarged base box. It equals $1 + O(u + H + \varrho)$. Thus Lemma 6.5 applies with $s = \varrho_0 = H$. Its exit and landing satisfy $\varrho_e \leq \varrho_l \leq 2\varrho_e$, with uniform derivatives, before the admissible endpoint.

If the thin exit is at $\varrho_e \geq 2M_0 H$, there is no positive boundary root before $M_0 H$. Indeed at a thin hit of $M_0 H$ the primary height is $Y = O_{M_0}(H)$. The projection, constructed on the fixed interval up to M_0, differs from the radius by $O_{M_0}(H)$ there. Since any first root is separated from M_0, a projected state below that root cannot give such a hit.

On these no-root parameters, transport the regular central height coordinate from x_L to M_0, using (4.21). The endpoint collars have fixed positive drift margins, although the action between them may diverge. In the height $y = z/H^3$, this gives

$$\Psi_o(y_o) = \mathrm{e}^{\mathcal{A}_+(M_0)/H} \frac{h}{H^3} \Upsilon\left(\mathrm{e}^{\mathcal{A}_+(x_L)/H} \frac{h}{H^3}\right), \qquad \Psi_o(y) = y + O(y^2), \quad \Upsilon(0) = 1. \tag{7.6}$$

By (4.22), the units and height coordinates, together with their inverses, have uniform derivative bounds in the expanded input height.

There is a fixed positive boundary-action gap from M_0 to $2M_0$. Hence an exit beyond $2M_0 H$ makes the height at $M_0 H$, normalized by H^3, at most $CH^{-N}\mathrm{e}^{-\mathcal{A}_{\mathrm{gap}}/H}$. The unscaled exit radius is bounded, so its prefactor costs only a fixed power of $1/H$. Differentiating the regular units in (7.6) gives

$$\frac{d \log \widehat{z}(M_0 H)}{d \log h} = 1 + O(\mathrm{e}^{-\mathcal{A}_{\mathrm{gap}}/(2H)}).$$

This is the bound on $d\lambda_{\mathrm{in}}/d\log h$ required in (6.15), with $\varrho_i = M_0 H$. The tail condition there holds at the landing: $\varrho_l \geq M_0 H$ gives $|q_0|/(m_{\mathrm{land}}\varrho_l^2) + q_1/\sqrt{m_{\mathrm{land}}\varrho_l^2} = O(H)$, and $|q_2| \leq H$.

To compare with the positive endpoint slope, return to the negative branch. It retains a fixed positive entry-time gap; its projected state lies in $[d_0, \beta)$, regardless of the large positive exit. The independently chosen negative endpoint bounds in (7.4) therefore give

$$0 < m_- = hI_{-,h} \leq CHE \leq CH^3.$$

Thus Corollary 6.6 applies, with $s = H$, $C_2 = 1$ and C_1 the constant just obtained, to the central heights in $\mathcal{H}_p$ with $\varrho_e \geq 2M_0 H$, an interval since the exit radius decreases with h. Fix $M_1 \geq 2M_0$ as required by the corollary; this gives the large-exit interval $\varrho_e \geq M_1 H$.

7.1.4. *Covering the central heights.* The bounded-radius estimate allows any fixed M, whereas the large-exit threshold M_1 uses only the independently fixed negative endpoint bound. We may therefore choose M sufficiently large after M_1. We show that every height outside the bounded-radius range satisfies the large-exit estimate. If the positive half-orbit of a central height enters the adaptive range thin and has not exited when its radius reaches MH, its eventual exit is larger. If it has exited, its radius before cone landing is at most $2\varrho_e$. After landing the cone estimates give, whenever the fixed output height is reached,

$$U_+(H^2Y_f) \le C\varrho_e^2 + CH^2Y_f.$$

If the half-orbit was already above the thin threshold at M_0H but below the cone, Lemma 6.5 bounds its landing radius by $2M_0H$, and the same inequality holds with $\varrho_e = M_0H$. If already in the cone, its continuation gives this bound directly. Thus, for a sufficiently large fixed M, every central height outside the bounded-radius range must have $\varrho_e \ge M_1H$.

The early interval (6.2), the bounded-radius interval, and the large-exit interval cover $\mathcal{H}_p$, with fixed overlaps. On these intervals the same function $\mathcal{L}$ of T satisfies $\mathcal{L} < 0$ in the early and large-exit ranges, and $\mathcal{L}'' < 0$ on the bounded-radius interval, including the shared cuts. Corollary 3.2 gives at most three zeros of Δ_p. This proves Proposition 7.1.

7.2. **Negative displaced contact: the weak root on the linear scale.** The preceding estimate loses its normalized constant-coefficient margin when that coefficient becomes small. If $q_2/\sqrt{q_1}$ remains bounded, the approaching positive boundary root gives the weak-root scale. Proposition 6.9 reduces the count to an estimate for the negative branch beyond the weak-root radius. We verify this estimate on the linear scale.

Proposition 7.2 (Negative displaced contact)**.** *For every fixed $A_0 < \infty$, there are fixed $a_{\mathrm{w}}, \kappa_{\mathrm{w}} > 0$ such that the return displacement has at most two distinct zeros on*

$$q_1 = \rho_1^2 > 0, \quad q_2 = \bar{c}\rho_1, \quad |\bar{c}| \le A_0, \quad q_0 = -a\rho_1^3, \quad 0 < a \le a_{\mathrm{w}}, \quad 0 < \kappa = \rho_1/a \le \kappa_{\mathrm{w}},$$

for all sufficiently small ρ_1 and original parameters.

Proof. To apply Proposition 6.9, set $\ell = \rho_1 a = |q_0|/q_1$. Then (6.16) holds with $\hat{r} = 1$ and $\epsilon_2 = -\bar{c}a$, $|\epsilon_2| \le A_0 a$, and the common logarithmic height coordinate is $T = -\rho_1 \log h = -\kappa a \log h$. At the radius $\ell_d = \rho_1 d$ the negative drift is

$$K_d(x) = a_d + x - \bar{c}dx^2 + d^2x^3R_q(-\ell_d x),$$

so (6.18) holds for a small fixed $d_{\max} = d_0$ depending on A_0.

It remains to verify (6.19) for the negative graphs with $z < Y_b|\eta|^2$ at $|\eta| = \rho_1 d_0$. Return to the linear scale ρ_1, with $|\eta| = \rho_1 X$. Since $\ell/a = \rho_1$, the negative drift on this scale is K_1 from Corollary 6.8:

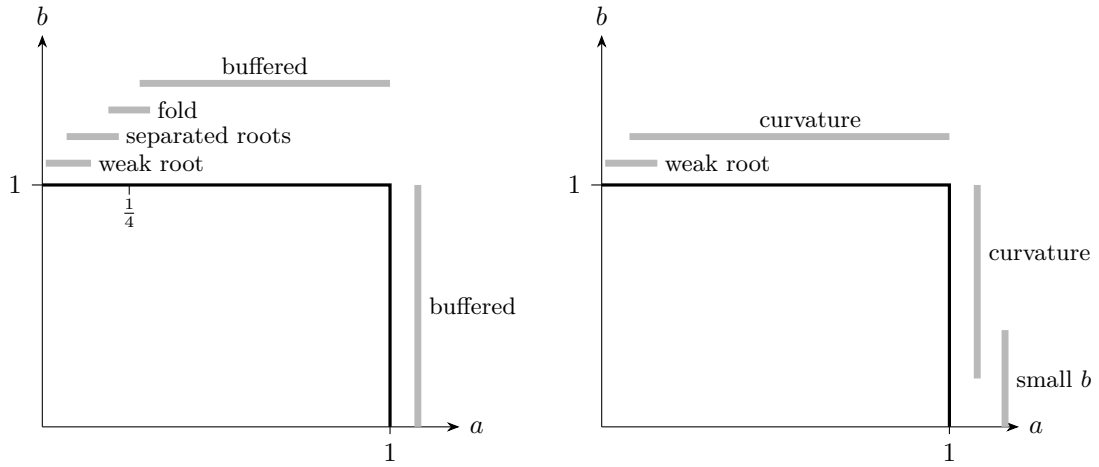
$$K_-^{\rho_1}(X) = a + X - \bar{c}X^2 + X^3R_q(-\rho_1 X).$$

Its first positive root β stays in a fixed compact interval. At $a = \rho_1 = 0$, it is the positive root of $1 - \bar{c}X - X^2 = 0$; the bound $|\bar{c}| \le A_0$ gives uniform separation and a simple-root margin. On $[d_0/2, \beta]$, $(K_-^{\rho_1})'(X)/X \ge -C_{A_0,d_0}$.

Apply Corollary 6.8 with $d_{\max} = d_0$ and with β^* the right end of that compact interval. It gives $\mathfrak{H}_- \ge -C_{A_0,d_0} - 1$ and $|I_{-,T}| = o(1)$ for these graphs, including the limiting negative root. Thus (6.19) holds. Proposition 6.9 now gives at most two distinct zeros of Δ_p, with a_{w} and κ_{w} equal to the constants $a_\circ$ and $\kappa_\circ$ of that proposition. □

8. The secondary scale and positive quadratic drift

When the normalized ratios of q_0 and q_1 to the scale set by q_2 are small, we use the secondary scales s, r of Section 3. For $q_2 > 0$, we compare the projected boundary flows near their first roots and continue the half-orbits to the fixed output windows. We use curvature away from the symmetric limit and the exterior contraction in root-distance coordinates near it. The case $q_2 < 0$ is treated in Sections 9 and 10: we first estimate the annular passage and the negative branch, then treat the positive branch in each coefficient region. For fixed parameters, the early, annular or root, and large-exit ranges form overlapping subintervals of $\mathcal{H}_p$. Their estimates give crossing directions or critical-point curvature for $\mathcal{L}$, which are combined by Corollary 3.2. Thus the coefficient region selects a passage argument, while the ordered height cover combines the estimates within that argument. Figure 4 shows how these regions cover the normalized coefficients of both signs of q_2.



(a) $q_2 < 0$ (b) $q_2 > 0$

Figure 4. Schematic of the compact coefficient covers on the secondary charts (9.1) and (8.1); thresholds are not to scale. The normalized coefficients lie on the curve $\max\{\sqrt{a}, b\} = 1$, and the bars show the overlapping regions. (a) For $q_2 < 0$: the buffered region $b^2 - 4a \le -\nu$ (Proposition 9.5), the fold $|b^2 - 4a| \le 2\nu$ (Proposition 10.2), separated roots $b^2 - 4a \ge \nu$, $a \ge a_0^-$ (Proposition 10.3), and the weak root $a \le a_-$ (Proposition 10.4). (b) For $q_2 > 0$: curvature for $a \ge a_0^+$, $b \ge b_{*,+}/2$, root-distance coordinates for small b, and the weak root $a \le a_+$ (Proposition 8.1). Near $a = 0$, the cover combines the weak-root regions, which require $s/a \le \kappa_\pm$, with the contact strip of Proposition 5.1.

8.1. **Positive quadratic-drift estimates.** Each branch on this chart has a first boundary root. When the central coefficient keeps these roots away from the central line, we compare the endpoint slopes either by curvature or, near the symmetric limit, by distances to the roots. A further rescaling treats the root that approaches the central line.

Suppose $q_2 = s > 0$, $q_0 < 0$, $q_1 > 0$, and write

$$r = \max\{\sqrt{|q_0|/s^3}, q_1/s^2\}, \quad q_0 = -as^3r^2, \quad q_1 = bs^2r, \quad \max\{\sqrt{a}, b\} = 1. \tag{8.1}$$

The small parameters $s, r > 0$ are independent. Put $t_0 = s^2r^2$, $Q = st_0$, and $X = |\eta|/(sr)$. Here sr is the radius scale, $Q = s^3r^2$ the central height scale, and t_0 the fixed-output height scale; the common logarithmic height coordinate used below is $T = -s \log h$. The outward drifts are

$$K_\sigma(X) = a - \sigma bX - X^2 - \sigma rX^3 R_q(\sigma srX), \qquad F_\sigma = K_\sigma/X. \tag{8.2}$$

Proposition 8.1 (Positive quadratic drift)**.** *There is $\kappa_+ > 0$ such that, for all sufficiently small independent $s, r > 0$ and every $L > 1/\kappa_+$, the return displacement has at most three distinct zeros on the part of (8.1) outside the strip $|q_0| \le Lq_1^2$.*

8.1.1. *Constant coefficient bounded away from zero.* First fix $a \ge a_0^+ > 0$. This lower bound keeps both roots in a compact positive radius interval, where we can prepare the endpoint maps uniformly. The roots are simple and satisfy

$$\alpha_\sigma = \frac{\sqrt{b^2+4a} - \sigma b}{2} + O(r), \qquad c \le \alpha_\sigma \le C, \qquad K_\sigma'(\alpha_\sigma) < -c.$$

At height $z = t_0 Y$,

$$X_{\log Y} = sF_\sigma(X) + YB(\sigma srX, t_0 Y)/X.$$

The boundary-flow coordinate applies with $\delta = s$, since the error in B is $O(sr) = O(s)$. At a fixed small output height $Y_f > 0$ the coefficient ratios are $O(s)$, and Lemma 4.9 with $\ell = sr$ and $\delta = s$ gives

$$-I_{\sigma,w} = EA_\sigma w V_\sigma(w), \qquad E = t_0(z_*/(t_0 Y_f))^{-2s} \to 0, \qquad V_\sigma = 1 + O_{C^3}(s), \tag{8.3}$$

on fixed intervals of positive projected states up to the roots. Here $A_\sigma \asymp 1$ and $A_-/A_+ \le \exp(2C_{\mathrm{cone}}\sqrt{z_*} + Cs)$.

Use the shifted coordinate $T_Q = T + s\log Q$, as in (3.2). The phases are

$$\begin{aligned} w_\sigma &= \Phi_{F_\sigma}^{T_Q + C_\sigma + s\Xi_\sigma(\xi_\sigma)}(x_{L,\sigma}), & T_Q &= -s\log(h/Q), \\ \xi_\sigma &= \mathrm{e}^{\mathcal{A}_\sigma(x_{L,\sigma})/s} h/Q, & C_\sigma &= s\log(Y_f/s) - \mathcal{A}_\sigma(x_{L,\sigma}), \\ \mathcal{A}_\sigma(x) &= \int_0^x u/K_\sigma(u)\,du. \end{aligned} \tag{8.4}$$

Choose both regular entry actions below one quarter of the minimum positive terminal action. Then $w_+ \ge \alpha_+/2$ forces a fixed advance on both legs. The negative state lies in a fixed interval away from zero; by Remark 4.4 it stays below α_-, and the inverse projection continues the negative graph.

Suppose also $b \ge \underline{b} > 0$. The drifts (8.2) have $K_\sigma'(0) = -\sigma b$ and, in the notation of (4.29), $R_{K_\sigma}(w) = -\sigma r R_q(\sigma srw) = O_{C^2}(r)$. Hence (4.29) gives, on these compact intervals,

$$\frac{wK_+'' - K_+'}{w^2} = \frac{b}{w^2} + O(r) > c, \qquad \frac{wK_-'' - K_-'}{w^2} = -\frac{b}{w^2} + O(r) < -c.$$

Since the negative state stays in a fixed interval away from zero, Lemma 4.10 gives $\mathcal{L}'' < 0$ on this root interval, including all approaches to its ends.

To extend this comparison to larger central heights, join the root interval to the regular prefix. Apply Lemma 6.4 with $\delta = s$, $\ell = sr$, $\epsilon_3 = r$, comparison drifts $a - \sigma bx - x^2$, $R = 3\alpha_+/4$ and $w_a = \alpha_+/2$. The positive drift margin and the gap $R - w_a$ are fixed, and the boundary-flow inverse is defined up to α_+ at Y_f. Since $\mathcal{L}_{TT} < 0$ on this root range, the lemma gives at most three zeros of Δ_p on $\mathcal{H}_p$.

8.1.2. *Small linear coefficient: comparison in root-distance coordinates.* For small b, the curvature comparison loses its fixed margin. We use linearizing coordinates for the two scalar boundary flows to obtain a power relation between the distances to the roots, then combine it with the exterior contraction. Put

$$k_\sigma = -F_\sigma'(\alpha_\sigma) = 2 + O(b + r) > c.$$

There is a positive root-distance coordinate $\ell_\sigma(w)$, decreasing to zero, such that

$$F_\sigma(w)\ell_\sigma'(w) = -k_\sigma \ell_\sigma(w), \qquad \ell_\sigma'(\alpha_\sigma) = -\alpha_\sigma. \tag{8.5}$$

To construct this coordinate, factor out $\alpha_\sigma - w$. The logarithmic derivative of the remaining unit is $-k_\sigma/F_\sigma + 1/(\alpha_\sigma - w)$, whose pole is removable. Integration and normalization at the root give

uniform derivative bounds and a regular inverse. Extend to the regular entry using the same scalar equation. At $b = r = 0$ the formulas are

$$F_\sigma = (a - w^2)/w, \qquad k_\sigma = 2, \qquad \ell_\sigma = (a - w^2)/2.$$

The constants

$$c_\sigma^{\mathrm{lin}} = \ell_\sigma(x_{L,\sigma}) \exp[k_\sigma \mathcal{A}_\sigma(x_{L,\sigma})]$$

are independent of the entry chosen, by (8.5). They have positive bounded values and smooth dependence on the compact coefficients. The phase identities give

$$\ell_\sigma(w_\sigma) = c_\sigma^{\mathrm{lin}} (Y_f/s)^{-sk_\sigma} \mathrm{e}^{-k_\sigma T_Q} \mathrm{e}^{-sk_\sigma \Xi_\sigma(\xi_\sigma)}. \tag{8.6}$$

Put $x = \ell_+(w_+)$, $y = \ell_-(w_-)$, and $\nu_k = k_-/k_+$. Since $\nu_k k_+ = k_-$,

$$\frac{y}{x^{\nu_k}} = \frac{c_-^{\mathrm{lin}}}{(c_+^{\mathrm{lin}})^{\nu_k}} \exp(-sk_-[\Xi_-(\xi_-) - \Xi_+(\xi_+)]) .$$

The exponential factors involving T_Q and Y_f/s cancel, giving

$$y = u_p x^{\nu_k}(1 + \rho_p(x)), \qquad \nu_k = k_-/k_+ = 1 + O(b + r), \qquad u_p = c_-^{\mathrm{lin}}/(c_+^{\mathrm{lin}})^{\nu_k} = 1 + O(b + r). \tag{8.7}$$

At $b = r = 0$, $c_+^{\mathrm{lin}} = c_-^{\mathrm{lin}} = a/2$ and $\nu_k = 1$, so the leading unit is exactly one. Its estimate is uniform because $a \geq a_0^+$.

On a sufficiently small fixed collar $0 < x < x_c$, a fixed root distance forces a positive phase gap on both legs. The argument of Lemma 4.8, now using $F_\sigma \asymp \ell_\sigma$ near the simple roots, gives

$$|\partial_T^j \xi_\sigma| \leq \varepsilon_{N,j}(s) \ell_\sigma(w_\sigma)^N, \qquad \varepsilon_{N,j}(s) \to 0$$

for every fixed N, j. The two distances are comparable through fixed positive powers by (8.6), and $d \log x/dT = -k_+ \partial_T \tau_+$ stays bounded away from zero. Thus substitution in (8.7) gives

$$|(x\partial_x)^j \rho_p(x)| \leq C_j x^\vartheta, \qquad 0 \leq j \leq 2,$$

with fixed $\vartheta > 0$, uniformly as $h \downarrow 0$.

In these root-distance coordinates the endpoints are

$$I_\sigma(\ell_\sigma) = b_\sigma + E\mathcal{F}_\sigma(\ell_\sigma), \qquad \mathcal{F}_\sigma(0) = 0, \quad c \leq \mathcal{F}'_\sigma \leq C, \quad \|\mathcal{F}_\sigma\|_{C^3} \leq C.$$

The normalization $\ell'_\sigma(\alpha_\sigma) = -\alpha_\sigma$ and (8.3) give

$$\mathcal{F}'_-(0)/\mathcal{F}'_+(0) \leq \exp(2C_{\mathrm{cone}}\sqrt{z_*} + C(s + b + r)).$$

If some central height in the collar lies in $\mathcal{H}_p$, the endpoints on the collar, including the reference endpoints b_σ, differ from admissible endpoints by $O(E)$, so they lie in $\widetilde{W}_\pm$ by Lemma 2.5. In these coordinates, the logarithmic slope ratio is

$$\mathcal{L}(x) = \log \left[\Pi'_p(I_-(y(x))) \frac{\mathcal{F}'_-(y(x))y'(x)}{\mathcal{F}'_+(x)} \right] = c_p + (\nu_k - 1) \log x + e_p(x),$$

where

$$c_p = \log \left[\Pi'_p(b_-) \frac{\mathcal{F}'_-(0)}{\mathcal{F}'_+(0)} u_p \nu_k \right] < -c_0, \qquad |e_p| + |xe'_p| \leq Cx^\vartheta.$$

Choose $b_{*,+}$, then r, s, for this strict margin. The nonlinear exterior contributes $O(Ey(x))$ with the same Euler estimate. Since $dx/dh = I_{+,h}/(E\mathcal{F}'_+(x)) > 0$, the common factor dx/dh cancels from the displayed quotient, and $x_T < 0$. Lemma 3.4 shows that every zero in a sufficiently small fixed collar satisfies $x\mathcal{L}_x < 0$, and that such zeros occur only if $\nu_k < 1$. Consequently $\mathcal{L}_T > 0$ at every collar zero.

Choose $R = (1 - \epsilon_R)\alpha_+$, with fixed small $\epsilon_R > 0$, strictly inside the prepared collar. The positive drift margin is fixed. Apply Lemma 6.4 on the same scales, choosing $w_a < R$ inside this collar with

a fixed gap. Every collar zero has $\mathcal{L}_T > 0$, so Δ_p has at most two zeros for $0 \le b \le b_{*,+}$. Taking $b \ge b_{*,+}/2$ in the curvature argument gives an overlap and covers the compact a region.

8.1.3. *The central weak root.* It remains to allow the constant coefficient to approach zero. For small $a > 0$, normalization in (8.1) forces $b = 1$. Put $\kappa = s/a$. With $\ell = sra$, the scales (6.16) hold with $\hat{r} = r$ and $\epsilon_2 = -a$. At the radius $\ell_d = srd$ the negative drift is

$$K_d(x) = a_d + x - dx^2 + rd^2x^3R_q(-\ell_d x),$$

so (6.18) holds for small fixed $d_{\max} = d_0$ and $r \le r(d_0)$.

For (6.19), consider the negative graphs with $z < Y_b|\eta|^2$ at $|\eta| = srd_0$. In the secondary radius $X = |\eta|/(sr)$ the negative drift is

$$K_-(X) = a + X - X^2 + rX^3R_q(-srX),$$

which is the drift K_1 of Corollary 6.8, since $\ell/a = sr$. Its first root is $\beta = 1 + O(a + r) \le 2$, and $K_-'/X \ge -C$ on $[d_0/2, \beta]$. Corollary 6.8, with $d_{\max} = d_0$ and $\beta^* = 2$, gives $\mathfrak{H}_- \ge -C - 1$ and $|I_{-,T}| = o(1)$ for these graphs, up to the limiting root. Proposition 6.9 gives at most two zeros of Δ_p for $0 < a \le a_+$ and $0 < \kappa \le \kappa_+$, where a_+ and κ_+ are its constants $a_\circ$ and $\kappa_\circ$.

The complement of $\kappa \le \kappa_+$ in the region $0 < a \le a_+$ has $|q_0|/q_1^2 = a/s < 1/\kappa_+$, and lies in the strip $|q_0| \le Lq_1^2$ for every $L > 1/\kappa_+$. The compact-a argument, applied down to a threshold $a_0^+ < a_+$, overlaps the weak-root region. The curvature, root-distance, and weak-root comparisons give bounds three, two, and two, respectively, so their maximum is three outside that strip. The thresholds for s, r are those of the three regions and are independent of L. This proves Proposition 8.1.

9. Negative quadratic drift: annulus and buffered region

For $q_2 < 0$, a half-orbit can pass from the secondary scale to the primary scale through an annulus. The two endpoint rates have the same leading term there; their comparison therefore depends on the first radial corrections and on errors that vanish with the radial variables. We first express these rates in the common central height. We then extend the negative-branch estimate to its primary root and, for positive half-orbits that cross the annulus, extend the positive-branch estimate to larger radii. The buffered region below uses these estimates to count zeros on the whole height interval; the root regimes of Section 10 use them to compare the two branches near positive roots and their degeneracies.

9.1. **The annulus for negative quadratic drift.** For the rest of the secondary analysis put

$$q_0 = -as^3r^2, \qquad q_1 = bs^2r, \qquad q_2 = -s, \qquad 0 < a \le 1, \quad 0 \le b \le 1, \quad \max\{\sqrt{a}, b\} = 1. \tag{9.1}$$

The positive scales s, r tend to zero independently. For a signed radius $\eta = \sigma s\rho$, define

$$u = r/\rho, \quad v = \rho, \quad D_\sigma(\rho) = -q(\sigma s\rho)/s^3, \quad K_\sigma(u, v) = 1 + au^2 - \sigma bu - \sigma vR_q(\sigma sv).$$

The variable u decreases as the radius moves away from the secondary scale, whereas v increases toward the primary scale. For the uniform estimates we regard u, v as independent base variables, including at zero. On the curve $u = r/\rho,\ v = \rho > 0$, we have $K_\sigma(r/\rho, \rho) = D_\sigma(\rho)/\rho^2$. Choose a large fixed K_0 and a small fixed $v_1 > 0$. On a slightly enlarged box $0 \le u \le 1/K_0$, $0 \le v \le 2v_1$, the functions K_σ have uniform positive bounds and uniform derivative bounds. Each branch entering the thin annular box therefore has positive outward radial flux. The regional arguments below identify the central heights in $\mathcal{H}_p$ whose half-orbits reach this box. The radial Euler derivative is

$$\mathcal{E} = \rho\partial_\rho = -u\partial_u + v\partial_v =: \mathcal{E}_{uv}.$$

Here $\mathcal{E}$ acts on the exit radius, and ∂_T, with $T = -s\log h$, on the central height. With $\kappa_3 = -R_q(0) = 1 + O(s)$, Taylor's formula gives

$$K_\sigma = 1 - \sigma bu + \sigma\kappa_3 v + O_{\mathcal{E}_{uv}^j}((u + v)^2). \tag{9.2}$$

The notation $O_{\mathcal{E}^j_{uv}}$ includes every fixed derivative order required below, with uniform constants.

9.1.1. *Transport and central height.* We begin by integrating the height growth through the annulus. The resulting first-exit equation will relate the exit radius to the central height and determine their order. Put $y = z/(s^3\rho^2)$, and use the fast time $d\tau = d\log\rho/s$. The extended equations are

$$u_\tau = -su, \qquad v_\tau = sv, \qquad y_\tau = (K_\sigma^{-1} - 2s)y + y^2\mathcal{N}_\sigma,$$
$$\mathcal{N}_\sigma = -\frac{\mathcal{B}_\sigma}{K_\sigma(K_\sigma + \mathcal{B}_\sigma y)}, \quad \mathcal{B}_\sigma = B(\sigma sv, s^3v^2y).$$

On $0 \le y \le 2\delta_{\mathrm{an}}$, with $\delta_{\mathrm{an}} > 0$ fixed small, the linear coefficient $K_\sigma^{-1} - 2s$, the logarithmic rate y_τ/y, and the derivative $\partial_y y_\tau$ are bounded below by positive constants. We take δ_{an} also so small that $y_0 = \delta_{\mathrm{an}}$ is an admissible thin threshold in Lemma 6.5 for every drift factor $\mathcal{K}$ with $1/8 \le \mathcal{K} \le 2$; this smallness depends only on these bounds and on B. Since $y_\tau/y = 1/(K_\sigma + \mathcal{B}_\sigma y) - 2s$,

$$\frac{d\log y}{d\log\rho} = \frac{1}{s(K_\sigma + \mathcal{B}_\sigma y)} - 2. \tag{9.3}$$

At $y = \delta_{\mathrm{an}}$ the right side is positive wherever $0 < s(K_\sigma + \mathcal{B}_\sigma\delta_{\mathrm{an}}) < 1/2$, and graphs cross this level only upward there. This holds for both signs on the annular box, for $\sigma = -$ also on $\rho \le 0.9$, where K_- keeps a positive margin, and for $\sigma = +$ wherever K_+ has a positive margin and $sK_+ = O(s + |\eta|)$ is small. To integrate this growth exactly, we construct a height coordinate with uniform bounds on its required derivatives,

$$\Psi_\sigma = y\exp\Theta_\sigma, \qquad \Theta_\sigma = y\widetilde{\Theta}_\sigma, \qquad \Psi_{\sigma,\tau} = (K_\sigma^{-1} - 2s)\Psi_\sigma, \qquad c < \Psi_{\sigma,y} < C.$$

As in Lemma 6.5, extend the coefficients by bounded smooth retractions into the enlarged positive box, preserving the height-rate margins. The backward base flow is (ue^{st}, ve^{-st}), with order-j derivatives bounded by e^{jst}. Thus Lemma 4.6 applies with $c_x = s$, after reducing s, and supplies all the stated height and inverse bounds.

Consequently, for $\widehat{z} = s^3\rho^2\Psi_\sigma$,

$$\widehat{z}(\rho) = \exp\left(\frac{1}{s}\int_{\rho_i}^{\rho}\frac{dt}{tK_\sigma(r/t,t)}\right)\widehat{z}(\rho_i), \qquad \rho_i = K_0 r. \tag{9.4}$$

Suppose the central prefix reaches ρ_i with adapted height

$$\widehat{z}(\rho_i) = he^{\mathcal{A}_{\sigma,i}/s}\Upsilon_\sigma(\xi_\sigma), \qquad \xi_\sigma = e^{\mathcal{A}_{\sigma,i}/s}h/(s^3r^2), \qquad \Upsilon_\sigma(0) = 1. \tag{9.5}$$

The positive units Υ_σ, their reciprocals, and their required height derivatives are uniformly bounded for small ξ_σ. After reducing this fixed interval, the maps $\xi \mapsto \xi\Upsilon_\sigma(\xi)$ have uniformly positive derivatives and uniformly bounded inverse derivatives. On regular prefixes this preparation follows from Lemma 4.7. Across a bottleneck it follows from the composition (4.22); this case is carried out in Section 10.2. To separate the possibly unbounded prefix action $\mathcal{A}_{\sigma,i}$ from the annular increment, write

$$\mathcal{A}_\sigma(\rho) = \mathcal{A}_{\sigma,i} + \int_{\rho_i}^{\rho}\frac{dt}{tK_\sigma(r/t,t)}.$$

At the first thin exit after ρ_i, $y = \delta_{\mathrm{an}}$,

$$s\log(h/s^3) = -\mathcal{A}_\sigma(\rho) + 2s\log\rho + s\log\Psi_\sigma(u,v,\delta_{\mathrm{an}}) - s\log\Upsilon_\sigma(\xi_\sigma). \tag{9.6}$$

Take $u \le u_0 < 1/K_0$, with a sufficiently large fixed gap from the inner cut, and write

$$\mathcal{J}_\sigma(\rho) = \mathcal{A}_\sigma(\rho) - \mathcal{A}_{\sigma,i} = \int_{K_0 r}^{\rho}\frac{dt}{tK_\sigma(r/t,t)}.$$

This action increment satisfies $\mathcal{J}_\sigma \ge c\log(\rho/(K_0 r))$. The equation for the central height becomes

$$\xi_\sigma\Upsilon_\sigma(\xi_\sigma) = \mathfrak{b}_\sigma(\rho), \qquad \mathfrak{b}_\sigma(\rho) := u^{-2}e^{-\mathcal{J}_\sigma(\rho)/s}\Psi_\sigma(u,v,\delta_{\mathrm{an}}) > 0.$$

Since $\mathcal{A}_{\sigma,i}$ is constant under $\mathcal{E}$, for every prescribed $j \geq 1$,

$$\mathcal{E}^j \mathcal{J}_\sigma = \mathcal{E}_{uv}^{j-1}(K_\sigma^{-1}), \qquad |\mathcal{E}^j \mathcal{J}_\sigma| \leq C_j.$$

Thus $\mathcal{E} \log \mathfrak{b}_\sigma = 2 - (sK_\sigma)^{-1} + \mathcal{E}_{uv} \log \Psi_\sigma$, and all higher fixed Euler derivatives of its logarithm are $O_j(s^{-1})$. Differentiating its exponential yields

$$|\mathcal{E}^j \mathfrak{b}_\sigma| \leq C_j s^{-j} \mathfrak{b}_\sigma, \qquad \mathfrak{b}_\sigma \leq C u^{c/(2s)-2}, \qquad j \geq 0, \tag{9.7}$$

after decreasing the fixed u_0.

The uniformly regular map $\mathcal{C}_\sigma(\xi) = \xi \Upsilon_\sigma(\xi)$ fixes zero. For small s, $\mathfrak{b}_\sigma \leq 1$ lies in its inverse domain; the chain rule and (9.7) give $|\mathcal{E}^j \mathcal{C}_\sigma^{-1}(\mathfrak{b}_\sigma)| \leq C_j s^{-j} \mathfrak{b}_\sigma$. Since $\log \Upsilon_\sigma(0) = 0$, the same estimate holds for its composition. Thus, for every fixed N, j, after reducing s,

$$|\mathcal{E}^j \xi_\sigma| + |\mathcal{E}^j \log \Upsilon_\sigma(\xi_\sigma)| \leq C_{N,j} u^N. \tag{9.8}$$

Indeed the excess power of $u \leq u_0 < 1$ in (9.7) absorbs s^{-j}. The small expanded incoming height and the outward increase of unscaled height show that the reconstructed prefix is thin.

The event equation gives

$$s\partial_{\log \rho} \log[s^3 \rho^2 \Psi_\sigma(u, v, \delta_{\mathrm{an}})] - 1/K_\sigma = -(y_\tau|_{y=\delta_{\mathrm{an}}})\, \partial_y \log \Psi_\sigma(u, v, \delta_{\mathrm{an}}) < -c.$$

Hence the central height decreases strictly with the exit radius, with inverse derivatives $O(s^j)$ in logarithmic incoming height. Scalar order makes the first-exit domains and radius subintervals into intervals of central heights.

Define the positive central-height rate $G_\sigma = -s\mathcal{E} \log h_\sigma$, and put $g_0 = 1 - 2s$. From (9.6),

$$\begin{aligned} G_\sigma &= K_\sigma^{-1} - 2s - s\mathcal{E}_{uv} \log \Psi_\sigma + s\mathcal{E} \log \Upsilon_\sigma \\ &= g_0 + \sigma b u - \sigma \kappa_3 v + O_{\mathcal{E}^j}((u+v)^2 + s(u+v)), \qquad c < G_\sigma < C. \end{aligned} \tag{9.9}$$

For $j \geq 1$, the bound $\mathcal{E}_{uv}^j \log \Psi_\sigma = O(u+v)$ follows from uniform base derivatives and the vanishing coefficients of $\mathcal{E}_{uv}$.

9.1.2. *The endpoint slope and its vanishing error.* The first-exit equation controls differentiation in central height. We now follow the graph from the thin exit to the fixed section, retaining the radial weight in the slope error. This weight is needed when the leading terms of the two rates cancel.

Lemma 9.1 (Annular endpoint and central-height rates)**.** *Under (9.5), consider a thin exit in the smaller annular box just specified. Let I_σ be its endpoint and $M_\sigma = -\mathcal{E} I_\sigma$. Then $M_\sigma > 0$, and*

$$\mathcal{E} \log M_\sigma = 2g_0 + e_\sigma, \qquad |\mathcal{E}^j e_\sigma| \leq C_j s(1 + |\log s|)(u+v), \quad 0 \leq j \leq 2, \qquad M_\sigma \leq C(s\rho)^{2g_0}. \tag{9.10}$$

In the common logarithmic height coordinate $T = -s \log h$, put $m_\sigma = h I_{\sigma,h} = sM_\sigma/G_\sigma$ and $\mathfrak{H}_\sigma = \partial_T \log m_\sigma$. Then

$$\begin{aligned} \mathfrak{H}_\sigma &= 2 + \sigma(-\gamma_s b u + \delta_s \kappa_3 v) + \varepsilon_\sigma, \\ \gamma_s &= (1 - 4s)/g_0^2, \qquad \delta_s = (3 - 4s)/g_0^2, \\ |\varepsilon_\sigma| + |\partial_T \varepsilon_\sigma| &\leq C(u+v)\{u + v + s(1 + |\log s|)\}. \end{aligned} \tag{9.11}$$

The two signs may have unrelated exit radii.

Proof. After the thin exit use $\mathcal{Y} = z/|\eta|^2$ as the independent variable, and write $\mathfrak{l} = \log(|\eta|/(s\rho))$. Until a small fixed $\mathcal{Y} = m_{\mathrm{land}}$, its equation is

$$\frac{d\mathfrak{l}}{d\mathcal{Y}} = \frac{sK_\sigma(u\mathrm{e}^{-\mathfrak{l}}, v\mathrm{e}^{\mathfrak{l}}) + B_\sigma \mathcal{Y}}{\mathcal{Y}[1 - 2sK_\sigma(u\mathrm{e}^{-\mathfrak{l}}, v\mathrm{e}^{\mathfrak{l}}) - 2B_\sigma \mathcal{Y}]}, \quad \mathfrak{l}(s\delta_{\mathrm{an}}) = 0, \quad B_\sigma = B(\sigma s v \mathrm{e}^{\mathfrak{l}}, s^2 v^2 \mathrm{e}^{2\mathfrak{l}} \mathcal{Y}). \tag{9.12}$$

The bracketed factor in the denominator has a fixed positive margin. Integration gives

$$0 \leq \mathfrak{l} \leq C_B m_{\mathrm{land}} + C_K s \log \frac{m_{\mathrm{land}}}{s\delta_{\mathrm{an}}}.$$

Here C_B, C_K use only the coefficient bounds on the enlarged drift box and the denominator margin. Choose the common landing level so that $C_B m_{\mathrm{land}} < \frac{1}{2}\log(0.95/0.9)$, then reduce s, with δ_{an} fixed, so that the second term is below the same bound. Thus $\mathfrak{l} < \log(0.95/0.9) < \log 2$. A stopped bootstrap keeps the graph in the enlarged positive annulus. For the negative-branch extension to exits $\rho \le 0.9$ below, use the fixed enlarged band $v < 0.95$: $K_- = 1 + au^2 + bu - v + O(s)$ has a positive margin there for small s, and the stricter growth bound keeps the landing inside that band. It lands in the cone at

$$t = m_{\mathrm{land}} s^2 \rho^2 \mathrm{e}^{2\mathfrak{l}}, \qquad U(t) = t/(2m_{\mathrm{land}}).$$

Admissible graphs land before their endpoint: that endpoint has $z/|\eta|^2$ bounded below, and m_{land} is chosen smaller.

At $u = v = 0$, $K = 1$ and $B_\sigma = B_0(p)$, so both signs have the landing shift

$$\mathfrak{l}_0 = \frac{s}{g_0}\log\frac{m_{\mathrm{land}}}{s\delta_{\mathrm{an}}} - \frac{1}{2g_0}\log\frac{g_0 - 2B_0 m_{\mathrm{land}}}{g_0 - 2B_0 s\delta_{\mathrm{an}}}.$$

Write $R(u, v, \mathfrak{l}, \mathcal{Y})$ for the right side of (9.12), and put $\epsilon_s = s(1 + |\log s|)$. Every positive-order mixed $(u, v, \mathfrak{l})$ derivative of B_σ has a factor s, while K_σ occurs multiplied by s. The denominator margin therefore gives

$$|\partial^\alpha_{u,v,\mathfrak{l}} R| \le C_\alpha(s/\mathcal{Y} + s), \qquad 1 \le |\alpha| \le 4.$$

The difference from the equation at the base origin has this bound times $u + v$. The finite-order variational argument used for (6.14) now has fixed initial height $s\delta_{\mathrm{an}}$, zero initial base variations, and

$$\int_{s\delta_{\mathrm{an}}}^{m_{\mathrm{land}}} (s/\mathcal{Y} + s)\, d\mathcal{Y} \le C\epsilon_s.$$

It gives $\|\partial^\alpha_{u,v}\mathfrak{l}\| \le C_\alpha\epsilon_s$ for $1 \le |\alpha| \le 4$, and therefore $|\mathfrak{l} - \mathfrak{l}_0| \le C\epsilon_s(u + v)$. At positive orders, $\mathcal{E}^j_{uv}$ is a finite sum of operators $c_{i_1 i_2} u^{i_1} v^{i_2} \partial_u^{i_1} \partial_v^{i_2}$, with $1 \le i_1 + i_2 \le j$; each coefficient vanishes at the base origin. Thus

$$|\mathcal{E}^j_{uv}(\mathfrak{l} - \mathfrak{l}_0)| \le C_j s(1 + |\log s|)(u + v), \qquad 0 \le j \le 4. \tag{9.13}$$

In particular $\mathcal{E}\log t = 2 + 2\mathcal{E}_{uv}\mathfrak{l} > 0$.

Choose $c_{\mathrm{land}} = 1/(2m_{\mathrm{land}}) > 4\sup B$. The moving-cone tail in Lemma 4.1 gives

$$\frac{dU(z_*; c_{\mathrm{land}} t)}{dt} = (z_*/t)^{2s}\tilde{a}_\sigma(t), \qquad c < \tilde{a}_\sigma < C,$$

where, for $1 \le j \le 3$,

$$|(t\partial_t)^j \log \tilde{a}_\sigma(t)| \le C_j(\sqrt{t} + q_1/\sqrt{t} + |q_0|/t) \le C_j s(v + bu + u^2).$$

At this landing the coefficients satisfy the cone smallness condition. Hence

$$M_\sigma = 2(1 + \mathcal{E}_{uv}\mathfrak{l})\, z_*^{2s} t^{\,1-2s}\tilde{a}_\sigma(t). \tag{9.14}$$

Put $\mathfrak{d} = \mathcal{E}_{uv}\mathfrak{l}$, $b_t = 2(1 + \mathfrak{d})$, and $A_j = (t\partial_t)^j \log\tilde{a}_\sigma$. Logarithmic differentiation of (9.14) gives

$$e_\sigma = 2g_0\mathfrak{d} + \frac{\mathcal{E}_{uv}\mathfrak{d}}{1 + \mathfrak{d}} + b_t A_1, \qquad \mathcal{E}A_j = b_t A_{j+1}.$$

Here $\mathfrak{d}, \mathcal{E}_{uv}\mathfrak{d}, \mathcal{E}^2_{uv}\mathfrak{d}, \mathcal{E}^3_{uv}\mathfrak{d}$ are $O(\epsilon_s(u + v))$, whereas $A_1, A_2, A_3 = O(s(u + v))$. Two further Euler derivatives of the displayed error involve derivatives of $\mathfrak{l}$ only through order four and of $\log\tilde{a}_\sigma$ only through order three; every term still contains one of these vanishing factors. The denominators $1 + \mathfrak{d}$ stay bounded away from zero. This proves the error bounds in (9.10); the bound on M_σ follows from (9.14) and $t \asymp (s\rho)^2$.

To express the endpoint estimate in the common central height, use $\partial_T = \mathcal{E}/G_\sigma$. Thus

$$\mathfrak{H}_\sigma = \frac{\mathcal{E}\log M_\sigma - \mathcal{E}\log G_\sigma}{G_\sigma}.$$

Insert (9.9) and (9.10). The constant term is exactly $2g_0/g_0 = 2$; the two linear coefficients are those in (9.11). Differentiation using $u_T = -u/G_\sigma$ and $v_T = v/G_\sigma$ preserves the stated weighted error. □

For comparable exit radii $\omega = \Omega\rho$, with fixed positive bounds on Ω,

$$\log \frac{M_-(\omega)}{M_+(\rho)} \le 2g_0 \log \Omega + 2C_{\mathrm{cone}}\sqrt{z_*} + Cs(1 + |\log s|)(u + v). \tag{9.15}$$

Indeed the shifts in (9.14) have the same base value $\mathfrak{l}_0$, and the moving-section factor is

$$c_{\mathrm{land}} - \mathcal{G}_\sigma(c_{\mathrm{land}}t, t) = c_{\mathrm{land}}g_0 - B_0 + O(s(u + v)).$$

The remaining factor of $\tilde{a}_\sigma$ is the tail variation (4.7) divided by ϖ_t, that is, the slope $a_{t,p}$ of (4.3) at $\mathcal{U}_\circ = c_{\mathrm{land}}$. This landing energy lies in the common cone, so on each branch (4.4) bounds the logarithm of that factor by $C_{\mathrm{cone}}(\sqrt{z_*} + q_1/\sqrt{t})$, and $q_1/\sqrt{t} \le Csu$. This proves (9.15).

Corollary 9.2 (Two annular exits)**.** *Under (9.5) on both branches, let the thin exits from an interval of $\mathcal{H}_p$ lie in the smaller annular box $u_\sigma \le u_0$, $v_\sigma \le v_0$. In $T = -s \log h$,*

$$\begin{aligned}(\mathfrak{H}_- - \mathfrak{H}_+)' = &-\frac{\gamma_s b(u_- + u_+) + \delta_s\kappa_3(v_- + v_+)}{g_0} \\ &+ O\left(\sum_\sigma (u_\sigma + v_\sigma)\{u_\sigma + v_\sigma + s(1 + |\log s|)\}\right).\end{aligned} \tag{9.16}$$

The negative annular branch alone gives

$$|\partial_T^2[(\log \Pi_p')(I_-)]| \le C(sv_-)^{2-4s} \le Cs^{3/2}v_-, \qquad s < 1/8, \quad v_- < 1. \tag{9.17}$$

For any fixed $b_{\min} > 0$, choosing the annular margins small relative to $b_{\min}$, then reducing s, gives $\mathcal{L}'' < 0$ when $b \ge b_{\min}$.

Proof. Differentiate (9.11) using $\partial_T = \mathcal{E}_\sigma/G_\sigma$, $u_{\sigma,T} = -u_\sigma/G_\sigma$, $v_{\sigma,T} = v_\sigma/G_\sigma$, and $G_\sigma = g_0 + O(u_\sigma + v_\sigma)$; this gives (9.16). By Lemma 9.1, $I_{-,T} = -M_-/G_-$, and $|I_{-,T}| + |I_{-,TT}| \le C(sv_-)^{2-4s}$. Equation (4.28) therefore gives the first bound in (9.17); the second follows from $(sv_-)^{2-4s}/v_- = s^{2-4s}v_-^{1-4s} \le s^{3/2}$. For $b \ge b_{\min}$, the leading negative term dominates both the weighted error and the exterior contribution, proving $\mathcal{L}'' < 0$. □

9.1.3. *The negative branch beyond the annulus.* Beyond the annulus, negative outputs may lie in compact primary intervals or near the primary root close to $X = 1$. We continue the annular calculation on the compact intervals. Near the root, where the drift tends to zero, the boundary-flow coordinate gives a negative endpoint rate and estimates its derivative with the same drift weight as the endpoint derivatives. These weights allow the exterior contribution to be absorbed in the regional comparisons.

For the negative branch fix the regular primary entry at $\rho_{\mathrm{pr}} = 0.8$, take 0.85 as the lower bound for the projected far-root state w_n, and use $\rho = 0.9$ as the last primary thin-exit test. These thresholds give a fixed phase advance and overlapping ranges of central heights. In primary coordinates $|\eta| = sX$, $z = s^2Y$, the negative boundary drift is

$$K_n(X) = ar^2 + brX + X^2 + X^3R_q(-sX), \qquad F_n = K_n/X.$$

It has a simple first far root $X_* = 1 + O(r + s)$, with $K_n'(X_*) < 0$, and is positive below it. Construct the boundary-flow coordinate on a band from below 0.8 to beyond X_*, with small fixed output height $Y_n > 0$. Its far-root interval is

$$w_n = \mathcal{W}_n(X(s^2Y_n; h), Y_n) \ge 0.85, \qquad w_n < X_*. \tag{9.18}$$

Lemma 9.3 (Negative branch)**.** *Fix $M \ge K_0$. Suppose that the negative graph of a central height in $\mathcal{H}_p$ is thin at $\rho = Mr$, with the prefix preparation (9.5) at $\rho_i = K_0r$.*

(i) *If its first thin exit occurs at $\rho \le 0.9$, then $\rho > Mr$ and, in $T = -s\log h$,*

$$\begin{aligned} \mathfrak{H}_- &= 2 - 3\rho + O(r/\rho + s(1 + |\log s|)), \\ \mathfrak{H}'_- &= -3\rho(1-\rho) + O(r + s(1 + |\log s|)) \quad \textit{on fixed compact primary intervals,} \end{aligned} \tag{9.19}$$

$|I_{-,T}| + |I_{-,TT}| \le C(s\rho)^{2-4s} = o(1)$.

(ii) *If the graph is still thin at $\rho = 0.9$, it satisfies (9.18) with $w_n > 0.89$. For every central height with (9.18),*

$$\begin{gathered} \mathfrak{H}_- = 2 - 3w_n + O(r + s + \varepsilon_{n,s}) < -c_n < 0, \qquad \mathfrak{H}'_- = F_n(w_n)\{Q_n(w_n) + o(1)\}, \\ Q_n \le -c_n, \qquad |I_{-,T}| + |I_{-,TT}| \le CE_nF_n(w_n), \end{gathered} \tag{9.20}$$

where $Q_n = (XK_n'' - K_n')/X^2$, $E_n = s^2(z_*/(s^2Y_n))^{2s} = O(s^2)$, and $\varepsilon_{n,s} \to 0$, uniformly up to the far root.

The exit radius in (i) and the state w_n in (ii) increase with T. The constants are uniform on the chart (9.1).

Proof. Part (i). Since the graph is thin at Mr and, by (9.3), crosses $y = \delta_{\mathrm{an}}$ only upward on the annulus, a first thin exit before $\rho = 0.9$ has $\rho > Mr$. The negative annular coordinate extends to $\rho \le 0.9$: its drift factor has a uniform positive margin there. Its equation for the central height gives

$$G_- = \frac{1}{1-\rho} + O_{\mathcal{E}^j}(r/\rho + s).$$

The endpoint argument of Lemma 9.1 on this enlarged compact base box gives

$$\mathcal{E}\log M_- = 2 - 4s + O_{\mathcal{E}^j}\left(s(1 + |\log s|)(r/\rho + \rho)\right).$$

The weighted error follows from the landing equation: the comparison at the base origin has $K = 1$, and the derivatives remain bounded on $0 \le \rho \le 0.9$. Inserting these expressions into $(\mathcal{E}\log M_- - \mathcal{E}\log G_-)/G_-$ gives (9.19); the last estimate there uses $I_{-,T} = -M_-/G_-$ and its logarithmic rate.

Part (ii). Every branch still thin at 0.9 reaches the output $Y = Y_n$ inside the band. Its initial projected state is $0.9 + O(s\delta_{\mathrm{an}}) > 0.89$ and lies below X_*. The projected flow increases toward the root, and the inverse projection differs by $O(Y_n)$, remaining in the band. The inverse-flux identity gives continuation to Y_n.

At $\rho_{\mathrm{pr}} = 0.8$, write

$$\mathcal{J}_b = \int_{K_0 r}^{\rho_{\mathrm{pr}}} \frac{dt}{tK_-(r/t,t)}, \qquad \mathcal{A}_b = \mathcal{A}_{-,i} + \mathcal{J}_b,$$

$$\xi_n = \mathrm{e}^{\mathcal{A}_b/s}\frac{h}{s^3}.$$

The secondary and primary expanded inputs are related by

$$\xi_- = \beta_b\xi_n, \qquad \beta_b = r^{-2}\mathrm{e}^{-\mathcal{J}_b/s}.$$

The factor r^{-2} comes from changing the scale s^3r^2 of the central height to s^3. The uniform upper bound for the annular drift gives $\mathcal{J}_b \ge c\log(\rho_{\mathrm{pr}}/(K_0 r))$. After fixing smaller upper bounds for r and then s, this implies

$$0 < \beta_b \le r^{c/(2s)-2} \le 1.$$

For $Y_i = z/s^2$, write $y_i = Y_i/s = z/s^3 = \rho_{\mathrm{pr}}^2 y_a(\rho_{\mathrm{pr}})$, where $y_a = z/(s^3\rho^2)$, and set

$$\mathcal{C}_b(y_i) = \rho_{\mathrm{pr}}^2\Psi_-(r/\rho_{\mathrm{pr}}, \rho_{\mathrm{pr}}, y_i/\rho_{\mathrm{pr}}^2).$$

This height coordinate and its inverse have uniform bounds on the required derivatives, with $\mathcal{C}_b'(0) = 1$. Annular transport gives

$$\mathcal{C}_b(y_i) = \xi_n\Upsilon_-(\beta_b\xi_n).$$

As in (4.22), inversion gives $Y_i = s\xi_n V_n(\xi_n)$, with $V_n(0) = 1$. Since $\beta_b \le 1$, the unit, its reciprocal, and the map inverse have uniform derivative bounds on a fixed small ξ_n-interval.

For central heights with (9.18), the projected state at $X = \rho_{\rm pr}$ satisfies $w_i = \mathcal{W}_n(\rho_{\rm pr}, Y_i) \le \rho_{\rm pr}$. The boundary flow must advance from at most 0.8 to at least 0.85 before the output Y_n. Its bounded speed and the projection identity give

$$s\log(Y_n/Y_i) \ge c_b > 0, \qquad y_i = Y_i/s \le (Y_n/s)\mathrm{e}^{-c_b/s}.$$

Thus the graph is thin at $\rho_{\rm pr}$. By (9.3), it has no downward crossing of $y_a = \delta_{\rm an}$ on the preceding annular segment, so the whole segment is thin. Annular transport gives

$$\frac{\widehat{z}(K_0 r)}{s^3 r^2} = \beta_b \mathcal{C}_b(y_i).$$

This lies in the uniform small output domain of the inner prefix. Its inverse recovers $\xi_- = O(\beta_b y_i)$; uniqueness of the flow identifies the recovered central height with h. Thus $\xi_n = \xi_-/\beta_b \le C y_i$ lies in the required small domain. At the fixed output $z = s^2 Y_n$, the entry $X = \rho_{\rm pr} = 0.8$ has $F_n(\rho_{\rm pr})$ bounded below, and the advance to $w_n \ge 0.85$ supplies a fixed positive phase gap. Thus Lemma 4.8 applies with $\delta = s$ in the same coordinate $T = -s\log h$. We use its drift-weighted bound (4.20) with $N = 1$ and $j = 2$ for the curvature estimate below; in particular,

$$(w_n)_T = \theta_n F_n(w_n), \qquad |\theta_n - 1| + |\theta_n'| + |\theta_n''| \le \varepsilon_{n,s} \longrightarrow 0.$$

Lemma 4.9, with $\ell = \delta = s$, gives

$$-I_{-,w_n} = E_n A_n w_n U_n(w_n), \qquad E_n = s^2(z_*/(s^2 Y_n))^{2s} = O(s^2), \qquad U_n = 1 + O_{C^3}(s).$$

The rate identity yields the first estimate in (9.20), uniformly up to the root, since $K_n'(w)/w = 2 - 3w + O(r+s)$. By (4.29), with $K_n'(0) = br$ and $R_K(X) = R_q(-sX)$, one has $Q_n(X) = -br/X^2 + 3R_q(-sX) + O(s) < -c_n$ on the band, since $R_q = -1 + O(s)$. The weighted phase estimate and (4.27) give the remaining assertions of (9.20). Finally $\rho_T = \rho/G_- > 0$, and the projected far-root state increases along the boundary flow. □

9.1.4. *The positive branch beyond the annulus.* A positive half-orbit may leave compact ρ-intervals and reach unscaled radii $\varrho \gg s$, where the drift grows cubically. The adaptive coordinate of Lemma 6.5 follows this transition from the inner cut and gives the endpoint rate in both the buffered and no-root regions, using their respective preparations at the inner cut.

The annular action controls the expanded incoming height and its derivatives with respect to the exit radius. This control makes the endpoint-rate comparison uniform as the positive exit moves from the quadratic to the cubic regime.

On the positive branch write $\varrho = \eta = s\rho > 0$, $Q = s^3 r^2$, and

$$\varrho^+ = s + \varrho, \quad u = \frac{sr}{\varrho}, \quad v = \frac{\varrho}{s+\varrho}, \quad K_a = 1 + (1-v)(au^2 - bu) - v\{R_q(\varrho) + 1\}. \tag{9.21}$$

Here $u = r/\rho$ retains its annular meaning. In K_a and the following lemma, v is the adaptive variable of (6.10); the annular coordinate Ψ_+ still has arguments $(r/\rho, \rho)$. With $\varrho_i = K_0 sr$, put

$$\mathcal{J}_a(\varrho) = \int_{\varrho_i}^{\varrho} \frac{x\,dx}{-q(x)},$$

so that the action increment of the annulus is $\mathcal{J}_+(\rho) = s\mathcal{J}_a(s\rho)$. At every radius the annular and adaptive normalized heights satisfy

$$\frac{z}{s\varrho^2} = \Big(1 + \frac{\varrho}{s}\Big)\frac{z}{\varrho^2(s+\varrho)}, \tag{9.22}$$

so, for a half-orbit thin at ϱ_i, the annular thin exit at the same threshold $\delta_{\rm an}$ precedes the adaptive exit of part (iii) below. At an exit radius ϱ_e we write u_e, v_e, ϱ_e^+ for the values of u, v, ϱ^+. The operator $\mathcal{E} = \varrho_e \partial_{\varrho_e}$ varies the exit radius at fixed parameters, and $T = -s\log h$.

Lemma 9.4 (Positive branch beyond the annulus)**.** *Let $\mathcal{J} \subset \mathcal{H}_p$ be an interval of central heights whose positive half-orbits are thin at ϱ_i, that is, $z(\varrho_i; h) < \delta_{\mathrm{an}} K_0^2 Q$, and satisfy*

$$z(\varrho_i; h) = Q\,\xi\mathcal{V}(\xi), \qquad \xi = \mathrm{e}^{\mathcal{A}_i/s} h/Q. \tag{9.23}$$

Here $\mathcal{A}_i$ depends only on p, ξ lies in a fixed interval $[0, \bar{\xi}]$, $\mathcal{V}(0) = 1$, $\mathcal{V}$ and $1/\mathcal{V}$ have uniformly bounded derivatives of the required orders there, and $\xi \mapsto \xi\mathcal{V}(\xi)$ has a uniformly positive derivative and an inverse with uniformly bounded derivatives.

(i) *One has $-q(\varrho) = \varrho^2(s+\varrho)K_a$. On the box $0 \le u \le 3/(2K_0)$, $0 \le v \le 1$, with small unscaled radius and small s, K_a lies between $1/8$ and 2 and has uniform derivative bounds, and $K_a - 1$ and its fixed Euler derivatives are $O(u + \varrho^+)$. Thus Lemma 6.5 applies with $\mathcal{K} = K_a$, $\varrho_0 = sr$ and thin threshold $y_0 = \delta_{\mathrm{an}}$; let Ψ_a be its coordinate and $\widehat{z}_a$ its adapted height.*

(ii) *Put $u_i = 1/K_0$, $v_i = K_0 r/(1 + K_0 r)$ and $c_i = K_0^2(1 + K_0 r)$, and define Υ_a by*

$$\xi\Upsilon_a(\xi) = c_i \Psi_a(u_i, v_i, \varrho_i, \xi\mathcal{V}(\xi)/c_i). \tag{9.24}$$

Then $\Upsilon_a(0) = 1$, and Υ_a, its reciprocal and the inverse of $\xi \mapsto \xi\Upsilon_a(\xi)$ have uniformly bounded derivatives on the interval of those $\xi \le \bar{\xi}$ with $\xi\mathcal{V}(\xi) \le \delta_{\mathrm{an}} K_0^2$. Moreover $\widehat{z}_a(\varrho_i) = Q\xi\Upsilon_a(\xi)$, so that $\lambda_{\mathrm{in}} = \log \widehat{z}_a(\varrho_i)$ satisfies $d\lambda_{\mathrm{in}}/d\log h = 1 + O(\xi)$.

(iii) *The adaptive exit, the first radius $\varrho_e > \varrho_i$ with $z = \delta_{\mathrm{an}}\varrho^2(s+\varrho)$, satisfies*

$$\xi\Upsilon_a(\xi) = \frac{\varrho_e^2 \varrho_e^+}{Q} \Psi_a(u_e, v_e, \varrho_e, \delta_{\mathrm{an}}) \mathrm{e}^{-\mathcal{J}_a(\varrho_e)}. \tag{9.25}$$

Its landing height $t = m_{\mathrm{land}} \varrho_l^2$ satisfies $\sqrt{t} + q_1/\sqrt{t} + |q_0|/t \le C(\varrho_e + s u_e)$, and $|q_2| = s$; in particular (4.1) holds there for small s.

(iv) *Suppose in addition that $\varrho_e \ge s\rho_{\mathrm{g}}$ on $\mathcal{J}$ and $\mathcal{J}_+(\rho_{\mathrm{g}}) \ge c_{\mathrm{g}} \log(1/r)$, for fixed $\rho_{\mathrm{g}}, c_{\mathrm{g}} > 0$, and that $r \le 1/2$. Then, for every fixed j and small s,*

$$|\mathcal{E}^j \xi| + |\mathcal{E}^j \log \Upsilon_a(\xi)| \le C_j s^{-3-j} r^{c_{\mathrm{g}}/s - 2}, \tag{9.26}$$

which is $O(s^N)$ for every fixed N. The rate $\mathfrak{g} = -\varrho_e^+ \mathcal{E}\log h$ satisfies

$$\mathfrak{g} = 1 + O(u_e + \varrho_e^+), \qquad \mathcal{E}\log\mathfrak{g} = O(u_e + \varrho_e^+), \qquad \mathcal{E}T = (s/\varrho_e^+)\mathfrak{g} > 0. \tag{9.27}$$

For small fixed m_{land} and radius box, and after reducing r and then s, the endpoint rate $\mathfrak{H}_+ = \partial_T \log m_+$ satisfies

$$\mathfrak{H}_+ = \frac{\varrho_e^+}{s\mathfrak{g}} [\, 2 - 4s + v_e + O(\epsilon_e + u_e + \varrho_e^+) \,] > c\varrho_e^+/s > 0, \tag{9.28}$$

with ϵ_e as in (6.14).

The constants depend on $\mathcal{V}$ only through its assumed bounds. The constants in (9.27) and (9.28), apart from terms $O(s^N)$, are independent of $\mathcal{V}$, $\mathcal{A}_i$, ρ_{g} and c_{g}.

Proof. (i) Since $(s+\varrho)(1-v) = s$ and $(s+\varrho)v = \varrho$,

$$\varrho^2(s+\varrho)K_a = s\varrho^2 + as^3r^2 - bs^2 r\varrho - \varrho^3 R_q(\varrho) = -q(\varrho).$$

Because $\kappa_3 = -R_q(0) = 1 + O(s)$, one has $R_q(\varrho) + 1 = O(s+\varrho)$ with its fixed Euler derivatives; here $\mathcal{E}u = -u$, $\mathcal{E}v = v(1-v)$ and $\mathcal{E}\varrho = \varrho$. On the box, $|(1-v)(au^2 - bu)| \le u < 3/4$, since $0 < a \le 1$, $0 \le b \le 1$ and $K_0 > 2$. This gives the bounds on K_a and on $K_a - 1$. Along the passage from ϱ_i, u decreases from $1/K_0$ while v and ϱ increase, so the base point stays in the box, and $y_0 = \delta_{\mathrm{an}}$ is admissible by the choice of δ_{an} in Section 9.1.

(ii) At the inner section, $\varrho_i^2(s + \varrho_i) = Qc_i$, so the adapted height is exactly $Q\xi\Upsilon_a(\xi)$ by (9.24). Here $K_0^2 \le c_i \le 2K_0^2$, and the stated thin interval gives $\xi\mathcal{V}(\xi)/c_i \le \delta_{\mathrm{an}}$. The uniform regularity and positive height derivative of Ψ_a, together with the assumed bounds for $\xi\mathcal{V}(\xi)$, give all the unit and inverse bounds. The normalization $\Psi_{a,y}(\cdot, 0) = \mathcal{V}(0) = 1$ gives $\Upsilon_a(0) = 1$, and logarithmic differentiation gives the last claim.

(iii) Thinness at ϱ_i allows Lemma 6.5 to supply the exit and landing before the admissible endpoint. Its exact action identity, evaluated at the exit and combined with (ii), is (9.25). Since $\varrho_e \le \varrho_l \le 2\varrho_e$,

$$\sqrt{t} \le 2\varrho_e, \qquad q_1/\sqrt{t} \le Cbsu_e, \qquad |q_0|/t \le Casu_e^2.$$

With $u_e \le 1/K_0$, these are the claimed tail bounds.

(iv) Denote the right side of (9.25) by $\mathfrak{b}_a$, a function of ϱ_e; through this equation ξ and h are functions of the exit radius. By (i), $-q > 0$ on the passage, so $\mathcal{J}_a$ increases; since $s\mathcal{J}_a(s\rho_{\mathrm{g}}) = \mathcal{J}_+(\rho_{\mathrm{g}})$, we get $\mathrm{e}^{-\mathcal{J}_a(\varrho_e)} \le r^{c_{\mathrm{g}}/s}$. Since $\mathcal{E}\mathcal{J}_a = 1/(\varrho_e^+ K_a)$, $\varrho_e^+ \ge s$, and the Euler derivatives of K_a are uniformly bounded, each fixed radial derivative of $\mathfrak{b}_a$ costs at most another factor s^{-1}: $|\mathcal{E}^j \mathfrak{b}_a| \le C_j s^{-j}\mathfrak{b}_a$, as in (9.7). In the fixed radius box $\mathfrak{b}_a \le Cs^{-3}r^{c_{\mathrm{g}}/s-2}$; since $r \le 1/2$, this is $O(s^N)$ for every fixed N, and in particular $\mathfrak{b}_a \le 1$ for small s. The chain-rule step before (9.8), applied to $\xi = (\xi\Upsilon_a)^{-1}(\mathfrak{b}_a)$ with the inverse of (ii), bounds $|\mathcal{E}^j\xi| + |\mathcal{E}^j \log \Upsilon_a(\xi)|$ by $C_j s^{-j}\mathfrak{b}_a$, which gives (9.26).

Logarithmic differentiation of (9.25), with $\log \xi = \mathcal{A}_i/s + \log(h/Q)$, $\mathcal{E}\log(\varrho_e^2\varrho_e^+) = 2 + v_e$ and $\varrho_e^+(2+v_e) = 2s + 3\varrho_e$, gives

$$\mathfrak{g} = K_a^{-1} - (2s + 3\varrho_e) - \varrho_e^+ \mathcal{E}\log \Psi_a + \varrho_e^+ \mathcal{E}\log \Upsilon_a(\xi),$$

with K_a and $\Psi_a(\cdot, \delta_{\mathrm{an}})$ evaluated at the exit. By (i), the uniform Euler derivatives of Ψ_a, and (9.26), this gives (9.27).

At the cone landing, (6.14) gives $\mathcal{E}\log t = 2 + O(\epsilon_e)$ and $\mathcal{E}^2 \log t = O(\epsilon_e)$. By (iii), the moving-cone part of Lemma 4.1 applies at the landing, as before (6.15). With $q_2 = -s$, the endpoint energy has derivative $(z_*/t)^{2s}a(t)$, and (4.5) gives

$$|t\partial_t \log a(t)| \le C(\sqrt{t} + q_1/\sqrt{t} + |q_0|/t) \le C(\varrho_e + su_e + su_e^2).$$

Therefore $M_a = -\mathcal{E}I_+$ satisfies

$$M_a = t(z_*/t)^{2s}a(t)\,\mathcal{E}\log t, \qquad \mathcal{E}\log M_a = 2 - 4s + O(\epsilon_e + \varrho_e + su_e + su_e^2).$$

Since $m_+ = hI_{+,h} = \varrho_e^+ M_a/\mathfrak{g}$, $\mathcal{E}\log \varrho_e^+ = v_e$ and $\partial_T = \mathcal{E}/\mathcal{E}T$, (9.27) gives the identity in (9.28). The factor K_a and its derivatives have the common bounds of (i), while Υ_a contributes only terms $O(s^N)$; this gives the stated dependence of the constants. At the exits $u_e \le r/\rho_{\mathrm{g}}$. Hence, for small m_{land} and radius box, after reducing r and then s, the bracket in (9.28) exceeds one and $\mathfrak{g} < 2$. □

9.2. **Negative quadratic drift away from the secondary fold.** We first treat the part of (9.1) on which

$$b^2 - 4a \le -\nu < 0. \tag{9.29}$$

The fixed margin implies $a \ge \nu/4$. Both secondary passages therefore have regular prefixes. The remaining comparison must match their exits at the same central height, even when the individual actions become large.

Proposition 9.5 (Buffered negative quadratic drift)**.** *For every fixed $\nu > 0$, the return displacement Δ_p on (9.29) has at most three distinct zeros, for all sufficiently small independent $s, r > 0$ and all central heights in $\mathcal{H}_p$. Moreover, the logarithmic slope ratio $\mathcal{L}$ has at most two distinct zeros on $\mathcal{H}_p$.*

9.2.1. *Actions and the regular prefix.* On a small fixed interval $0 \le t \le v_1$,

$$D_\sigma(t) = ar^2 - \sigma brt + t^2 - \sigma t^3 R_q(\sigma st) \asymp r^2 + t^2.$$

Indeed the two quadratic forms have a uniform positive margin by (9.29), and the cubic term is absorbed by reducing v_1. Set

$$\mathcal{A}_\sigma(\rho) = \int_0^\rho t/D_\sigma(t)\,dt. \tag{9.30}$$

The positive action is defined up to any fixed positive primary radius; we use the negative action first on compact subintervals below its far root near $\rho = 1$. At $\rho_i = K_0 r$, the central scale $\ell = sr$, $Q = s^3 r^2$ has drifts

$$a - \sigma b x + x^2 - \sigma r x^3 R_q(\sigma s r x), \qquad 0 \le x \le K_0.$$

They have a fixed positive margin for sufficiently small r. The central-height construction and composition with the annular coordinate therefore give (9.5), with $\mathcal{A}_{\sigma,i} = \mathcal{A}_\sigma(K_0 r)$.

At a common small primary radius,

$$|\mathcal{A}_+(\rho) - \mathcal{A}_-(\rho)| \le C \int_0^\rho \frac{brt^2 + t^4}{(r^2 + t^2)^2}\, dt \le C(b + \rho). \tag{9.31}$$

The substitution $t = rx$ gives an integrable density for the first term, and the second integrand is bounded. This bounds the difference even when the individual actions grow like $\log(1/r)$.

9.2.2. *The intermediate annulus.* We prove that there are fixed $M_0 > 2K_0$ and $v_0 > 0$ such that the interval of central heights in $\mathcal{H}_p$ with positive thin exit in

$$M_0 r \le \rho \le v_0 \tag{9.32}$$

contains at most three zeros of Δ_p. We first construct the negative exit for each of these central heights. It has the form $\omega = \Omega\rho$, where

$$0 < \Omega_0 \le \Omega \le \Omega_1, \qquad |\log \Omega| \le C(b + \rho + s). \tag{9.33}$$

By (9.1), $0 \le b \le 1$. Fix the lower action-slope margin $c_{\mathrm{a}} > 0$ on the whole regular range $K_0 r \le t \le v_1$. Choose fixed factors $0 < \Omega_0 < 1 < \Omega_1$ so far apart that

$$c_{\mathrm{a}} \min\{|\log \Omega_0|, \log \Omega_1\} > 8C_{\mathrm{act}}, \tag{9.34}$$

where C_{act} is a fixed bound for the b-dependent part of the full $\Phi(1;\rho)$ comparison, including the action and bounded height-factor terms, when $0 \le b \le 1$ and $0 < \rho \le 1$. The terms proportional to $s \log \Omega$ are handled after the fixed trial factors have been chosen. Then choose $M_0 > 2K_0/\Omega_0$ and $v_0 > 0$ so small that $\Omega_1 v_0 < v_1$. On $[\Omega_0\rho, \Omega_1\rho]$, with $M_0 r \le \rho \le v_0$, the positive margin in (9.29) gives

$$0 < c_{\mathrm{a}} \le \frac{t^2}{D_-(t)} \le C_{\mathrm{a}}, \qquad \left| \frac{t^2}{D_-(t)} - \frac{t^2}{D_+(t)} \right| \le C(b + t), \tag{9.35}$$

after reducing v_0. Let $h_\sigma(\rho)$ denote the central height obtained from (9.6) at the first thin exit ρ on branch σ. The event equation and (9.9) imply that h_σ is C^3 and strictly decreasing, and that

$$\Phi(\Omega;\rho) := s \log \frac{h_-(\Omega\rho)}{h_+(\rho)}, \qquad \partial_{\log\Omega}\Phi = -\frac{(\Omega\rho)^2}{D_-(\Omega\rho)} + O(s) \le -\frac{1}{2} c_{\mathrm{a}}. \tag{9.36}$$

At $\Omega = 1$, the action difference estimate (9.31), together with the bounded height factors in (9.6), gives

$$|\Phi(1;\rho)| \le C_{\mathrm{act}} + C(v_0 + s), \tag{9.37}$$

with C_{act} as chosen above. First fix the trial separation in (9.34); then reduce v_0 and finally s. The bound in (9.37) is then smaller than $\frac{1}{4} c_{\mathrm{a}} \min\{|\log \Omega_0|, \log \Omega_1\}$ uniformly in $b \in [0,1]$. Integrating (9.36) then gives $\Phi(\Omega_0;\rho) > 0 > \Phi(\Omega_1;\rho)$, uniformly for $M_0 r \le \rho \le v_0$. The intermediate value theorem therefore gives an exit $\omega = \Omega\rho$ with $h_-(\omega) = h_+(\rho)$. Strict decrease in (9.36) gives uniqueness of the matching radius, and each passage has a unique first exit by (9.9).

At the matching radius, combine the equal-height equation with (9.31). The lower bound in (9.36) yields

$$|\log \Omega| \le C(b + \rho + s), \tag{9.38}$$

after absorbing the $O(s)$ height factors, proving the second bound in (9.33) for sufficiently small parameters. Differentiation of the equation of equal central heights gives

$$\mathcal{E}\log\omega = G_+(\rho)/G_-(\omega), \qquad \mathcal{E}\log\Omega = O(u+v).$$

For $b \geq b_{*,-} > 0$, where the threshold will be fixed by the small-b estimate below, the preceding prefixes and comparable exit radii meet Corollary 9.2 after choosing the annular margins small relative to $b_{*,-}$, then reducing s. It gives $\mathcal{L}'' < 0$ in $T = -s\log h$, and at most three zeros of Δ_p.

For small b, curvature need not have a uniform margin. We bound the endpoint slope quotient and use the exterior contraction. The quotient is

$$\frac{I_{-,h}}{I_{+,h}} = \frac{M_-(\omega)}{M_+(\rho)}\frac{G_+(\rho)}{G_-(\omega)}.$$

The logarithm of this quotient has, by (9.15), (9.33), and (9.9), the upper bound

$$C\{b + u_0 + v_0 + s(1 + |\log s|)\} + 2C_{\mathrm{cone}}\sqrt{z_*}.$$

Choose $b_{*,-} > 0$ and the annular margins small to make this less than $-\log\Lambda_+$ for $b \leq b_{*,-}$, after reducing s. Then $\Pi_p' I_{-,h}/I_{+,h} < 1$, so Δ_p has at most one zero. Shrink the annular margins further for the curvature argument with this same $b_{*,-}$, then reduce s. This proves the three-zero bound on (9.32).

9.2.3. *Exits on a compact primary interval.* Fix $0 < \rho_{\mathrm{lo}} < \rho_{\mathrm{hi}} < \infty$. Suppose the positive first exit at $y = z/(s|\eta|^2) = \delta_{\mathrm{an}}$ satisfies $\rho_{\mathrm{lo}} \leq \rho \leq \rho_{\mathrm{hi}}$. On this compact interval we first match the two exits, then prepare their endpoints and the return displacement. The matching equation gives a rational leading relation with a controlled remainder. Choose a small fixed reference radius $v_b > 0$, and set

$$L_+(\rho) = \log\frac{\rho}{1+\rho}, \qquad L_-(\rho) = \log\frac{\rho}{1-\rho}, \qquad \mathfrak{c}_\sigma(p) = \mathcal{A}_\sigma(v_b) - L_\sigma(v_b).$$

On compact positive intervals away from zero, and for the negative sign away from one,

$$\mathcal{A}_\sigma(\rho) = \mathfrak{c}_\sigma + L_\sigma(\rho) + O_{C^j}(r+s). \tag{9.39}$$

Subtract the integrands on the interval between v_b and ρ. Their denominators have fixed positive margins, and $R_q(\sigma s t) = -1 + O_{C^j}(s)$; integration and differentiation give this estimate. Equation (9.31) implies

$$\Lambda = \exp(\mathfrak{c}_+ - \mathfrak{c}_-), \qquad 0 < \underline{\Lambda}_{\mathrm{act}} \leq \Lambda \leq \overline{\Lambda}_{\mathrm{act}} < \infty.$$

The annular height construction extends to any such fixed compact primary interval with positive drift margin. For these exits, the increase of action from $K_0 r$ is at least $c_1\log(\rho/(K_0 r))$. The equation for the central height therefore makes the expanded prefix input and its first j radius derivatives $O(s^{-j}\mathrm{e}^{-c_2/s})$, uniformly in small r. Both signs have thin central prefixes: the positive action gap can be made larger than their fixed central action difference by reducing r. Thus the equation for the central height is

$$s\log(h/s^3) = -\mathcal{A}_\sigma(\rho_\sigma) + sM_\sigma^0(\rho_\sigma), \qquad \|M_\sigma^0\|_{C^4} \leq C. \tag{9.40}$$

Here $M_\sigma^0 = 2\log\rho_\sigma + \log\Psi_\sigma - \log\Upsilon_\sigma$. Equating the two equations for the central height gives

$$L_-(\omega) - L_+(\rho) = \log\Lambda + O_{C^3}(r+s).$$

For each value of the parameter-dependent multiplier Λ, the leading scalar relation has the solution

$$f_\Lambda(\rho) = \frac{\Lambda\rho}{1 + (1+\Lambda)\rho}.$$

For $\rho_{\mathrm{lo}} \leq \rho \leq \rho_{\mathrm{hi}}$, $f_\Lambda(\rho)$ lies in a fixed compact subset of $(0,1)$. Comparing the central-height equations at slightly wider negative exits gives opposite signs; first-exit order gives the matching negative passage. The lower bound on L_-' then gives, by implicit differentiation,

$$\omega = f_\Lambda(\rho) + O_{C^3}(r+s). \tag{9.41}$$

The rational profile is the inverse, after a sign change and positive rescaling, of the base-point relation of Huzak–Kristiansen [8, (2.13) and (6.8)]. For this scalar relation, the substitutions

$$\rho = -\beta_{\mathrm{HK}} x_{\mathrm{out}}^{b_{\mathrm{HK}}}, \qquad \omega = \beta_{\mathrm{HK}} x_{\mathrm{in}}^{b_{\mathrm{HK}}}, \qquad \Lambda = \mathrm{e}^{-K_{\mathrm{HK}}}, \qquad K_{\mathrm{HK}} = \frac{\lambda_1^{\mathrm{HK}}\pi}{\sqrt{-4\lambda_0^{\mathrm{HK}} - (\lambda_1^{\mathrm{HK}})^2}}$$

give $\omega = f_\Lambda(\rho)$; their domain becomes $\rho > 0$, $0 < \omega < \Lambda/(1+\Lambda)$. The two-scale C^3 remainder in (9.41) follows by matching the two central-height equations above.

For the endpoints apply Lemma 4.2 with $\ell = \delta = s$, $x = \rho_\sigma$ and $\zeta_0(x) = \delta_{\mathrm{an}} x^2$. The coefficient ratios are bounded, and $K_\sigma(x) = -s^{-3} q(\sigma s x) = D_\sigma(x)$ has a fixed positive margin on the compact exit intervals above. The entrance factor is positive with bounded C^3 norm, including its moving height. Choose a common $Z_1 = Z_0$ and put $b_\sigma^e = b_\sigma$, $a_\sigma^e = \varpi_{s^2 Z_0} A_\sigma$. Since $q_2 = -s$, the multiplier is $\varpi_{s^2 Z_0} = 1 + O(s(1 + |\log s|))$. Thus a_σ^e has fixed positive bounds, and

$$I_\sigma(\rho_\sigma) = b_\sigma^e - \tfrac{1}{2} s^2 a_\sigma^e \rho_\sigma^2 + s^2 O_{C^3}(s(1 + |\log s|)). \tag{9.42}$$

The endpoint preparation now turns the matching relation into an expansion of the return displacement. Substituting (9.41) and (9.42) into the exterior map gives

$$\frac{2\Delta}{s^2 a_+^e} = D(p) + \rho^2 - \lambda(p) f_\Lambda(\rho)^2 + O_{C^3}(r + s(1 + |\log s|)), \qquad \lambda = \Pi_p'(b_-^e) a_-^e / a_+^e. \tag{9.43}$$

Here $D(p)$ depends only on p, and λ has fixed positive bounds. For every $h \in \mathcal{H}_p$ whose positive first exit lies in $[\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}]$, the reference endpoints b_σ^e differ from the corresponding admissible endpoints by $O(s^2)$. Thus, whenever this set of heights is nonempty, the reference endpoints lie in the enlarged exterior windows. The exterior Taylor remainder adds $O_{C^3}(s^2)$ after normalization. For the principal part $\widehat{\Delta}_{\mathrm{m}} = D + \rho^2 - \lambda f_\Lambda^2$,

$$\left(\frac{\widehat{\Delta}_{\mathrm{m}}'}{2\rho}\right)' = \frac{3\lambda\Lambda^2(1+\Lambda)}{[1+(1+\Lambda)\rho]^4} > c_* > 0.$$

Apply Lemma 3.6 with $\phi = \rho^2$, $\psi = f_\Lambda^2$, and $\widehat{\Delta} = 2\Delta/(s^2 a_+^e)$. The compact positive radius interval gives $\phi' \geq 2\rho_{\mathrm{lo}}$; the displayed margin therefore persists under the remainder in (9.43). It gives at most two zeros on this interval and

$$\left(\frac{\widehat{\Delta}_\rho}{2\rho}\right)_\rho > 0.$$

For the crossing direction, use the same endpoint identity as in (6.9), here in the exit coordinate:

$$\frac{\widehat{\Delta}_\rho}{2\rho} = \frac{I_{+,\rho}}{s^2 a_+^e \rho}(\mathrm{e}^{\mathcal{L}} - 1), \qquad I_{+,\rho} < 0.$$

At a slope zero it gives $\mathcal{L}_\rho < 0$. Since (9.40) gives $T_\rho = \mathcal{A}_+' - s(M_+^0)' > 0$, every such zero has $\mathcal{L}_T < 0$.

9.2.4. *Large positive exits.* We now treat the central heights in $\mathcal{H}_p$ whose positive half-orbit is thin at ρ_i, with the prefix (9.5), and whose adaptive exit, at $z = \delta_{\mathrm{an}} \varrho^2 (s + \varrho)$ in the notation of Lemma 9.4, has radius $\varrho_e = s\rho_e$ with $\rho_e \geq 1$. The adaptive exit is the unique upward crossing of that level, so $\rho_e \geq 1$ means $z(s;h) \leq 2\delta_{\mathrm{an}} s^3$. By Lemma 3.3, this condition and thinness at ρ_i each define an interval; their intersection is the interval considered here. For each such height, the negative annular thin exit at $z/(s|\eta|^2) = \delta_{\mathrm{an}}$ remains in a fixed compact interval strictly below its far root, and for large ρ_e the return contracts.

At ρ_i, (9.5) reads

$$K_0^2 \Psi_+(1/K_0, K_0 r, y_i) = \xi_+ \Upsilon_+(\xi_+), \qquad y_i = \frac{z(\varrho_i; h)}{K_0^2 Q}, \qquad \varrho_i = K_0 s r.$$

Inverting this height coordinate gives

$$V_{\mathrm{an}}(\xi)=\frac{K_0^2}{\xi}\Psi_+^{-1}(1/K_0,K_0r;\xi\Upsilon_+(\xi)/K_0^2),\qquad z(\varrho_i;h)/Q=\xi_+V_{\mathrm{an}}(\xi_+).$$

Thinness keeps the inverse in its fixed domain. Its derivative at zero is one, so $V_{\mathrm{an}}(0)=1$, and the annular coordinate bounds supply the unit, reciprocal and inverse bounds required by (9.23). Thus Lemma 9.4 applies with $\mathcal{V}=V_{\mathrm{an}}$ and $\mathcal{A}_i=\mathcal{A}_{+,i}$.

For $1\le\rho\le\varrho_*/s$, where ϱ_* is a small fixed unscaled radius, subtraction of the positive action integrands gives the value estimate

$$\mathcal{A}_+(\rho)=\mathfrak{c}_++L_+(\rho)+O\{r+s(1+\log(1+\rho))\}.$$

Indeed $D_+(t)\asymp t^2(1+t)$ and $R_q(st)+1=O(s+st)$; the difference is bounded by $Cr/t^2+Cs/(1+t)$ for $t\ge v_b$. The action increment from K_0r to 1 gives the gap in part (iv) of the lemma, with $\rho_{\mathrm{g}}=1$: since $D_+>0$, the annular bound $\mathcal{J}_+(\rho)\ge c\log(\rho/(K_0r))$ at $\rho=v_1$ gives $\mathcal{J}_+(1)\ge\mathcal{J}_+(v_1)\ge\frac{c}{2}\log(1/r)$ for small r. Taking logarithms in (9.25), with $\mathcal{A}_{+,i}+s\mathcal{J}_a(s\rho_e)=\mathcal{A}_+(\rho_e)$ and $\varrho_e^2(s+\varrho_e)/Q=\rho_e^2(1+\rho_e)/r^2$, gives the equation for the central height at the adaptive exit:

$$s\log(h/s^3)=-\mathcal{A}_+(\rho_e)+s\log[\rho_e^2(1+\rho_e)\Psi_a(u_e,v_e,\varrho_e,\delta_{\mathrm{an}})]-s\log\Upsilon_a(\xi_+).$$

By (9.26), ξ_+ and $\log\Upsilon_a(\xi_+)$ are at most $Cs^{-3}r^{c/(2s)-2}$. After reducing r, the negative input $\xi_-=\mathrm{e}^{(\mathcal{A}_{-,i}-\mathcal{A}_{+,i})/s}\xi_+$ is exponentially small as well, since (9.31) bounds the difference of the central actions. The remaining height prefactor $\rho_e^2(1+\rho_e)\Psi_a$ is at least $c\delta_{\mathrm{an}}$ and, since the unscaled exit radius is bounded, at most a fixed power of s^{-1}. Thus

$$s\log(h/s^3)=-\mathfrak{c}_+-L_+(\rho_e)+O(r+s(1+|\log s|)).$$

Since $f_\Lambda(\rho_e)$, for $\rho_e\ge1$, lies in a fixed compact subset of $(0,1)$, comparison with (9.40) at two wider negative endpoints proves existence of the negative exit. Keep these negative comparison endpoints fixed when choosing the large-exit threshold below.

Differentiation of the equation for the central height on the negative branch gives

$$\frac{d\omega}{d\log h}=-\frac{s}{\mathcal{A}_-'(\omega)-s(M_-^0)'(\omega)},\qquad 0<hI_{-,h}\le Cs^3$$

by (9.42). On the positive branch, parts (ii) and (iv) of Lemma 9.4 give $d\lambda_{\mathrm{in}}/d\log h=1+O(\xi_+)=1+O(s^N)$. By part (iii), the landing satisfies (4.1). These are the hypotheses of (6.15). Since $q_2=-s$, the condition $q_2\le C_2s$ of Corollary 6.6 holds with $C_2=0$; take C_1 the constant of the negative bound above, and fix $R_0\ge1$ satisfying the condition on M_1 there. The corollary gives $\mathcal{L}<0$, and at most one zero of Δ_p, for $\rho_e\ge R_0$. Writing ρ for the radius of the annular thin exit and ρ_e for that of the adaptive exit, we have $\rho\le\rho_e$ by (9.22). On the interval $\rho\ge\rho_{\mathrm{lo}}>0$, any zero of $\mathcal{L}$ must have $\rho_e<R_0$, because the large-exit estimate gives $\mathcal{L}<0$ otherwise. Such a central height has $\rho_{\mathrm{lo}}\le\rho<R_0$, so the compact-primary comparison proves $\mathcal{L}_T<0$ at that zero. Thus every zero of $\mathcal{L}$ on this entire later interval of central heights is a strict downward crossing.

9.2.5. *Covering all central heights.* We now join the comparisons in their central-height order. On the overlaps they concern the same slope ratio, so their crossing directions give one zero count on the whole interval. Use the same sufficiently small thin threshold δ_{an} at $z_e=s^3r^2\delta_{\mathrm{an}}$. The quadratic-prefix lemma, with δ there replaced by s, $\ell=sr$, and comparison drifts $a-\sigma bx+x^2$, gives at most two zeros on

$$h\ge z_e,\qquad\text{or}\qquad h<z_e,\quad|\eta_+(z_e;h)|/(sr)\le M_0.$$

The compact negative inverse follows from the first assertion of Lemma 6.4, since $a\ge\nu/4$ and the positive margin holds up to M_0. Every other central height has its positive half-orbit below height z_e at all radii up to M_0sr. For $K_0r\le\rho\le M_0r$, it therefore satisfies

$$y(\rho)<\delta_{\mathrm{an}}(r/\rho)^2<\delta_{\mathrm{an}}.$$

Its positive thin exit occurs beyond $M_0 r$. Such an exit exists before the admissible endpoint: the boundary coefficient satisfies $q(\eta) < 0$ on the positive branch throughout the small box in the coordinates (η, z), whereas the endpoint has $y \asymp s^{-1} > \delta_{\mathrm{an}}$. At a crossing of $y = \delta_{\mathrm{an}}$, the right side of (9.3) is positive, because $sK_+ = O(s + |\eta|)$ past the inner cut. Thus every crossing is upward, and the first exit is unique.

Take $\rho_{\mathrm{lo}} = v_0/2$ in the compact-primary comparison and reduce r so that $M_0 r < \rho_{\mathrm{lo}}$. In increasing T, the early central heights, the annular interval $M_0 r \le \rho \le v_0$, and the later interval $\rho \ge \rho_{\mathrm{lo}}$ can be cut within their overlaps to give an ordered cover of $\mathcal{H}_p$. Their order follows from the decrease of the central height with the exit radius. The first join has an overlap: an annular exit at $M_0 r$ has height $s^3 r^2 M_0^2 \delta_{\mathrm{an}} > z_e$, whereas at the height threshold of the early interval the positive half-orbit has height z_e at that radius. The second overlap is $v_0/2 \le \rho \le v_0$. These assertions persist after intersection with $\mathcal{H}_p$, including empty end pieces.

On the early interval, Lemma 6.3 gives $\mathcal{L}_T > 0$ at every zero of $\mathcal{L}$. On the middle interval, either $\mathcal{L}'' < 0$, for $b \ge b_{*,-}$, or $\mathcal{L} < 0$, for $b \le b_{*,-}$. In the latter case merge the nonvanishing middle interval with the initial interval; all zeros on their union cross upwards. On the later interval every zero has $\mathcal{L}_T < 0$, by the compact-primary and large-exit comparisons above. Lemma 3.1 therefore gives at most two zeros of $\mathcal{L}$ on the whole interval $\mathcal{H}_p$, and Corollary 3.2 gives at most three zeros of Δ_p. This proves Proposition 9.5.

10. Fold, separated roots, and the weak root

The buffered estimate uses a fixed positive margin for the secondary drift. Near a positive root or a minimum tending to zero, we instead compare the drift-weighted positive passage with the negative branch of Section 9. At the fold this controls bounded outputs, but on the no-root side the positive passage can continue through the annulus after a long bottleneck. We treat that continuation separately, then handle separated roots and the root approaching the central line to complete the coefficient cover.

10.1. **The boundary-drift fold and bounded outputs.** Continue with (9.1). Take $1/2 \le b \le 1$, set $x_{\mathrm{f}} = b/2$, and choose a small fold width $\epsilon_{\mathrm{f}} > 0$ before fixing the parameter neighborhood. Consider

$$|a - b^2/4| \le \epsilon_{\mathrm{f}}^2/100, \qquad r \ll \epsilon_{\mathrm{f}}^2. \tag{10.1}$$

This region is specified by the quadratic comparison. The exact fold is the parameter locus where the minimum of K_+ defined below vanishes. In positive secondary coordinates $x = \eta/(sr)$, $Z = z/(s^2 r^2)$, the drift is

$$K_+(x) = a - bx + x^2 - r x^3 R_q(srx), \qquad F_+(x) = K_+(x)/x.$$

It is strictly convex on the fixed fold box: $K_+'' = 2 + O(r) > 1$. Its minimum is at $x_{\mathrm{f}} + O(r)$. When roots are present, the first lies in $(x_{\mathrm{f}} - \epsilon_{\mathrm{f}}/4, x_{\mathrm{f}} + \epsilon_{\mathrm{f}}/4)$ and may be double. On $0 \le x \le x_L = x_{\mathrm{f}} - 2\epsilon_{\mathrm{f}}$ there is a fixed positive drift margin.

Fix a finite $B_* > x_{\mathrm{f}} + \epsilon_{\mathrm{f}}$, and a small positive output height Z_f. Construct the boundary-flow coordinate of Lemma 4.3 on a fixed positive box from x_L to beyond B_*, retaining the exact drift F_+. Put $t_0 = s^2 r^2$. The bounded-output interval of central heights is

$$x_{\mathrm{f}} - \epsilon_{\mathrm{f}} \le w = \mathcal{W}_+(x(t_0 Z_f; h), Z_f) \le B_*. \tag{10.2}$$

We consider the central heights of $\mathcal{H}_p$ in the positive-drift component reached from x_L. They form an interval, by Lemma 3.3 and $\mathcal{W}_x > 0$.

Proposition 10.1 (Bounded outputs at the fold)**.** *For each fixed B_*, one can choose ϵ_{f} sufficiently small so that Δ_p has at most three distinct zeros on (10.2), uniformly on both sides of the fold. If K_+ has a positive root, Δ_p has at most three distinct zeros on all of $\mathcal{H}_p$.*

10.1.1. *Positive passage in the fold region.* Reaching the bounded-output interval forces a fixed advance through the slow part of the boundary flow. We use this advance first to make the entry height small, then to apply the drift-weighted phase estimate. Since $\mathcal{W}_+ \le x$, for a central height in (10.2) the positive half-orbit reaches x_L below height Z_f and then crosses the projected interval $[x_{\mathrm{f}} - 2\epsilon_{\mathrm{f}}, x_{\mathrm{f}} - \epsilon_{\mathrm{f}}]$, where $0 < F_+ \le C\epsilon_{\mathrm{f}}^2$. Its incoming height at $x = x_L$ therefore satisfies

$$Z_i \le Z_f \mathrm{e}^{-\mathcal{A}_{\mathrm{gap}}/s}, \qquad \mathcal{A}_{\mathrm{gap}} \ge c_1/\epsilon_{\mathrm{f}}, \qquad h/(s^3 r^2) \le (Z_f/s)\mathrm{e}^{-\mathcal{A}_{\mathrm{gap}}/s}. \tag{10.3}$$

The projected entry is at most x_L, so its action is at least the displayed increment. The central prefix is therefore thin and has the preparation $Z_i = s\xi\Upsilon(\xi)$, with $\xi = k_p h$, $\Upsilon(0) = 1$, and uniform derivative bounds.

Use $T = -s\log h$ at the fixed output $z = t_0 Z_f$. The regular entry margin and (10.3) verify Lemma 4.8 with $\delta = s$, $N = 1$, $j = 2$. Thus $w_T = \theta_+ F_+(w)$, with $\theta_+ - 1, \theta'_+, \theta''_+ = o(1)F_+(w)$. The constants may depend on fixed $\epsilon_{\mathrm{f}}, B_*$. Remark 4.4 keeps the projected state below its first root and the inverse graph outward, also at a double root.

The coefficient ratios at the output height $t_0 Z_f$ are $O(s)$, and Lemma 4.9 with $\ell = sr$ and $\delta = s$ gives

$$-I_{+,w} = E_f A_p w\, U_p(w), \qquad E_f = t_0(z_*/(t_0 Z_f))^{2s} \longrightarrow 0, \qquad U_p = 1 + O_{C^3}(s), \quad A_p \asymp 1. \tag{10.4}$$

Compute the rate $\mathfrak{H}_+ = \partial_T \log(hI_{+,h})$ and its derivative from (4.26) with $K = K_+$ and $\theta = \theta_+$. On the fixed enlarged interval, the coefficient expansion and (4.29), with $K'_+(0) = -b$ and $R_K(x) = -rR_q(srx)$, give

$$\frac{K'_+(w)}{w} = 2 - \frac{b}{w} + 3\kappa_3 r w + O(sr^2), \qquad wK''_+ - K'_+ = b + 3\kappa_3 r w^2 + O(sr^2) > b/2.$$

Constants in these remainders may depend on B_*. Since $w_T = \theta_+ F_+(w) > 0$, the rate and its derivative therefore satisfy

$$-C\epsilon_{\mathrm{f}} - o(1) \le \mathfrak{H}_+ \le 2 - \frac{b_{\min}}{2B_*}, \qquad \mathfrak{H}'_+ \ge -o(1), \qquad b_{\min} = 1/2. \tag{10.5}$$

The constant C is independent of ϵ_{f}; the errors tend to zero after $\epsilon_{\mathrm{f}}, B_*$ are fixed, uniformly as $K_+(w)$ tends to zero.

10.1.2. *The negative branch.* The positive action gap also makes the negative branch thin at a fixed secondary cut. This places it in the range of the negative-branch estimates from the preceding section. The negative secondary drift

$$a + bX + X^2 + rX^3 R_q(-srX)$$

has a fixed positive margin on every fixed prefix. Choose a large fixed M_n, and then choose ϵ_{f} so that $\mathcal{A}_{\mathrm{gap}}$ in (10.3) exceeds the negative action to $X = M_n$ by a fixed positive amount. The inverse central-height coordinate puts the negative branch strictly below $y = z/(s^3\rho^2) = \delta_{\mathrm{an}}$ at $\rho = M_n r$, with the regular prefix preparation (9.5). Hence Lemma 9.3 applies with $M = M_n$.

10.1.3. *Counting and the root-present cover.* Let $\rho_{\mathrm{f}} = b_{\min}/(12B_*)$, reduced if necessary so that $\rho_{\mathrm{f}} < 0.5$. Choose M_n large enough to absorb the r/ρ errors, then ϵ_{f} small for the action gap and the lower bound in (10.5). The positive projected output stays in the bounded interval just fixed. We divide according to the negative output: its small-radius and far-root rates give opposite signs for the first derivative of the slope ratio, while the compact primary range controls its critical points. For the slope ratio $\mathcal{L}$ of (3.1), the following three intervals cover the bounded-output central heights:

(i) If the negative thin exit has $\rho \le \rho_{\mathrm{f}}$, then $\mathcal{L}' > 0$, because the negative-branch rate $2 - 3\rho + o(1)$ exceeds the upper bound for the positive-branch rate in (10.5).

(ii) For $\rho_{\mathrm{f}}/2 \le \rho \le 0.9$, (9.19) and (10.5) give $\mathcal{L}'' < 0$.

(iii) On (9.18), $\mathcal{L}' < 0$, by (9.20) and the lower bound for the positive-branch rate. This includes every branch still thin at 0.9.

The endpoint estimates and bounded derivatives of Π_p make the exterior derivatives $o(1)$ in these comparisons. By Lemma 9.3, the three ranges are ordered intervals of central heights after intersection with the bounded-output interval. The last two overlap, and the first two overlap on $[\rho_{\mathrm{f}}/2, \rho_{\mathrm{f}}]$. Their inequalities for $\mathcal{L}(T)$ hold also at the cuts. The first and third comparisons exclude critical points in their respective ranges; the second controls their curvature. Thus, on the whole bounded-output interval,

$$\mathcal{L}' = 0 \quad \Longrightarrow \quad \mathcal{L}'' < 0. \tag{10.6}$$

By Corollary 3.2, Δ_p has at most three distinct zeros on the bounded-output interval.

To cover the early heights as well when K_+ has a positive root, apply Lemma 6.4 with $\delta = s$, $\ell = sr$, $\epsilon_3 = r$, comparison drifts $a - \sigma bx + x^2$, $R_p = x_{\mathrm{f}} - \epsilon_{\mathrm{f}}/2$, $w_a = x_{\mathrm{f}} - 3\epsilon_{\mathrm{f}}/4$ and the first root $x_* \in (x_{\mathrm{f}} - \epsilon_{\mathrm{f}}/4, x_{\mathrm{f}} + \epsilon_{\mathrm{f}}/4)$. The positive margin up to R_p is fixed, and the inverse projection continues in the fold box up to Z_f, including when x_* is double. Its root range lies in (10.2), where (10.6) holds. The lemma therefore gives the root-present bound of three.

10.2. **Passage on the no-root side of the fold.** Suppose now that the exact minimum of K_+ in (10.1) is positive. The normalized drift K_+ is then positive on the whole positive unscaled radius interval: outside the fixed fold collar its quadratic part has a positive margin, and its cubic term is positive. As this minimum tends to zero, we need estimates uniform in the possibly divergent bottleneck action. The bounded-output result compares half-orbits whose projected states at a fixed secondary height stay bounded. For larger projected outputs, we isolate the bottleneck action in the exact exponential transport, prepare the outgoing height uniformly, and compare exits on the primary and cubic scales.

Proposition 10.2 (Zero bound across the fold)**.** *For fixed sufficiently small $\epsilon_{\mathrm{f}} > 0$, Δ_p has at most three distinct zeros throughout* (10.1)*, including the fold and both its sides, for all sufficiently small independent $s, r > 0$.*

10.2.1. *Bottleneck transport and escaped central heights.* Fix the postfold cut $K_0 > 2$ from the annular construction, and later a larger fixed $M > K_0$. A central height is called escaped if its positive half-orbit is still thin at $x = M$:

$$z(Msr; h)/(s^3 M^2 r^2) \le \delta_{\mathrm{an}}. \tag{10.7}$$

This is one interval of central heights. Use a single boundary-flow coordinate on a fixed positive box from $x_L = x_{\mathrm{f}} - 2\epsilon_{\mathrm{f}}$ to beyond K_0. The bottleneck formula (4.21) gives

$$\Psi_{K_0}(y_{K_0}) = \mathrm{e}^{\mathcal{A}_f/s}\Psi_L(y_L), \qquad \mathcal{A}_f = \int_{x_L}^{K_0} \frac{x\,dx}{K_+(x)}.$$

Here $Q = s^3 r^2$ and $y_j = z(jsr; h)/Q$ for $j = x_L, K_0$. The endpoint coordinates have uniform bounds because their integrals of $1/F_+$ lie on regular collars. The full bottleneck action remains in the explicit exponential factor; the composition below is differentiated in the expanded height, where the inverse bounds stay uniform. The construction following (4.21) applies with $x_R = K_0$, $\delta = s$, $\beta = \beta_f$, and the regular central prefix of Lemma 4.7, of action $\mathcal{A}_L = \int_0^{x_L} x/K_+(x)\,dx$, where

$$\mathcal{A}_{K_0} = \mathcal{A}_L + \mathcal{A}_f, \qquad \beta_f = \mathrm{e}^{-\mathcal{A}_f/s}, \qquad \xi = \mathrm{e}^{\mathcal{A}_{K_0}/s} h/Q.$$

By (4.22),

$$y_{K_0} = \xi V_p(\xi), \qquad V_p(0) = 1, \tag{10.8}$$

on a fixed small ξ-interval whose image contains a fixed small output interval; the unit, its reciprocal, and the map inverse have uniformly bounded ξ-derivatives. Composing with the annular coordinate gives (9.5) with action $\mathcal{A}_{K_0}$.

We next check that every escaped central height lies in the fixed domain of this inverse construction. For such a height, put $y_a(\rho) = z(s\rho; h)/(s^3\rho^2)$. For fixed $M > K_0$ and sufficiently small r, the segment $K_0 r \le \rho \le Mr$ lies in the positive annular box, where (9.3) excludes a downward crossing of $y_a = \delta_{\mathrm{an}}$. The escaped condition at Mr implies $y_a \le \delta_{\mathrm{an}}$ on the whole preceding segment. The annular transport (9.4) yields

$$\frac{\widehat{z}(K_0 r)}{Q} = \mathrm{e}^{-\mathcal{J}_{i,M}/s}\frac{\widehat{z}(Mr)}{Q} \le CM^2\delta_{\mathrm{an}}\mathrm{e}^{-\mathcal{J}_{i,M}/s},$$
$$\mathcal{J}_{i,M} = \int_{K_0 r}^{Mr} \frac{dt}{tK_+(r/t,t)} \ge c\log(M/K_0) > 0.$$

At the inner cut, $y_{K_0} = K_0^2 y_a(K_0 r)$ is uniformly comparable to $\widehat{z}(K_0 r)/Q$. Thus y_{K_0} lies in the small output domain of the bottleneck and central-prefix inverse coordinates. Their composition gives a small ξ, and flow uniqueness identifies the recovered central height with h. Hence (9.5) applies throughout the interval of escaped central heights.

For every escaped central height, the positive half-orbit crosses the fold collar below a bounded secondary height. The action gap gives

$$h/(s^3 r^2) \le C_{M,\delta_{\mathrm{an}}}\mathrm{e}^{-\mathcal{A}_{\mathrm{gap}}/s}, \qquad \mathcal{A}_{\mathrm{gap}} \ge c/\epsilon_{\mathrm{f}}. \tag{10.9}$$

Fix the negative secondary cuts needed below, then choose ϵ_{f} small so that $\mathcal{A}_{\mathrm{gap}}$ exceeds their regular actions by a fixed gap. At each such cut the expanded negative input is exponentially small by (10.9). The regular inverse central-height coordinate therefore constructs a thin negative prefix; flow uniqueness identifies it with the branch from the original h. Both signs now satisfy the hypotheses of Lemma 9.1 whenever their exits have small primary radius, and the negative branch satisfies those of Lemma 9.3.

10.2.2. *Both exits on the small primary annulus.* Take small fixed annular margins $u_0, v_0 > 0$, with u_0 eventually much smaller than v_0. Choose M and the action gap so that both inner ratios $u_\sigma = r/\rho_\sigma$ are at most u_0. If both thin exits satisfy $v_\sigma = \rho_\sigma \le v_0$, the prefixes constructed above meet Corollary 9.2. Since $b \ge 1/2$, small annular margins followed by small s give $\mathcal{L}'' < 0$ in $T = -s\log h$.

10.2.3. *The negative primary and far-root outputs.* When the positive exit is at most v_0, its annular rate satisfies

$$\mathfrak{H}_+ \ge 2 - Cu_0 - Cv_0^2 - Cs(1 + |\log s|).$$

For a negative exit in $v_0/2 \le \rho_- \le 0.9$, (9.19) gives $\mathfrak{H}_- = 2 - 3\rho_- + o(1)$. First choose v_0 small, then u_0 much smaller than v_0, and finally reduce r, s. The rate difference $\mathfrak{H}_- - \mathfrak{H}_+$ is strictly negative. The same sign holds if the negative graph is in its far-root interval, by (9.20). The exterior contribution to $\mathcal{L}'$ tends to zero with the negative endpoint derivative. Hence $\mathcal{L}' < 0$ on both ranges.

It remains to locate the negative branch when the positive exit has left the small annulus. For an escaped central height, a positive annular thin exit with $\rho_+ \ge v_0$ places the negative graph in the far-root interval, as the following comparison of central heights shows. Choose first the compact negative comparison cut 0.9 and the annular cut v_0, and let C_{th} bound all fixed endpoint and prefix terms in the two central-height equations. Then reduce the fold width ϵ_{f}, after the preceding choices, so that $\mathcal{A}_{\mathrm{gap}} > 2C_{\mathrm{th}}$. Compare the central height for which the positive exit is at v_0 with the central height for which the negative height at 0.9 equals the thin threshold. Their actions satisfy

$$\begin{aligned} \mathcal{A}_+(v_0) &= \mathcal{A}_{K_0} + \log\frac{v_0}{K_0 r} + O(K_0^{-1} + v_0), \\ \mathcal{A}_-(0.9) &= \log(1/r) + O(1). \end{aligned} \tag{10.10}$$

The first estimate follows by integrating the annular factor $K_+(r/\rho, \rho)^{-1} - 1 = O(r/\rho + \rho)$ after $K_0 r$. For the second, the negative secondary prefix has no bottleneck, and the remaining primary

action has fixed endpoint margins. Since $\mathcal{A}_{K_0} \geq \mathcal{A}_{\mathrm{gap}}$, the displayed estimates and the order of choices give, uniformly for small r,

$$\mathcal{A}_+(v_0) - \mathcal{A}_-(0.9) \geq \mathcal{A}_{\mathrm{gap}} - C_{\mathrm{th}} \geq C_{\mathrm{th}} > 0. \tag{10.11}$$

Writing $h_+^{v_0}$ and $h_-^{0.9}$ for the two central heights (with $h_-^{0.9}$ defined by the thin threshold at the fixed radius 0.9), (9.6) and the uniform boundedness of its height factors give

$$s \log \frac{h_+^{v_0}}{h_-^{0.9}} = -\big[\mathcal{A}_+(v_0) - \mathcal{A}_-(0.9)\big] + O(s) < 0 \tag{10.12}$$

after reducing s. Thus $h_+^{v_0} < h_-^{0.9}$. The exit-height maps are strictly decreasing in their radii by the event equation, so for any $\rho_+ \geq v_0$ its central height satisfies $h \leq h_+^{v_0} < h_-^{0.9}$. The negative graph at that same h is consequently still thin at 0.9: by Lemma 3.3, its height at this fixed radius is smaller than the thin threshold attained by $h_-^{0.9}$. Part (ii) of Lemma 9.3 now gives $w_n > 0.89$, hence (9.18).

10.2.4. *Positive continuation through the cubic regime.* For escaped central heights whose positive annular exit is at least v_0, which form an interval, we apply Lemma 9.4. Its prefix hypothesis (9.23) is (10.8), with $\mathcal{V} = V_p$ and $\mathcal{A}_i = \mathcal{A}_{K_0}$. Escaped central heights are thin at $K_0 r$, and their inverse estimate makes ξ uniformly small, so it lies in the domain of V_p after reducing s. By (9.22), the adaptive exit follows the annular thin exit; hence $\varrho_e \geq s v_0$. By the first estimate in (10.10), $\mathcal{J}_+(v_0) = \mathcal{A}_+(v_0) - \mathcal{A}_{K_0} \geq \log(1/r) - C$, which is at least $\frac{1}{2}\log(1/r)$ after fixing a sufficiently small upper bound for r. Thus part (iv) of the lemma applies with $\rho_{\mathrm{g}} = v_0$ and $c_{\mathrm{g}} = 1/2$. It gives (9.28), $\mathfrak{H}_+ > c\varrho_e^+/s$, at these exits. The bounds on V_p in (4.22) make this estimate uniform in the bottleneck action. The negative graph is in the far-root interval by (10.10). Its strictly negative rate and the small exterior term imply $\mathcal{L}' < 0$ here as well.

We now combine the exit comparisons on the interval of escaped heights. On the interval where both exits are at most v_0, we have $\mathcal{L}'' < 0$. Within the escaped central heights, the condition $y_-(v_0/2; h) \leq \delta_{\mathrm{an}}$ defines a second interval, because the height at a fixed radius increases with h. The negative primary, negative far-root, and positive adaptive estimates cover this interval, including their fixed overlaps. Thus $\mathcal{L}' < 0$ throughout the second interval. These two intervals cover (10.7): a positive exit at least v_0 forces the negative far-root interval; otherwise either both exits are small or the negative graph is already in the second interval. The estimates for $\mathcal{L}(T)$ include both cuts $v_0/2$ and v_0. Consequently, on the entire escaped interval,

$$\mathcal{L}' = 0 \quad \Longrightarrow \quad \mathcal{L}'' < 0. \tag{10.13}$$

10.2.5. *Covering the height interval.* Set $B_* = 4M$ in Proposition 10.1. Its bounded-output estimates give (10.6). The quadratic-prefix lemma applies on the early interval

$$h \geq z_e = s^3 r^2 \delta_e, \qquad \text{or} \qquad h < z_e, \quad x_+(z_e; h) \leq x_{\mathrm{f}} - \epsilon_{\mathrm{f}}/2.$$

Its regular-prefix hypotheses hold on the no-root side. In T, the early heights form an initial interval, and each zero of $\mathcal{L}$ on it satisfies $\mathcal{L}' > 0$.

For every other central height, the projected state of the positive half-orbit at its hit of $x = x_{\mathrm{f}} - \epsilon_{\mathrm{f}}/2$ is greater than $x_{\mathrm{f}} - 3\epsilon_{\mathrm{f}}/4$. Follow the half-orbit to the first event $Z = Z_f$ or $x = 3M$. If the height is reached first, the projected output lies between $x_{\mathrm{f}} - \epsilon_{\mathrm{f}}$ and B_*, and is counted by the bounded-output proposition. If $x = 3M$ is reached first, the central height satisfies (10.7). Indeed, between $x = M$ and that event the half-orbit lies in the coordinate box of Section 10.1, which extends beyond $B_* = 4M$, and $B_0 Z_f \leq M^2$ since Z_f is small. For small s, Lemma 4.5 with $\delta = s$, $X_1 = M$, $X_2 = 3M$ and $C_F = \sup|F_+|$ on that box bounds the height at $x = M$ by $Z_f \mathrm{e}^{-M/(C_F s)} < s M^2 \delta_{\mathrm{an}}$. Every admissible half-orbit under consideration reaches one of these events before its fixed endpoint.

The complement of the early set is the single interval

$$J = \{T : h \in \mathcal{H}_p,\ h < z_e, \quad x_+(z_e; h) > x_{\mathrm{f}} - \epsilon_{\mathrm{f}}/2\}.$$

Every point of J satisfies a bounded-output or escaped estimate. Their common conclusion is the critical-point inequality $\mathcal{L}' = 0 \Rightarrow \mathcal{L}'' < 0$, by (10.6) and (10.13), for the same smooth function on J. The early cut belongs to the initial interval, where (6.9) includes the terminal radius. Thus the initial interval and J cover the whole height interval $\mathcal{H}_p$. Apply Corollary 3.2, with J as the middle piece and an empty final piece. It gives the bound of three on the no-root side. The same bound on the root-present side follows from Proposition 10.1.

10.3. **Separated roots on the negative quadratic-drift scale.** Even for separated roots, the limiting endpoint rates can agree. We compare derivatives of these rates in the common logarithmic height coordinate. Retaining their drift weights makes the curvature comparison strict at every interior height.

Fix $a_0^-, \underline{b}, \nu > 0$, and consider

$$a \geq a_0^-, \qquad b \geq \underline{b}, \qquad b^2 - 4a \geq \nu \tag{10.14}$$

in (9.1). The first positive secondary root is simple:

$$\alpha = \frac{b - \sqrt{b^2 - 4a}}{2} + O(r), \qquad c < \alpha < C, \qquad K_+'(\alpha) < -c.$$

Set $x_L = \alpha/4,\ w_a = \alpha/2,\ R_p = 3\alpha/4$. The exact drift and its quadratic comparison have fixed positive margins up to R_p.

Proposition 10.3 (Separated secondary roots)**.** *In* (10.14)*, Δ_p has at most three zeros for all sufficiently small independent $s, r > 0$, including parameters at which the two limiting characteristic rates agree.*

Proof. Construct the positive boundary-flow coordinate at a fixed small output $z = t_0 Z_f,\ t_0 = s^2 r^2$. The root-approach interval of central heights is

$$w = \mathcal{W}_+(x(t_0 Z_f; h), Z_f) \in [w_a, \alpha). \tag{10.15}$$

Advancing from a projected entry at most x_L to w_a requires a fixed positive F_+-flow time τ_*. Thus each central height in this interval satisfies

$$h/(s^3 r^2) \leq (Z_f/s)\mathrm{e}^{-\tau_*/s}.$$

The regular prefix gives (4.19) at the fixed output $z = t_0 Z_f$. The positive drift margin at x_L and the advance τ_* verify Lemma 4.8 with $\delta = s$, $N = 1$ and $j = 2$. Thus, in $T = -s \log h$, the phase errors are $o(1)F_+(w)$ through two T-derivatives. Lemma 4.9, with $\ell = sr$ and $\delta = s$, gives $-I_{+,w} = E_f A_p w(1 + O_{C^3}(s))$, with $E_f = t_0(z_*/(t_0 Z_f))^{2s} \to 0$. Apply (4.27):

$$\mathfrak{H}_+' = F_+(w)\{Q_+(w) + o(1)\}, \qquad Q_+(w) = \frac{wK_+'' - K_+'}{w^2}.$$

By (4.29), with $K_+'(0) = -b$ and $R_K(x) = -rR_q(srx)$, one has $Q_+(w) = b/w^2 + O(r) > c_Q > 0$, since $b \geq \underline{b}$ and $w < \alpha < C$. Consequently $\mathfrak{H}_+' \geq cF_+(w) > 0$ on the root interval.

The negative output at the same central height may lie on either scale. We follow it from bounded secondary outputs through the annulus and the compact primary interval to its far root, retaining the endpoint estimates needed to absorb the exterior curvature in each range. Choose a small fixed $x_c > 0$ so that the negative boundary action to $2x_c$ is less than $\tau_*/4$. This is possible because the central coefficient is at least a_0^-. The negative prefix coordinate gives exponentially small height at this cut. The negative secondary drift is

$$K_b(X) = a + bX + X^2 + rX^3 R_q(-srX), \qquad F_b = K_b/X.$$

Choose a large fixed M, and a small fixed output height $Z_b > 0$. The boundary-flow coordinate on a box containing $[x_c/4, 5M]$ shows that every negative output with $X \leq 4M$ at $t_0 Z_b$ has projected state $w_b \in [x_c, 4M]$. At the regular entry $x_c/2$, the drift F_b has a fixed positive lower bound. The

advance from that entry to $w_b \ge x_c$, at the fixed output $z = t_0 Z_b$, has a fixed positive phase gap. Hence Lemma 4.8 applies with $\delta = s$, $N = 1$ and $j = 2$, in the same T, giving phase errors $o(1)F_b(w_b)$ through two T-derivatives. By Lemma 4.9, its endpoint has amplitude $E_b = t_0(z_*/(t_0 Z_b))^{2s} \to 0$. By (4.29), with $K_b'(0) = b$ and $R_K(X) = rR_q(-srX)$, one has $Q_b(X) = -b/X^2 + O(r) < -c_M < 0$ for $x_c \le X \le 4M$. The endpoint rate identities and the weighted phase estimate then give

$$\mathfrak{H}_-' \le -cF_b(w_b), \qquad |I_{-,T}| + |I_{-,TT}| \le CE_bF_b(w_b).$$

If the radius X of the negative graph at $t_0 Z_b$ exceeds $4M$, the graph is thin at $X = M$. Indeed, the outward graph then reaches $X = 4M$ below Z_b, it lies in the coordinate box between $X = M$ and $X = 4M$, and $8B_0 Z_b \le 15M^2$ since Z_b is small. For small s, Lemma 4.5 with $\delta = s$, $X_1 = M$, $X_2 = 4M$ and $C_F = \sup|F_b|$ on the box bounds its secondary height at $X = M$ by $Z_b \mathrm{e}^{-3M/(2C_F s)} < sM^2\delta_{\mathrm{an}}$.

For a thin exit in $Mr \le \rho \le v_0$, the negative regular prefix allows us to apply Lemma 9.1, with $u = r/\rho$, $v = \rho$. Choose M large and v_0, s small. Since $b \ge \underline{b}$, (9.11) gives

$$\mathfrak{H}_-' \le -c(u+v), \qquad |I_{-,T}| + |I_{-,TT}| \le C(s\rho)^{2-4s}.$$

For $v_0/2 \le \rho \le 0.9$, Lemma 9.3(i) instead gives $\mathfrak{H}_-' < -c_{v_0} < 0$, with the same endpoint size estimate.

Every graph still thin at 0.9 satisfies (9.18), and Lemma 9.3(ii) gives, uniformly up to the far root,

$$\mathfrak{H}_-' \le -cF_n(w_n), \qquad |I_{-,T}| + |I_{-,TT}| \le CE_nF_n(w_n).$$

These ranges cover every negative-branch output. We now compare the exterior term with the curvature margin in each range. Equation (4.28) gives exterior contributions $o(1)F_b$ and $o(1)F_n$ on the bounded secondary and far-root intervals and $o(1)$ on the compact primary interval. On the small primary annulus, (9.17) gives $o(\rho)$. Each exterior contribution is absorbed by its negative curvature margin. Thus

$$\mathcal{L}'' = \mathfrak{H}_-' - \mathfrak{H}_+' + \partial_T^2[(\log \Pi_p')(I_-)] < 0$$

throughout (10.15). The positive drift weights give strict negativity at every interior central height, although both may tend to zero at the endpoint. Thus the curvature remains strictly negative at each interior height even when the limiting rates agree. In the leading model the characteristic rates agree at $\alpha = b/3$, $a = 2b^2/9$, which is included here.

Finally apply Lemma 6.4 with $\delta = s$, $\ell = sr$, $\epsilon_3 = r$, comparison drifts $a - \sigma bx + x^2$ and the radii w_a, R_p, α fixed above. The positive margin and the gap $R_p - w_a$ are fixed, and the inverse boundary-flow coordinate is defined up to α at Z_f. The root range is (10.15), where $\mathcal{L}_{TT} < 0$. Hence Δ_p has at most three zeros on $\mathcal{H}_p$. □

10.3.1. *The compact secondary coefficient cover.* For any fixed $a_0^- > 0$, the normalized coefficients with $a \ge a_0^-$ now have a bound of three. Choose the fixed fold width ϵ_{f} from Proposition 10.2, and let $\nu = \epsilon_{\mathrm{f}}^2/50$. The three overlapping regions are

$$b^2 - 4a \le -\nu, \qquad |b^2 - 4a| \le 2\nu, \qquad b^2 - 4a \ge \nu.$$

They are covered respectively by Propositions 9.5, 10.2 and 10.3. For the last two regions normalization forces $b = 1$: otherwise $a = 1$ would give $b^2 - 4a \le -3$. On the middle region $|a - b^2/4| \le \epsilon_{\mathrm{f}}^2/100$, as required for the fold. Each of the three propositions bounds the zeros on the whole height interval. The minimum of their scale thresholds therefore gives a uniform bound of three on this coefficient set.

10.4. **The secondary root approaching the central line.** The remaining part of the coefficient cover has $a \downarrow 0$. Normalization in (9.1) then forces $b = 1$. To treat the positive root approaching the central line, apply Proposition 6.9 with

$$\kappa = s/a, \qquad \ell = sra, \qquad \hat{r} = r, \qquad \epsilon_2 = a.$$

These choices give (6.16) and the logarithmic height coordinate $T = -s \log h$. The remaining hypothesis concerns the negative branch outside the local weak-root region. We verify it by separating bounded secondary outputs from graphs that reach the annulus while still thin. The first group has a positive endpoint rate; for the second, the annular and far-root estimates give the uniform lower bound required by the weak-root comparison.

Proposition 10.4 (Negative quadratic drift at the weak root)**.** *There are fixed $a_-, \kappa_- > 0$ such that Δ_p has at most two zeros on*

$$0 < a \le a_-, \qquad 0 < s/a \le \kappa_-$$

for all sufficiently small independent $s, r > 0$. Consequently, for all sufficiently small independent $s, r > 0$ and every $L > 1/\kappa_-$, Δ_p has at most three zeros on the part of (9.1) outside the strip $|q_0| \le Lq_1^2$.

Proof. At the radius $\ell_d = srd$, the outward drift on the negative branch is

$$K_d(x) = a_d + x + dx^2 + rd^2x^3R_q(-\ell_d x).$$

For a fixed large M and $r \le r(M)$, (6.18) holds with $d_{\max} = M$, with constants depending on M. We verify (6.19) for the negative graphs with $z < Y_b|\eta|^2$ at $|\eta| = srM$.

Let $X = |\eta|/(sr)$, and follow such a graph to the height $\ell_M^2 Y_b$. At $d = M$, the local radius in the weak-root estimate is $x = X/M$. If $X \le 4M$ at this height, the corresponding projected state lies in $[0.9, 4]$, and Lemma 6.7(ii) with $d = M$ gives $\mathfrak{H}_- \ge c/M$ and $|I_{-,T}| = o(1)$.

If $X > 4M$ at that height, the graph is thin at the annular cut $X = M$:

$$z(-srM; h)/(s^3r^2M^2) < \delta_{\mathrm{an}},$$

where δ_{an} is the thin threshold of Section 9.1. To obtain this estimate, use $|\eta| = \ell_M x$, $z = \ell_M^2 Y$ with the same $d = M$. The outward graph reaches $x = 4$ below Y_b and lies in the box between $x = 1$ and $x = 4$. Also, $8B_0Y_b \le 15$ since Y_b is small. For small s, apply Lemma 4.5 in x, with $\delta = s/M$, $X_1 = 1$, $X_2 = 4$ and $C_F = \sup|K_M/x|$ on the box. The height at $x = 1$ is at most $Y_b \mathrm{e}^{-3M/(2C_F s)} < s\delta_{\mathrm{an}}$, as required.

For this thin input, take M large enough for the annular margins on the negative branch. In the secondary coordinate $X = |\eta|/(sr)$, the negative drift is comparable to $a + X$ and its derivative is bounded below on this fixed box. Here $\delta = s$ and the weak-root condition gives $a/s \ge 1/\kappa_-$; Lemma 4.7 supplies the height unit at a fixed secondary cut below M, uniformly as $a \to 0$. Lemma 9.3, whose constants are uniform down to $a = 0$, gives, for exits of the negative branch up to $\rho = 0.9$,

$$\mathfrak{H}_- = 2 - 3\rho + O(r/\rho + s(1 + |\log s|)) \ge -C_n, \qquad |I_{-,T}| \le C(s\rho)^{2-4s} = o(1).$$

Graphs still thin at $\rho = 0.9$ satisfy $\mathfrak{H}_- \ge -C_n$ and $|I_{-,T}| \le Cs^2$. Thus (6.19) holds in both cases. Proposition 6.9 gives at most two zeros of Δ_p, with a_- and κ_- equal to its constants $a_\circ$ and $\kappa_\circ$.

To obtain the assertion outside the contact strip, combine the weak-root bound with the compact coefficient cover of Section 10.3, which gives the bound three on $a \ge a_-/2$, overlapping the weak-root region. On the remaining small-a part, where $s/a > \kappa_-$, normalization gives $b = 1$ and $|q_0|/q_1^2 = a/s < 1/\kappa_-$. This part lies in $|q_0| \le Lq_1^2$ for every $L > 1/\kappa_-$. Hence the weak-root and compact-coefficient regions cover the parameter set outside the contact strip. Their bounds on the whole interval $\mathcal{H}_p$ give at most three zeros there. The thresholds for s, r are those of these two regions and are independent of L. □

11. The finite parameter cover and the geometric conclusion

The regional estimates bound the displacement on the whole interval of admissible central heights. It remains to show that their parameter regions cover a full neighborhood of the organizing value, with common sections and compatible constant choices. The resulting zero bound is the maximum of the regional bounds. Geometric capture then gives the upper bound for periodic orbits. The table below summarizes the regions; the proof fixes their thresholds and checks their overlaps and boundary cases.

Table 3. Parameter regions, the comparison that bounds the zeros of Δ_p on the whole height interval $\mathcal{H}_p$, and the resulting bound. After the first row, $q_1 > 0$. The constants $L, \kappa_{\mathrm{p}}, m_{\mathrm{cov}}, A_0, a_{\mathrm{w}}, \kappa_{\mathrm{w}}, r_{\mathrm{d}}$ are chosen in the proof below, in Section 11, which also checks the overlaps and the boundary axes.

Parameter region	Obstruction	Comparison	Section	Zero bound
$q_1 \le 0$	Vanishing radial flux at the central point	Signed paired contraction	2.8	1
$\lvert q_0\rvert \le Lq_1^2$, arbitrary q_2	Contact and low flux	Scattering ratio and endpoint bounds	5.1	2
$q_0 > 0$, $q_1/\sqrt{q_0} \le \kappa_{\mathrm{p}}$	Interior saddle	Dulac slopes and exterior contraction	5.2	2
$q_0 < 0$, $\lvert q_0\rvert/H^3 \ge m_{\mathrm{cov}}$	Merging boundary roots	Drift-weighted curvature; early central heights	7.1	3
$q_0 < 0$, $\lvert q_2\rvert/\sqrt{q_1} \le A_0$, $\lvert q_0\rvert/q_1^{3/2} \le a_{\mathrm{w}}$, $q_1^2/\lvert q_0\rvert \le \kappa_{\mathrm{w}}$	Weak boundary root	Weak-root comparison	7.2	2
$q_0 < 0$, $q_2 > 0$, $r < r_{\mathrm{d}}$, outside the contact strip	Near-symmetric roots	Curvature or power comparison	8.1	3
$q_0 < 0$, $q_2 < 0$, $r < r_{\mathrm{d}}$, outside the contact strip	Unequal exits; long bottleneck	Radially weighted rates and exact boundary action	9–10	3

Proposition 11.1 (Uniform zero bound)**.** *For each fixed $C_0 > 1$, there is a full parameter neighborhood of $(1, C_0, 0, 0, 0)$ such that, with*

$$N_{\mathrm{dir}} = 3,$$

Δ_p has at most N_{dir} distinct zeros on the height interval $\mathcal{H}_p$.

Proof. Common sections and the contraction margin. First fix a preliminary box in the coordinates (η, z) for this C_0, the common cone of Section 4, and the fixed adaptive and annular landing levels. The organizing endpoints satisfy $\eta_\pm^2/(2z_*) \to 1$ as $z_* \downarrow 0$, by Lemma 2.5. Hence their windows can be chosen with a fixed lower bound $z_*/\eta_\pm^2 \ge c_{\mathrm{end}} > 0$ for all sufficiently small sections; choose each landing level below $c_{\mathrm{end}}/2$. The other landing restrictions use the enlarged compact drift domains before the later coefficient and height thresholds.

Choose Λ_* with $\Lambda_0 < \Lambda_* < 1$, and reserve a fixed fraction of $-\log \Lambda_*$ for tail errors. By Lemma 2.5 with $\Lambda_+ = \Lambda_*$, we can then fix z_* small and choose the enlarged exterior windows $\widetilde{W}_\pm$ of that lemma. Thus (2.17) holds with $\Lambda_+ = \Lambda_*$ in every regional estimate.

The same tail bound must fit the reserved contraction margin in every region. The half-orbits enter the common cone after a finite scaled continuation, as in Lemma 6.1, in Sections 5.1 and 5.2, and by (4.2). From a common height t in that cone, (4.4) bounds the tail contribution to the logarithm of the endpoint derivative ratio by $2C_{\mathrm{cone}}(\sqrt{z_*} + \lvert q_1\rvert/\sqrt{t})$, as in Lemma 4.9. The coefficient of $\sqrt{z_*}$ in every such upper bound is therefore at most $2C_{\mathrm{cone}}$, independently of L, of coefficient margins such as ν, and of the regional cuts.

The continuation constants and higher exterior derivatives may depend on the cuts and on z_*, but enter only the final parameter thresholds. Choose z_* also so that $2C_{\mathrm{cone}}\sqrt{z_*}$ is smaller than the reserved fraction. This choice is made after the common cone and before all later cuts and thresholds, and meets every regional smallness requirement on z_*. If $q_1 \le 0$, Proposition 2.9 gives at most one zero. Henceforth assume $q_1 > 0$.

Secondary negative offsets. For $q_0 < 0$, $q_2 \neq 0$, let s, r be the scales of Section 3. We first fix thresholds for these scales; the contact width will be chosen after the remaining regions have been compared. Proposition 8.1 for $q_2 > 0$ and the second assertion of Proposition 10.4 for $q_2 < 0$ give constants κ_+ and κ_-, respectively, and thresholds for s, r independent of L. Below these thresholds, for every $L > \max\{\kappa_+^{-1}, \kappa_-^{-1}\}$, Δ_p has at most three zeros outside the strip $|q_0| \le Lq_1^2$, for either sign of q_2. Consequently a single secondary threshold $0 < r_{\mathrm{d}} < 1/4$ for r can be fixed for both signs before the final contact constant L is selected.

The primary and linear scales. For the remaining negative offsets, including $q_2 = 0$, the primary and linear scales must overlap the secondary region. We choose the thresholds so that small normalized constant coefficient, outside the secondary region, forces the normalized linear coefficient to stay bounded away from zero. Put

$$b_{\mathrm{cov}} := r_{\mathrm{d}}/2, \qquad A_0 := 2/\sqrt{b_{\mathrm{cov}}}.$$

For this fixed A_0, the primary displaced-contact estimate Proposition 7.2 gives constants $a_{\mathrm{w}}, \kappa_{\mathrm{w}} > 0$. Choose $m_{\mathrm{cov}} > 0$ so that

$$2m_{\mathrm{cov}} < 1, \qquad \sqrt{2m_{\mathrm{cov}}} < b_{\mathrm{cov}}, \qquad 2m_{\mathrm{cov}}/b_{\mathrm{cov}}^{3/2} < a_{\mathrm{w}}. \tag{11.1}$$

With H and ρ_1 of Section 3, put $\bar{a} = |q_0|/H^3$ and $\bar{b} = q_1/H^2$. If $\bar{a} \ge m_{\mathrm{cov}}$, Proposition 7.1 gives at most three zeros. On the remaining region $\bar{a} < m_{\mathrm{cov}}$ (and hence also $\bar{a} \le 2m_{\mathrm{cov}}$), use the secondary estimates when $q_2 \neq 0$ and $r < r_{\mathrm{d}}$. For the complementary part, it suffices to consider $q_2 = 0$ or $r \ge r_{\mathrm{d}}/2 = b_{\mathrm{cov}}$. On this part we claim

$$\bar{b} \ge b_{\mathrm{cov}}, \qquad |q_2|/\rho_1 \le 1/\sqrt{b_{\mathrm{cov}}} < A_0, \qquad |q_0|/\rho_1^3 \le 2m_{\mathrm{cov}}/b_{\mathrm{cov}}^{3/2} < a_{\mathrm{w}}. \tag{11.2}$$

To prove the overlap, consider which term realizes the maximum defining H. If $H = \rho_1$, the first assertion is immediate. The possibility $H = |q_0|^{1/3}$ would give $\bar{a} = 1$, excluded by (11.1). In the remaining case $H = |q_2|$,

$$r = \max\{\sqrt{\bar{a}}, \bar{b}\}.$$

Because $\sqrt{\bar{a}} < b_{\mathrm{cov}}$ and $r \ge b_{\mathrm{cov}}$, one has $\bar{b} \ge b_{\mathrm{cov}}$. The other assertions in (11.2) follow from $\rho_1 = H\sqrt{\bar{b}}$.

If also $q_1^2/|q_0| \le \kappa_{\mathrm{w}}$, these inequalities put the parameter in Proposition 7.2, giving at most two zeros. Within this part of coefficient space, the complementary case satisfies $|q_0|/q_1^2 < 1/\kappa_{\mathrm{w}}$ and lies in bounded contact. The bands $m_{\mathrm{cov}} \le \bar{a} \le 2m_{\mathrm{cov}}$ and $r_{\mathrm{d}}/2 \le r < r_{\mathrm{d}}$ provide fixed overlaps of these coefficient regions.

Positive offsets and the contact strip. If $q_0 > 0$, Proposition 5.3 gives at most two zeros whenever $q_1/\sqrt{q_0} \le \kappa_{\mathrm{p}}$, for a fixed $\kappa_{\mathrm{p}} > 0$. Its complement satisfies $q_0/q_1^2 < \kappa_{\mathrm{p}}^{-2}$. We can now choose the contact width to include all the remaining parameters:

$$L > 2\max\{\kappa_{\mathrm{p}}^{-2}, \kappa_{\mathrm{w}}^{-1}, \kappa_+^{-1}, \kappa_-^{-1}, 1\}. \tag{11.3}$$

Proposition 5.1 gives at most two zeros on this strip for sufficiently small parameters.

Table 4 records the global order of choices. Within each regional proof, its fixed coefficient and height cuts precede the final scale reduction, as specified there.

Boundary cases and a full neighborhood. The regions now have fixed overlaps. We check that they also include their boundary cases. If $q_1 = 0$, signed contraction applies, including $q_0 = q_2 = 0$ and the axes $H = 0$ or $r = 0$. If $q_1 > 0$ and $q_0 = 0$, the contact row applies.

TABLE 4. Global order of fixed choices. Each row may depend on preceding rows; the original parameter neighborhood is reduced last.

Choice	Purpose and permitted dependence
Fixed C_0; preliminary box, common cone and landing level m_{land}; $\Lambda_* = \Lambda_+$, then z_*, exterior windows and capture data	Reserve the exterior margin using Λ_0, and fix $2C_{\mathrm{cone}}\sqrt{z_*}$ below a fixed fraction of it, before any coefficient or height threshold. The landing level is small for Lemma 6.5 and (9.28), which use only bounds of the drift factors.
For $q_2 > 0$: a_+, κ_+, then the compact thresholds. For $q_2 < 0$: the fold width ϵ_{f} and $\nu = \epsilon_{\mathrm{f}}^2/50$, a_-, κ_-, then the compact thresholds. Then r_{d}	Both signs of q_2; the complements of the weak-root regions lie in bounded contact strips.
$b_{\mathrm{cov}} = r_{\mathrm{d}}/2$, A_0, a_{w}, κ_{w}, m_{cov}	Fixed primary–secondary overlaps in (11.1)–(11.2).
Positive-offset threshold κ_{p}, then contact width L	(11.3) includes the complements of all weak-root and saddle regions.
Contact height thresholds, compact passing sets and matching height; final original parameter neighborhood	Cut-dependent finite-height errors and exterior derivative bounds are absorbed by parameter reduction with z_* fixed.

If $q_0 < 0$ and $q_2 = 0$, then $\bar{a} < m_{\mathrm{cov}}$ puts $H = \rho_1$, so $\bar{b} = 1$ and the primary–contact overlap above applies. Its weak-root condition puts the point in the displaced row; the complementary condition puts it in the contact row.

For $q_0 < 0$, $\bar{a} < m_{\mathrm{cov}}$ and $q_2 \neq 0$, the cases $r < r_{\mathrm{d}}/2$, $r_{\mathrm{d}}/2 \le r < r_{\mathrm{d}}$, and $r \ge r_{\mathrm{d}}$ are covered by the secondary row, the doubled overlap, and the primary–contact overlap. The primary row includes $\bar{a} = m_{\mathrm{cov}}$; the overlap and secondary rows include $r = r_{\mathrm{d}}/2$, and the overlap includes $r = r_{\mathrm{d}}$.

Equality $|q_0| = Lq_1^2$ belongs to contact. For $q_0 > 0$, the saddle row and its contact complement are exhaustive. Equality in the saddle condition is assigned to the saddle row, and both rows allow $q_2 = 0$. Thus every parameter belongs to at least one row of Table 3. The maximum of the listed zero bounds is $N_{\mathrm{dir}} = 3$.

Finally, choose the smallest of the finitely many remaining parameter thresholds, keeping the sections fixed. The coefficient comparison in Appendix A gives a full neighborhood in the original five parameters on which all these conditions hold. All estimates are uniform for (A, C) in the fixed neighborhood, independently of H, s, r and ρ_1. This proves the proposition. □

Proof of Theorem 1.1. Fix $C_0 > 1$ and use the fixed sections, exterior windows, and full parameter neighborhood constructed in the proof of Proposition 11.1. Choose the whole-graphic capture data of Proposition 2.1 for these windows, reduced as in the paragraph before Lemma 2.5 and in (2.28). Let V be the intersection of the parameter neighborhood with V_{cap}, and choose $\varepsilon \le \rho_{\mathrm{cap}}$; by Proposition 2.1(ii), every periodic orbit within Hausdorff distance ε of $\Gamma(C_0)$ is then captured.

Let $p \in V$, and let γ be a periodic orbit with $d_{\mathrm{H}}(\overline{\gamma}, \Gamma(C_0)) < \varepsilon$. By Lemma 2.6, it has exactly one transverse central crossing. The interval construction in Proposition 2.7 assigns it a zero of Δ_p, injectively. The uniform zero bound therefore gives at most $N_{\mathrm{dir}} = 3$ such orbits.

It remains to check isolation. The interior-window construction in Proposition 2.7 places the height of every captured orbit in $\operatorname{int} \mathcal{H}_p$. The finite zero bound makes its displacement zero isolated in an open central-height neighborhood. The correspondence in that proposition preserves isolation, so every periodic orbit counted above is a limit cycle. Consequently $N(V, \varepsilon) \le 3$ in (1.4), and $\mathrm{Cycl}(\Gamma(C_0)) \le 3$. □

12. SHARPNESS AND THREE HYPERBOLIC LIMIT CYCLES

To attain the upper bound, we need three displacement zeros whose orbits approach the whole graphic. Two arise on a fixed compact interval of normalized exit radii: along a buffered coefficient

path, the difference of the boundary actions gives a profile with a strict negative minimum, and varying A sets its additive constant to a small positive value. This produces two zeros. The third comes from the organizing contraction at a larger central height, beyond the compact exit interval. We must then show that this third height also tends to zero and that all three orbits converge to the whole graphic. These are the final steps of the proof below.

12.1. **A buffered path and its action difference.** Choose the coefficient path

$$\nu = \min\{1, (C_0-1)/64\}, \qquad a = (1+\nu)/4, \qquad q_0 = -as^5, \quad q_1 = s^3, \quad q_2 = -s, \qquad s > 0. \tag{12.1}$$

It lies in the buffered region (9.29) with $b = 1$, $r = s$, and $b^2 - 4a = -\nu < 0$. The coefficient change (A.1) determines $\mu(s) = (\mu_1(s), \mu_2(s), \mu_3(s)) \to 0$. We keep $C = C_0$ and write

$$p_s(A) = (A, C_0, \mu(s)).$$

This path fixes the boundary drift through μ, while A remains free to adjust the connection splitting. In particular, the boundary actions are independent of this adjustment.

Besides the limiting exterior multiplier Λ_0 in (2.16), put

$$\Lambda_{\mathrm{b}} = \exp(\pi/\sqrt{\nu}).$$

The choice of ν gives the multiplier margin

$$\Lambda_0 \Lambda_{\mathrm{b}}^2 \geq \mathrm{e}^{3\pi/2} > 100. \tag{12.2}$$

Indeed, the logarithm of the left-hand side is $12\pi/\sqrt{C_0 - 1}$ when $C_0 - 1 \leq 64$, and $2\pi - 4\pi/\sqrt{C_0 - 1}$ otherwise.

The following lemma identifies Λ_{b} as the limit of the boundary-action multiplier. In the comparison with [8, (2.13) and (6.8)] in Section 9.2.3, their $(\lambda_0^{\mathrm{HK}}, \lambda_1^{\mathrm{HK}}) = (-a, -1)$ give $\Lambda_{\mathrm{b}} = \mathrm{e}^{-K_{\mathrm{HK}}}$.

Lemma 12.1 (Buffered action limit)**.** *Along* (12.1)*, the constants* $\mathfrak{c}_\sigma$ *in* (9.39) *satisfy*

$$\mathfrak{c}_+(s) - \mathfrak{c}_-(s) \longrightarrow \frac{\pi}{\sqrt{\nu}}, \qquad \Lambda_s := \mathrm{e}^{\mathfrak{c}_+(s) - \mathfrak{c}_-(s)} \longrightarrow \Lambda_{\mathrm{b}}.$$

They are independent of A*.*

Proof. Here D_σ, $\mathcal{A}_\sigma$, L_σ, $\mathfrak{c}_\sigma$ and the reference radius v_b are those of Section 9.2 with $r = s$ and $b = 1$; thus $D_\sigma(t) = as^2 - \sigma st + t^2 - \sigma t^3 R_q(\sigma st)$. The fixed positive margin ν gives $D_\sigma(t) \geq c_\nu(s^2 + t^2)$ for $0 \leq t \leq v_b$. The same bound holds after replacing R_q by -1. Since $R_q(\sigma st) = -1 + O(s)$, this replacement changes each action by at most

$$Cs \int_0^{v_b} \frac{t^4}{(s^2 + t^2)^2}\, \mathrm{d}t = O(s).$$

We separate the remaining integral at a fixed $0 < \varepsilon < v_b$ to distinguish the action accumulated near the central section from the outer reference terms. On $[0, \varepsilon]$, deleting the remaining cubic term changes each action by at most $C\varepsilon$. With $t = sx$, the difference of the resulting inner actions tends to

$$\begin{aligned} c_\infty &= \int_0^\infty x \left(\frac{1}{x^2 - x + a} - \frac{1}{x^2 + x + a} \right) \mathrm{d}x \\ &= \int_0^\infty \frac{2x^2}{x^4 + (2a-1)x^2 + a^2}\, \mathrm{d}x = \frac{\pi}{\sqrt{4a - 1}}. \end{aligned} \tag{12.3}$$

For the last equality, substitute $x = \sqrt{a}\, u$, use $u \mapsto 1/u$ to equate the integrals with numerators u^2 and 1, and apply $y = u - 1/u$ to their sum. Both ends of the integral converge.

On $[\varepsilon, v_b]$, the action difference converges to

$$(L_+ - L_-)(v_b) - (L_+ - L_-)(\varepsilon).$$

Subtract the reference terms in $\mathfrak{c}_+ - \mathfrak{c}_-$, first let $s \to 0$, and then let $\varepsilon \to 0$. The remaining outer term vanishes because $(L_+ - L_-)(\varepsilon) = \log((1-\varepsilon)/(1+\varepsilon)) \to 0$. This proves the result. □

12.2. **Half-orbits with a prescribed exit.** The action comparison must be realized by half-orbits with a common central height. We first construct each half-orbit backward from its exit; monotonicity of the resulting height then gives the matching. Fix $J_+ = [\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}] \Subset (0,\infty)$ and $J_- \Subset (0,1)$ whose interior contains $f_{\Lambda_{\mathrm{b}}}(J_+)$. For a terminal radius $x_e \in J_\sigma$, prescribe $\eta = \sigma s x_e$, $z = s^3\delta_{\mathrm{an}}x_e^2$ and integrate toward the central section. In the local variables $t = |\eta|/s$ and $y = z/s^3$, the exact radial equation is

$$y_t = \frac{ty}{s\{D_\sigma(t) + B(\sigma st, s^3 y, p_s(A))y\}}, \qquad y(x_e) = \delta_{\mathrm{an}}x_e^2.$$

The quadratic margin and the upper end of J_- give $c(s^2+t^2) \le D_\sigma(t) \le C(s^2+t^2)$ on these fixed intervals. The backward solution remains positive and bounded, and reaches $t = 0$ with $h_\sigma(x_e) = s^3 y(0) > 0$. Its radius and height remain in the normal-rate box for small s, and $q - zB < 0$ identifies it with the corresponding half-orbit of the original field.

The correction to its boundary action is exactly

$$\mathscr{E}_\sigma(x_e) = \int_0^{x_e} \frac{tBy}{sD_\sigma(D_\sigma + By)}\,dt = \int_{y(0)}^{y(x_e)} \frac{B}{D_\sigma}\,dy \ge 0,$$

with the coefficients evaluated along the half-orbit. Choose a fixed $\kappa > 0$ with $2\kappa < \inf(J_+ \cup J_-)$. The logarithmic growth rate on $[\kappa, 2\kappa]$ is at least c/s, so $y(\kappa) \le C\mathrm{e}^{-c/s}$. The action correction up to κ is at most $Cs^{-2}y(\kappa)$; thereafter it is at most $Cy(x_e)$, since $D_\sigma \ge c\kappa^2$. Thus $\mathscr{E}_\sigma$ is uniformly bounded and

$$s\log(h_\sigma(x_e)/s^3) = -\mathcal{A}_\sigma(x_e) + sM^0_\sigma(x_e), \qquad M^0_\sigma = \log(\delta_{\mathrm{an}}x_e^2) + \mathscr{E}_\sigma, \qquad \|M^0_\sigma\|_{C^0} \le C. \tag{12.4}$$

All constants may depend on $J_\pm, C_0$ and the fixed A-interval; δ_{an} is held fixed before reducing s.

At $t = K_0 s$ the normalized height y/t^2 is exponentially small. At any later crossing of $y/t^2 = \delta_{\mathrm{an}}$,

$$\frac{d\log(y/t^2)}{d\log t} = \frac{t^2}{s(D_\sigma + B\delta_{\mathrm{an}}t^2)} - 2 > 0$$

for small s, so the prescribed point is the first upward crossing after $t = K_0 s$. The central height decreases strictly with its exit radius: the radial solution has positive derivative in its initial height, whereas its terminal radial derivative exceeds the derivative of $\delta_{\mathrm{an}}x_e^2$ by the same inequality.

On $J_\pm$, subtraction of the action integrands from their limits gives $\mathcal{A}_\sigma = \mathfrak{c}_\sigma + L_\sigma + O(s)$. The strict endpoint margins of J_-, Lemma 12.1, and (12.4) therefore bracket $h_+(\rho)$ between the two negative endpoint heights. Continuity and strict decrease give a unique negative exit ω with the same central height. Equating their actions and using the positive lower bound on L'_- gives

$$\omega = f_{\Lambda_s}(\rho) + O_{C^0}(s), \qquad f_\Lambda(\rho) = \frac{\Lambda\rho}{1 + (1+\Lambda)\rho}. \tag{12.5}$$

Uniqueness at the regular central point makes these two half-orbits parts of one orbit. The continuation below places their endpoints in the fixed windows.

12.3. **A transverse organizing splitting.** Write $p_0(A) = (A, C_0, 0, 0, 0)$. The tuning argument requires a strict sign for the derivative of the organizing splitting. We first construct the two boundary graphs and their dependence on A. The variational equation along the parabola gives the sign of their splitting in an affine coordinate; the section change has positive derivative and preserves this sign in the endpoint coordinates.

Lemma 12.2 (Transversality of the organizing splitting)**.** *For A close to 1, the graph equation (2.21) at $p_0(A)$ has unique solutions*

$$U_\sigma^0(z;A) = z + O(z^{3/2}), \qquad \sigma = \pm,$$

which depend C^1 on A. Set

$$b_\sigma^0(A) = z_* - U_\sigma^0(z_*;A), \qquad \mathcal{S}_0(A) = \Pi_{p_0(A)}(b_-^0(A)) - b_+^0(A).$$

Then $\mathcal{S}_0(1) = 0$ and $\mathcal{S}_0'(1) < 0$.

Proof. Boundary graphs and their variations. At $\mu = 0$, both q and $\eta = v + v^2 + C_0 vz$ are independent of A. Differentiating the exact projective field gives

$$B_A = -v(1 + 2v + C_0 z), \qquad B(0, z, p_0(A)) = 1 + C_0 z. \tag{12.6}$$

On $U \asymp z$, uniformly for A near 1, the graph right side $\mathcal{G}_{\sigma,p_0(A)}$ therefore satisfies

$$\mathcal{G}_{\sigma,p_0(A)} = 1 + O(\sqrt{z}), \qquad |\mathcal{G}_{\sigma,U}| \le Cz^{-1/2}, \qquad |\mathcal{G}_{\sigma,A}| \le C\sqrt{z},$$

where the middle bound is (4.6) at $\mu = 0$. The Volterra operator

$$\mathcal{T}_A U(z) = \int_0^z \mathcal{G}_{\sigma,p_0(A)}(U(t),t)\,\mathrm{d}t$$

maps $|U(z)/z - 1| \le 1/2$ into itself. In the norm $\|U\|_* = \sup_{0<z\le z_*} |U(z)|/z$, its Lipschitz constant is at most $C\sqrt{z_*} < 1$. The operator is C^1 in U, A on a slightly wider positive cone, so differentiation of the convergent iteration gives the asserted C^1 fixed point. Its parameter derivative satisfies

$$\begin{gathered}(\partial_A U_\sigma^0)' = \mathcal{G}_U \partial_A U_\sigma^0 + B_A, \qquad \partial_A U_\sigma^0(0) = 0,\\ \partial_A U_\sigma^0(z) = O(z^{3/2}).\end{gathered} \tag{12.7}$$

Comparison along the organizing parabola. For the entire $p_0(A)$-family, the function H_0 of (2.1) satisfies the exact identity

$$\dot H_0 = 2xH_0 - (A-1)x^2. \tag{12.8}$$

Continue the left boundary graph forward from the negative section, and the right boundary graph backward from the positive section, to the same point x on the organizing parabola. Denote their A-derivatives at $A = 1$, measured in the H_0-coordinate at fixed x, by $\mathfrak{k}_\mathrm{L}$ and $\mathfrak{k}_\mathrm{R}$, respectively. Their difference is the signed variation of the splitting at that point. Changing from fixed-x to fixed-z sections adds a tangential variation along the organizing parabola, whose contribution to the variation of H_0 at $A = 1$ is zero.

Use x as time, as in the proof of Lemma 2.5. The last term of (12.8) gives the forcing in the variational equation. With $P_\mathrm{aff}(x) = x^2 + 2x + C_0 > 0$, both variations satisfy

$$\mathfrak{k}' = \frac{4x}{P_\mathrm{aff}(x)}\mathfrak{k} - \frac{2x^2}{P_\mathrm{aff}(x)}.$$

The boundary conditions follow from (12.7): at fixed z,

$$\partial_A v = \frac{\sigma \partial_A U_\sigma^0}{\sqrt{2U_\sigma^0}(1 + 2v + C_0 z)} = O(z), \qquad \partial_A H_0 = -\frac{v\,\partial_A v}{z^2} = O(z^{-1/2}) = O(|x|).$$

With

$$W(x) = \exp\left(-\int_0^x \frac{4t}{P_\mathrm{aff}(t)}\,\mathrm{d}t\right) > 0,$$

direct integration gives $W(x) \asymp |x|^{-4}$ at both ends. Hence $W\mathfrak{k}_\mathrm{L} \to 0$ as $x \to -\infty$ and $W\mathfrak{k}_\mathrm{R} \to 0$ as $x \to +\infty$. Variation of constants with these two endpoint conditions yields

$$\mathfrak{k}_\mathrm{L}(x) - \mathfrak{k}_\mathrm{R}(x) = -W(x)^{-1}\int_{-\infty}^{\infty} \frac{2W(t)t^2}{P_\mathrm{aff}(t)}\,\mathrm{d}t < 0. \tag{12.9}$$

The integral is absolutely convergent and strictly positive.

The splitting in endpoint coordinates. To relate this variation to the displacement, let $\chi_\pm$ be the H_0-coordinates on the two fixed-z sections. The section changes are regular, so $\chi'_\pm(I^0_\pm) > 0$, and the chain rule at the positive section gives

$$\mathcal{S}'_0(1) = \frac{1}{\chi'_+(I^0_+)}\big(\mathfrak{k}_{\mathrm{L}}(x_+) - \mathfrak{k}_{\mathrm{R}}(x_+)\big), \qquad \chi'_+(I^0_+) > 0. \tag{12.10}$$

Here x_+ is the organizing point on the positive section, where both variations are evaluated. The finite exterior propagation is included in $\mathfrak{k}_{\mathrm{L}}(x_+)$, and $1/\chi'_+(I^0_+)$ is the positive H_0-to-I endpoint conversion factor computed in Lemma 2.5. Thus (12.9) and (12.10) give $\mathcal{S}'_0(1) < 0$. □

12.4. **Tuning the splitting and obtaining three cycles.**

Proof of Proposition 1.2. Endpoint preparation. We first continue the half-orbits from the compact exit intervals to the fixed sections, where their endpoint maps can be composed with the exterior transition. Use Lemma 4.2 with $\ell = \delta = s$ and $\zeta_0(\rho_\sigma) = \delta_{\mathrm{an}}\rho_\sigma^2$, as in (9.42). On the path (12.1), the coefficient ratios are bounded, and the fixed compact exit intervals have positive drift margins, uniformly for A near 1. Choose the large height $Z_1 = Z_0$ of that lemma uniformly in A, and put $t_s = s^2 Z_0$. The continued and reference energies lie in the common cone. Let U^{ref}_σ be the tail with initial energy $U^{\mathrm{ref}}_\sigma(t_s) = B_0(p)t_s$, and define

$$\begin{aligned} b^e_\sigma(p) &= z_* - U^{\mathrm{ref}}_\sigma(z_*), \\ a^e_\sigma(p) &= \partial_u U_\sigma(z_*; t_s, u, p)|_{u=B_0(p)t_s} > 0. \end{aligned} \tag{12.11}$$

The reference endpoint and the tail-energy derivative are smooth in A for fixed $s > 0$. They are the lemma's b_σ and $\varpi_{t_s} A_\sigma$, respectively. Since $\varpi_{t_s} = 1 + O(s(1 + |\log s|))$, the coefficients a^e_σ have uniform positive bounds, and (4.11) gives

$$I_\sigma(\rho_\sigma; p) = b^e_\sigma - \tfrac{1}{2}s^2 a^e_\sigma \rho_\sigma^2 + s^2 O_{C^3}(s(1 + |\log s|)). \tag{12.12}$$

Domains of the exterior transition. To place the reference and continued endpoints in the exterior windows for small s, uniformly on a fixed small A-interval, compare their tails to $U^0_\sigma(z; A)$. On the common cone, the difference of the graph right sides at $p_s(A)$ and $p_0(A)$ is bounded by

$$C(s + s^5/z + s^3/\sqrt{z}),$$

whereas, by (4.6), a Lipschitz coefficient for the difference is

$$2s/z + C(z^{-1/2} + s^3 z^{-3/2}). \tag{12.13}$$

Its integral over $[t_s, z_*]$ is uniformly bounded. The initial energy difference is $O(s^2)$, and the integrated forcing is $O(s)$. Gronwall's inequality gives, uniformly in A and the fixed exit ranges,

$$b^e_\sigma(A, s) \longrightarrow b^0_\sigma(A), \qquad I_\sigma(\rho_\sigma; A, s) \longrightarrow b^0_\sigma(A). \tag{12.14}$$

First restrict A so that $b^0_\sigma(A) \in \operatorname{int} W_\sigma$. Uniform convergence then puts, for small s, all the endpoints in (12.14), including those at the wider comparison exits, in $\operatorname{int} W_\sigma$. The continued half-orbits thus give admissible central heights in $\mathcal{H}_p$. All these endpoints also lie in the enlarged windows, where the exterior map and its derivatives are defined for the displacement expansion and splitting comparison below.

The normalized displacement. We separate the constant term, which will be adjusted by A, from the coefficient of the compact exit profile. Define the continuous splitting and slope coefficient

$$\mathcal{S}_s(A) = \Pi_{p_s(A)}(b^e_-) - b^e_+, \qquad \lambda_s(A) = \Pi'_{p_s(A)}(b^e_-)\frac{a^e_-}{a^e_+}. \tag{12.15}$$

By (12.5) and (12.12), the prepared displacement (9.43) holds with $r = s$, $\Lambda = \Lambda_s$, $\lambda(p) = \lambda_s(A)$ and $D(p) = 2\mathcal{S}_s(A)/(s^2 a_+^e)$:

$$\frac{2\Delta_{p_s(A)}(h(\rho; p_s(A)))}{s^2 a_+^e} = \frac{2\mathcal{S}_s(A)}{s^2 a_+^e} + g_{\lambda_s(A),\Lambda_s}(\rho) + O_{C^0}(s(1 + |\log s|)), \tag{12.16}$$

where

$$g_{\lambda,\Lambda}(\rho) = \rho^2 - \lambda f_\Lambda(\rho)^2 = \rho^2 - \frac{\lambda\Lambda^2\rho^2}{[1 + (1+\Lambda)\rho]^2}.$$

The three test radii must work throughout the tuning interval in A. We therefore bound the slope coefficient uniformly first. The common factor ϖ_{t_s} cancels from the quotient, so (4.4) gives $\log(a_-^e/a_+^e) = O(\sqrt{z_*}) + o_s(1)$. Also, $\Pi_0'(I_-^0) = \Lambda_0\{1 + O(\sqrt{z_*})\}$ by Lemma 2.5. As in Section 4, the coefficient of $\sqrt{z_*}$ depends only on the common cone; constants depending on the fixed exit ranges enter only the $o_s(1)$ term. Thus we may first choose the fixed section, then restrict A and s, so that

$$\lambda_{\mathrm{lo}} := \Lambda_0/2 \le \lambda_s(A) \le \lambda_{\mathrm{hi}} := (1 + \Lambda_0)/2 < 1. \tag{12.17}$$

Three test radii. The multiplier margin makes the model negative at an interior minimum; a small positive additive constant will give the two outer positive signs. To choose radii with fixed sign margins, put

$$M_{\mathrm{b}} = \sqrt{\lambda_{\mathrm{lo}}}\Lambda_{\mathrm{b}} > 1, \qquad K_{\mathrm{b}} = 1 + \Lambda_{\mathrm{b}}.$$

The strict negative minimum of $g_{\lambda_{\mathrm{lo}},\Lambda_{\mathrm{b}}}$ occurs at

$$\rho_{\mathrm{b}} = \frac{M_{\mathrm{b}}^{2/3} - 1}{K_{\mathrm{b}}}, \qquad g_{\lambda_{\mathrm{lo}},\Lambda_{\mathrm{b}}}(\rho_{\mathrm{b}}) = -\frac{(M_{\mathrm{b}}^{2/3} - 1)^3}{K_{\mathrm{b}}^2} < 0. \tag{12.18}$$

This follows directly from

$$g'_{\lambda,\Lambda}(\rho) = 2\rho\left(1 - \frac{\lambda\Lambda^2}{[1 + (1+\Lambda)\rho]^3}\right).$$

Choose $0 < D_* < -g_{\lambda_{\mathrm{lo}},\Lambda_{\mathrm{b}}}(\rho_{\mathrm{b}})/4$. Since g decreases in λ, $D_* + g_{\lambda,\Lambda_{\mathrm{b}}}(\rho_{\mathrm{b}}) < -3D_*$ throughout (12.17). Take $0 < \rho_{\mathrm{a}} < \rho_{\mathrm{b}}$ so small that

$$\sup_{\lambda_{\mathrm{lo}} \le \lambda \le \lambda_{\mathrm{hi}}} |g_{\lambda,\Lambda_{\mathrm{b}}}(\rho_{\mathrm{a}})| < D_*/4,$$

and then take $\rho_{\mathrm{c}} > \rho_{\mathrm{b}}$ with $\rho_{\mathrm{c}}^2 > \lambda_{\mathrm{hi}} + 1$. Since $0 < f_\Lambda < 1$, the last choice gives $g_{\lambda,\Lambda}(\rho_{\mathrm{c}}) > 1$. By Lemma 12.1, these choices imply, uniformly in A, for small s,

$$\begin{aligned} D_* + g_{\lambda_s(A),\Lambda_s}(\rho_{\mathrm{a}}) &> D_*/2, \\ D_* + g_{\lambda_s(A),\Lambda_s}(\rho_{\mathrm{b}}) &< -2D_*, \\ D_* + g_{\lambda_s(A),\Lambda_s}(\rho_{\mathrm{c}}) &> 1 + D_*. \end{aligned} \tag{12.19}$$

All three radii and their negative comparison intervals are fixed before the final reduction of s. Apply the compact construction above with J_+ containing $[\rho_{\mathrm{a}}, \rho_{\mathrm{c}}]$ and $f_{\Lambda_{\mathrm{b}}}(J_+) \subset \operatorname{int} J_-$. Figure 5 shows the model profile and the three radii for $C_0 = 65$.

Realizing the additive constant. The choices above fix the desired normalized constant D_*. To realize it, we apply the intermediate value theorem to the unscaled splitting. By (12.14),

$$\mathcal{S}_s(A) \longrightarrow \mathcal{S}_0(A)$$

uniformly on a fixed small A-interval. By Lemma 12.2, choose its endpoints $A_{\mathrm{lo}} < 1 < A_{\mathrm{hi}}$ so that $\mathcal{S}_0$ has opposite signs there and its only zero in the interval is 1. The continuous function

$$\Phi_s(A) = \mathcal{S}_s(A) - \tfrac{1}{2}s^2 a_+^e(A, s)D_*$$

converges uniformly to $\mathcal{S}_0$. The intermediate value theorem therefore gives $A_s \in (A_{\mathrm{lo}}, A_{\mathrm{hi}})$ with

$$\mathcal{S}_s(A_s) = \tfrac{1}{2}s^2 a_+^e(A_s, s)D_*. \tag{12.20}$$

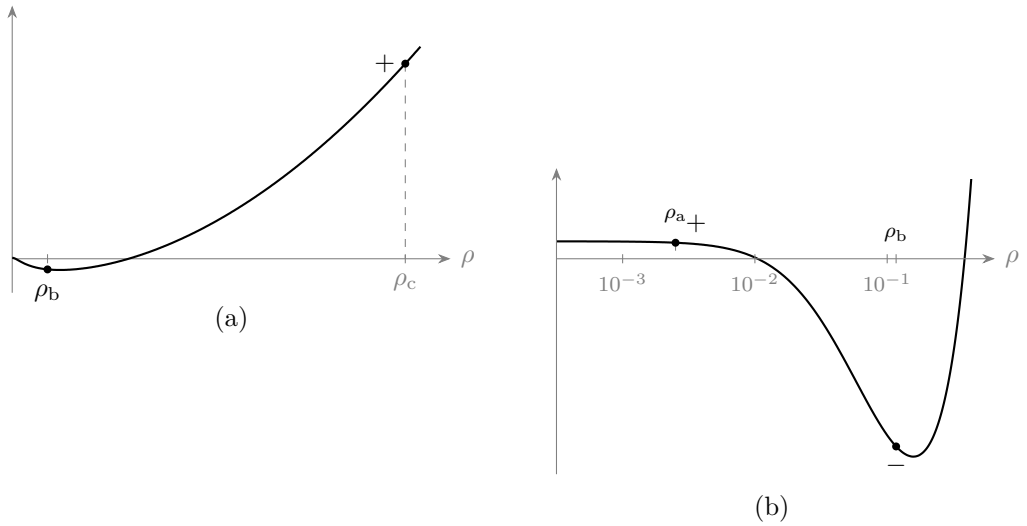


FIGURE 5. The model profile $D_* + g_{\lambda,\Lambda}(\rho)$ of (12.16) for $C_0 = 65$, computed from its formula with $\lambda = \Lambda_0 = \mathrm{e}^{-\pi/2}$ and $\Lambda = \Lambda_b = \mathrm{e}^{\pi}$. The radii $\rho_a < \rho_b < \rho_c$ and the constant D_* are chosen as in the proof. (a) Linear scale in ρ. (b) Logarithmic scale in ρ near ρ_a and ρ_b, with the vertical scale enlarged. The two model zeros on the compact exit interval lie between the three marked radii. The fourth sign at a larger central height in (12.21) gives the third displacement zero.

Every accumulation point of any such choices is a zero of $\mathcal{S}_0$, so $A_s \to 1$. Put $p_s = p_s(A_s)$.

Three isolated zeros. Let $h_s(\rho)$ be the common central height of (12.4) at p_s. By (12.16), (12.19), and (12.20), with the remainder made smaller than the fixed margins, Δ_{p_s} is positive at $h_s(\rho_a)$ and $h_s(\rho_c)$, and negative at $h_s(\rho_b)$. On the fixed radius interval, h_s is strictly decreasing by the radial construction above. Moreover $\mathcal{A}_+(\rho) \geq c\log(1/s) - C$, so these three central heights tend to zero.

The three test radii give two zeros. For a third zero, we need a negative displacement at a larger central height. We obtain it from the organizing contraction at a fixed height. At the organizing parameter, Proposition 2.9 gives $\Delta_0'(h) < 0$, while $\Delta_0(h) \to 0$ as $h \downarrow 0$. Choose a fixed small $\bar{h} > 0$ whose endpoints lie in $\operatorname{int} W_\pm$. Then $\Delta_0 < 0$ on $(0, \bar{h}]$. For each fixed $0 < \widetilde{h} < \bar{h}$, regular dependence on the parameters at positive central heights (Lemma 2.4, applied to the field extended by $\dot{p} = 0$) gives $\Delta_{p_s} \to \Delta_0$ uniformly on $[\widetilde{h}, \bar{h}]$. Hence $\Delta_{p_s} < 0$ on $[\widetilde{h}, \bar{h}]$ for small s; in particular $\Delta_{p_s}(\bar{h}) < 0$.

All four central heights belong to $\mathcal{H}_{p_s}$, by Proposition 2.7, and satisfy

$$h_s(\rho_c) < h_s(\rho_b) < h_s(\rho_a) < \bar{h}, \qquad \operatorname{sgn} \Delta_{p_s} = (+,-,+,-). \tag{12.21}$$

Since the admissible heights form an interval, there is a zero in each of the three intervening open intervals.

By Proposition 2.7, Δ_{p_s} is real analytic on $\operatorname{int} \mathcal{H}_{p_s}$, and the four signs in (12.21) exclude an identically zero displacement. The three selected zeros are therefore isolated and, by the correspondence in that proposition, give three distinct isolated periodic orbits.

Let their central heights be $h_{j,s}$, $j = 1, 2, 3$. The first two are trapped between test heights tending to zero. For the third, negativity of Δ_{p_s} on every fixed $[\widetilde{h}, \bar{h}]$ for small s excludes a positive limiting height. Thus all three heights tend to zero, including the zero between $h_s(\rho_a)$ and the fixed height $\bar{h}$.

Convergence of the orbit segments. To pass from vanishing central heights to convergence of the orbits, we compare their graphs near infinity and along the two organizing branches. Regular flow dependence then treats the finite connecting arc. The half-orbits of the smallest test central height

$h_{\mathrm{low},s} = h_s(\rho_c)$ have radius $O(s)$ in the thin passage, energy $O(s^2)$ at t_s, and thereafter lie in the common cone. Hence, for each fixed small $\zeta > 0$, their graphs satisfy

$$0 \le U_\sigma(z; h_{\mathrm{low},s}) \le C\zeta \quad (h_{\mathrm{low},s} \le z \le \zeta),$$

and converge uniformly to $U_\sigma^0(z;1)$ on $[\zeta, z_*]$, by (12.13). Every selected zero lies above $h_{\mathrm{low},s}$. The exact graph order (2.25) therefore bounds its energy above by the graph for $h_{\mathrm{low},s}$. Its portion with $z \le \zeta$ lies in a neighborhood shrinking to the point at infinity as $\zeta \to 0$.

To obtain convergence along both organizing branches away from infinity, fix $0 < \varepsilon < \zeta$. Once $h_{j,s} < \varepsilon$, the same order gives, for $\zeta \le z \le z_*$,

$$U_\sigma(z; \varepsilon, p_s) \le U_\sigma(z; h_{j,s}, p_s) \le U_\sigma(z; h_{\mathrm{low},s}, p_s).$$

By regular dependence, the left graph converges to the organizing-parameter graph with central height ε. As $\varepsilon \to 0$, that graph converges uniformly on $[\zeta, z_*]$ to $U_\sigma^0(z;1)$: at $z = K\varepsilon$, for fixed large K, both graphs lie in the common cone with energy difference $O(\varepsilon)$, and the integrable variation coefficient $O(z^{-1/2})$ propagates this difference to every fixed height. The two bounds give uniform convergence of both branches of each periodic orbit away from the point at infinity.

In particular, the endpoints converge to the organizing endpoints. The exterior passage lies in finitely many regular flow boxes, so endpoint convergence and regular dependence of the flow give uniform convergence of the entire finite connecting arc in the compactified phase space.

Hausdorff convergence to the whole graphic. We now express these segment estimates in the fixed metric on $\mathbb{S}^2$. Let $\mathscr{P}_p$ denote the compactification map from the parameter-dependent (η, z)-chart to $\mathbb{S}^2$, and let $q_\infty = \mathscr{P}_{p_0}(0,0)$ be the nilpotent point of the graphic. Uniform smoothness of these charts gives, on a smaller chart, a constant $C_{\mathbb{S}}$ such that

$$d_{\mathbb{S}}\big(\mathscr{P}_p(\eta, z), q_\infty\big) \le C_{\mathbb{S}}\big(|\eta| + |z| + |p - p_0|\big). \tag{12.22}$$

Because $\eta_\sigma = \sigma\sqrt{2U_\sigma}$, the portions of the three orbits with $0 \le z \le \zeta$ are at distance at most $C'_{\mathbb{S}}\sqrt{\zeta} + o_s(1)$ from q_∞, uniformly in s after the fixed ζ is chosen. The same estimate applies to the corresponding part of $\Gamma(C_0)$ (with $p = p_0$).

For the reverse approximation near infinity, the central section point $\mathscr{P}_{p_s}(0, h_{j,s})$ belongs to each periodic orbit and tends to q_∞. Thus every point of $\Gamma(C_0)$ with $0 \le z \le \zeta$ is within $C'_{\mathbb{S}}\sqrt{\zeta} + o_s(1)$ of that orbit. On the remaining portions, the graph convergence on $[\zeta, z_*]$ and the regular-flow-box convergence of the finite connecting arcs give both inclusions in the Hausdorff estimate. Therefore, for every fixed $\zeta > 0$,

$$\limsup_{s \downarrow 0} d_{\mathrm{H}}(\overline{\gamma_{j,s}}, \Gamma(C_0)) \le C'_{\mathbb{S}}\sqrt{\zeta}, \qquad j = 1, 2, 3.$$

Letting $\zeta \downarrow 0$ proves

$$d_{\mathrm{H}}(\overline{\gamma_{j,s}}, \Gamma(C_0)) \longrightarrow 0, \qquad j = 1, 2, 3.$$

Taking any sequence $s_n \downarrow 0$ completes the proof. □

Proof of Corollary 1.3. By Theorem 1.1, $\mathrm{Cycl}(\Gamma(C_0)) \le 3$. Let V be a neighborhood of $(1, C_0, 0, 0, 0)$ and let $\varepsilon > 0$. For large n, the parameter p_n of Proposition 1.2 lies in V, and its three limit cycles $\gamma_{j,n}$ satisfy $d_{\mathrm{H}}(\overline{\gamma_{j,n}}, \Gamma(C_0)) < \varepsilon$. Hence $N(V, \varepsilon) \ge 3$ in (1.4), and $\mathrm{Cycl}(\Gamma(C_0)) \ge 3$. □

12.5. **Hyperbolicity and separation of central heights.**

Corollary 12.3 (Three hyperbolic limit cycles)**.** *For sufficiently small positive s, the three periodic orbits constructed above are hyperbolic. In increasing order of their central heights, they are attracting, repelling, and attracting. For each such s, all three persist, with these stabilities, on an open neighborhood of p_s in the original five-parameter space. Such open sets occur arbitrarily close to the organizing parameter, with all three orbits arbitrarily close to the whole graphic.*

Proof. We obtain simplicity from the zero bound for the logarithmic slope ratio: the two gaps between the three displacement zeros already require two slope zeros. By Proposition 9.5, which applies to p_s by (12.1), $\mathcal{L}$ has at most two distinct zeros on all of $\mathcal{H}_{p_s}$. By (3.1), where $I_{+,h} > 0$, $\Delta_{p_s,h}$ has the sign of $\mathrm{e}^{\mathcal{L}} - 1$. Write the three displacement zeros as $h_1 < h_2 < h_3$. Rolle's theorem gives a zero of $\mathcal{L}$ in each of (h_1, h_2) and (h_2, h_3). If $\Delta_{p_s,h}(h_i) = 0$ for any i, then h_i would be a third distinct zero of $\mathcal{L}$. Thus all three zeros are simple. The same argument excludes a fourth displacement zero.

The signs in (12.21) now imply

$$\Delta_{p_s,h}(h_1) < 0, \qquad \Delta_{p_s,h}(h_2) > 0, \qquad \Delta_{p_s,h}(h_3) < 0.$$

At each fixed point, differentiation of $\mathcal{P}_{p_s} = I_+^{-1} \circ \Pi_{p_s} \circ I_-$ gives

$$\mathcal{P}'_{p_s}(h_i) = \mathrm{e}^{\mathcal{L}(h_i)}. \tag{12.23}$$

The two negative signs give multipliers in $(0, 1)$, and the positive sign gives a multiplier greater than one. This proves hyperbolicity and the stated stabilities.

For fixed $s > 0$, regular dependence (Lemma 2.4, applied to the field extended by $\dot{p} = 0$) makes these finite transverse returns smooth in all five parameters, with central height as the section coordinate. Apply the implicit function theorem near the three simple fixed points and intersect the resulting parameter neighborhoods. This gives an open set $V_s \ni p_s$ on which the three distinct hyperbolic orbits persist. Their stability is unchanged after reducing V_s.

Given fixed small parameter and whole-graphic neighborhoods, first take s small by the lower-bound construction, then reduce V_s to retain the geometric margins. Within the neighborhood and the Hausdorff condition of Theorem 1.1, these are all the periodic orbits counted there. □

Corollary 12.4 (Separation of the third central height)**.** *For the parameters p_s constructed in the proof of Proposition 1.2, write the three central heights as $h_{1,s} < h_{2,s} < h_{3,s}$. Then*

$$\lim_{s \downarrow 0} s \log \frac{h_{3,s}}{h_s(\rho_{\mathrm{a}})} = +\infty. \tag{12.24}$$

The same limit holds with $h_s(\rho_{\mathrm{a}})$ replaced by $h_{1,s}$ or $h_{2,s}$.

Proof. We first place the third zero outside each fixed compact exit interval. Fix $0 < \rho_{\mathrm{lo}} < \rho_{\mathrm{a}}$. The central-height construction (12.4) extends to $[\rho_{\mathrm{lo}}, \rho_{\mathrm{c}}]$; it is strictly decreasing there and $h_s(\rho_{\mathrm{lo}}) \to 0$. Hence, for sufficiently small s,

$$h_s(\rho_{\mathrm{c}}) < h_s(\rho_{\mathrm{lo}}) < \bar{h}.$$

Both outer central heights belong to $\mathcal{H}_{p_s}$, as shown in the proof of Proposition 1.2. Since this is an interval, it contains the image of $[\rho_{\mathrm{lo}}, \rho_{\mathrm{c}}]$ under h_s.

By (12.1), the compact-primary estimate in Section 9.2.3 applies to the chosen p_s on $[\rho_{\mathrm{lo}}, \rho_{\mathrm{c}}]$, after reducing s with ρ_{lo} fixed. It gives at most two zeros on the corresponding height interval. That interval contains $h_{1,s}$ and $h_{2,s}$, so

$$h_{3,s} > h_s(\rho_{\mathrm{lo}}).$$

The count includes endpoints; equivalently, one can first apply it on a slightly larger fixed compact interval.

We now quantify this separation. Subtracting the equations for the two central heights and using (9.39) gives

$$s \log \frac{h_s(\rho_{\mathrm{lo}})}{h_s(\rho_{\mathrm{a}})} = L_+(\rho_{\mathrm{a}}) - L_+(\rho_{\mathrm{lo}}) + O_{\rho_{\mathrm{lo}}}(s).$$

The common action constant cancels, and the terms sM_+^0 contribute $O_{\rho_{\mathrm{lo}}}(s)$. Given $K > 0$, first fix ρ_{lo} so that $L_+(\rho_{\mathrm{a}}) - L_+(\rho_{\mathrm{lo}}) > K + 1$, and then take s small. The preceding inequalities give $s \log(h_{3,s}/h_s(\rho_{\mathrm{a}})) > K$, proving (12.24). Finally, $h_{1,s}, h_{2,s} < h_s(\rho_{\mathrm{a}})$, which proves the last assertion. □

Remark 12.5 (A numerical example)**.** Take $C_0 = 65$, so that $\nu = 1$ and $a = 1/2$ in (12.1), and take $s = 0.05$ and $A = 1$, with μ given by (A.1). We computed the return map to the central section numerically, by an adaptive Runge–Kutta integration of the field in the chart (2.3), with $\log z$ as height variable, and of (1.1) between two crossings of a fixed section, with relative tolerance 10^{-11}.

The accompanying numerical supplement contains a standalone Python script, its fixed parameters and tolerances, execution instructions, and the reference table. It uses a Dormand–Prince 5(4) integrator, logarithmic-height bisection, and the same affine field between section crossings. For central heights with $-120 \le \log h \le -3.5$, the computation finds three fixed points:

j	$\log h_j$	multiplier
1	-93.45	0.095
2	-34.94	9.25
3	-11.39	0.289

Here the multiplier is the derivative of the return map at the fixed point. In increasing central height the orbits are attracting, repelling and attracting, as in Corollary 12.3, and the largest central height is much larger than the other two; compare Corollary 12.4. The computation illustrates the construction at these moderate parameter values.

Appendix A. Coefficient scales

The parameter cover is expressed in boundary coefficients, so we need both a local coordinate change from μ_1, μ_2, μ_3 to q_0, q_1, q_2 and a comparison of their weighted scales. We compute the coordinate change first and then prove the scale bounds directly.

On $z = 0$, let v_{bd} be the small root of $\mu_3 + v_{\mathrm{bd}} + v_{\mathrm{bd}}^2 = 0$, and put $t = 1 + 2v_{\mathrm{bd}}$. Writing $v = v_{\mathrm{bd}} + u$ gives

$$\eta = tu + u^2, \qquad u = \eta/t - \eta^2/t^3 + O(\eta^3).$$

The boundary identity (2.10) reads

$$q(\eta) = (t + 2u)\{P(v_{\mathrm{bd}}) + P'(v_{\mathrm{bd}})u - 3v_{\mathrm{bd}}u^2 - u^3\}.$$

It follows that

$$q_0 = tP(v_{\mathrm{bd}}), \qquad q_1 = P'(v_{\mathrm{bd}}) + 2P(v_{\mathrm{bd}})/t, \qquad q_2 = -3v_{\mathrm{bd}}/t + q_1/t^2 - 4q_0/t^4. \tag{A.1}$$

Differentiating (A.1) at the organizing point gives the Jacobian and its inverse:

$$D_\mu(q_0, q_1, q_2)\big|_0 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 1 & 3 \end{pmatrix}, \qquad D_q(\mu_1, \mu_2, \mu_3)\big|_0 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4/3 & -1/3 & 1/3 \end{pmatrix}.$$

Thus the coefficient change is locally invertible.

Set

$$H = \max\{|q_0|^{1/3}, \sqrt{|q_1|}, |q_2|\}, \qquad H_\mu = \max\{|\mu_1|^{1/3}, \sqrt{|\mu_2|}, |\mu_3|\}.$$

These scales are comparable on a fixed neighborhood. Indeed (A.1) gives $|v_{\mathrm{bd}}| \le CH$ when $1/2 < t < 3/2$, and the inverse identities

$$\mu_2 = q_1 - 2q_0/t^2 + 3v_{\mathrm{bd}}^2, \qquad \mu_1 = q_0/t - \mu_2 v_{\mathrm{bd}} + v_{\mathrm{bd}}^3, \qquad \mu_3 = -v_{\mathrm{bd}} - v_{\mathrm{bd}}^2$$

give $H_\mu \le CH$. Conversely $v_{\mathrm{bd}} = O(H_\mu)$, $P(v_{\mathrm{bd}}) = O(H_\mu^3)$, $P'(v_{\mathrm{bd}}) = O(H_\mu^2)$, and (A.1) imply $H \le CH_\mu$.

For $H>0$, put $\bar{q}_j = q_j/H^{3-j}$. Uniformly on this compact normalized coefficient set,

$$\frac{v_{\mathrm{bd}}}{H} = -\frac{\bar{q}_2}{3} + O(H), \qquad \frac{\mu_3}{H} = \frac{\bar{q}_2}{3} + O(H), \qquad \frac{\mu_2}{H^2} = \bar{q}_1 + \frac{\bar{q}_2^2}{3} + O(H),$$

$$\frac{\mu_1}{H^3} = \bar{q}_0 + \frac{\bar{q}_1\bar{q}_2}{3} + \frac{2\bar{q}_2^3}{27} + O(H), \qquad H^{-3}q(HX) = \bar{q}_0 + \bar{q}_1 X + \bar{q}_2 X^2 - X^3 + O_{C^k}(H).$$

The last estimate, for every fixed k on every fixed X-interval, follows from analyticity and the fact that the boundary drift depends only on (μ_1,μ_2,μ_3); A, C remain independent parameters in the positive-height coefficient B.

Appendix B. Geometric capture and saddle preparation

B.1. **The isolating block and whole-graphic capture.** We prove Proposition 2.1 in two steps. An outward transverse boundary around the finite equilibrium gives the isolating block. Near the graphic, flow boxes force crossings of local section segments, and intersection numbers in a tubular annulus show that each section is crossed exactly once.

At the finite equilibrium, the linear part of the organizing field $\mathcal{X}_0$ is

$$\mathsf{L}_0 = \begin{pmatrix} 1 & -1 \\ C_0 & 0 \end{pmatrix}.$$

The symmetric matrix

$$\mathsf{Q}_0 = \begin{pmatrix} (C_0+1)/2 & -1/2 \\ -1/2 & (C_0+2)/(2C_0) \end{pmatrix}$$

is positive definite and satisfies $\mathsf{L}_0^T\mathsf{Q}_0 + \mathsf{Q}_0\mathsf{L}_0 = \mathsf{I}$. Hence, for $\mathsf{Q}(u) = u^T\mathsf{Q}_0 u$,

$$\dot{\mathsf{Q}}(u) = |u|^2 + O(|u|^3),$$

where the dot denotes the derivative along $\mathcal{X}_0$. Choose a sufficiently small disk $B_{\mathrm{eq}} = \{\mathsf{Q} \le \rho_{\mathrm{eq}}^2\}$, disjoint from $\Gamma(C_0)$, for which the vector field points strictly out of B_{eq}. Then $B_{\mathrm{iso}} = \mathbb{S}^2 \setminus \operatorname{int} B_{\mathrm{eq}}$ has a boundary along which the field points strictly inward. This property persists in a full parameter neighborhood and proves (i).

The periodic orbits of (1.1) lie in the open upper hemisphere of $\mathbb{S}^2$. The graphic $\Gamma(C_0)$ is a smooth simple closed curve in $\mathbb{S}^2$. Away from the nilpotent point it is the parabola (1.3). In the chart (2.3), whose coordinate z takes both signs on the sphere, the part of the parabola with $y>0$ is the graph

$$z = \frac{\sqrt{1+C_0v^2}-1}{C_0} = \frac{v^2}{2} + O(v^4), \qquad v \neq 0,$$

and its closure passes smoothly through the nilpotent point $(v,z)=(0,0)$. Hence $\Gamma(C_0)$ has closed tubular neighborhoods in $\operatorname{int} B_{\mathrm{iso}}$; each of them is a compact annulus, since $\mathbb{S}^2$ is orientable.

Take Σ_0 on the affine orbit and the two projective faces $\Sigma_{\mathrm{in}}, \Sigma_{\mathrm{out}}$ on its regular branches. Choose a closed tubular annulus $B_{\mathrm{cap}} \Subset \operatorname{int} B_{\mathrm{iso}}$ so thin that

$$S_j = \Sigma_j \cap B_{\mathrm{cap}}, \qquad j \in \{0, \mathrm{in}, \mathrm{out}\},$$

are disjoint proper spanning arcs in the cyclic order of $\Gamma(C_0)$. Choose a smaller tube $\overline{\mathcal{N}'} \Subset B_{\mathrm{cap}}$ for which $T_j = S_j \cap \overline{\mathcal{N}'}$ are compact transverse cores. Around the marked points $\xi_j = \Gamma(C_0)\cap S_j$, choose nested, pairwise disjoint flow boxes $F_j' \Subset F_j \Subset B_{\mathrm{cap}}$, with $T_j \subset F_j$, such that every trajectory through F_j' meets T_j before leaving F_j, in positive or negative time. All crossings of each T_j have the organizing orientation. These properties persist for nearby parameters by transversality and smooth dependence of the flow boxes.

For sufficiently small $\delta>0$,

$$N_\delta(\Gamma(C_0)) \subset \mathcal{N}', \qquad N_\delta(\xi_j) \subset F_j',$$

where N_δ denotes the metric neighborhood. If $d_{\mathrm{H}}(\overline{\gamma}, \Gamma(C_0)) < \delta$, one Hausdorff inclusion puts γ in $\mathcal{N}'$, and the other supplies a point in each F_j'. Hence γ crosses every T_j at least once, and all its intersections with S_j lie in that core and have the same sign.

Let $\pi : B_{\mathrm{cap}} \to \Gamma(C_0)$ be the tubular projection. The algebraic intersection with a spanning arc is $I(\gamma, S_j) = \deg(\pi|_\gamma)$, up to orientation, and is nonzero. Thus γ, an embedded circle when taken with minimal period, is essential in the annulus. Such a circle represents a primitive generator of $H_1(B_{\mathrm{cap}}; \mathbb{Z})$; its intersection number with each spanning arc has absolute value one. Since all crossings have the same sign, there is exactly one per arc. Disjoint spanning arcs are met in their cyclic order, and the flow fixes that order to be the direction of $\Gamma(C_0)$. This proves (ii).

B.2. **Uniform preparation of the interior saddle.** For (5.23), we first straighten the stable and unstable graphs with uniform finite-order bounds, then normalize the equations on the two axes. The parameter-dependent stable manifold theorem [13, Theorem 9.6] supplies the graphs through the following amplitude comparison.

Fix a finite order k, with extra derivatives for the coordinate changes below. In the scaled parameters $\widehat{p} = (A, C, \kappa, q_2, \ell)$, where $q_0 = \ell^2$ and $q_1 = \kappa\ell$, the field extends smoothly to a neighborhood of a compact parameter box including $\ell = 0$. Factoring $B(\ell X, \ell^2 S, p) - B(0, 0, p)$ by ℓ writes it as

$$\mathcal{X}_{\widehat{p}} = \mathcal{X}^0_{B_0(p)} + \kappa\mathcal{X}_1 + q_2\mathcal{X}_2 + \ell\mathcal{X}_{3,\widehat{p}}, \qquad \mathcal{X}_1 = (X, 0), \quad \mathcal{X}_2 = (X^2, 0),$$

where $\mathcal{X}^0_{B_0}$ is (5.24) and $\mathcal{X}_{3,\widehat{p}}$ is smooth with uniformly bounded state derivatives on the fixed state box. Replace the displayed coefficients κ, q_2, ℓ by independent small amplitudes $\alpha = (\alpha_1, \alpha_2, \alpha_3)$, keeping $\widehat{p}$ fixed. At $\alpha = 0$, the comparison saddle has eigenvalues $1, -1$; reducing the amplitude box gives a common positive spectral margin. The cited theorem on a finite cover of the compact $(\widehat{p}, \alpha)$ set gives stable and unstable graphs on a common smaller state box, with bounded state derivatives and one additional amplitude derivative. The mean value theorem from $\alpha = 0$ to $\alpha = (\kappa, q_2, \ell)$ gives graph differences $O_{C^k}(\kappa + |q_2| + \ell)$. Smooth eigen-coordinates centered at each saddle have the same bounds; straightening the graphs gives

$$x' = xg_{\mathrm{u}}(x, y), \qquad y' = -yg_{\mathrm{s}}(x, y), \qquad g_{\mathrm{u}}, g_{\mathrm{s}} > c > 0.$$

An orbital time change makes $x' = x,\ y' = -yg(x, y)$, where $g(0, 0) = \nu_p$.

Normalize the $x = 0$ equation by the regular coordinate

$$\psi(y) = y \exp \int_0^y \left(\frac{\nu_p}{g(0, t)} - 1 \right) \frac{dt}{t}.$$

The new equation is $y' = -y[\nu_p + xc_1(x, y)]$. Multiplication of y by $\exp \int_0^x c_1(t, 0)\, dt$ then gives exactly the prepared equation (5.28).

Appendix C. Notation and terminology index

Table 5: Notation used in more than one section.

Symbol	Meaning
C_0; $A, C, \mu_1, \mu_2, \mu_3$; p, $\mathcal{X}_p$	Fixed organizing value; parameters of (1.1); the parameter point $p = (A, C, \mu_1, \mu_2, \mu_3)$ and its vector field.
$\Gamma(C_0)$, $\mathbb{S}^2$, d_{H}	The graphic I_2^1; the Poincaré sphere; Hausdorff distance on $\mathbb{S}^2$.
ε, V, $N(V, \varepsilon)$	Hausdorff radius, parameter neighborhood and the supremum over $p \in V$ of the number of limit cycles satisfying the Hausdorff condition in (1.4).
(v, z), P, η	Projective chart (2.3); $P(v) = \mu_1 + \mu_2 v - v^3$; normal-rate coordinate (2.7).
q, q_0, q_1, q_2, R_q; B, B_0, b_0, b_1, b_2	Boundary drift $q = q_0 + q_1\eta + q_2\eta^2 + \eta^3 R_q$; positive coefficient in (2.8), with $B_0 = B(0, 0, p)$, and the margins $B \geq b_0$, $B_\eta \geq b_1$, $3R_q + \eta\partial_\eta R_q \leq -b_2$ in (2.12).

Table 5: Notation used in more than one section (continued).

Symbol	Meaning
z_*, $W_\pm$, $\widetilde{W}_\pm$, Π_p, I	Height of the fixed section; capture windows and the enlarged windows of Lemma 2.5; exterior map; section coordinate $I = z_* - \eta^2/2$.
Λ_0, $\Lambda_\pm$, Λ_*	Limit of the organizing exterior multiplier as $z_* \downarrow 0$; bounds $\Lambda_- \le \Pi_p' \le \Lambda_+$ of Lemma 2.5; the value of Λ_+ fixed in Section 11.
h, $U_\sigma(z;h)$, η_σ, $\mathcal{G}_{\sigma,p}$	Central height; energy $\eta_\sigma^2/2$ of the half-orbit of sign σ; right side of (2.21).
I_σ, $\mathcal{H}_p$, Δ_p, $\mathcal{D}_p$	Endpoints; interval of admissible central heights; displacement; domain of the central return.
H, s, r, ρ_1	Coefficient scales of Table 2; $\rho_1 = \sqrt{q_1}$ is distinct from the annular exit radius ρ.
a, b, $\bar{c}$	Normalized constant, linear and quadratic coefficients of the drift in each chart.
$\mathcal{L}$, m_σ, M_σ	Logarithmic slope ratio (3.1); endpoint slopes $m_\sigma = hI_{\sigma,h}$ and, in the annulus, $M_\sigma = -\mathcal{E}I_\sigma$, where $\mathcal{E} = \rho\partial_\rho$ and $m_\sigma = sM_\sigma/G_\sigma$ with $G_\sigma = -s\mathcal{E}\log h_\sigma > 0$.
T, T_Q, c_T	Logarithmic height coordinate (3.2), with all parameter scales fixed; $T = -c_T\log(h/h_{\mathrm{ref}})$, $T_Q = T + s\log Q$ in the positive quadratic region.
$c_\pm$, C_{cone}, ϖ_t	Common cone and its first-variation constant; tail multiplier $\varpi_t(z) = (z/t)^{-2q_2}$ of Lemma 4.1.
c_σ^{lin}, $\mathfrak{c}_\sigma(p)$	Root-distance normalization constants in (8.6); boundary-action constants in (9.39).
δ, ϵ_3, F, K	Small parameter and drift in (4.12); outward boundary drift K, with $F = K/X$; $\epsilon_3 = \ell/\delta$ on the quadratic-prefix scales.
$\mathcal{W}$, w, χ	Boundary-flow coordinate, projected state and inverse projection of Lemma 4.3.
$\mathcal{U}$, Θ, Υ	Scaled energy; logarithmic height correction; positive prefix unit, with their scales and arguments specified in each construction.
ℓ, E, A_p, U_p	Radius scale, amplitude, amplitude constant and dimensionless positive unit factor in an endpoint derivative, Lemma 4.9; in Section 5.2, $\ell = \sqrt{q_0}$.
$\mathcal{A}$, $\mathfrak{H}$, Q_σ	Boundary action integral; logarithmic endpoint rate in a common logarithmic height coordinate; curvature coefficient $(wK_\sigma'' - K_\sigma')/w^2$.
$\widehat{h}$, $\widehat{h}_+$	Rescaled central height h/q_1^2 in Section 5.1; h/q_0 in Section 5.2.
κ, $a_\circ$, $\kappa_\circ$	Weak-root scale of (6.16); the constants of Proposition 6.9.
ϱ, ρ	Positive unscaled radius; normalized annular radius, with $\varrho = s\rho$ as a coordinate relation in the negative quadratic chart.
ϱ_0, ϱ^+; ρ_e, ϱ_e	Fixed coefficient in $u = \varrho_0/\varrho$; $\varrho^+ = s + \varrho$; normalized and unscaled adaptive exit radii, $\varrho_e = s\rho_e$. The annular exit has unscaled radius $s\rho$; at the same thin threshold, $\rho \le \rho_e$ in the buffered comparison of Section 9.2.
δ_{an}, ν, ϵ_{f}	Thin threshold of Section 9.1; margin of the buffered region (9.29); fold width in (10.1).
$\mathcal{Y}$, m_{land}	Height-to-radius ratio $\mathcal{Y} = z/\varrho^2$; fixed geometric landing level.
$a_0^{\mathrm{pr}}, a_0^+, a_0^-$, $b_{*,+}, b_{*,-}$, $\underline{b}$	Regional coefficient thresholds in the primary, positive-secondary and negative-secondary arguments; see Table 4.
κ_{p}; $a_{\mathrm{w}}, \kappa_{\mathrm{w}}$; $a_\pm, \kappa_\pm$; r_{d}, b_{cov}, A_0, m_{cov}; L	Fixed thresholds, margins and overlaps of the parameter cover; their choice order is given in Table 4.
Λ, Λ_s, λ_s	Boundary-action multiplier; its value on the path of Section 12; slope coefficient in the normalized displacement (12.16).
c, C	Positive constants whose values may change from line to line, with or without indices. The family parameter C appears only in (1.1), in p, in the formulas of Section 2, and as (A, C) or $C = C_0$.

Passage terminology. A central height is *admissible* if it belongs to $\mathcal{H}_p$; the associated half-orbits reach the windows $W_\pm$. A *captured* periodic orbit is one satisfying the Hausdorff condition of Proposition 2.1. Its central height is admissible by Proposition 2.7.

A *prefix* runs from the central point to a prescribed radius, and a *tail* from an incoming height to z_*. The *projected state* is the boundary-flow coordinate w. A half-orbit is *thin* while its stated normalized height is below the chosen threshold; its *thin exit* is the first outward crossing of that threshold. The subsequent *landing* is the first hit of $z = m_{\mathrm{land}}|\eta|^2$.

A *weak root* approaches the central line relative to the current coefficient scale. The *fold* is a double root of the positive secondary drift; a small positive minimum on its no-root side is the *bottleneck*. The *far root* is the negative branch's first primary root $X_* = 1+O(r+s)$ in Section 9.1.3.

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References

1. Peter De Maesschalck and Freddy Dumortier, *Canard cycles in the presence of slow dynamics with singularities*, Proceedings of the Royal Society of Edinburgh Section A: Mathematics **138** (2008), 265–299.
2. ———, *Detectable canard cycles with singular slow dynamics of any order at the turning point*, Discrete and Continuous Dynamical Systems **29** (2011), no. 1, 109–140.
3. Freddy Dumortier, Yulij Ilyashenko, and Christiane Rousseau, *Normal forms near a saddle-node and applications to finite cyclicity of graphics*, Ergodic Theory and Dynamical Systems **22** (2002), no. 3, 783–818.
4. Freddy Dumortier, Robert Roussarie, and Christiane Rousseau, *Hilbert's 16th problem for quadratic vector fields*, Journal of Differential Equations **110** (1994), no. 1, 86–133.
5. Freddy Dumortier and Christiane Rousseau, *Study of the cyclicity of some degenerate graphics inside quadratic systems*, Communications on Pure and Applied Analysis **8** (2009), no. 4, 1133–1157.
6. Maite Grau, Francesc Mañosas, and Jordi Villadelprat, *A Chebyshev criterion for Abelian integrals*, Transactions of the American Mathematical Society **363** (2011), no. 1, 109–129, Author version: arXiv:0805.1140v2.
7. Renato Huzak, *Cyclicity of degenerate graphic DF_{2a} of Dumortier–Roussarie–Rousseau program*, Communications on Pure and Applied Analysis **17** (2018), no. 3, 1305–1316.
8. Renato Huzak and Kristian Uldall Kristiansen, *On entry–exit formulas for degenerate turning point problems in planar slow–fast systems*, Nonlinearity **39** (2026), no. 8, 085026, Author version: arXiv:2510.02770v1.
9. David Marín and Jordi Villadelprat, *Asymptotic expansion of the Dulac map and time for unfoldings of hyperbolic saddles: coefficient properties*, Journal of Differential Equations **404** (2024), 43–107.
10. Robert Roussarie, *Bifurcations of planar vector fields and Hilbert's sixteenth problem*, Progress in Mathematics, vol. 164, Birkhäuser, Basel, 1998.
11. Robert Roussarie and Christiane Rousseau, *Finite cyclicity of some center graphics through a nilpotent point inside quadratic systems*, Transactions of the Moscow Mathematical Society **76** (2015), 181–218.
12. Christiane Rousseau, Chunhua Shan, and Huaiping Zhu, *Finite cyclicity of some graphics through a nilpotent point of saddle type inside quadratic systems*, Qualitative Theory of Dynamical Systems **15** (2016), 237–256.
13. Gerald Teschl, *Ordinary differential equations and dynamical systems*, Graduate Studies in Mathematics, vol. 140, American Mathematical Society, Providence, RI, 2012.

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